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ON THE CARTAN-HADAMARD CONJECTURE AND THE REGULARITY OF GENERALIZED SURFACES

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Uncertainty is the very condition to impel man to unfold his powers.
— Erich Fromm

RESUMO

Esta tese reúne os resultados de dois artigos desenvolvidos durante o doutorado do autor, dedicados às desigualdades isoperimétricas em variedades e à regularidade de varifolds. Esses trabalhos abordam questões distintas em análise geométrica e teoria geométrica da medida. O primeiro artigo [2] estabelece a conjectura generalizada de Cartan–Hadamard, também conhecida como conjectura de Aubin, para volumes suficientemente pequenos sob uma cota inferior para a curvatura de Ricci, em qualquer dimensão maior ou igual a dois. Também reduz a demonstração da conjectura geral à obtenção de rigidez no caso de igualdade. O segundo artigo [1] estabelece o teorema de ε-regularidade interior de Allard em espaços de Alexandrov não colapsantes com curvatura limitada superior e inferiormente e, em particular, em variedades de classe $\mathcal{C}^2$.

Palavras-chave: Teoria geométrica da medida; desigualdades isoperimétricas; conjectura de Cartan–Hadamard; conjectura de Aubin; teoria da regularidade; varifolds; espaços de Alexandrov.

ABSTRACT

This thesis brings together the results of two papers developed during the author's doctoral studies, concerning isoperimetric inequalities on manifolds and the regularity of varifolds. These works address distinct questions in geometric analysis and geometric measure theory. The first paper [2] establishes the generalized Cartan–Hadamard conjecture, also known as the Aubin conjecture, for sufficiently small volumes under a lower Ricci curvature bound, in every dimension greater than or equal to two. It also reduces the proof of the general conjecture to establishing rigidity in the equality case. The second paper [1] establishes Allard's interior ε-regularity theorem in non-collapsed Alexandrov spaces with curvature bounded above and below, and, in particular, in manifolds of class $\mathcal{C}^2$.

Keywords: Geometric measure theory; isoperimetric inequalities; Cartan–Hadamard conjecture; Aubin conjecture; regularity theory; varifolds; Alexandrov spaces.

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1 INTRODUCTION

1.1 AN OVERVIEW OF THE PAPERS IN THIS PHD THESIS

Part II and Section 1.2 address the generalized Cartan–Hadamard conjecture, also known as the Aubin conjecture. The main results establish the sharp model-space isoperimetric comparison for small volumes under a lower Ricci curvature bound and reduce the conjecture for arbitrary volumes to a rigidity statement for the equality case in a broader class. These results were obtained in collaboration with Márcio Fabiano da Silva, Stefano Nardulli, and Reinaldo Resende and are presented in the following paper:

> [2]: M. Agnoletto, M. F. da Silva, S. Nardulli, and R. Resende. *A proof of the Cartan–Hadamard conjecture for small volumes under a Ricci curvature lower bound*. arXiv:2609.17093, 2026.

Part III and Section 1.3 develop an intrinsic version of Allard's interior regularity theorem and extend it to non-collapsed Alexandrov spaces with two-sided curvature bounds. These results were obtained in collaboration with Júlio Cesar Correa Hoyos, Márcio Fabiano da Silva, and Stefano Nardulli and are presented in the following paper:

> [1]: M. Agnoletto, J. C. Correa Hoyos, M. F. da Silva, and S. Nardulli. *Allard's interior ε-regularity theorem in Alexandrov spaces*. arXiv:2504.10758, 2025.

There is a connection between the two papers: the intrinsic regularity theory developed in [1] provides the graphical descriptions used in [2], both in the small-volume argument and in the arbitrary volume regime; see Chapters 9 and 11.

1.2 THE GENERALIZED CARTAN–HADAMARD CONJECTURE

1.2.1 Historical background

The classical Euclidean isoperimetric inequality states that balls minimize perimeter among sets of prescribed volume. More precisely, every set $E \subset \mathbb{R}^n$ of finite perimeter and finite volume satisfies

$$P(E) \geq n\omega_n^{1/n}|E|^{(n-1)/n},$$

where $P(E)$ and $|E|$ denote the perimeter and volume of E, respectively, and ω_n is the volume of the Euclidean unit ball. The equality characterizes balls.

On a Riemannian manifold (M^n, g), the isoperimetric profile is defined for $0 < v < \mathrm{vol}_g(M)$ by

$$I_{(M,g)}(v) := \inf\{P_g(E) : E \subset M \text{ smooth, bounded and } \mathrm{vol}_g(E) = v\}, \tag{1}$$

where P_g denotes the Riemannian perimeter and vol_g the Riemannian volume. A set attaining this infimum is called an isoperimetric region. For instance, the hyperbolic paraboloid contains no isoperimetric region, so the infimum in (1) is never attained for this manifold; see Section 1.2.2 for further discussion.

Let $(\mathbb{M}^n_b, \delta^b)$ denote the n-dimensional simply connected model space form of constant sectional curvature $b \in \mathbb{R}$. This model is the Euclidean space when $b = 0$ and the hyperbolic space with curvature b when $b < 0$. We denote the Euclidean model by $(\mathbb{R}^n, \delta_{\mathrm{euc}})$ and write $I_{(\mathbb{R}^n,\delta_{euc})}$ and $I_{(\mathbb{M}^n_b,\delta^b)}$ for the Euclidean and hyperbolic isoperimetric profiles, respectively. The isoperimetric regions in these model spaces are geodesic balls, see [80] and [87].

A complete, simply connected Riemannian manifold with nonpositive sectional curvature is called a Cartan–Hadamard manifold. The Cartan–Hadamard conjecture asserts that every such manifold satisfies

$$I_{(M,g)}(v) \geq I_{(\mathbb{R}^n,\delta_{euc})}(v) \qquad \text{for every } v > 0.$$

More generally, if its sectional curvature satisfies $\mathrm{Sec}_g \leq \bar{k} \leq 0$, the generalized Cartan–Hadamard conjecture asserts that

$$I_{(M,g)}(v) \geq I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(v) \qquad \text{for every } v > 0.$$

In both conjectures, the associated rigidity assertion states that any region attaining equality in the corresponding comparison bound must be isometric to a geodesic ball in the model space.

Figure 1 summarizes the main milestones in the development of the Cartan–Hadamard conjecture and its generalized version, the recent developments of September 2026, and the contribution of this thesis. It highlights the dimension-specific breakthroughs.

The history begins with the two-dimensional problem. In 1926, Weil proved the Cartan–Hadamard conjecture in the case $n = 2$ [97, 23]. Beckenbach and Radó gave an independent proof in 1933 using the theory of subharmonic functions [21]. In 1941, Bol established the case $n = 2$ of the generalized Cartan–Hadamard conjecture [24].

Aubin formulated the conjecture in dimensions $n \geq 3$ in 1975 [18] and developed its relationship with Sobolev inequalities in 1976 [19]. The terminology “Aubin conjecture” reflects this analytical motivation. Related formulations subsequently appeared in the work of Gromov and in Burago and Zalgaller’s treatment of geometric inequalities [49, 26].

For many years, the higher-dimensional history was shaped by two proofs with markedly different mechanisms. In 1984, Croke [33], using an integral-geometric method, proved an Euclidean-type isoperimetric inequality in every dimension $n \geq 3$: $I_{(M^n,g)} \geq \gamma_n I_{(\mathbb{R}^n,\delta_{\mathrm{euc}})}$, where $\gamma_n \in (0, 1]$ is a dimensional constant. The constant obtained by his argument satisfies $\gamma_n = 1$ if and only if $n = 4$; consequently, Croke established the Cartan–Hadamard conjecture in dimension $n = 4$. This method

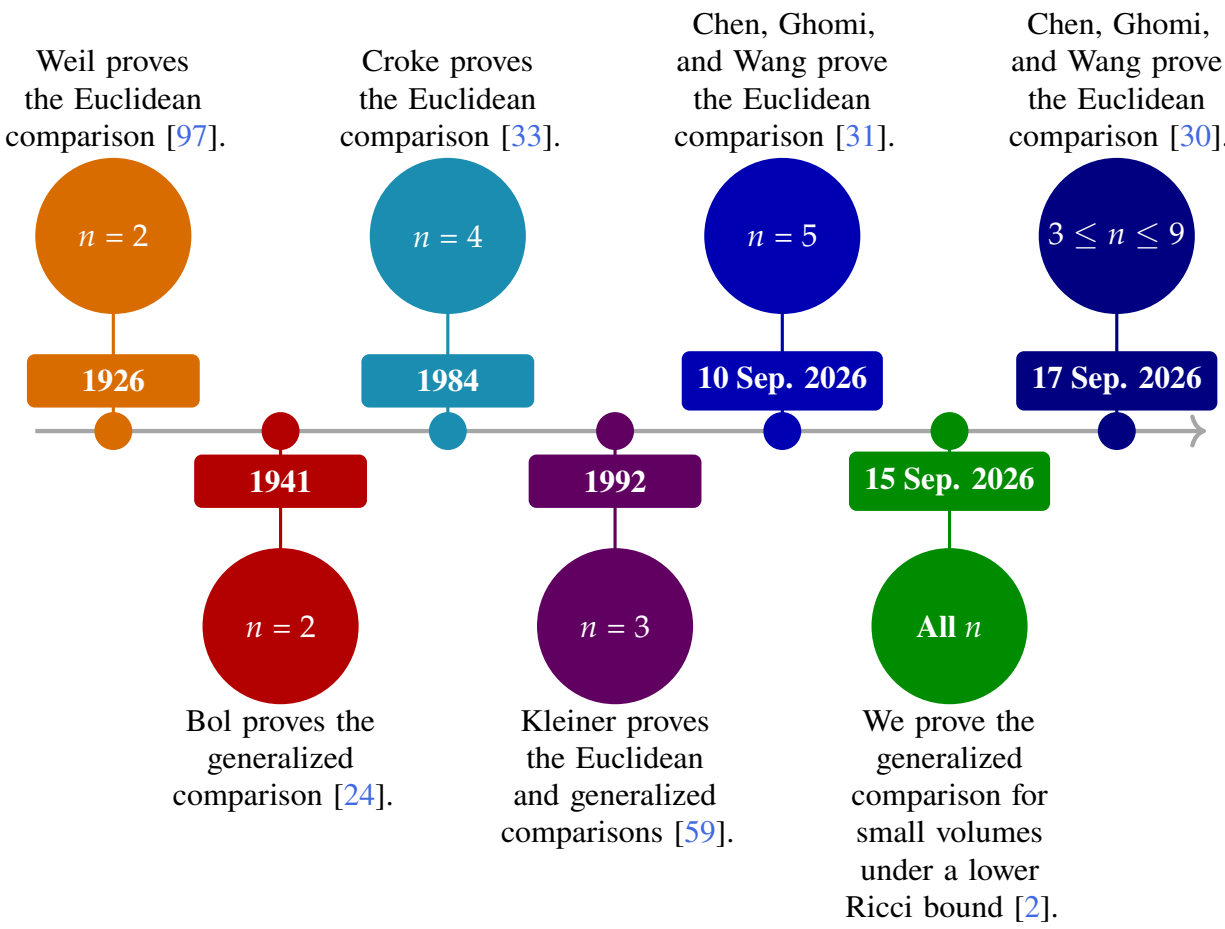


Figure 1: Selected milestones in the Cartan–Hadamard conjecture and its generalized version.

belongs to a broader program relating isoperimetric inequalities to geometric and spectral estimates [32]. An alternative proof in dimension $n = 4$ is available in [60].

In 1992, Kleiner established, in dimension $n = 3$, both the Cartan–Hadamard conjecture and its generalized version [59]. His proof relates isoperimetry to the total curvature of hypersurfaces and to the variation of the isoperimetric profile. The Gauss–Bonnet theorem supplies a decisive ingredient in dimension three. Alternative proofs are available in [88, 90]. It is somewhat surprising that Croke's proof in dimension $n = 4$ appeared in the literature before Kleiner's proof in dimension $n = 3$, and that the *non-optimal* inequality $I_{(M^3,g)} \geq \gamma_3 I_{(\mathbb{R}^3,\delta_{euc})}$ is due to Croke, whereas the *sharp* inequality $I_{(M^3,g)} \geq I_{(\mathbb{R}^3,\delta_{euc})}$ was later established by Kleiner. Neither proof offered an immediate extension to arbitrary dimension.

An important feature of Kleiner's proof is that, in dimension $n = 3$, it reduces the Cartan–Hadamard conjecture to a total-curvature inequality. The Gauss–Bonnet theorem then yields this inequality for the boundary of an isoperimetric region. This motivates a broader investigation of total curvature. For a sufficiently regular convex hypersurface Γ in an n-dimensional Cartan–Hadamard manifold, the total curvature conjecture asserts that

$$\int_\Gamma |GK|\, d\sigma \geq |\mathbb{S}^{n-1}|,$$

where GK denote the Gauss-Kronecker curvature of Γ and $d\sigma$ is the volume form of Γ. Ghomi and Spruck proved in every dimension that the total curvature conjecture implies the Cartan–Hadamard

conjecture [44, Theorem 7.1]. Their work also develops comparison formulas for total curvature, estimates for convex hulls, and regularity results for distance functions. The implication provides a major reduction, but it must be distinguished from an unconditional proof of either conjecture in every dimension.

We highlight the great results and very detailed discussions about the Cartan-Hadamard conjecture by Ghomi and Spruck in [44] and its generalized version in $n = 4$ by Kloeckner and Kuperberg in [60], as well as the following resources [37, 87]. In a different direction, Hass showed that isoperimetric regions in Cartan-Hadamard manifolds need not be connected [52].

Remark 1.2.1. Several further developments appeared in September 2026. Chen, Ghomi, and Wang [31] presented a proof of the conjecture in dimension $n = 5$. Seven days later, the same authors [30] put forward a proof of the conjecture in dimensions $3 \leq n \leq 9$.

We are not yet familiar with their entire proofs since they are extremely recent. However, we can state that our approach is completely different than theirs.

The small-volume problem has its own history. Morgan and Johnson established sharp small-volume comparisons under suitable curvature hypotheses on compact manifolds and showed that small minimizing boundaries are nearly round [71]. Druet then obtained sharp local isoperimetric inequalities from a strict scalar curvature comparison [36]. In the present notation, the relevant condition at a point is $\mathrm{Scal}_g < n(n-1)\kappa$, which yields the model comparison in a sufficiently small neighborhood. Nardulli developed the description of small-volume isoperimetric regions and the profile on noncompact manifolds with controlled geometry [75]; Nardulli and Osorio Acevedo proved sharp small-volume inequalities under bounded-geometry assumptions and a strict upper comparison for the supremum of the scalar curvature [78].

Two distinctions explain why these results do not settle the problem treated here. First, a comparison inequality established on a fixed neighborhood does not automatically produce a volume threshold that is uniform over the entire manifold. On a noncompact manifold, a set of small volume can be located anywhere, and the center of a minimizing sequence may escape to infinity; a threshold derived from data local to one neighborhood offers no guarantee once the sequence leaves that neighborhood. Second, $\mathrm{Sec}_g \leq \kappa$ yields only the non-strict inequality $\mathrm{Scal}_g \leq n(n-1)\kappa$. At the borderline, or along sequences where the scalar curvature approaches the model value, the first non-Euclidean term in an asymptotic expansion does not determine the sign of the isoperimetric deficit. A sharp theorem must handle this regime as well as the strict one.

The conjecture is important beyond the geometry of minimizing regions. By the coarea formula, the Euclidean isoperimetric comparison implies the sharp Sobolev inequality, together with its optimal constants – this was actually Aubin's original motivation for proposing the conjecture [19], and it has since fed a whole line of sharper Sobolev-type inequalities on Cartan-Hadamard manifolds [38, 63, 64]. Equality in the Sobolev inequality, and the question of when optimizers exist, also leads to rigidity questions about the ambient manifold, as studied by Muratori and Soave [74].

The same comparison gives the Faber–Krahn inequality: among domains of a given volume, a ball minimizes the first Dirichlet eigenvalue. Rayleigh conjectured this already in 1877, and Faber and Krahn proved it in the 1920s [86, 40, 61, 62]; the generalized isoperimetric comparison gives the same

statement with a ball in $\mathbb{M}^n_\kappa$ in place of the Euclidean ball. In a different direction, Kloeckner showed that a sharp isoperimetric inequality yields a lower bound on the isometric defect of a continuous map $\phi : M \to N$ between Riemannian manifolds, a quantitative measure of how far ϕ is from being an isometry [60].

For a broader account of the applications of isoperimetric inequalities in analysis and geometry, we refer the reader to [20, 81, 85, 28, 95, 44, 87].

1.2.2 A proof of the Cartan–Hadamard conjecture for small volumes under a Ricci curvature lower bound

In [2], we prove the generalized Cartan-Hadamard conjecture in every dimension $n \geq 2$ for small volumes. The only additional hypothesis we impose is a lower bound on the Ricci curvature. In this setting, our results are new even for the Euclidean inequality version of the conjecture.

Theorem 1.2.2. *Let (M^n, g) be a Cartan-Hadamard manifold with $\mathrm{Sec}_g \leq \bar{k} \leq 0$ and $\mathrm{Ric}_g \geq (n-1)\underline{k}$. Then there exists $v_0 = v_0(\underline{k}, \bar{k}, n) > 0$ such that*

$$I_{(M^n,g)}(v) \geq I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(v) \text{ for any } v \in (0, v_0]. \tag{2}$$

Moreover, if there exists a finite perimeter set $E \subset M$ with $\mathrm{vol}_g(E) = v \in (0, v_0]$ and $P_g(E) = I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(v)$, then E is isometric to a geodesic ball in $\mathbb{M}^n_{\bar{k}}$.

Remark 1.2.3 (Alexandrov spaces)**.** We actually prove Theorem 1.2.2 in the more general context of non-collapsing boundaryless Alexandrov spaces with two-sided curvature bounds, cf. Chapters 8 and 9.

In Chapter 11, we prove Theorem 1.2.4, which gives a major reduction that states the following: if one assumes the rigidity in the equality case at volumes up to which the inequality in the conjecture is already known (under a lower curvature bound, namely Assumption 1), then Theorem 1.2.2 holds for *every volume* without any lower curvature bound. The assumption and precise statement are displayed below, see Section 5.1 for the definitions regarding Alexandrov spaces.

Assumption 1 (Equality case)**.** Let (Z, d_Z) be a non-collapsing Alexandrov space with no Alexandrov boundary and curvature bounded from above by $\bar{k} \leq 0$ and from below by $\underline{k}$. Let $V > 0$ be such that

$$I_{(Z,d_Z)}(w) \geq I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(w) \quad \text{for every } w \in (0, V].$$

We assume that for every finite perimeter set $E \subset Z$ with $\mathcal{H}^n_{d_Z}(E) = V$ and $P_{d_Z}(E) = I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(V)$, it holds that (E, d_Z) is isometric to a geodesic ball in $\mathbb{M}^n_{\bar{k}}$.

Next, we state our result in the arbitrary volume regime which we prove in Chapter 11.

Theorem 1.2.4. *Let (M^n, g) be a Cartan-Hadamard manifold with $\mathrm{Sec}_g \leq \bar{k} \leq 0$. Assume in addition that Assumption 1 holds. Then we have*

$$I_{(M^n,g)}(v) \geq I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(v) \text{ for any } v \geq 0. \tag{3}$$

Moreover, if there exists a finite perimeter set $E \subset M$ with finite volume $v > 0$ and $P_g(E) = I_{(\mathbb{M}^n_{\bar{k}}, \delta^{\bar{k}})}(v)$, then E is isometric to a geodesic ball in $\mathbb{M}^n_{\bar{k}}$.

Remark 1.2.5. Notice that Assumption 1 is needed at the level of Alexandrov spaces to establish Theorem 1.2.4 for smooth Cartan-Hadamard manifolds. Furthermore, in Assumption 1, the space also admits a *lower* bound on the curvature. The latter fact comes from a reduction argument in Chapter 10 showing that it suffices to prove the Cartan-Hadamard conjecture under a Ricci lower bound. We actually prove Theorem 1.2.4 for Alexandrov spaces in Proposition 11.0.1.

Remark 1.2.6. By Theorem 1.2.2 for Alexandrov spaces (Theorem 8.0.1), the conclusion of Assumption 1 holds unconditionally when $V \leq v_0(v_{nc}, \underline{k}, \bar{k}, n)$, so its content is the case $V > v_0$.

It's known that isoperimetric regions in Cartan-Hadamard manifolds need not be connected, by work of Hass [52]. Establishing their boundedness is still an open problem that we settle under the assumptions of each theorem above. We prove this in Section 10.2.

Corollary 1.2.7. *Let (M^n, g) be a Cartan-Hadamard manifold with $\mathrm{Sec}_g \leq \bar{k} \leq 0$ and $E \subset M^n$ be an isoperimetric region. If (3) holds true, then E is bounded.*

Remark 1.2.8. By Theorems 1.2.2 and 1.2.4, if either (M^n, g) has a Ricci lower bound and $\mathrm{vol}_g(E) \leq v_0$, or Assumption 1 holds, the corollary applies.

What is new?

The approach developed in this paper is entirely different from the previous methods used to approach the conjecture. In particular, we neither assume nor prove the total curvature conjecture, and our argument does not pass through it whatsoever. Consequently, even in dimensions where the Cartan-Hadamard conjecture is already known, our result yields a *genuinely different proof*, under the additional assumption of a lower Ricci curvature bound. *Our proof in contrast with the known results in the literature is not dimension dependent*. Moreover, none of the previous approaches to the conjecture form part of the mechanism of our proof.

The lower Ricci curvature bound, and the reduction to this case, cf. Section 10, play a fundamental role in our argument through the generalized existence and compactness theory for isoperimetric regions developed by the third named author in [75, 73]. Without such a lower bound, the corresponding existence and compactness theory is known to fail. Indeed, one of the main difficulties in approaching the Aubin conjecture is that minimizing sequences—namely, sequences of sets of volume v whose perimeters converge to $I_{(M^n,g)}(v)$—may run off to infinity. This phenomenon can already be seen in the hyperbolic paraboloid (P, h) in $\mathbb{R}^3$, which satisfies $\mathrm{Sec}_h \leq 0$ and $\mathrm{Ric}_h \geq -1$, as it does not contain any isoperimetric region. We show how the lower Ricci curvature bound allows us to overcome this lack of genuine compactness using a generalized one.

Overview of the strategy of the proof

The small-volume case, Theorem 1.2.2, is proved through two major reductions: first, to generalized isoperimetric regions; and second, to regions represented as normal graphs. More precisely:

1. In Section 8, we use the lower Ricci curvature bound to invoke the generalized existence and compactness theory described above. This reduces the proof of Theorem 1.2.2 to Proposition 8.0.3, cf. Section 8. In other words, the lower curvature bound allows us to apply the generalized existence which reduces the proof of Theorem 1.2.2 to proving it at the limit spaces that contain an isoperimetric region. On the other hand, the generalized existence forces us to work in the setting of Alexandrov metric spaces.

2. Once this reduction has been established, we prove Proposition 8.0.3 in Section 9. The proof centers on showing that generalized isoperimetric regions of small volumes are represented by normal graphs over geodesic spheres. It then suffices to prove the conjecture for normal graphs over geodesic spheres, which we do by hand in Lemma 7.0.1.

The proof of Theorem 1.2.4 requires an additional step because no Ricci lower bound is assumed in that result. Specifically, in Section 10, we show that it suffices to prove Theorem 1.2.4 under the additional assumption $\mathrm{Ric}_g \geq (n-1)\underline{k}$. Since the third author and Muñoz-Flores show in [72] that the isoperimetric profile (5) can be defined by minimizing over bounded smooth sets, we use this boundedness to modify the metric g outside a sufficiently large ball, constructing a metric $\hat{g}$ with the desired Ricci curvature lower bound. The fact that the metric g and $\hat{g}$ coincide inside that very large ball then completes the reduction after a simple computation.

Once the reduction to the Ricci lower-bound setting is complete, we essentially follow the strategy outlined in steps (1) and (2) above.

1.3 ALLARD'S INTERIOR ε-REGULARITY THEOREM

1.3.1 Historical background

The regularity questions in the third part of the thesis arise from a basic feature of geometric variational problems. A sequence of smooth surfaces with controlled area need not converge to a smooth surface, and a convenient class of competitors must therefore include a notion of "generalized" surfaces. Geometric measure theory provides such classes together with compactness and lower semicontinuity properties.

Almgren introduced varifolds in the 1960s as a notion of "generalized" surfaces [6, 7]. An m-dimensional varifold on $\mathbb{R}^n$ can be understood as a Radon measure on the Grassmannian of m-planes. A smooth submanifold determines such a measure by integrating over its tangent planes, possibly with a multiplicity. The reduced boundary of a set of finite perimeter likewise determines a multiplicity-one varifold whose weight measure coincides with the perimeter measure of the set. This representation allows the boundaries of isoperimetric regions, as in Section 1.2, to be studied using Allard's ε-regularity theorem, provided that its quantitative hypotheses are satisfied. The definition also accommodates limits whose smoothness or even rectifiability is not known in advance. The central question is then

which geometric or variational assumptions force a varifold to recover the structure of a classical submanifold.

Allard's 1972 paper answered this question under quantitative hypotheses and established a foundational regularity theorem [5]. Roughly, if a varifold has a suitable positive density lower bound, its mass in a ball is close to that of a multiplicity-one plane, and its generalized mean curvature is sufficiently small in L^p with $p > m$, then its support is a $C^{1,\alpha}$ graph in a smaller ball, with $\alpha = 1 - m/p$. The hypotheses are quantitative, as is the conclusion: they control the scale on which the graph exists and its differentiability estimates. This is the meaning of ε-regularity:

A weak object that is sufficiently close to a regular model must itself be regular at a smaller scale. (★)

Allard's interior ε-regularity theorem naturally led to the question of regularity in curved ambient spaces. One possible strategy is to express the Riemannian area functional in local coordinates. Another classical approach is to isometrically embed the ambient manifold into a Euclidean space. This latter method is connected to Nash's isometric embedding theorem, but it introduces extrinsic geometry into the estimates.

The embedding approach also has a regularity limitation. A $\mathcal{C}^{k,\alpha}$ metric with $k + \alpha > 2$ admits an isometric embedding of the same class [54], whereas the existence of a $\mathcal{C}^2$ isometric embedding for metrics of class exactly $\mathcal{C}^2$ remains open; see [50, Outrageous $\mathcal{C}^2$-Immersion Conjecture, p. 177]. Moreover, the existence of an embedding alone does not provide the uniform extrinsic bounds needed in compactness problems involving varying ambient metrics. These issues become particularly significant for Alexandrov spaces, whose limiting metrics need not be smooth. They motivate an intrinsic theory in which the regularity scale is controlled directly by curvature bounds and non-collapsing data.

1.3.2 Allard's interior ε-regularity theorem in Alexandrov spaces

In [1], we give an affirmative answer in a broad and geometrically significant class: we prove Allard's Interior ε-Regularity Theorem in non-collapsed Alexandrov spaces with curvature bounded both from above and below. Our approach is intrinsic — built directly on the geometry of the ambient space — and crucially relies on the Approximation Theorem of Berestovskij and Nikolaev [22, Section 15.1]. This allows us to obtain explicitly constants depending only on the dimension of the varifold, the dimension of the ambient space, the sectional curvature bounds, and the harmonic radius.

Complete statements with explicit dependence of all constants are given in Theorem 14.0.3 for non-collapsed Alexandrov spaces with curvature bounded both from above and below, and in Theorem 13.6.8 for Riemannian manifolds with metric tensor of class $\mathcal{C}^2$. We state below a simplified version of Allard's Interior ε-Regularity Theorem for non-collapsed Alexandrov spaces with curvature bounded both from above and below, this is the precise version of (★). Precise definitions are given as follows: the space of varifolds $\mathbf{V}_m(X)$ (Definition 12.1.1), the weight measure $\|V\|$ (Definition 12.1.2), the density $\Theta^m(\cdot,\cdot)$ (Definition 12.1.3), and the generalized mean curvature vector $\overrightarrow{H_g}$ (Definition 12.2.9).

Theorem 1.3.1. *Let $m, n \in \mathbb{N}$ and $p \in \mathbb{R}$ such that $1 \leq m < n$ and $m < p < \infty$; in case $m = 1$, we require that $p \geq 2$, (X, g) be an n-dimensional non-collapsed Alexandrov space with curvature bounded both from above and below. Then there exist $\delta_0 > 0$ and $\rho_0 > 0$ such that, if $V \in \mathbf{V}_m(X)$ satisfies*

$$\Theta^m(\|V\|, x) \geq d > 0,$$

for $\|V\|$-a.e. $x \in B_{d_g}(\xi, \rho)$,

$$\frac{\|V\|(B_{d_g}(\xi, \rho))}{\omega_m \rho^m} \leq d(1 + \delta),$$

and

$$\|\overrightarrow{H_g}\|_{L^p_{loc}(\|V\|)} \leq \frac{d^{\frac{1}{p}}\delta}{\rho^{1-\frac{m}{p}}},$$

for some $0 < \delta < \delta_0$, for some $0 < d < \infty$, and for some $0 < \rho < \min\{\rho_0, \sqrt{\delta}\}$, then $\operatorname{supp}(\|V\|) \cap B_d(\xi, \varepsilon\rho)$ is a $C^{1,1-\frac{m}{p}}$ m-dimensional submanifold of X with scaling invariant $C^{1,1-\frac{m}{p}}$ estimates.

Before stating Corollary 1.3.3, a consequence of Theorem 1.3.1, let us define the regular and singular sets of a varifold.

Definition 1.3.2. Let (X, g) be an n-dimensional non-collapsed Alexandrov space with curvature bounded both from above and below, $V \in \mathbf{V}_m(X)$ and $\xi \in \operatorname{supp}(\|V\|)$. We say that ξ is a regular point of V if there exists $\rho > 0$ such that $B_{d_g}(\xi, \rho) \cap \operatorname{supp}(\|V\|)$ is an m-dimensional $\mathcal{C}^1$ embedded submanifold of X. Furthermore,

$$\operatorname{reg}(V) := \{\xi \in \operatorname{supp}(\|V\|) : \xi \text{ is a regular point of } V\} \quad \text{and} \quad \operatorname{sing}(V) := \operatorname{supp}(\|V\|) \setminus \operatorname{reg}(V).$$

With this definition in hand, Corollary 1.3.3 establishes that $\operatorname{reg}(V)$ is a relatively open dense subset of $\operatorname{supp}(\|V\|)$. We remark that this is essentially the best result currently available: already in $\mathbb{R}^n$, the optimal size of $\operatorname{sing}(V)$ — even its Hausdorff dimension, or whether $\mathcal{H}^m(\operatorname{sing}(V)) = 0$ — remains unknown for general stationary varifolds, and constitutes one of the main open problems in the regularity theory of varifolds.

Corollary 1.3.3. *Let $m, n \in \mathbb{N}$ and $p \in \mathbb{R}$ such that $1 \leq m < n$ and $m < p < \infty$; in case $m = 1$, we require that $p \geq 2$, (X, g) be an n-dimensional Alexandrov space with double-sided bounded intrinsic sectional curvature and $V \in \mathbf{V}_m(X)$ with generalized mean curvature vector $\overrightarrow{H_g} \in L^p_{\text{loc}}(X, \|V\|)$ and $\Theta^m(\|V\|, x) \in \mathbb{N} \setminus \{0\}$, for $\|V\|$-a.e. $x \in \operatorname{supp}(\|V\|)$. Then $\operatorname{reg}(V)$ is a relatively open dense subset of $\operatorname{supp}(\|V\|)$, i.e., $\operatorname{sing}(V)$ is nowhere dense in $\operatorname{supp}(\|V\|)$ and $\operatorname{supp}(\|V\|) = \overline{\operatorname{reg}(V)}$.*

Remark 1.3.1. Another consequence of Theorem 13.6.8 is that the hypothesis of $\mathcal{C}^4$ bounded geometry in [76] can be relaxed to $\mathcal{C}^2$ bounded geometry, at the cost of weaker control on the remainder terms of order strictly greater than two in the Puiseux expansion of the isoperimetric profile for small volumes.

What is new?

The main novelty is that the proof is intrinsic. Rather than embed the ambient manifold into a Euclidean space and apply an extrinsic form of Allard's theorem, we work directly with the Riemannian geometry of the ambient space. This avoids introducing extrinsic quantities into the estimates and makes the dependence of the regularity scale on curvature bounds and harmonic-radius data explicit.

This intrinsic formulation is particularly suited to varying ambient spaces and to Alexandrov limits. The limiting metric generally has lower regularity than the smooth approximating metrics, so estimates depending on higher derivatives of a chosen smoothing would not survive the approximation process. The uniform intrinsic estimates developed here provide precisely the control needed to pass to the limit.

A further consequence is that the $\mathcal{C}^4$ bounded-geometry hypothesis in [76] can be relaxed to $\mathcal{C}^2$ bounded geometry, at the cost of weaker control of remainder terms of order strictly greater than two in the Puiseux expansion of the small-volume isoperimetric profile.

Overview of the strategy of the proof

This paper is divided into three main parts. In Chapter 12, we establish the basic framework for the theory of intrinsic varifolds on the ambient of abstract Riemannian manifolds, fix notation, and recall fundamental definitions. Specifically we begin by defining varifolds intrinsically and obtaining the first variation of a varifold in a Riemannian manifold. The first place where the adaptations to the Riemannian setting become substantial is estimating $\operatorname{div}_S(u_y \nabla_g u_y)$ (see Lemma 12.3.8), which requires comparison geometry to be controlled in terms of sectional curvature bounds. Using these estimates, we prove the monotonicity formula (see Lemma 12.3.9) and derive its consequences, including the existence and upper semicontinuity of the density $\Theta^m(\|V\|, y)$ for varifolds with L^p_{loc}-bounded generalized mean curvature (see Corollary 12.3.14). Furthermore we prove the Rectifiability Theorem (see Theorem 12.4.1).

In Chapter 13, we develop the intrinsic notion of excess for varifolds in Riemannian manifolds and carry out the proof of the regularity theorem. We begin with a discussion of Fermi coordinates and harmonic m-submanifolds, the key new geometric ingredient of our approach. Since $\mathcal{C}^2$ metrics do not provide sufficient regularity for normal coordinates alone [55, Section 2.8], harmonic submanifolds offer better-controlled coordinate systems. We prove the Caccioppoli inequality (Lemma 13.2.1) with our intrinsic excess. The curvature of the harmonic submanifold contributes additional terms when estimating $\operatorname{div}_{T_x V}(u_\Sigma \nabla g(u_\Sigma))$ (see Equation (126)), which are controlled via the principal curvatures of the harmonic submanifold. We then introduce the main hypotheses on varifolds carried through the rest of the paper (Definition 13.3.1) and establish the density estimate (Lemma 13.3.4) to obtain a first decay estimate for the excess (Corollary 13.4.3) from harmonic submanifolds and the parameter ρ_0. Finally we introduce the notion of normal graph and prove the Lipschitz Approximation Lemma (Lemma 13.5.2).

We note that the proofs of many key results follow arguments parallel to those in [91], with two notable exceptions: the Caccioppoli inequality [91, Lemma 2.5, p. 131] (see Lemma 13.2.1), where the main difference arises from our intrinsic definition of excess (see Definition 13.1.15); and the Lipschitz

approximation [91, Lemma 2.11, p. 135] (see Lemma 13.5.2), where instead of the graph of a Lipschitz function one must work with normal graphs (see Definition 13.5.1).

In Chapter 14, we conclude the paper by proving Allard's Interior ε-Regularity Theorem in non-collapsed Alexandrov spaces with curvature bounded both from above and below. The argument proceeds by approximating the metric of the Alexandrov space by smoother Riemannian metrics via the Berestovskij–Nikolaev theorem, applying *a priori* estimates given by the intrinsic regularity theory developed in Chapter 13 to each approximation, and then passing to the limit. The explicit dependence of all constants on the geometric data, established throughout Chapter 13, is what makes this limiting argument viable.

Part I

BACKGROUND

2 ELEMENTS OF GEOMETRIC MEASURE THEORY

In Part I, we present the notation commonly used in the geometric measure theory literature. Since Parts II and III reproduce two separate papers, each part fixes its own notation independently, and we adapt the notation to each context as needed.

This chapter is devoted to presenting results from geometric measure theory on which our work is based. It is not intended to provide a complete overview; the standard references are Maggi's book [68] and Simon's lecture notes [91], while Federer's treatise [41] remains the classical reference. Throughout this work, the reader is assumed to be familiar with the content of a first graduate course in measure theory, as presented for instance in Folland [42]: σ-algebras, construction of measures from outer measures, Lebesgue integral, monotone and dominated convergence theorems, L^p spaces, Hölder's inequality, Radon–Nikodym and Riesz representation theorems.

2.1 RADON MEASURES ON $\mathbb{R}^n$

We work with measures in the usual sense in geometric measure theory, namely outer measures that are Borel regular.

Definition 2.1.1. An outer measure μ is a Radon measure on $\mathbb{R}^n$ if it is a locally finite, Borel regular measure on $\mathbb{R}^n$.

Remark 2.1.2. Notice that if μ is a Radon measure on $\mathbb{R}^n$ and $E \subset \mathbb{R}^n$ is a Borel set, then

$$\mu(E) = \inf\{\mu(A) : E \subset A, A \text{ is open}\} = \sup\{\mu(K) : K \subset E, K \text{ is compact}\}.$$

We adopt the outer-measure point of view, rather than requiring all sets to be measurable, for convenience: it allows us to restrict a measure to an arbitrary set without first checking measurability. For a Radon measure μ on $\mathbb{R}^n$ and an arbitrary set $A \subset \mathbb{R}^n$, the restriction $\mu \llcorner A$ is defined by $\mu \llcorner A(B) := \mu(A \cap B)$, for all $B \subset \mathbb{R}^n$. This restriction is again a Radon measure whenever A is Borel or has locally finite measure.

While the Lebesgue measure $\mathcal{L}^n$ measures full-dimensional subsets of $\mathbb{R}^n$, the Hausdorff measure provides the appropriate notion of measure for lower-dimensional sets, including curves, surfaces, and their generalized counterparts.

Definition 2.1.3. Given $m \in \mathbb{N}$, $\delta > 0$, the m-dimensional Hausdorff measure of step δ of a set $E \subset \mathbb{R}^n$ is defined as

$$\mathcal{H}^m_\delta(E) = \inf_{\mathcal{F}} \sum_{F\in\mathcal{F}} \omega_m \left(\frac{\operatorname{diam}(F)}{2}\right)^m,$$

where $\operatorname{diam}(F)$ denotes the diameter of $F \subset \mathbb{R}^n$, $\mathcal{F}$ is a countable covering of E by sets $F \subset \mathbb{R}^n$ such that $\operatorname{diam}(F) < \delta$, and $\omega_m = \mathcal{L}^m(B_{g_e}(0,1)))$ denotes the Lebesgue measure (volume) of the unit ball in $\mathbb{R}^m$ with respect to the Euclidean metric g_e. The m-dimensional Hausdorff measure of E is then

$$\mathcal{H}^m(E) = \sup_{\delta\in(0,\infty]} \mathcal{H}^m_\delta(E) = \lim_{\delta\downarrow 0} \mathcal{H}^m_\delta(E).$$

The normalizing constant ω_m is chosen so that $\mathcal{H}^n$ coincides with $\mathcal{L}^n$ on $\mathbb{R}^n$ and $\mathcal{H}^m$ agrees with the usual m-dimensional area measure on smooth m-dimensional submanifolds.

The Riesz Representation Theorem identifies (vector-valued or signed) Radon measures with continuous linear functionals on the spaces of compactly supported continuous functions. We record it in the following form, which underlies the definition of generalized mean curvature in Section 2.2.

Theorem 2.1.4 ([68, Theorem 4.7, p. 34])**.** *If $L : C^0_c(\mathbb{R}^n, \mathbb{R}^m) \to \mathbb{R}$ is a bounded linear functional, then its total variation $|L|$ is a Radon measure on $\mathbb{R}^n$, there exists a $|L|$-measurable function $g : \mathbb{R}^n \to \mathbb{R}^m$ with $|g| = 1$ $|L|$-a.e. on $\mathbb{R}^n$ and*

$$L(\varphi) = \int_{\mathbb{R}^n} \langle \varphi, g\rangle d|L|,$$

for all $\varphi \in C^0_c(\mathbb{R}^n, \mathbb{R}^m)$. Moreover, for every open set $A \subset \mathbb{R}^n$,

$$|L|(A) = \sup\left\{ \int_{\mathbb{R}^n} \langle \varphi, g\rangle d|L| \colon \varphi \in C^0_c(A, \mathbb{R}^m), |\varphi| \le 1 \right\}.$$

Remark 2.1.5. Notice that if μ is a Radon measure on $\mathbb{R}^n$, then the functional $L : C^0_c(\mathbb{R}^n) \to \mathbb{R}$ given by

$$L(\varphi) := \int_{\mathbb{R}^n} \varphi d\mu,$$

for all $\varphi \in C^0_c(\mathbb{R}^n)$, is linear and bounded on $C^0_c(\mathbb{R}^n)$, with the additional property of being monotone.

The compactness theory of geometric variational problems is formulated in terms of the weak-$*$ convergence of measures.

Definition 2.1.6. A sequence (μ_i) of Radon measures on $\mathbb{R}^n$ converges weakly-$*$ to a Radon measure μ, written $\mu_i \overset{*}{\rightharpoonup} \mu$, if

$$\int_{\mathbb{R}^n} \varphi d\mu_i \to \int_{\mathbb{R}^n} \varphi d\mu,$$

for every $\varphi \in C^0_c(\mathbb{R}^n)$.

For the purposes of this thesis, the main local invariant attached to a measure is its density.

Definition 2.1.7. Let μ be a Radon measure on $\mathbb{R}^n$ and $m \in (0, n]$, we define the m-dimensional upper (resp., lower) density of μ as

$$\Theta^{*m}(\mu, x) := \limsup_{r \downarrow 0} \frac{\mu(B_{g_e}(x, r))}{\omega_m r^m} \left(\text{resp., } \Theta^m_*(\mu, x) := \liminf_{r \downarrow 0} \frac{\mu(B_{g_e}(x, r))}{\omega_m r^m} \right), \quad x \in \mathbb{R}^n.$$

When the two coincide their common value is the density $\Theta^m(\mu, x)$.

Densities provide a comparison between an arbitrary measure and the Hausdorff measure.

Theorem 2.1.8 ([68, Theorem 6.4, p. 66])**.** *Let μ be a Radon measure on $\mathbb{R}^n$, $A \subset \mathbb{R}^n$ a Borel set, and $m \in (0, n)$.*

1. *If $1 \leq \Theta^{*m}(\mu, x)$ for all $x \in A$, then $\mathcal{H}^m(A) \leq \mu(A)$.*
2. *If $\Theta^{*m}(\mu, x) \leq 1$ for all $x \in A$, then $\mu(A) \leq 2^m \mathcal{H}^m(A)$.*

Let $m \leq n$. A smooth m-dimensional submanifold of $\mathbb{R}^n$ admits a well-defined tangent plane at each of its points. Rectifiable sets constitute the broadest class of sets admitting a measure-theoretic substitute for this tangent plane almost everywhere, and they provide the notion of surface underlying rectifiable varifolds.

Definition 2.1.9. Given a $\mathcal{H}^m$-measurable set $\Gamma \subset \mathbb{R}^n$, we say that Γ is countably m-rectifiable if there exist countable many Lipschitz maps $f_i : \mathbb{R}^m \to \mathbb{R}^n$ such that

$$\mathcal{H}^m \left(\Gamma \setminus \bigcup_{i \in \mathbb{N}} f_i(\mathbb{R}^m) \right) = 0;$$

we say that Γ is locally m-rectifiable provided $\mathcal{H}^m(K \cap \Gamma) < \infty$ for every compact set $K \subset \mathbb{R}^n$; finally, if $\mathcal{H}^m(\Gamma) < \infty$, then Γ is simply called m-rectifiable.

The following definitions introduce approximate tangent spaces for locally m-rectifiable sets and for Radon measures on $\mathbb{R}^n$, which serve as the measure-theoretic counterpart of tangent spaces.

Definition 2.1.10. Let Γ be a locally m-rectifiable set. We say that $S \in \mathrm{Gr}(m, n)$ is the approximate tangent space of Γ at $x \in \Gamma$, denoted by $\mathrm{Tan}^m(\Gamma, x)$, if

$$\mathcal{H}^m \llcorner \left(\frac{\Gamma - x}{r} \right) \overset{*}{\rightharpoonup} \mathcal{H}^m \llcorner S, \quad \text{as } r \downarrow 0,$$

where $\mathrm{Gr}(m, n)$ denotes the Grassmannian of m-planes through the origin in $\mathbb{R}^n$.

Definition 2.1.11. Let μ be a Radon measure on $\mathbb{R}^n$ and Γ a Borel set in $\mathbb{R}^n$ such that μ is concentrated on Γ. We say that $S \in \mathrm{Gr}(m, n)$ is the approximate tangent space of μ at $x \in \Gamma$ if

$$\frac{(\eta_{x,r})_\# \mu}{r^m} \overset{*}{\rightharpoonup} \mathcal{H}^m \llcorner S, \quad \text{as } r \downarrow 0,$$

where $\eta_{x,r}(y) = \frac{y-x}{r}$, $y \in \mathbb{R}^n$, and $(\eta_{x,r})_\# \mu$ denotes the push-forward of μ by $\eta_{x,r}$.

Now we can establish the equivalence between rectifiability and almost-everywhere existence of approximate tangent spaces.

Theorem 2.1.12 ([68, Theorem 10.2, p. 99, and Theorem 10.8, p. 102])**.** *If $\Gamma \subset \mathbb{R}^n$ is locally m-rectifiable, then for $\mathcal{H}^m$-a.e. $x \in \Gamma$ there exists a unique $S \in \mathrm{Gr}(m,n)$ which is the approximate tangent space of Γ at x. Conversely, if μ is a Radon measure on $\mathbb{R}^n$, Γ is a Borel set in $\mathbb{R}^n$ such that μ is concentrated on Γ, and, for every $x \in \Gamma$, there exists an $S \in \mathrm{Gr}(m,n)$ which is the approximate tangent space of μ at x, then $\mu = \mathcal{H}^m \llcorner \Gamma$ and Γ is locally m-rectifiable.*

2.2 VARIFOLDS ON $\mathbb{R}^n$

In the following, we introduce varifolds in $\mathbb{R}^n$, which will serve as the principal notion of generalized surfaces throughout this thesis.

Definition 2.2.1. We define an m-dimensional varifold in an open set $U \subset \mathbb{R}^n$ as a Radon measure V on the Grassmannian bundle $\mathrm{Gr}_m(U) := U \times \mathrm{Gr}(m,n)$. The space of such varifolds is denoted by $\mathbf{V}_m(U)$.

To every varifold one naturally associates a Radon measure on U, called its weight measure, which describes the distribution of its mass in the ambient space.

Definition 2.2.2. Let $V \in \mathbf{V}_m(U)$ and $\pi : \mathrm{Gr}_m(U) \to U$ the projection onto the first factor. We define the weight of V, denoted by $\|V\|$, as the measure given by the push-forward of V by π.

Among all varifolds, rectifiable varifolds play a central role. They are precisely those supported on a countably m-rectifiable set with a locally integrable multiplicity.

Definition 2.2.3. We say that $V \in \mathbf{V}_m(U)$ is a rectifiable m-varifold in U if there exist a countably m-rectifiable set $\Gamma \subset U$ and a positive, locally $\mathcal{H}^m$-integrable function θ on Γ such that

$$V(B) = V(\Gamma,\theta)(B) := \int_{\{x\in\Gamma:(x,\mathrm{Tan}^m(\Gamma,x))\in B\}} \theta d\mathcal{H}^m, \text{ for every Borel set } B \subset \mathrm{Gr}_m(U).$$

We denote the space of rectifiable m-varifolds in U by $\mathbf{RV}_m(U)$. If, in addition, θ is integer-valued, then V is called an integral varifold, and the space of integral m-varifolds in U is denoted by $\mathbf{IV}_m(U)$.

The additional structure that distinguishes a varifold from a Radon measure is its first variation, which encodes the infinitesimal variation of its weight under ambient deformations. To define it, let $S \in \mathrm{Gr}(m,n)$ and $X \in \mathcal{C}^1_c(U;\mathbb{R}^n)$. Denote by $P_S : \mathbb{R}^n \to S$ the orthogonal projection onto S. The tangential divergence of X along S is defined by $\mathrm{div}_S X := \mathrm{tr}(P_S \circ DX)$ or, equivalently, $\mathrm{div}_S X = \sum_{i=1}^m \langle e_i, D_{e_i} X\rangle$, where $\{e_1,\dots,e_m\}$ is any orthonormal basis of S.

Definition 2.2.4. The first variation of a varifold $V \in \mathbf{V}_m(U)$ is the linear functional $\delta V : \mathcal{C}^1_c(U;\mathbb{R}^n) \to \mathbb{R}$ defined by

$$\delta V(X) := \int_{\mathrm{Gr}_m(U)} \mathrm{div}_S X(x) dV(x,S).$$

Remark 2.2.5. Notice that the first variation of a varifold V coincides with the derivative of its weight under the flow generated by X. More precisely, if (Φ_t) denotes the flow generated by X, then $\delta V(X) = \frac{d}{dt}\Big|_{t=0} \|(\Phi_t)_\# V\|(U)$, where the push-forward varifold $(\Phi_t)_\# V$ is defined by

$$(\Phi_t)_\# V(B) := \int_{\{(x,S)\in \mathrm{Gr}_m(U):(\Phi_t(x),D\Phi_t(x)S)\in B\}} \mathrm{Jac}_S\, \Phi_t(x)\, dV(x,S),$$

for every Borel set $B \subset \mathrm{Gr}_m(U)$, and $\mathrm{Jac}_S\, \Phi_t(x)$ denotes the m-dimensional Jacobian of Φ_t at x along the subspace S.

We now define stationary varifolds and varifolds with locally bounded first variation.

Definition 2.2.6. We say that $V \in \mathbf{V}_m(U)$ has locally bounded first variation in U if, for every $W \subset\subset U$, there exists a constant C_W, which depends on W, such that $|\delta V(X)| \leq C_W \|X\|_{L^\infty}$ for all $X \in \mathcal{C}^1_c(W;\mathbb{R}^n)$. If $\delta V = 0$, then V is said to be stationary.

Note that if V has locally bounded first variation, then Theorem 2.1.4 applies to δV and yields a Radon measure $|\delta V|$ together with a unit-vector density. The Radon–Nikodym theorem then gives the decomposition of $|\delta V|$ with respect to $\|V\|$, as stated in the following proposition.

Proposition 2.2.7 ([91, Chapter 8, Section 2, p. 239])**.** *Let $V \in \mathbf{V}_m(U)$ have locally bounded first variation. Then $|\delta V|$ is a Radon measure. Furthermore, there exists a $\|V\|$-measurable function $\overrightarrow{H} : U \to \mathbb{R}^n$, a real constant $C_W > 0$ and $Z \subset U$ with $\|V\|(Z) = 0$, such that for every open subset $W \subset\subset U$ and $X \in \mathcal{C}^1_c(U;\mathbb{R}^n)$*

$$\delta V(X) = -\int_U \left\langle \overrightarrow{H}, X \right\rangle d\|V\| + \int_U \langle \nu, X\rangle\, d|\delta V|_{\mathrm{sing}},$$

and $\|\overrightarrow{H}\|_{L^1(W,\|V\|)} \leq C_W$, where ν is a $|\delta V|$-measurable function with $|\nu| = 1$, and $|\delta V|_{\mathrm{sing}} := |\delta V| \llcorner Z$.

We refer to $\overrightarrow{H}$ as the generalized mean curvature vector of V, to Z as the generalized boundary of V, to $|\delta V|_{\mathrm{sing}}$ as the generalized boundary measure of V, and to $\nu|_Z$ as the generalized unit conormal of V.

If instead of assuming that V has locally bounded first variation, we assume that δV satisfies an L^p-bound for some p, then, by an argument analogous to that of the Proposition 2.2.7, we obtain the following result.

Proposition 2.2.8 ([91, Chapter 8, Section 2, p. 240])**.** *Let $1 \leq p < \infty$, q the conjugate exponent of p, and $V \in \mathbf{V}_m(U)$. Then the following statements are equivalent:*

1. *For every open subset $W \subset\subset U$, there exists a real constant $C_W > 0$ (which depends of W) such that*

$$|\delta V(X)| \leq C_W \|X\|_{L^p(W,\|V\|)}$$

holds for every $X \in \mathcal{C}^1_c(W;\mathbb{R}^n)$.

2. *The total variation of δV is a Radon measure. Furthermore, there exists a $\|V\|$-measurable function $\overrightarrow{H} : U \to \mathbb{R}^n$ such that*

$$\delta V(X) = -\int_U \left\langle \overrightarrow{H}, X \right\rangle d\|V\|$$

and $\|\overrightarrow{H}\|_{L^q(W,\|V\|)} \leq C_W$, for every open subset $W \subset\subset U$ and $X \in \mathcal{C}^1_c(W;\mathbb{R}^n)$.

A fundamental consequence of a bound on the mean curvature is the monotonicity formula for the mass ratio, as follows.

Theorem 2.2.9 ([91, Chapter 4, Sections 3 and 4])**.** *Let $V \in \mathbf{V}_m(U)$ have locally bounded first variation in U. Then, for all $y \in U$ and $0 < \sigma_1 < \sigma_2$ such that $B_{g_e}(y,\sigma_2) \subset\subset U$,*

$$\frac{\|V\|(B_{g_e}(y,\sigma_2))}{\omega_m \sigma_2^m} - \frac{\|V\|(B_{g_e}(y,\sigma_1))}{\omega_m \sigma_1^m} = \int_{B_{g_e}(y,\sigma_2)\setminus B_{g_e}(y,\sigma_1)} \frac{|(x-y)^\perp|^2}{|x-y|^{m+2}} d\|V\| + \mathcal{R},$$

where $(x-y)^\perp$ is the component of $x-y$ orthogonal to the approximate tangent plane and $\mathcal{R}$ is an error controlled by $\overrightarrow{H}$. In particular, if V is stationary, then $\mathcal{R} = 0$ and, for all $y \in U$ and $0 < \sigma$ such that $B_{g_e}(y,\sigma) \subset\subset U$, the function

$$\sigma \mapsto \frac{\|V\|(B_{g_e}(y,\sigma))}{\omega_m \sigma^m}$$

is monotone non-decreasing.

Two consequences of Theorem 2.2.9 are that the mass ratio has a limit as $\sigma \downarrow 0$, and hence the density $\Theta^m(\|V\|, y)$ exists at every point $y \in U$. Moreover, the function $y \mapsto \Theta^m(\|V\|, y)$ is upper semicontinuous.

2.3 RECTIFIABILITY AND ALLARD'S INTERIOR ε-REGULARITY THEOREMS

We are ready to present the Euclidean version of our main results of Part III. The first is the Euclidean version of Theorem of Rectifiability for varifolds.

Theorem 2.3.1 ([91, Chapter 8, Section 5])**.** *Let $V \in \mathbf{V}_m(U)$ have locally bounded first variation in U and satisfy $\Theta^m(\|V\|, x) > 0$ for $\|V\|$-a.e. $x \in U$. Then $V \in \mathbf{RV}_m(U)$.*

The second is the Euclidean version of Allard's Interior ε-Regularity Theorem.

Theorem 2.3.2 ([91, Chapter 5, Section 5])**.** *Let $m, n \in \mathbb{N}$ and $p \in \mathbb{R}$ satisfy $1 \leq m < n$ and $m < p < \infty$; in the case $m = 1$, assume in addition that $p \geq 2$. Let $\xi \in \mathbb{R}^n$ and $\varepsilon \in (0,1)$. There exist $\delta_0 > 0$ and $\rho_0 > 0$, depending only on m, n, p, and ε, such that if $V \in \mathbf{V}_m(\mathbb{R}^n)$ satisfies*

$$\Theta^m(\|V\|, x) \geq 1,$$

for $\|V\|$-a.e. $x \in B_{g_e}(\xi, \rho)$,

$$\frac{\|V\|(B_{g_e}(\xi, \rho))}{\omega_m \rho^m} \leq 1 + \delta,$$

and

$$\|\overrightarrow{H}\|_{L^p(B_{g_e}(\xi,\rho),\|V\|)} \leq \frac{\delta}{\rho^{1-\frac{m}{p}}},$$

for some $0 < \delta \leq \delta_0$ and $0 < \rho \leq \rho_0$, then $\operatorname{supp}(\|V\|) \cap B_{g_e}(\xi, \varepsilon\rho)$ is the graph of a $\mathcal{C}^{1,1-\frac{m}{p}}$ function over the approximate tangent space at ξ. Moreover, this function has scaling-invariant estimates for its $\mathcal{C}^{1,1-\frac{m}{p}}$-norm, with constants depending only on m, n, p, and ε.

3 ELEMENTS OF RIEMANNIAN GEOMETRY

This chapter collects the background material on Riemannian manifolds that will reappear in Parts II and III. Throughout this work, the reader is assumed to have a working knowledge of the material typically covered in a first graduate course on Riemannian manifolds, as presented, for instance, in do Carmo [35], including smooth manifolds, tangent and cotangent bundles, vector fields and their flows, and the basic theory of differential forms, together with the standard foundations of Riemannian geometry. For the general theory and comparison results, we refer the reader to Petersen's book [82] and Chavel's book [27], while Gray's book [48] provides a reference for the theory of tubes, which is central to our approach.

3.1 BASICS AND COMPARISON GEOMETRY ON RIEMANNIAN MANIFOLDS

Definition 3.1.1. Let M^n be an n-dimensional manifold endowed with a $\mathcal{C}^3$ differentiable structure. A Riemannian metric of class $\mathcal{C}^2$ on M^n is a $\mathcal{C}^2$ assignment to each $x \in M^n$ of an inner product g_x on the tangent space T_xM^n, varying $\mathcal{C}^2$-smoothly with x. The pair (M^n, g) is called a Riemannian manifold of class $\mathcal{C}^2$. In local coordinates $(x^1, \dots, x^n)$, the metric is represented by its components

$$g_{ij} := g\left(\frac{\partial}{\partial x^i}, \frac{\partial}{\partial x^j}\right),$$

which form a symmetric positive-definite matrix of functions. The metric can be written locally as $g = g_{ij}dx^i dx^j$, where the summation convention is understood.

The length of a piecewise $\mathcal{C}^1$ curve $\gamma : [a, b] \to M^n$ is defined by

$$L_g(\gamma) := \int_a^b \sqrt{g_{\gamma(t)}(\gamma'(t), \gamma'(t))}\, dt.$$

The Riemannian distance $d_g(x, y)$ is defined as the infimum of $L_g(\gamma)$ over all piecewise $\mathcal{C}^1$ curves γ joining x to y. If (M^n, g) is connected then the manifold topology is induced by d_g. The metric also induces the Riemannian volume measure vol_g and, for each $m \leq n$, the m-dimensional Hausdorff measure $\mathcal{H}^m_{d_g}$ obtained from the construction in Definition 2.1.3, with the Euclidean distance replaced by the Riemannian distance d_g.

The differentiation of vector fields requires a connection and a Riemannian metric determines a canonical one.

Theorem 3.1.2 ([82, Theorem 2.2.2, p.53]). *On a Riemannian manifold (M^n, g), there exists a unique affine connection ∇^g, called the Levi-Civita connection, that is compatible with the metric, in the sense that*

$$Zg(X,Y) = g(\nabla^g_Z X, Y) + g(X, \nabla^g_Z Y),$$

for all vector fields X, Y, Z on M^n, and is torsion-free, in the sense that

$$\nabla^g_X Y - \nabla^g_Y X = [X,Y],$$

for all vector fields X, Y on M^n.

In local coordinates $(x^1, \dots, x^n)$, the connection is represented by the Christoffel symbols Γ^k_{ij}, defined by

$$\nabla^g_{\frac{\partial}{\partial x^i}} \frac{\partial}{\partial x^j} =: \Gamma^k_{ij} \frac{\partial}{\partial x^k},$$

where, by the Koszul formula,

$$\Gamma^k_{ij} = \frac{1}{2} g^{kl} \left(\frac{\partial}{\partial x^i} g_{jl} + \frac{\partial}{\partial x^j} g_{li} - \frac{\partial}{\partial x^l} g_{ij} \right).$$

The gradient of a differentiable function $f : M^n \to \mathbb{R}$ is the vector field $\nabla^g f$ uniquely determined by

$$g(\nabla^g f, X) = df(X) = X(f),$$

for every vector field X on M^n. The Hessian of a twice differentiable function f on M^n is the symmetric 2-tensor defined by

$$\mathrm{Hess}_g f(X,Y) = g(\nabla^g_X \nabla^g f, Y),$$

for all vector fields X and Y on M^n.

Let $m \leq n$, $\Sigma \subset M^n$ an m-dimensional submanifold and $x \in \Sigma$. The tangential divergence of a vector field X along Σ at x is defined by

$$\mathrm{div}_{T_x\Sigma} X = g_x(\nabla^g_{e_i} X, e_i),$$

where $\{e_i\}_{i=1}^n$ is an orthonormal basis of $T_x\Sigma$ and $P_{T_x\Sigma}$ denote the orthogonal projection onto $T_x\Sigma$.

Definition 3.1.3. A smooth curve γ on (M^n, g) is called a geodesic if $\nabla^g_{\gamma'} \gamma' \equiv 0$.

Geodesics are locally length-minimizing, and in local coordinates they are characterized as solutions of a second-order system of ordinary differential equations. Given $x \in M^n$ and $v \in T_x M^n$, there exists a unique geodesic γ_v satisfying $\gamma_v(0) = x$, and $\gamma_v'(0) = v$.

Definition 3.1.4. For $x \in M^n$, the exponential map at x is defined by $\exp_x(v) := \gamma_v(1)$ for v in a sufficiently small neighborhood of $0 \in T_x M^n$.

For sufficiently small $r > 0$, let $B_{g_x}(0,r) \subset T_p M$ denote the open ball of radius r centered at 0 with respect to the norm induced by g_x. The restriction $\exp_{x \restriction B_{g_x}(0,r)}$ is a diffeomorphism onto its image,

which is a neighborhood of $x \in M^n$. The largest scale for which this property holds is measured by the injectivity radius.

Definition 3.1.5. Let $x \in M$. The injectivity radius at x is defined by

$$\mathrm{inj}_g(x) := \sup\left\{r > 0 : \exp_x : B_{g_x}(0,r) \subset T_pM \to M \text{ is a diffeomorphism onto its image}\right\}.$$

And the injectivity radius of (M, g) is $\mathrm{inj}(M^n, g) := \inf_{x\in M} \mathrm{inj}_g(x)$.

Definition 3.1.6. For $x \in M^n$ and $0 < r < \mathrm{inj}_g(x)$, the geodesic ball of radius r centered at x is defined by $B_g(x,r) := \exp_x(B_{g_x}(0,r))$.

Let $y \in M^n$ and $0 < r < \mathrm{inj}_g(y)$. Then $\exp_y : B_{g_y}(0,r) \to B_g(y,r)$ is a diffeomorphism. Consequently, the distance function $u_y(\cdot) := d_g(y,\cdot)$ is smooth on $B_g(y,r) \setminus \{y\}$ and satisfies $|\nabla^g u_y|_g \equiv 1$. Moreover, the integral curves of $\nabla^g u_y$ are the unit-speed geodesics issuing from y.

Curvature measures the failure of second covariant derivatives to commute and, equivalently, the rate at which geodesics diverge or converge.

Definition 3.1.7. The Riemann curvature tensor is the $(1,3)$-tensor field R^g defined by

$$R^g(X,Y)Z := \nabla^g_X\nabla^g_Y Z - \nabla^g_Y\nabla^g_X Z - \nabla^g_{[X,Y]}Z,$$

for vector fields X, Y, Z on M^n. The associated $(0,4)$-tensor is $R^g(X,Y,Z,W) = g(R^g(X,Y)Z,W)$, for vector fields X, Y, Z, W on M^n.

Definition 3.1.8. Let $x \in M^n$. For a 2-dimensional subspace $\sigma \subset T_xM^n$ spanned by linearly independent vectors $X, Y \in T_xM^n$, the sectional curvature of σ is defined by

$$\mathrm{sec}_g(\sigma) := \frac{R^g(X,Y,Y,X)}{g(X,X)g(Y,Y) - g(X,Y)^2}.$$

Definition 3.1.9. The Ricci curvature tensor Ric_g is the $(0,2)$-tensor obtained by tracing R^g over its first and last arguments:

$$\mathrm{Ric}_g(Y,Z) := R^g(e_i,Y,Z,e_i)$$

for vector fields Y, Z on M^n, where $\{e_1,\dots,e_n\}$ is an orthonormal basis of T_xM^n.

Definition 3.1.10. The scalar curvature Scal_g at a point $x \in M^n$ is the trace of the Ricci tensor with respect to g:

$$\mathrm{Scal}_g(x) := \mathrm{Ric}_g(e_i,e_i),$$

where $\{e_1,\dots,e_n\}$ is an orthonormal basis of T_xM^n.

The geometric content of curvature bounds is rendered quantitative by comparison theory, by means of the model functions that solve the associated Jacobi and Riccati equations. For $b \in \mathbb{R}$, let

$s_b : [0, \pi/\sqrt{b}) \to \mathbb{R}$ (with the domain understood to be $[0, \infty)$ when $b \leq 0$) denote the solution of the Jacobi equation

$$s_b'' + b\, s_b = 0, \qquad s_b(0) = 0, \quad s_b'(0) = 1,$$

explicitly given by

$$s_b(t) = \begin{cases} \dfrac{1}{\sqrt{b}} \sin(\sqrt{b}\, t), & b > 0, \\ t, & b = 0, \\ \dfrac{1}{\sqrt{-b}} \sinh(\sqrt{-b}\, t), & b < 0. \end{cases}$$

For $b > 0$, s_b is positive on $(0, \pi/\sqrt{b})$ and vanishes at $t = \pi/\sqrt{b}$; for $b \leq 0$, s_b is positive on all of $(0, \infty)$. On this common domain of positivity, define the associated cotangent-type function

$$\cot_b(t) = \frac{s_b'(t)}{s_b(t)}, \tag{4}$$

which solves the Riccati equation $\cot_b' + \cot_b^2 + b = 0$. Finally, set

$$c(t, b) = t \cot_b(t).$$

Although $\cot_b(t) \sim 1/t$ as $t \to 0^+$, the product $c(t, b) = t \cot_b(t)$ extends continuously to $t = 0$ with $c(0, b) := 1$ for every $b \in \mathbb{R}$; this normalization is what makes $c(t, b) - 1$ vanishes at the origin and behave as a genuine curvature-correction term in the Riemannian version of Theorem 2.2.9.

The technical adaptation from the Euclidean to the Riemannian monotonicity formula consists in controlling the Hessian of the distance function under sectional curvature bounds.

Theorem 3.1.11 ([65, Theorem 11.7, p. 327])**.** *Let (M^n, g) be an n-dimensional Riemannian manifold of class C^2 with $|\sec_g| \leq K$. Let $y \in M^n$, $0 < r < \operatorname{inj}_g(y)$, and write $u_y(\cdot) = d_g(y, \cdot)$. Then, for all $x \in B_g(y, r) \setminus \{y\}$,*

$$\cot_K(u_y(x))\, g \leq \operatorname{Hess}_g u_y(x) \leq \cot_{-K}(u_y(x))\, g.$$

This is the standard two-sided Hessian comparison, with $\cot_K(u_y(x))$ and $\cot_{-K}(u_y(x))$ as the values attained on the model spaces of constant curvature K and $-K$, respectively. Taking the trace over an m-dimensional subspace $S \subset T_x M^n$ and including the radial term $u_y \operatorname{Hess}_g u_y(\nabla^g u_y, \nabla^g u_y)$ gives the two-sided estimate

$$m\, c(u_y(x), K) \leq \operatorname{div}_S(u_y \nabla^g u_y)(x) \leq m\, c(u_y(x), -K).$$

Since $c(t, \pm K) \to 1$ as $K \to 0$, all curvature corrections vanish in the flat limit, and one recovers the Euclidean identities.

3.2 SUBMANIFOLDS

In a curved space we use submanifolds to measure excess instead of Euclidean planes. In this section we introduce the terminology and basic results on submanifold geometry.

Definition 3.2.1. Let $\Sigma \subset M^n$ be an m-dimensional $\mathcal{C}^2$ submanifold and $x \in \Sigma$. The shape operator with respect to a unit normal vector $\nu \in (T_x\Sigma)^\perp$ is the self-adjoint map $S_{\nu,x} : T_x\Sigma \to T_x\Sigma$ given by $S_{\nu,x}(X) := P_{T_x\Sigma}(\nabla^g_X \nu)$, where ν is extended to a local unit normal field near x. The second fundamental form is the symmetric bilinear form $II_{\nu,x}(X,Y) := g(S_{\nu,x}(X), Y)$, for all $X, Y \in T_x\Sigma$.

Since $S_{\nu,x}$ is self-adjoint, its eigenvalues $\kappa_1, \ldots, \kappa_m$ are real; they are called the principal curvatures of Σ at x in the direction ν, and the corresponding eigenvectors the principal curvature directions. For an affine plane $\Sigma \subset \mathbb{R}^n$, $\kappa_i = 0$ for all i, so the curvature-dependent terms of the Riemannian Caccioppoli inequality (see Lemma 13.2.1) vanish identically in the Euclidean case.

The deviation of a varifold from a model submanifold Σ is measured using Fermi coordinates on a neighborhood of Σ (see [48]) together with the distance function $u_\Sigma(\cdot) := \operatorname{dist}_g(\cdot, \Sigma)$.

Definition 3.2.2. Let $\Sigma \subset M^n$ be an m-dimensional $\mathcal{C}^2$ submanifold. Consider the normal exponential map of Σ,

$$\begin{aligned} E : N\Sigma &\to M^n \\ (x, v) &\mapsto E(x, v) := \exp_x(v), \end{aligned}$$

where $N\Sigma := \{(x, v) : x \in \Sigma,\ v \in (T_x\Sigma)^\perp\}$. The normal injectivity radius of Σ is

$$\operatorname{inj}^\perp(\Sigma) := \sup \Big\{ r \geq 0 : E \text{ restricted to } \{(x, v) \in N\Sigma : |v|_g < r\} \text{ is a diffeomorphism onto its image} \Big\}.$$

For $r < \operatorname{inj}^\perp(\Sigma)$, the restriction of E to the disk bundle $\{(x, v) \in N\Sigma : |v|_g \leq r\}$ is injective, and its image is a tube.

Definition 3.2.3. Let $\Sigma \subset M^n$ be an m-dimensional $\mathcal{C}^2$ submanifold and $0 < r < \operatorname{inj}^\perp(\Sigma)$. The tube of radius r around Σ is

$$\operatorname{Tube}(\Sigma, r) := \bigcup_{x \in \Sigma} \{\exp_x(v) : v \in (T_x\Sigma)^\perp,\ |v|_g \leq r\}.$$

For $0 < t \leq r$, the tubular hypersurface at distance t is $\Sigma_t := \{x \in \operatorname{Tube}(\Sigma, r) : \operatorname{dist}_g(x, \Sigma) = t\}$.

A chart on Σ and a orthonormal frame normal do Σ induce coordinates on $\operatorname{Tube}(\Sigma, r)$ as follows.

Definition 3.2.4. Let $\Sigma \subset M^n$ be an m-dimensional $\mathcal{C}^2$ submanifold, $x \in \Sigma$, and let $(y^1, \ldots, y^m)$ be a coordinate system on an open set $V \subset \Sigma$ containing x, together with orthonormal sections $\nu_{m+1}, \ldots, \nu_n$

of $(T\Sigma)^\perp$ over V. The Fermi coordinate system $(y^1, \dots, y^m, z^{m+1}, \dots, z^n)$ on a neighborhood $U \subset \mathrm{Tube}(\Sigma, r)$ of x, relative to $(y^1, \dots, y^m)$ and $\nu_{m+1}, \dots, \nu_n$, are defined by

$$y^a\left(\exp_{x'}\left(\sum_{j=m+1}^{n} t^j \nu_j(x')\right)\right) = y^a(x') \qquad (a = 1, \dots, m),$$

$$z^i\left(\exp_{x'}\left(\sum_{j=m+1}^{n} t^j \nu_j(x')\right)\right) = t^i \qquad (i = m+1, \dots, n),$$

for $x' \in V$, provided $t^{m+1}, \dots, t^n$ are small enough that $\sum_{j=m+1}^n t^j \nu_j(x') \in (T_{x'}\Sigma)^\perp$ has norm less than $\mathrm{inj}^\perp(\Sigma)$.

The normal block $(z^{m+1}, \dots, z^n)$ of this coordinate system defines a radial function σ and a radial vector field N on the tube.

Definition 3.2.5. Let $(y^1, \dots, y^m, z^{m+1}, \dots, z^n)$ be the Fermi coordinate system for an m-dimensional $\mathcal{C}^2$ submanifold $\Sigma \subset M^n$. For $\sigma > 0$ we set

$$\sigma^2 = \sum_{i=m+1}^{n} (z^i)^2 \qquad \text{and} \qquad N = \sum_{i=m+1}^{n} \frac{z^i}{\sigma} \frac{\partial}{\partial z^i}.$$

The next two lemmas identify σ and N with u_Σ and $\nabla^g u_\Sigma$, respectively.

Lemma 3.2.6 ([48, Lemma 2.7, p. 23])**.** *Let $\Sigma \subset M^n$ be an m-dimensional $\mathcal{C}^2$ submanifold, $0 \le r < \mathrm{inj}^\perp(\Sigma)$ and $x \in \mathrm{Tube}(\Sigma, r)$. Suppose there is a unique unit-speed geodesic γ from Σ to x meeting Σ orthogonally. Then*

$$\sigma(x) = u_\Sigma(x)$$

and, for all $0 < s \le r$,

$$N(\gamma(s)) = \gamma'(s).$$

Consequently, σ is defined on $\mathrm{Tube}(\Sigma, r)$, and N is defined on $\mathrm{Tube}(\Sigma, r) \setminus \Sigma$.

Lemma 3.2.7 ([48, Lemma 2.11, p. 25])**.** *Let $\Sigma \subset M^n$ be an m-dimensional $\mathcal{C}^2$ submanifold and $0 \le r < \mathrm{inj}^\perp(\Sigma)$. Then, on $\mathrm{Tube}(\Sigma, r) \setminus \Sigma$, we have $N = \nabla^g u_\Sigma$.*

The shape operator $S(t)$ of the tubular hypersurface Σ_t along a normal geodesic is governed by the following Riccati equation,

$$S'(t) = -S^2(t) - R^g(t), \qquad \kappa_j'(t) = -\kappa_j^2(t) - \sec_g(F_j(t), N) = 0,$$

with $R^g(t) := R^g(\cdot, N)N$, and $F_1(t), \dots, F_{n-1}(t)$ a parallel orthonormal frame of eigenvectors of $S(t)$, with corresponding eigenvalues $\kappa_1(t), \dots, \kappa_{n-1}(t)$. Each $\kappa_j(t)$ thus solves a scalar Riccati equation with prescribed initial value $\kappa_j(0)$, in contrast with $\cot_b$ of Equation (4), which is pinned to the singular initial behavior $\cot_b(t) \sim 1/t$ as $t \to 0^+$. This requires the more general model function $\cot_{K,\kappa}$, solving the same Riccati equation with an arbitrary initial condition $\kappa \in \mathbb{R}$ in place of that singularity.

For $\kappa \in \mathbb{R}$ and $K > 0$, and for $t \in \left[0, \frac{1}{\sqrt{K}}\left(\pi - \cot^{-1}\left(\frac{\kappa}{\sqrt{K}}\right)\right)\right[$ set

$$\cot_{K,\kappa}(t) := \sqrt{K}\cot\left(\sqrt{K}\,t + \cot^{-1}\left(\frac{\kappa}{\sqrt{K}}\right)\right).$$

By construction, $\cot_{K,\kappa}$ solves $\cot'_{K,\kappa}(t) = -\cot^2_{K,\kappa}(t) - K$ with $\cot_{K,\kappa}(0) = \kappa$. As $\kappa \to \infty$, $\cot_{K,\kappa}$ converges point-wise to $\cot_K$ of Equation (4), recovering the point-comparison model as the case in which Σ degenerates to a point and the initial principal curvature diverges.

3.3 HARMONIC RADIUS AND HARMONIC COORDINATES

Working at the regularity $\mathcal{C}^2$ requires coordinates in which the metric retains sufficient analytical control. Harmonic coordinates do that.

Definition 3.3.1. A coordinate system $(x^1, \dots, x^n)$ on a Riemannian manifold (M^n, g) is harmonic if $\Delta_g x^k = 0$ for all $k = 1, \dots, n$.

Since the manifolds under consideration are only assumed to be of class $\mathcal{C}^2$, it is convenient to fix from now on a Hölder exponent $\alpha \in (0,1)$ on which the harmonic radius is based. This is legitimate because the embedding $\mathcal{C}^2 \subset \mathcal{C}^{1,\alpha}$ for every $\alpha \in (0,1)$ (see [39, Chapter 5] or [47, Chapter 4]). The endpoint $\alpha = 1$ is excluded, since Hebey–Herzlich's harmonic radius is only defined for $\alpha \in (0,1)$, the Schauder theory underlying its construction (in particular the regularity theorem below) failing at the borderline exponent $\alpha = 1$.

Definition 3.3.2. Let (M^n, g) be a Riemannian manifold of class $\mathcal{C}^2$ without boundary, $x \in M^n$, $Q > 1$, and $\alpha \in (0,1)$. The harmonic radius at x with constants (Q, α) is the largest number $r_H = r_H(Q,\alpha)(x)$ such that, on the geodesic ball $B_g(x, r_H)$, there is a harmonic coordinate system in which the metric tensor is $C^{1,\alpha}$-controlled. Namely, if g_{ij}, $i, j = 1, \dots, n$, are the components of g in these coordinates, then

1. $Q^{-1}\delta_{ij} \le g_{ij} \le Q\,\delta_{ij}$ as bilinear forms,
2. $r_H \sup_y |\partial g_{ij}(y)| + r_H^{1+\alpha} \sup_{y \neq z} \dfrac{|\partial g_{ij}(y) - \partial g_{ij}(z)|}{\mathrm{dist}_g(y,z)^\alpha} \le Q - 1.$

The harmonic radius $r_H(Q,\alpha)(M^n, g)$ of (M^n, g) is defined by $r_H(Q,\alpha)(M^n, g) := \inf_{x \in M^n} r_H(Q,\alpha)(x)$.

Fixing $Q > 1$ and $\alpha \in (0,1)$ from now on, we write $r_H(M^n, g) := r_H(Q,\alpha)(M^n, g)$. Curvature bounds control this radius from below as follows.

Theorem 3.3.3 ([53, Theorem 6, p.577])**.** *Let (M^n, g) be a complete $\mathcal{C}^2$ Riemannian manifold satisfying $|\mathrm{Ric}_g| \le K_1$ (a consequence of $|\mathrm{sec}_g| \le K$) and $\mathrm{inj}_g \ge i > 0$. Then $r_H(M^n, g) > 0$, with a lower bound depending only on the dimension n, the Ricci bound K_1, and the injectivity radius bound i.*

In harmonic coordinates the metric components satisfy elliptic equations with favorable structure, so that elliptic regularity theory [39, Chapter 5] improves the regularity of geometric quantities constructed from g. This is what permits the construction, in Part III, of the harmonic m-dimensional submanifolds — the curved substitutes for affine planes — and the control of their principal curvatures by $o(1)$ as the scale tends to zero.

4 THEORY OF GENERALIZED EXISTENCE OF ISOPERIMETRIC REGIONS

This chapter collects the background material on sets of finite perimeter, synthetic Ricci curvature bounds, and generalized existence theory for isoperimetric regions on which Part II is based. The standard references are Ambrosio–Fusco–Pallara's book [10] for sets of finite perimeter and functions of bounded variation, Morgan's book [70] for the classical theory of isoperimetric problems, and Villani's book [96], together with the foundational papers of Lott–Villani [66], Sturm [92, 93], and Ambrosio–Gigli–Savaré [12], for the synthetic theory of Ricci curvature bounds on metric measure spaces. The reader is assumed to have the same background in measure theory as in Chapter 2, together with basic familiarity with metric measure spaces and Gromov–Hausdorff convergence.

4.1 SETS OF FINITE PERIMETER AND THE ISOPERIMETRIC PROFILE

Throughout this section (X, d, m) denotes a complete, separable metric space endowed with a Radon measure m, referred to as a metric measure space. We first recall the notion of perimeter, which extends the classical Euclidean notion of surface area to sets that need not have smooth (or even rectifiable) boundary, via a relaxation procedure.

Definition 4.1.1. Let (X, d, m) be a metric measure space, $\Omega \subset X$ an open set, and $E \subset X$ a Borel set. The perimeter of E in Ω is defined by

$$P(E, \Omega) := \inf \left\{ \liminf_{i \to \infty} \int_\Omega \operatorname{Lip} u_i \, d\mathrm{m} \ : \ u_i \in \operatorname{Lip}_{\mathrm{loc}}(\Omega),\ u_i \to \chi_E \text{ in } L^1_{loc}(\Omega, \mathrm{m}) \right\},$$

where $\operatorname{Lip} u_i(x) := \limsup_{y \to x} \dfrac{|u_i(y) - u_i(x)|}{d(x, y)}$ denotes the local Lipschitz constant of u_i at x. We say that E has finite perimeter in Ω if $P(E, \Omega) < \infty$, and write $P(E) := P(E, X)$.

Remark 4.1.2. When $(X, d, \mathrm{m}) = (\mathbb{R}^n, d_{g_e}, \mathcal{L}^n)$, this definition of perimeter recovers the classical notion of perimeter of a set of finite perimeter in the sense of De Giorgi, and $P(E, \Omega)$ coincides with the total variation $|D\chi_E|(\Omega)$ of the distributional gradient of χ_E; when ∂E is smooth, $P(E, \Omega) = \mathcal{H}^{n-1}(\partial E \cap \Omega)$.

The functional $\Omega \mapsto P(E, \Omega)$ extends to a Borel measure on X whenever E has finite perimeter, called the perimeter measure of E, and denoted by $P(E, \cdot)$; this is the metric-measure analogue of the Gauss–Green measure of a set of finite perimeter in $\mathbb{R}^n$. We say a sequence (E_i) of sets of

finite perimeter converges to E in the sense of sets of finite perimeter if $\chi_{E_i} \to \chi_E$ in $L^1_{loc}(X, \mathrm{m})$ and $P(E_i, \cdot) \overset{*}{\rightharpoonup} P(E, \cdot)$; lower semicontinuity of the perimeter under L^1_{loc}-convergence, together with a suitable compactness statement for the class of sets of finite perimeter with uniformly bounded perimeter and volume, underlies the direct method used in existence proofs for isoperimetric regions.

The isoperimetric problem consists in minimizing perimeter among sets of prescribed measure. The relevant invariant is the isoperimetric profile function.

Definition 4.1.3. Let (X, d, m) be a metric measure space with $\mathrm{m}(X) = \infty$. For $v > 0$, the isoperimetric profile of (X, d, m) is defined by

$$I_{X,d}(v) := \inf\{P(E) : E \subset X \text{ Borel}, \mathrm{m}(E) = v\}.$$

A Borel set $E \subset X$ with $\mathrm{m}(E) = v$ and $P(E) = I_{X,d}(v)$ is called an isoperimetric region of volume v.

The isoperimetric profile of a general metric measure space need not be achieved by a single set, nor need a minimizing sequence converge, even along subsequences, to a set of finite perimeter of the same volume; this failure of compactness is precisely the obstruction that generalized existence theory is designed to overcome.

4.2 ALEXANDROV, CD AND RCD SPACES

Curvature-dimension conditions provide a synthetic, purely metric-measure-theoretic formulation of a lower bound on the Ricci curvature and an upper bound on the dimension, which reduces to the classical notion on smooth Riemannian manifolds and is stable under measured Gromov–Hausdorff convergence.

Before advancing to the RCD condition, it is worth recalling briefly the notion of curvature-dimension condition. The idea of encoding a lower Ricci curvature bound synthetically, in terms of convexity properties of entropy functionals along Wasserstein geodesics, goes back to the independent contributions of Sturm [92, 93] and of Lott–Villani [66], who introduced the $\mathrm{CD}(K, N)$ condition for general metric measure spaces. Because of our purposes, we just outline the main references on the subject and refer the interested reader to the survey of Ambrosio [8] and the references therein. We adopt here the definition of CD space as in [8, Definition 5.4, p. 18] and of RCD space as in [8, Definition 7.4, p. 24].

Remark 4.2.1. A smooth, complete Riemannian manifold (M^n, g), equipped with the Riemannian distance d_g and the volume measure vol_g, satisfies $\mathrm{RCD}(K, n)$ if and only if $\mathrm{Ric}_g \geq Kg$.

The dimensional bound in the definition of $\mathrm{RCD}(K, N)$ is compatible with a well-defined notion of essential dimension, and yields a rigidity statement for the volume of small metric balls.

Definition 4.2.2. An $\mathrm{RCD}(K, N)$ space (X, d, m), with $N < \infty$, is said to be non-collapsing if $\mathrm{m} = \mathcal{H}^N_d$.

We first recall the notions of pointed Gromov–Hausdorff convergence and pointed measured Gromov–Hausdorff convergence and two basic facts concerning RCD(K, N) spaces. We will then be in a position to state the behavior of non-collapsed RCD(K, n) spaces under Gromov–Hausdorff convergence.

Definition 4.2.3. A pointed metric space is a triple (X, d, p) where (X, d) is a metric space and $p \in X$ is a distinguished basepoint. A pointed metric measure space is a quadruple (X, d, m, p) where (X, d, m) is a metric measure space and $p \in X$.

Definition 4.2.4. A sequence $\{(X_i, d_i, p_i)\}_{i\in\mathbb{N}}$ of pointed metric spaces is said to converge in the pointed Gromov–Hausdorff topology, pGH sense for short, to a pointed metric space $(X_\infty, d_\infty, p_\infty)$ if there exist a complete separable metric space (Z, d_Z) and isometric embeddings

$$\Psi_i : (X_i, d_i) \to (Z, d_Z), \qquad \forall i \in \mathbb{N},$$

$$\Psi : (X_\infty, d_\infty) \to (Z, d_Z),$$

such that for any $\varepsilon, R > 0$ there is $i_0(\varepsilon, R) \in \mathbb{N}$ such that

$$\Psi_i(B_R^{X_i}(p_i)) \subset [\Psi(B_R^{X_\infty}(p_\infty))]_\varepsilon, \qquad \Psi(B_R^{X_\infty}(p_\infty)) \subset [\Psi_i(B_R^{X_i}(p_i))]_\varepsilon,$$

for any $i \geq i_0$, where $[A]_\varepsilon := \{z \in Z : d_Z(z, A) \leq \varepsilon\}$ for any $A \subset Z$.

Let m_i and m_∞ be given in such a way that $(X_i, d_i, \mathrm{m}_i, p_i)$ and $(X_\infty, d_\infty, \mathrm{m}_\infty, p_\infty)$ are pointed metric measure spaces. If, in addition to the previous requirements, we also have

$$(\Psi_i)_\# \mathrm{m}_i \overset{*}{\rightharpoonup} \Psi_\# \mathrm{m}_\infty$$

with respect to the duality with continuous bounded functions on Z with bounded support, then the convergence is said to hold in the pointed measured Gromov–Hausdorff topology, pmGH sense for short.

Remark 4.2.5. We recall that, whenever it exists, a pmGH limit of a sequence $\{(X_i, d_i, \mathrm{m}_i, x_i)\}_{i\in\mathbb{N}}$ of (pointed) RCD(K, N) spaces is still an (pointed) RCD(K, N) metric measure space.

Remark 4.2.6. Here we recall the synthetic variant of Gromov's precompactness theorem for RCD spaces, see [34, Equation (2.1)]. Let $\{(X_i, d_i, \mathrm{m}_i, x_i)\}_{i\in\mathbb{N}}$ be a sequence of RCD(K_i, N) spaces with $N \in [1, +\infty)$, $\mathrm{supp}(\mathrm{m}_i) = X_i$ for every $i \in \mathbb{N}$, $\mathrm{m}_i(B_1(x_i)) \in [v, v^{-1}]$ for some $v \in (0, 1)$ and for every $i \in \mathbb{N}$, and $K_i \to K \in \mathbb{R}$. Then there exists a subsequence pmGH-converging to some RCD(K, N) space (X, d, m, x) with $\mathrm{supp}(\mathrm{m}) = X$.

This closure property ensures that the pieces produced by the generalized existence argument described below remain within the same class of spaces on which the isoperimetric problem is originally posed.

Alongside these synthetic lower Ricci bounds, we will also make use of Alexandrov's notion of a sectional curvature bound, formulated in terms of comparison inequalities for geodesic triangles. We refer the reader to [25, Chapter 4 and 10] for the precise formulation.

Definition 4.2.7. Let $0 \geq \bar{k} \geq \underline{k}$ be real numbers and (X, d) a complete and locally compact metric space. We say that:

1. (X, d) is an Alexandrov space with curvature bounded from above by $\bar{k}$ if it is CAT($\bar{k}$) and bounded from below by $\underline{k}$ if it is CBB($\underline{k}$);

2. $(X, d, \mathcal{H}^n_d)$ is non-collapsing if there exists a real number c_0 such that $\inf_{x\in X} \mathcal{H}^n_d(B_d(x, 1)) \geq c_0 > 0$.

Remark 4.2.8. Both notions of non-collapsing express the same idea, that the reference measure does not degenerate with respect to the Hausdorff measure of the correct dimension, but they do so in different ways, reflecting the different roles the measure plays in the two theories. In the RCD(K, N) setting the measure m is part of the data of the space, a priori only compatible with the CD(K, N) condition and not assumed equal to a Hausdorff measure, so non-collapsing is stated as the global identity $\mathrm{m} = \mathcal{H}^N_d$; by the Bishop–Gromov inequality, this identity already forces local uniform control on the volume of unit balls. In the Alexandrov setting, by contrast, the measure is fixed to be $\mathcal{H}^n_d$ from the outset, so non-collapsing cannot be expressed as such an identity and is instead formulated directly as the quantitative requirement $\inf_{x\in X} \mathcal{H}^n_d(B_d(x, 1)) \geq c_0 > 0$, playing the same degeneration-excluding role that $\mathrm{m} = \mathcal{H}^N_d$ plays automatically in the RCD framework.

The synthetic Ricci and sectional curvature conditions are compatible, in the following precise sense, which allows results proved under an RCD lower bound to be applied to Alexandrov spaces with a two-sided curvature bound.

Theorem 4.2.9 ([83, 57])**.** *Let (X, d) be a non-collapsing n-dimensional Alexandrov space with curvature bounded from below by $\underline{k}$ and from above by $\bar{k}$. Then $(X, d, \mathcal{H}^n_d)$ is a non-collapsing* RCD$((n-1)\underline{k}, n)$ *space.*

4.3 GENERALIZED EXISTENCE OF ISOPERIMETRIC REGIONS

We now turn to describing the behavior of perimeter minimizing sequences for the isoperimetric problem on noncompact RCD(K, N) spaces, following [14]. We first recall the notion of L^1-strong and L^1_{loc} convergence of sets defined on a sequence of metric measure spaces converging in the pmGH sense, which is needed to state the results below.

Definition 4.3.1 (L^1-strong and L^1_{loc} convergence)**.** Let $\{(X_i, d_i, \mathrm{m}_i, x_i)\}_{i\in\mathbb{N}}$ be a sequence of pointed metric measure spaces converging in the pmGH sense to a pointed metric measure space (Y, d_Y, μ, y) and let (Z, d_Z) be a realization as in Definition 4.2.4.

We say that a sequence of Borel sets $E_i \subset X_i$ such that $\mathrm{m}_i(E_i) < +\infty$ for any $i \in \mathbb{N}$ converges in the L^1-strong sense to a Borel set $F \subset Y$ with $\mu(F) < +\infty$ if $\mathrm{m}_i(E_i) \to \mu(F)$ and $\chi_{E_i}\mathrm{m}_i \to \chi_F\mu$ with respect to the duality with continuous bounded functions with bounded support on Z.

We say that a sequence of Borel sets $E_i \subset X_i$ converges in the L^1_{loc}-sense to a Borel set $F \subset Y$ if $E_i \cap B_R(x_i)$ converges to $F \cap B_R(y)$ in L^1-strong for every $R > 0$.

Building on the mass splitting techniques developed in [75, 15], the following description of the behavior of perimeter minimizing sequences for the isoperimetric problem on RCD spaces is obtained, generalizing and improving previous results.

Theorem 4.3.2 ([14]). *Let $K \leq 0$ and $N \geq 2$. Let $(X, d, \mathcal{H}^N)$ be a noncompact* RCD(K, N) *space. Assume there exists $v_0 > 0$ such that $\mathcal{H}^N(B_1(x)) \geq v_0$ for every $x \in X$. Let $V > 0$. For every minimizing (for the perimeter) sequence of bounded sets $\Omega_i \subset X$ of volume V, up to passing to a subsequence, there exist a nondecreasing bounded sequence $\{N_i\}_{i\in\mathbb{N}} \subseteq \mathbb{N}$, disjoint finite perimeter sets $\Omega_i^c, \Omega_{i,j}^d \subset \Omega_i$, and points $p_{i,j}$, with $1 \leq j \leq N_i$ for any i, such that the following claims hold*

- $\lim_i d(p_{i,j}, p_{i,\ell}) = \lim_i d(p_{i,j}, o) = +\infty$, *for any $j \neq \ell \leq \overline{N}$ and any $o \in X$, where $\overline{N} := \lim_i N_i < +\infty$;*
- Ω_i^c *converges to $\Omega \subset X$ in the sense of finite perimeter sets, $\mathcal{H}^N(\Omega_i^c) \to_i \mathcal{H}^N(\Omega)$, and $P(\Omega_i^c) \to_i P(\Omega)$. Moreover Ω is an isoperimetric region in X;*
- *for every $0 < j \leq \overline{N}$, $(X, d, \mathcal{H}^N, p_{i,j})$ converges in the pmGH sense to a pointed* RCD(K, N) *space $(X_j, d_j, \mathcal{H}^N, p_j)$. Moreover there are isoperimetric regions $Z_j \subset X_j$ such that $\Omega_{i,j}^d \to_i Z_j$ in L^1-strong and $P(\Omega_{i,j}^d) \to_i P(Z_j)$;*
- *it holds that*

$$I_X(V) = P(\Omega) + \sum_{j=1}^{\overline{N}} P(Z_j), \qquad V = \mathcal{H}^N(\Omega) + \sum_{j=1}^{\overline{N}} \mathcal{H}^N(Z_j).$$

Roughly speaking, the mass of a minimizing sequence splits into at most finitely many pieces, and it is totally recovered by finitely many isoperimetric regions sitting in spaces "located at infinity" with respect to the original ambient space. The preceding theorem provides a general tool for treating the isoperimetric problem by direct method, from which one can recover the main existence and nonexistence results previously known in the literature, suitably extended to the nonsmooth RCD setting.

This asymptotic mass decomposition result follows from a new generalized compactness theorem for sequences of sets with uniformly bounded volume and perimeter, which we now state.

Theorem 4.3.3 ([14]). *Let $K \in \mathbb{R}$ and $N \geq 2$. Let $(X_i, d_i, \mathcal{H}^N)$ be a sequence of* RCD(K, N) *spaces, and let $E_i \subset X_i$ be bounded sets of finite perimeter such that $\sup_i P(E_i) + \mathcal{H}^N(E_i) < +\infty$. Assume there exists $v_0 > 0$ such that $\mathcal{H}^N(B_1(x)) \geq v_0$ for every $x \in X_i$, and for every i.*

Then, up to subsequence, there exist a nondecreasing, possibly unboundend, sequence $\{N_i\}_{i\in\mathbb{N}} \subseteq \mathbb{N}_{\geq 1}$, points $p_{i,j} \in X_i$, with $1 \leq j \leq N_i$ for any i, and pairwise disjoint subsets $E_{i,j} \subset E_i$ such that

- $\lim_i d_i(p_{i,j}, p_{i,\ell}) = +\infty$, *for any $j \neq \ell < \overline{N} + 1$, where $\overline{N} := \lim_i N_i \in \mathbb{N} \cup \{+\infty\}$;*
- *For every $1 \leq j < \overline{N} + 1$, the sequence $(X_i, d_i, \mathcal{H}^N, p_{i,j})$ converges in the pmGH sense to a pointed* RCD(K, N) *space $(Y_j, d_{Y_j}, \mathcal{H}^N, p_j)$ as $i \to +\infty$;*

- *there exist sets $F_j \subset Y_j$ such that $E_{i,j} \to_i F_j$ in L^1-strong and there holds*

$$\lim_i \mathcal{H}^N(E_i) = \sum_{j=1}^{\overline{N}} \mathcal{H}^N(F_j),$$

$$\sum_{j=1}^{\overline{N}} P(F_j) \le \liminf_i P(E_i).$$

Moreover, if E_i is an isoperimetric set in X_i for any i, then F_j is an isoperimetric set in Y_j for any $j < \overline{N} + 1$ and

$$P(F_j) = \lim_i P(E_{i,j}),$$

for any $j < \overline{N} + 1$.

We observe that in the first part of the previous theorem, a priori minimality property is not required on the sequence of sets E_i.

Exploiting the properties of the isoperimetric profile of $\mathrm{RCD}(K, N)$ spaces established in [17], one can moreover obtain a more explicit upper bound on the number $\overline{N}$ appearing in the asymptotic mass decomposition theorem, which turns out to be bounded from above linearly in terms of the volume V.

Corollary 4.3.4 ([14])**.** *Let $K \le 0$, $N \ge 2$, and $v_0 > 0$. Then there exists $\varepsilon = \varepsilon(K, N, v_0) > 0$ such that the following holds.*

Let $(X, d, \mathcal{H}^N)$ be a noncompact $\mathrm{RCD}(K, N)$ space such that $\mathcal{H}^N(B_1(x)) \ge v_0$ for every $x \in X$. Let $V > 0$ and let $\Omega_i \subset X$ be a minimizing (for the perimeter) sequence of bounded sets of volume V. Letting $\overline{N}$ be given by the asymptotic mass decomposition theorem, then

$$\overline{N} \le 1 + \frac{V}{\varepsilon}.$$

Finally, employing the mass splitting result for minimizing sequences, one can also give an equivalent condition for a minimizing sequence for the isoperimetric problem to converge to an isoperimetric set without losing mass at infinity. This abstract condition involves a function I_X^∞, which can be interpreted as the isoperimetric profile at infinity.

Theorem 4.3.5 ([14])**.** *Let $K \le 0$, and let $N \ge 2$. Let $(X, d, \mathcal{H}^N)$ be a noncompact $\mathrm{RCD}(K, N)$ space. Assume that the isoperimetric profile I_X is continuous.*

Let $V \in (0, \mathcal{H}^N(X))$. Then the following are equivalent.

(1) $I_X(V) < I_X(V_1) + I_X^\infty(V_2)$ for all $V_1 + V_2 = V$ with $V_2 \in (0, V]$.

(2) For any sequence $\Omega_k \subset X$ such that $\mathcal{H}^N(\Omega_k) \to V$ and $P(\Omega_k) \to I_X(V)$, there exists a subsequence converging in $L^1(X)$.

Item (2) (or, equivalently, item (1)) is a much stronger property than mere existence of isoperimetric regions, and it fails on $\mathbb{R}^N$ or, more generally, on noncompact spaces endowed with a strong homogeneity structure, such as simply connected models of constant sectional curvature $K \le 0$. On the other

hand, this equivalent criterion is expected to be applicable on spaces with no homogeneity invariance satisfying suitable asymptotic assumptions.

Part II

THE CARTAN-HADAMARD CONJECTURE FOR SMALL VOLUMES UNDER RICCI BOUND

5 NOTATION

For each $b \in \mathbb{R}$, let $\mathbb{M}^n_b$ denote the n-dimensional simply connected model space of constant curvature b. We let $\delta_{euc} := \delta^0$ and $\omega_{n-1} := \mathrm{vol}_\sigma(\partial B^n_1(0))$ where σ is the round metric on the sphere.

Given a metric space (X, d), we denote by $\mathcal{H}^s_d$ the s-dimensional Hausdorff measure induced by d. For a set $A \subset X$, its diameter is $\mathrm{diam}_d(A) := \sup\{d(x, y) : x, y \in A\}$. The symmetric difference of two sets $E, F \subset X$ is denoted by $E \,\Delta\, F := (E \setminus F) \cup (F \setminus E)$, where for any given sets A, B we define $A \setminus B$ as the set theoretic difference of A and B, i.e., $A \setminus B := \{a \in A : a \notin B\}$.

For a Riemannian manifold (M, g), we denote by vol_g its Riemannian volume measure and denote the sectional curvature of g by Sec_g. Given $k_1, k_2 \in \mathbb{R}$, the notation $k_1 \leq \mathrm{Sec}_g \leq k_2$ means that $k_1 \leq \mathrm{Sec}_g(\sigma) \leq k_2$ for every two-dimensional subspace $\sigma \subset T_xM$ and every $x \in M$. For $x \in M$ and $R > 0$, we denote by $B^g_R(x)$ the geodesic ball with respect to the metric g centered at x of radius R.

Let (M, g) be a Riemannian manifold and $\Sigma \subset M$ be a smooth embedded hypersurface with trivial normal bundle, and let ν be a choice of unit normal vector field along Σ. Given a function $u : \Sigma \to \mathbb{R}$ such that $u(x)\nu(x)$ lies in the domain of the normal exponential map for every $x \in \Sigma$, the normal graph of u over Σ is defined by

$$\mathrm{graph}^g_\Sigma(u) := \left\{\exp^g_x(u(x)\nu(x)) : x \in \Sigma\right\},$$

where $\exp^g$ denotes the Riemannian exponential map.

Let (X, d, μ) be a metric measure space. A measurable set $E \subset X$ is a set of finite perimeter if $\mathbf{1}_E \in BV(X)$. Its perimeter in an open set $U \subset X$ is denoted by $P_d(E; U) := |D\mathbf{1}_E|(U)$, and we set $P_d(E) := P_d(E; X)$. In a Riemannian manifold (M, g), we use the notations $P_g(E; U) := P_{d_g}(E; U)$ and $P_g(E) := P_{d_g}(E)$, where d_g is the canonical path length metric on M induced by the Riemannian metric tensor g.

A sequence of measurable sets $\{E_j\}_j$ converges to E in $L^1_{\mathrm{loc}}(X, \mu)$ if $\mu((E_j \,\Delta\, E) \cap K) \longrightarrow 0$ for every compact set $K \subset X$. It converges to E in $L^1(X, \mu)$ if $\mu(E_j \,\Delta\, E) \longrightarrow 0$.

In view of [72, Definition 1.2 and Theorem 1], the isoperimetric profile of a complete Riemannian manifold can equivalently be defined by minimizing over sets of finite perimeter. Accordingly, we use the following formulation.

The *isoperimetric profile of a Riemannian manifold* (M, g) is the function

$$I_{(M,g)}(v) := \inf\{P_g(E) : E \subset M \text{ is a set of finite perimeter and } \mathrm{vol}_g(E) = v\}, \tag{5}$$

defined for any $v \in (0, \mathrm{vol}_g(M))$. A finite perimeter set $E \subset M$ realizing this infimum is called an isoperimetric region of volume v. In (1) we take the infimum over smooth, bounded sets, whereas

in (5) we take the infimum over sets of finite perimeter; these two definitions are equivalent by [72, Theorem 1].

A *minimizing sequence for* $I_{(M,g)}(v)$ is a sequence of sets of finite perimeter $\{E_j\}_j$ such that

$$\mathrm{vol}_g(E_j) = v \quad \text{for every } j, \qquad P_g(E_j) \longrightarrow I_{(M,g)}(v).$$

For the following notions, we refer the reader to [9, Section 1.3]. We write

$$(X_j, d_j, \mu_j, x_j) \xrightarrow{\text{pmGH}} (X, d, \mu, x)$$

to denote pointed measured Gromov-Hausdorff convergence. More precisely, this means that the spaces can be isometrically embedded into a common complete metric space (Z, d_Z) in such a way that the base points converge, bounded balls converge in the Hausdorff sense, and the pushforwards of the measures converge weakly against functions in $C_c(Z)$, cf. [16, Definition 2.12].

Under such a realization, we say that measurable sets $E_j \subset X_j$ *converge* L^1*-strongly to a measurable set* $E \subset X$ if

$$(\iota_j)_\#(\mathbf{1}_{E_j}\mu_j) \rightharpoonup \iota_\#(\mathbf{1}_E\mu)$$

weakly as measures on Z and

$$\mu_j(E_j) \longrightarrow \mu(E),$$

where $\iota_j : X_j \to Z$ and $\iota : X \to Z$ are the corresponding isometric embeddings, cf. [16, Definition 2.15].

5.1 ALEXANDROV SPACES.

We define and recall some basic properties of these spaces, we refer the reader to [3, 57] for the notions of Alexandrov curvatures and definitions of metric classes as CAT and CBB, and to [46, 11] for the theory of RCD spaces.

Definition 5.1.1. Let $\bar{k} \geq \underline{k}$ be real numbers and (X, d) a complete and locally compact metric space. We say that:

(i) (X, d) is an Alexandrov space with curvature bounded from above by $\bar{k}$ if it is CAT($\bar{k}$) and bounded from below by $\underline{k}$ if it is CBB($\underline{k}$);

(ii) $(X, d, \mathcal{H}^n_d)$ is non-collapsing if there exists a real number v_{nc} such that

$$\inf_{x \in X} \mathcal{H}^n_d(B^d_1(x)) \geq v_{nc} > 0;$$

(iii) we refer the reader to [25, §10.10] for the definition of Alexandrov boundary, the specific definition is not relevant to this article as we only use it as an assumption to apply results in the literature.

In the following proposition, we collect groundbreaking results obtained for Alexandrov spaces that will be vital in this article.

Proposition 5.1.2 (Properties of Alexandrov spaces)**.** *Let (X, d) be an Alexandrov space with curvature bounded from above by $\bar{k} \leq 0$ and bounded from below by $\underline{k}$ with no Alexandrov boundary. Then*

(i) *(X, d) is simply-connected;*

(ii) *$(X, d, \mathcal{H}^n_d)$ is a $\mathrm{RCD}((n-1)\underline{k}, n)$ space;*

(iii) *(X, d) admits a $C^{3,\alpha}$ atlas and a $C^{1,\alpha}$ metric g, for any $\alpha \in (0,1)$, whose induced distance on X coincides with d. Moreover, the metric g is actually of class $W^{2,p}$ for every $p \geq 1$;*

(iv) *there exists a sequence of Riemannian metrics $g_i \in C^\infty$ on X that converge to g in $C^{1,\alpha}$-topology for any $\alpha \in (0,1)$ or in $W^{2,p}$-topology for any $p \geq 1$. Furthermore, let d_i be the distance induced by g_i, then each (X, d_i) is $\mathrm{CAT}(\bar{k}_i)$ and $\mathrm{CBB}(\underline{k}_i)$ with*

$$\underline{k} \leq \liminf_i \underline{k}_i \leq \limsup_i \bar{k}_i \leq \bar{k}.$$

Moreover, $\bar{k}_i$ can be taken strictly smaller than $\bar{k}$.

Remark 5.1.3. In view of [36], we observe the following. Let (M^n, g) be a Cartan-Hadamard manifold, and g_i given by (iv), which satisfy $\mathrm{Sec}_{g_i} < 0$. By [36, Theorem 1], each $x \in M$ admits a Druet radius $r_i(x) > 0$ such that $P_{g_i}(E) > I_{(\mathbb{R}^n, \delta_{euc})}(\mathrm{vol}_{g_i}(E))$ for every set $E \subset B^{g_i}_{r_i(x)}(x)$ of finite and positive volume. If $r(x) := \liminf_i r_i(x) > 0$, continuity allows us to pass to the limit and obtain $P_g(E) \geq I_{(\mathbb{R}^n, \delta_{euc})}(\mathrm{vol}_g(E))$ for every such set $E \Subset B^g_{r(x)}(x)$. However, the available estimates ensuring a uniform positive lower bound for the Druet radii require a uniform lower Ricci curvature bound in general. This brings us back to the setting of Theorem 1.2.2, which is more general because it only requires the volume to be small.

Moreover, the Druet radius admits a lower bound in terms of the harmonic radius. By [84, Proposition 5.20], the harmonic radius is infinite only if the manifold is isometric to the Euclidean space.

Proof. Item (i) follows since Alexander, Kapovich, and Petrunin, in [3, Corollary 2.2.6] showed that (X, d) is contractible. Item (ii) follows from work of Petrunin in [83, Main theorem] which ensures that it satisfies the $\mathrm{CD}((n-1)\underline{k}, n)$ and then we can apply a result by Kapovich and Ketterer in [57, Theorem 1]. Item (iii) is a famous theorem proved by Berestovskij and Nikolaev in [22, Theorem 14.1]. Item (iv) is stated in [22, Theorem 15.1, Remark 1] by Berestovskij and Nikolaev except for the moreover part.

We are then left to prove the moreover part of item (iv). This is a routine argument.

First, assume that $\bar{k} < 0$. By the approximation theorem cited above [22, Theorem 15.1, Remark 1], choose complete smooth metrics $h_j \to g$ with $\underline{k} - \delta_j \leq \mathrm{Sec}_{h_j} \leq \bar{k} + \delta_j$, where $\delta_j \downarrow 0$ and $\delta_j < -\bar{k}$. Define $\lambda_j := \frac{\bar{k}+\delta_j}{\bar{k}-\delta_j} \in (0,1)$ and $g_j := \lambda_j h_j$. Since sectional curvature scales by the inverse of the metric scaling factor, we obtain $\frac{\underline{k}-\delta_j}{\lambda_j} \leq \mathrm{Sec}_{g_j} \leq \bar{k} - \delta_j < \bar{k}$. As $\lambda_j \to 1$, the metrics g_j are complete and have the required convergences, and the lower and upper curvature bounds tend to $\underline{k}$ and $\bar{k}$, respectively.

We now assume that $\bar{k} = 0$.

Fix $o \in X$ and put $r = d(o,\cdot)$, $\rho = 1 + r^2$. We have that $\rho \in C^{1,1}_{\mathrm{loc}}(X)$ and $|D\rho|_g = 2r$ by [58, Proposition 4.1 and 1.6]. By adapting the argument in [58, Section 2.4] using the *global* condition of X being CAT(0) and [4, Theorem 9.25] using the *global* condition CBB($\underline{k}$), we obtain the following global inequality almost everywhere

$$2g \le \mathrm{Hess}_g \rho \le 2(1 + \sqrt{-\underline{k}}r)\, g. \tag{6}$$

Fix a locally finite atlas $(U_\nu)_{\nu\ge 1}$ and a partition of unity (χ_ν) with $K_\nu = \operatorname{supp}\chi_\nu \subset U_\nu$ compact. In U_ν, $\mathrm{Hess}_g(\rho * \eta_\sigma) - (\mathrm{Hess}_g \rho) * \eta_\sigma \to 0$ uniformly on K_ν for a mollifier $\eta_\sigma \ge 0$, because the Christoffel symbols Γ and $D\rho$ are continuous. Positivity of η_σ therefore preserves (6) up to small errors. Choose $\rho_\nu := \rho * \eta_{\sigma_\nu}$ with value, gradient and Hessian errors at most $2^{-\nu}/(8 + 8\sup_{K_\nu}(|\mathrm{Hess}_g \chi_\nu|_g + 2|D\chi_\nu|_g))$ on K_ν, and put

$$f = \sum_\nu \chi_\nu \rho_\nu.$$

Since $\sum D\chi_\nu = 0 = \sum \mathrm{Hess}_g \chi_\nu$, then $\mathrm{Hess}_g f = \sum_\nu \chi_\nu \mathrm{Hess}_g \rho_\nu + Err$ where Err only involves term with $\rho_\nu - \rho$ and $D(\rho_\nu - \rho)$. Hence $|f - \rho| \le \frac{1}{2}$, $|Df - D\rho|_g \le 1$ and, by (6), $g \le \mathrm{Hess}_g f \le (3 + 2\sqrt{-\underline{k}}r)g$. As $f \ge \frac{1}{2} + r^2$, for constants $A_0, B_0 > 0$ depending only on $\underline{k}$, we obtain

$$g \le \mathrm{Hess}_g f \le A_0 \sqrt{f}\, g \text{ and } |Df|^2_g \le B_0 f.$$

Let h_j be the metric given by item (iv), $A := 3A_0$, and $B := 2B_0$, by the $W^{2,p}$ convergence of h_j to g and the estimates above, we readily obtain

$$\underline{k} - \varepsilon_j \le \mathrm{Sec}_{h_j} \le \varepsilon_j, \qquad \frac{1}{2} g \le h_j \le \frac{3}{2} g, \tag{7}$$

$$\frac{1}{2} h_j \le \mathrm{Hess}_{h_j} f \le A\sqrt{f}\, h_j, \qquad |Df|^2_{h_j} \le Bf. \tag{8}$$

Let $g_j = e^{4\varepsilon_j f} h_j$, and let P be a 2-plane with h_j-orthonormal basis e_1, e_2. By a direct computation, we get

$$e^{4\varepsilon_j f}\mathrm{Sec}_{g_j}(P) = \mathrm{Sec}_{h_j}(P) - 2\varepsilon_j \sum_{\ell=1}^{2} \mathrm{Hess}_{h_j} f(e_\ell, e_\ell) - 4\varepsilon_j^2 |\mathrm{proj}_{P^\perp} \nabla^{h_j} f|^2_{h_j}.$$

Upper bound. By (7) and (8), the right-hand side is at most $\varepsilon_j - 2\varepsilon_j$, so $\mathrm{Sec}_{g_j} \le -\varepsilon_j e^{-4\varepsilon_j f} < 0$. Thus the lower Hessian bound absorbs the possibly positive curvature error of h_j.

Lower bound. By (7) and (8), the right-hand side is at least $\underline{k} - \varepsilon_j - 4A\varepsilon_j\sqrt{f} - 4B\varepsilon_j^2 f$. We have $\underline{k} - \varepsilon_j \le 0$, $f \ge 0$, $\sqrt{\varepsilon_j f}\, e^{-4\varepsilon_j f} \le (8e)^{-1/2}$, and $\varepsilon_j f\, e^{-4\varepsilon_j f} \le (4e)^{-1}$, this implies

$$\mathrm{Sec}_{g_j} \ge \underline{k} - \frac{\sqrt{2}A}{\sqrt{e}}\sqrt{\varepsilon_j} - \Big(1 + \frac{B}{e}\Big)\varepsilon_j \ge \underline{k} - C\sqrt{\varepsilon_j}.$$

This shows that we can choose $\bar{k}_j < 0$ as desired. The corresponding convergences follow trivially by the construction. □

5.2 LIMITS OF ALEXANDROV SPACES

We recall a couple of definitions about the limiting behavior of spaces with lower Ricci curvature bounds and in particular of Alexandrov spaces.

Definition 5.2.1. Let $\alpha \in (0,1)$. A Riemannian manifold (M,g) has $C^{1,\alpha}$-locally asymptotic bounded geometry if it has Ricci curvature bounded below, it is non-collapsing, and if for every diverging sequence of points (p_j), there exist a subsequence (p_{j_l}) and a pointed manifold $(M_\infty, g_\infty, p_\infty)$, with g_∞ of class $C^{1,\alpha}$, such that $(M, g, p_{j_l}) \to (M_\infty, g_\infty, p_\infty)$ in the pointed $C^{1,\alpha}$-topology

Later we use the fact that manifolds with two-sided bounds on the sectional curvature admit particularly good properties for their pointed limits, as we remark below.

Remark 5.2.2. If (Z, g_Z) is a Cartan-Hadamard manifold with $\mathrm{Ric}_{g_Z} \geq (n-1)\underline{k}$, by [13, Theorem 1.1 and Remark 2.4(i)], we have that (Z, g_Z) has $C^{1,\alpha}$-locally asymptotic bounded geometry.

The next is the well-known smooth version of the pointed Gromov-Hausdorff convergence which will be used below, cf. [82, Section 11.3.2, pg. 414].

Definition 5.2.3. Let $\alpha \in (0,1)$, $\{(M_i, g_i, p_i)\}_{i\in\mathbb{N}}$ be a sequence of pointed smooth Riemannian manifolds, and $E_i \subset M_i$ be locally finite perimeter sets for all $i \in \mathbb{N}$. Suppose that $(M_i, g_i, p_i) \to (M_\infty, g_\infty, p_\infty)$ in the pointed $C^{1,\alpha}$-flat topology, and let $E_\infty \subset M_\infty$ be a locally finite perimeter set.

We say that (E_i, g_i, p_i) converges to $(E_\infty, g_\infty, p_\infty)$ in the pointed $C^{1,\alpha}$-flat topology if, for every $R > 0$, there exist a domain Ω_R such that $B_R^{g_\infty}(p_\infty) \subset \Omega_R \subset M_\infty$, an index $i_R \in \mathbb{N}$, and $C^{2,\alpha}$ embeddings $\Psi_{i,R}\colon \Omega_R \to M_i$ such that for all $i \geq i_R$ we have that $B_R^{g_i}(p_i) \subset \Psi_{i,R}(\Omega_R)$, $\Psi_{i,R}(p_\infty) = p_i$, $\Psi_{i,R}^* g_i \to g_\infty$ in $C^{1,\alpha}(\Omega_R)$, and

$$\mathrm{vol}_{g_\infty}\left(B_R^{g_\infty}(p_\infty) \cap (\Psi_{i,R}^{-1}(E_i) \,\Delta\, E_\infty)\right) \longrightarrow 0.$$

6 PRELIMINARY RESULTS

In [71, Theorem 3.4], Morgan and Johnson prove an isoperimetric inequality for compact manifolds; [71, Section 4.6] extends this to noncompact manifold with compact quotient by the isometry group. In [69, Prop. 3.2], Mondino and Nardulli prove a similar result for noncompact M^n satisfying a Ricci curvature lower bound. Both results apply to space forms and readily give the next lemma. However, since the proof is substantially simpler in the case of space forms, we present it here.

Lemma 6.0.1. *If $a \leq A \leq 0$, then $I_{(\mathbb{M}^n_a,\delta^a)} \geq I_{(\mathbb{M}^n_A,\delta^A)}$.*

Remark 6.0.2. As it is easily seen from the proof, when $a < A$, we get $I_{(\mathbb{M}^n_a,\delta^a)} > I_{(\mathbb{M}^n_A,\delta^A)}$.

Proof. For each $b \leq 0$, by [87, Theorem 4.35 and Theorem 4.39], the isoperimetric regions in $\mathbb{M}^n_b$ are geodesic balls. Therefore, for all $v \geq 0$, $I_{(\mathbb{M}^n_b,\delta^b)}(v) = P_b(r_b)$, where $P_b(r)$ denote the perimeter of a geodesic ball of radius r in $\mathbb{M}^n_b$ and r_b is such that the volume of the geodesic ball of radius r_b in $\mathbb{M}^n_b$ is equal to v. For $b \leq 0$, define

$$s_b(r) := \begin{cases} r, & b = 0, \\ \dfrac{\sinh\left(\sqrt{-b}\, r\right)}{\sqrt{-b}}, & b < 0. \end{cases}$$

Then, for all $r \geq 0$,

$$P_b(r) = \omega_{n-1}\left(s_b(r)\right)^{n-1} \text{ and } V_b(r) = \omega_{n-1}\int_0^r \left(s_b(t)\right)^{n-1} dt,$$

Since $V_b'(r) > 0$ for all $r > 0$, the function V_b is invertible. Using the inverse function theorem, we obtain

$$I'_{(\mathbb{M}^n_b,\delta^b)}(v) = P_b'(R_b(v))R_b'(v) = \frac{P_b'(R_b(v))}{V_b'(R_b(v))},$$

for all $v \geq 0$, where $R_b(v) := V_b^{-1}(v)$. Using that $V_b'(r) = P_b(r)$ for any $r > 0$ and the last displayed equation, we straightforwardly see that for $b < 0$

$$I'_{(\mathbb{M}^n_b,\delta^b)}(v) = (n-1)\frac{s_b'(R_b(v))}{s_b(R_b(v))} = (n-1)\sqrt{-b}\coth(\sqrt{-b}\, R_b(v)), \tag{9}$$

and, for $b = 0$

$$I'_{(\mathbb{M}^n_0,\delta^0)}(v) = (n-1)\frac{s_0'(R_0(v))}{s_0(R_0(v))} = (n-1)\frac{1}{R_0(v)}. \tag{10}$$

Since the conclusion is immediate when $a = A$, it only remains to treat two cases. Firstly, suppose that $a < A < 0$. Since the function in the right-hand side of (9) is nondecreasing with respect to b and $a < A$, we derive that

$$I'_{(\mathbb{M}^n_a,\delta^a)}(v) \geq (n-1)\frac{s'_A(R_a(v))}{s_A(R_a(v))}. \tag{11}$$

Moreover, by the Bishop–Gromov Volume Comparison Theorem [65, Theorem 11.19], we have that $R_a(v) \leq R_A(v)$. Combining this with the fact that the right-hand side of (11) is nonincreasing in the radius, we deduce that, for all $v \geq 0$,

$$I'_{(\mathbb{M}^n_a,\delta^a)}(v) \geq (n-1)\frac{s'_A(R_A(v))}{s_A(R_A(v))} \overset{(9)}{=} I'_{(\mathbb{M}^n_A,\delta^A)}(v).$$

Integrating this inequality and using that both isoperimetric profiles vanish at $v = 0$, we conclude this case. Secondly, suppose that $a < A = 0$. Since for every $r > 0$ and every $b < 0$ we have

$$(n-1)\sqrt{-b}\coth(\sqrt{-b}r) \geq (n-1)\frac{1}{r},$$

it follows from (9) that

$$I'_{(\mathbb{M}^n_a,\delta^a)}(v) \geq (n-1)\frac{1}{R_a(v)}.$$

Combining the last displayed equation with the inequality $R_a(v) \leq R_0(v)$, we obtain that

$$I'_{(\mathbb{M}^n_a,\delta^a)}(v) \geq (n-1)\frac{1}{R_0(v)} \overset{(10)}{=} I'_{(\mathbb{M}^n_0,\delta^0)}(v).$$

Integrating this inequality and using that both isoperimetric profiles vanish at $v = 0$, we conclude the proof. □

We will use some properties of the isoperimetric profile of model spaces that we prove below.

Lemma 6.0.3. *For all $a \leq 0$, $I_{(\mathbb{M}^n_a,\delta^a)}$ is strictly concave, and hence strictly subadditive.*

Proof. Arguing exactly as in the proof of Lemma 6.0.1, by Equation (9), we straightforwardly see that for $b < 0$

$$I''_{(\mathbb{M}^n_b,\delta^b)}(v) = -\frac{(n-1)(-b)^{\frac{n-1}{2}}}{\omega_{n-1}(\sinh(\sqrt{-b}R_b(v)))^{n+1}} < 0,$$

and, for $b = 0$

$$I''_{(\mathbb{M}^n_0,\delta^0)}(v) = -\frac{n-1}{\omega_{n-1}(R_0(v))^{n+1}} < 0.$$

The two last displayed equations implies that $I_{(\mathbb{M}^n_a,\delta^a)}$ is concave for all $a \leq 0$. Fix $a \leq 0$ arbitrarily and define $f(v) := I_{(\mathbb{M}^n_a,\delta^a)}(v)$. Since f is strictly concave and $f(0) = 0$, we obtain for any $v \neq 0$ that

$$f(tv) = f(tv + (1-t)0) > tf(v), \text{ for any } t \in (0,1). \tag{12}$$

Thus, for any $v, w \neq 0$, we get

$$\begin{aligned} f(v) + f(w) &= f\left((v+w)\frac{v}{v+w}\right) + f\left((v+w)\frac{w}{v+w}\right) \\ &\overset{(12)}{>} f(v+w)\frac{v}{v+w} + f(v+w)\frac{w}{v+w} \\ &= f(v+w). \end{aligned}$$

Since a was taken arbitrarily, we conclude that $I_{(\mathbb{M}_a^n, \delta^a)}$ is subadditive for all $a \leq 0$. □

7 THE CONJECTURE FOR SMALL NORMAL GRAPHS

In Lemma 7.0.1, we show that the conjecture holds true for C^1 small normal graphs over a g-geodesic ball in the Euclidean space with a Cartan-Hadamard metric g. To the best of our knowledge, this was not known for arbitrary $\bar{k}$. For $\bar{k} = 0$, a version of this was proved by Ghomi and Stavroulakis in [45].

Before stating the lemma, we need to set some notation. In polar coordinates, the metric of the model space of curvature $\bar{k}$ is $dr^2 + s_{\bar{k}}(r)^2\sigma$, where σ is the round metric on $\mathbb{S}^{n-1}$. Let $(\mathbb{R}^n, g)$ be a Cartan-Hadamard manifold with $\underline{k} \leq \mathrm{Sec}_g \leq \bar{k} \leq 0$, $r = d_g(0, \cdot)$ and, in polar geodesic coordinates around 0, we can write

$$g = dr^2 + s_{\bar{k}}(r)^2 Q_r. \tag{13}$$

A straightforward computation shows that $\partial_r(s_{\bar{k}}^2 Q_r) = 2\mathrm{Hess}_g r$ on $T\mathbb{S}^{n-1}$[1]. By $\mathrm{Sec}_g \leq \bar{k} \leq 0$ and the Hessian comparison theorem (cf. [82, Chapter 6]), we have that $\partial_r(s_{\bar{k}}^2 Q_r) \geq 2s'_{\bar{k}} s_{\bar{k}} Q_r$ and thus $\partial_r Q_r \geq 0$. It is easy to see that $Q_r \longrightarrow \sigma$ as $r \downarrow 0$ and therefore

$$Q_r \geq \sigma. \tag{14}$$

Similarly, from Hessian comparison and $\mathrm{Sec}_g \geq \underline{k}$, we obtain $\partial_r(s_{\bar{k}}^2 Q_r) \leq 2s'_{\underline{k}} s_{\bar{k}}^2 Q_r / s_{\underline{k}}$ and this gives $\partial_r(s_{\bar{k}}^2 s_{\underline{k}}^{-2} Q_r) \leq 0$. Therefore, we conclude

$$Q_r \leq \frac{s_{\underline{k}}(r)^2}{s_{\bar{k}}(r)^2}\,\sigma. \tag{15}$$

Define

$$q(r, \theta) := \sqrt{\det_\sigma Q_r(\theta)}, \tag{16}$$

equivalently $d\mathrm{vol}_{Q_r} = q(r, \theta) d\mathrm{vol}_\sigma$.

Lemma 7.0.1. *Let $(\mathbb{R}^n, g)$ be a Cartan-Hadamard manifold with $\underline{k} \leq \mathrm{Sec}_g \leq \bar{k} \leq 0$. For every $R_0 > 0$, there exist $\epsilon \in (0, \frac{1}{2}]$ and $c > 0$, depending only on $n, \underline{k}, \bar{k}, R_0$, with the following property.*

Let $0 < R \leq R_0$ and let $E \subset \mathbb{R}^n$ be a domain such that ∂E is given by the normal graph of $u \in C^1(\partial B_R^g, \mathbb{R})$ with respect to g and $\|u/R\|_{L^\infty} + \|\nabla u\|_{L^\infty} \leq \epsilon$. Then

$$P_g(E) - I_{(\mathbb{M}^n_{\bar{k}}, \delta^{\bar{k}})}(\mathrm{vol}_g(E)) \geq cR^{n-1} \int_{\mathbb{S}^{n-1}} (q(R + u(\theta), \theta) - 1)\, d\sigma \geq 0, \tag{17}$$

1 Let $\{\partial_{\theta_i}\}$ be a orthonormal frame for $T\mathbb{S}^{n-1}$, by metric compatibility of the Levi-Civita connection and its torsion-free property, we get $\partial_r g(\partial_{\theta_i}, \partial_{\theta_j}) = g(\nabla_{\partial_r}\partial_{\theta_i}, \partial_{\theta_j}) + g(\partial_{\theta_i}, \nabla_{\partial_r}\partial_{\theta_j}) = g(\nabla_{\partial_{\theta_i}}\partial_r, \partial_{\theta_j}) + g(\partial_{\theta_i}, \nabla_{\partial_{\theta_j}}\partial_r) = \mathrm{Hess}_g r(\partial_{\theta_i}, \partial_{\theta_j}) + \mathrm{Hess}_g r(\partial_{\theta_j}, \partial_{\theta_i}) = 2\mathrm{Hess}_g r(\partial_{\theta_j}, \partial_{\theta_i})$.

Furthermore, if $P_g(E) = I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(\mathrm{vol}_g(E))$, *then* $g \simeq \delta^{\bar{k}}$ *on* E.

Proof. By (14), we have $q(r,\theta) \geq 1$, and $r \mapsto q(r,\theta)$ is nondecreasing. The eigenvalues of Q_r relative to σ are $\mu_i \geq 1$ by (14), and then the eigenvalues of $(\det_\sigma Q_r)Q_r^{-1}$ are $\prod_{j\neq i}\mu_j \geq 1$.

Let $E_{\bar{k}} \subset \mathbb{M}^n_{\bar{k}}$ be the region with the same radial graph $\partial E = \{(R+u(\theta),\theta) : \theta \in \mathbb{S}^{n-1}\}$ in the model space. Using the monotonicity of q, we obtain

$$\begin{aligned} 0 \leq \mathrm{vol}_g(E) - \mathrm{vol}_{\delta^{\bar{k}}}(E_{\bar{k}}) &= \int_{\mathbb{S}^{n-1}}\int_0^{R+u(\theta)} s_{\bar{k}}(r)^{n-1}(q(r,\theta)-1)\,dr\,d\sigma \\ &\leq \int_{\mathbb{S}^{n-1}}(q(R+u(\theta),\theta)-1)\int_0^{R+u(\theta)} s_{\bar{k}}(r)^{n-1}\,dr\,d\sigma. \end{aligned} \tag{18}$$

The metric induced by (13) on the graph $r = R+u(\theta)$ is $s_{\bar{k}}(R+u)^2 Q_{R+u} + du\otimes du$. The matrix determinant lemma therefore gives

$$\begin{aligned} P_g(E) &= \int_{\mathbb{S}^{n-1}} \sqrt{\det(s_{\bar{k}}(R+u)^2 Q_{R+u} + du\otimes du)}\,d\sigma \\ &= \int_{\mathbb{S}^{n-1}} s_{\bar{k}}(R+u)^{n-1} q(R+u,\theta)\sqrt{1+s_{\bar{k}}(R+u)^{-2}Q^{-1}_{R+u}(du,du)}\,d\sigma. \end{aligned}$$

Since $q(R+u,\theta)^2 Q^{-1}_{R+u}(du,du) \geq |du|^2_\sigma$, it follows that

$$P_g(E) \geq \int_{\mathbb{S}^{n-1}} s_{\bar{k}}(R+u)^{n-1}\sqrt{q(R+u,\theta)^2 + \frac{|du|^2_\sigma}{s_{\bar{k}}(R+u)^2}}\,d\sigma.$$

On the other hand,

$$P_{\delta^{\bar{k}}}(E_{\bar{k}}) = \int_{\mathbb{S}^{n-1}} s_{\bar{k}}(R+u)^{n-1}\sqrt{1+\frac{|du|^2_\sigma}{s_{\bar{k}}(R+u)^2}}\,d\sigma.$$

For $z \geq 0$ and $q \geq 1$, we have that $\sqrt{q^2+z^2} - \sqrt{1+z^2} = \int_1^q \frac{t}{\sqrt{t^2+z^2}}\,dt \geq \frac{q-1}{\sqrt{1+z^2}}$. Therefore

$$P_g(E) - P_{\delta^{\bar{k}}}(E_{\bar{k}}) \geq \int_{\mathbb{S}^{n-1}} s_{\bar{k}}(R+u)^{n-1}\,\frac{q(R+u(\theta),\theta)-1}{\sqrt{1+s_{\bar{k}}(R+u)^{-2}|du|^2_\sigma}}\,d\sigma. \tag{19}$$

By Lemma 6.0.3, $I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}$ is concave and increasing, and by (9) and (10), we recall that $I'_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(V_{\bar{k}}(r)) = (n-1)s'_{\bar{k}}(r)/s_{\bar{k}}(r)$, which is nonincreasing in r. Since $\|u\|_{L^\infty} \leq \epsilon R$, we have $\mathrm{vol}_{\delta^{\bar{k}}}(E_{\bar{k}}) \geq V_{\bar{k}}((1-\epsilon)R)$, and hence (18) gives

$$\begin{aligned} &I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(\mathrm{vol}_g(E)) - I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(\mathrm{vol}_{\delta^{\bar{k}}}(E_{\bar{k}})) \\ &\quad\leq (n-1)\frac{s'_{\bar{k}}((1-\epsilon)R)}{s_{\bar{k}}((1-\epsilon)R)}\int_{\mathbb{S}^{n-1}}(q(R+u(\theta),\theta)-1)\int_0^{R+u(\theta)} s_{\bar{k}}(r)^{n-1}\,dr\,d\sigma. \end{aligned} \tag{20}$$

Combining (19), (20), and $P_{\delta^{\bar{k}}}(E_{\bar{k}}) \geq I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(\mathrm{vol}_{\delta^{\bar{k}}}(E_{\bar{k}}))$, we conclude that

$$P_g(E) - I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(\mathrm{vol}_g(E)) \geq \int_{\mathbb{S}^{n-1}} (q(R+u(\theta),\theta)-1)\,\Gamma(\theta)\,d\sigma, \tag{21}$$

where

$$\Gamma(\theta) := \frac{s_{\bar{k}}(R+u)^{n-1}}{\sqrt{1+s_{\bar{k}}(R+u)^{-2}|du|^2_\sigma}} - (n-1)\frac{s'_{\bar{k}}((1-\epsilon)R)}{s_{\bar{k}}((1-\epsilon)R)}\int_0^{R+u(\theta)} s_{\bar{k}}(r)^{n-1}\,dr.$$

It remains to show that $\Gamma \geq cR^{n-1}$. By $\|u\|_{L^\infty} \leq \epsilon R$, $(1-\epsilon)R \leq R+u \leq (1+\epsilon)R$. The metric induced by g on ∂B^g_R is $s_{\bar{k}}(R)^2 Q_R$, by $\|u\|_{C^1} \leq \epsilon$, we get (15) gives $|du|_\sigma \leq s_{\underline{k}}(R)|\nabla^{s^2_{\bar{k}}Q_r} u|_{s^2_{\bar{k}}Q_r} \leq \epsilon s_{\underline{k}}(R)$. Since $s_{\bar{k}}(r) \geq r$, $r \mapsto s_{\underline{k}}(r)/r$ is nondecreasing, and $R+u \geq (1-\epsilon)R \geq R/2$, we get $s_{\bar{k}}(R+u)^{-1}|du|_\sigma \leq 2\epsilon s_{\underline{k}}(R_0)/R_0 =: \epsilon C_0$.

Using the monotonicity of $s_{\bar{k}}$ and $s_{\bar{k}}(Rt) = Rs_{\bar{k}R^2}(t)$, $s'_{\bar{k}}(Rt) = s'_{\bar{k}R^2}(t)$, we obtain $\Gamma \geq R^{n-1}\gamma(\bar{k}R^2,\epsilon)$, where

$$\gamma(b,\epsilon) := \frac{s_b(1-\epsilon)^{n-1}}{\sqrt{1+C_0^2\epsilon^2}} - (n-1)\frac{s'_b(1-\epsilon)}{s_b(1-\epsilon)}\int_0^{1+\epsilon} s_b(t)^{n-1}\,dt.$$

We claim that $\gamma(b,0) > 0$ for every $b \leq 0$. Indeed, write $s = s_b$ and let

$$h(t) := s(t)^n - (n-1)s'(t)\int_0^t s(\tau)^{n-1}\,d\tau \text{ and } h(1) = s(1)\gamma(b,0).$$

Using $s'' = -bs$ and $(s')^2 + bs^2 = 1$, we get $h' = s\varphi$ and $\varphi' = (n-2)s^{n-3}$, where $\varphi(t) := s(t)^{n-2}s'(t) + (n-1)b\int_0^t s(\tau)^{n-1}d\tau$. If $n \geq 3$, then $\varphi(0) = 0$, and if $n = 2$, then $\varphi \equiv 1$. In both cases $\varphi > 0$ on $(0,\infty)$, hence $h > 0$ on $(0,\infty)$ since $h(0) = 0$, which proves the claim. Since γ is continuous on $[\bar{k}R_0^2, 0] \times [0, \frac{1}{2}]$ and $\bar{k}R^2 \in [\bar{k}R_0^2, 0]$, by compactness we can choose $\epsilon \in (0, \frac{1}{2}]$ and $c > 0$, depending only on $n, \underline{k}, \bar{k}, R_0$, such that $\gamma(b,\epsilon) \geq c$ for every $b \in [\bar{k}R_0^2, 0]$. Thus $\Gamma \geq cR^{n-1}$, and (21) gives (17).

For the rigidity statement, suppose that $P_g(E) = I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(\mathrm{vol}_g(E))$. Then (17) and the continuity of q imply that $q(R+u(\theta),\theta) = 1$ for every θ, and by the monotonicity of q we get $q(r,\theta) = 1$ for $0 < r \leq R+u(\theta)$. The eigenvalues of Q_r relative to σ are at least one by (14), while their product is $q(r,\theta)^2 = 1$. Thus every eigenvalue is equal to one and $Q_r = \sigma$ for $0 < r \leq R+u(\theta)$. Hence

$$g = dr^2 + s_{\bar{k}}(r)^2\sigma \qquad \text{on } E. \tag{22}$$

Moreover, by (18), (19) and (22), we have $\mathrm{vol}_g(E) = \mathrm{vol}_{\delta^{\bar{k}}}(E_{\bar{k}})$ and $P_g(E) = P_{\delta^{\bar{k}}}(E_{\bar{k}}) = I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(\mathrm{vol}_{\delta^{\bar{k}}}(E_{\bar{k}}))$. Therefore $E_{\bar{k}}$ is an isoperimetric region in $(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})$, and hence it is a geodesic ball. Consequently, by (22), (E,g) is isometric to a geodesic ball in $(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})$, which proves the rigidity statement. □

8 MAIN REDUCTION

In this chapter, we address one of the main difficulties: the Cartan-Hadamard manifold may fail to admit isoperimetric regions, even under a lower Ricci curvature bound. We show that the problem can be reduced to studying an isoperimetric region in an Alexandrov space with two-sided curvature bounds.

The result below is our main theorem (Theorem 1.2.2) in the more general context of Alexandrov spaces, cf. Remark 8.0.2.

Theorem 8.0.1. *Let (X, d) be a Alexandrov space with no Alexandrov boundary, curvature bounded from above by $\bar{k} \leq 0$ and below by $\underline{k}$, and noncollapsing constant $v_{nc} > 0$ as in Definition 5.1.1. There exists $v_0 = v_0(v_{nc}, \underline{k}, \bar{k}, n) > 0$ such that*

$$I_{(X,d)}(v) \geq I_{(\mathbb{M}^n_{\bar{k}}, \delta^{\bar{k}})}(v) \text{ for any } v \in (0, v_0].$$

Moreover, if there exists a finite perimeter set $E \subset X$ with $\mathcal{H}^n_d(E) = v \in (0, v_0]$ and $P_d(E) = I_{(\mathbb{M}^n_{\bar{k}}, \delta^{\bar{k}})}(v)$, then E is isometric to a geodesic ball in $\mathbb{M}^n_{\bar{k}}$.

Remark 8.0.2. Theorem 1.2.2 follows from Theorem 8.0.1 since any Cartan-Hadamard manifold satisfies the global condition of being CAT($\bar{k}$). Moreover, any pointed GH-limit of a Cartan-Hadamard manifold with $\underline{k} \leq \mathrm{Sec}_g \leq \bar{k} \leq 0$ is an Alexandrov space with no Alexandrov boundary and curvature bounded above by $\bar{k}$ and below by $\underline{k}$. Indeed, any pointed GH-limit of (X, d, p_i) is locally geodesically complete by [67, Example 4.3] and [67, §2.2]. This and an application of [56, Proposition 3.13(iii)] conclude the proof.

Fix a volume $v > 0$. Since (X, d) is a RCD$((n-1)\underline{k}, n)$ space by Proposition 5.1.2(ii), we are under the assumptions of [16, Theorem 1.1] which gives a minimizing sequence $\Omega_i \subset X$, for any $i \in \mathbb{N}$, that can be split into diverging and converging pieces (Ω_i^d and Ω_i^c, respectively) as we give a precise description below. We have

- $\Omega_i^c \to \Omega_0 \subset X$ in the sense of sets of finite perimeter and Ω_0 is an (possibly empty) isoperimetric region for its own volume $0 \leq \mathcal{H}^n_d(\Omega_0) \leq v$.

There exist a positive natural number $\bar{N}$ and points $p_{i,j} \in X$, for $j \in \{1, \dots, \bar{N}\}$, such that $(X, d, \mathcal{H}^n_d, p_{i,j}) \to (X_j, d_j, \mathcal{H}^n_{d_j}, p_j)$ in the pmGH sense, where each space $(X_j, d_j, \mathcal{H}^n_{d_j})$ is an RCD$((n-1)\underline{k}, n)$ space which is also non-collapsing, see [34, Theorem 1.2]. We have

- $\Omega^d_{i,j} \to \Omega^\infty_j \subset X_j$ in the sense of sets of finite perimeter, $P_d(\Omega^d_{i,j}) \to P_{d_j}(\Omega^\infty_j)$, and Ω^∞_j is an isoperimetric region in (X_j, d_j) for any $j \in \{1 \dots, \bar{N}\}$.

Moreover, the isoperimetric profile and the volume split also as follows:

- $I_{(X,d)}(v) = P_d(\Omega_0) + \sum_{j=1}^{\bar{N}} P_{d_j}(\Omega_j^\infty) = I_{(X,d)}(\mathcal{H}^n_d(\Omega_0)) + \sum_{j=1}^{\bar{N}} I_{(X_j,d_j)}(\mathcal{H}^n_{d_j}(\Omega_j^\infty))$, and
- $v = \mathcal{H}^n_d(\Omega_0) + \sum_{j=1}^{\bar{N}} \mathcal{H}^n_{d_j}(\Omega_j^\infty)$.

Next, since (X, d) is CAT($\bar{k}$), we apply [3, Proposition 2.1.1] to derive that (X_j, d_j) is CAT($\bar{k}$) as well for each $j \in \{1, \dots, \bar{N}\}$. Since each $(X_j, d_j, \mathcal{H}^n_{d_j})$ is also an RCD$((n-1)\underline{k}, n)$ space, we can apply [57, Theorem 5.1(a)] to obtain that each (X_j, d_j) is an Alexandrov space with the same curvature bounds of (X, d) and the noncollapsing constants.

We show how to conclude the proof of Theorem 8.0.1 using the proposition below whose proof is done in Chapter 9.

Proposition 8.0.3. *Let (Z, d_Z) be an Alexandrov space with no Alexandrov boundary, curvature bounded from above by $\bar{k} \leq 0$ and below by $\underline{k}$, and noncollapsing constant $v_{nc} > 0$ as in Definition 5.1.1. Then there exists $v_0 = v_0(v_{nc}, \underline{k}, \bar{k}, n) > 0$ such that, for every isoperimetric region $E \subset Z$ of volume $v \in (0, v_0]$, we have*

$$P_{d_Z}(E) \geq I_{(\mathbb{M}^n_{\bar{k}}, \delta^{\bar{k}})}(v).$$

Moreover, if equality holds, then E is isometric to a geodesic ball in $\mathbb{M}^n_{\bar{k}}$.

By the construction above and this proposition, we derive that

$$I_{(X,d)}(v) = P_d(\Omega_0) + \sum_{j=1}^{\bar{N}} P_{d_j}(\Omega_j^\infty) \geq I_{(\mathbb{M}^n_{\bar{k}}, \delta^{\bar{k}})}(\mathcal{H}^n_d(\Omega_0)) + \sum_{j=1}^{\bar{N}} I_{(\mathbb{M}^n_{\bar{k}}, \delta^{\bar{k}})}(\mathcal{H}^n_{d_j}(\Omega_j^\infty)).$$

Using that $I_{(\mathbb{M}^n_{\bar{k}}, \delta^{\bar{k}})}$ is subadditive by Lemma 6.0.3, and

$$v = \mathcal{H}^n_d(\Omega_0) + \sum_{j=1}^{\bar{N}} \mathcal{H}^n_{d_j}(\Omega_j^\infty),$$

we obtain $I_{(X,d)}(v) \geq I_{(\mathbb{M}^n_{\bar{k}}, \delta^{\bar{k}})}(v)$ which concludes the proof of Theorem 8.0.1 assuming the validity of Proposition 8.0.3.

The rigidity of the equality case is already included in Proposition 8.0.3.

9 PROOF FOR SMALL VOLUMES

Here we prove Proposition 8.0.3 (which establishes Theorem 1.2.2, as explained in Chapter 8), under the small volume assumption.

9.1 THE INEQUALITY.

Let g_Z be the $C^{1,\alpha}$ Riemannian metric on Z given by Proposition 5.1.2(iii). We prove that there exists $v_0 = v_0(v_{nc}, \underline{k}, \bar{k}, n) > 0$ such that if $0 < v \leq v_0$ then the proposition holds. Let E be an isoperimetric region of volume v.

By [76, Lemma 3.2][1], up to choosing v_0 small enough depending on $v_{nc}, \underline{k}, \bar{k}, n$, there exist $p \in Z$ and $R = R(v) > 0$ such that ∂E is given by the normal graph of a $C^{2,\alpha}$ function u defined on $\partial B_R^{g_Z}(p)$, and $\|u/R\|_{C^{2,\alpha}} \to 0$ as $v \to 0$. We also recall that, by [17, Proposition 4.21] (see also [78, Lemma 4.9]), there exist v_1 and C depending on $v_{nc}, \underline{k}, \bar{k}, n$ such that, if $v_0 \leq v_1$, we obtain

$$\operatorname{diam}_{g_Z}(E) \leq C v^{1/n}. \tag{23}$$

Since $B^{g_Z}_{R/2}(p) \subset E$, (23) gives $R \leq C v^{1/n}$. Up to choosing v_0 smaller, we have $R \leq 1$. Let $\epsilon = \epsilon(n, \underline{k}, \bar{k}, 1)$ be given by Lemma 7.0.1 with $R_0 = 1$.

Let g_i be the C^∞ approximants of Proposition 5.1.2(iv) that converge to g_Z in $C^{1,\alpha}$ and that satisfy $2\underline{k} \leq \operatorname{Sec}_{g_i} \leq \bar{k} \leq 0$. Apply Lemma 7.0.1 with $\tilde{E}_i := (\exp_p^{g_i})^{-1}(E)$ and $\tilde{g}_i = ((\exp_p^{g_i})^{-1})_* g_Z$

$$P_{g_Z}(E) = \lim_i P_{\tilde{g}_i}(\tilde{E}_i) \geq \lim_i I_{(\mathbb{M}^n_{\bar{k}}, \delta^{\bar{k}})}(\operatorname{vol}_{\tilde{g}_i}(\tilde{E}_i)) = I_{(\mathbb{M}^n_{\bar{k}}, \delta^{\bar{k}})}(\operatorname{vol}_{g_Z}(E)),$$

where we used the $C^{1,\alpha}$ convergence of g_i to g_Z and the fact that the perimeter and volume only depend on the Riemannian metric in C^0 topology. This concludes the proof of the inequality.

1 The C^4-regularity assumption on the metric in [76, Lemma 3.2] is solely due to an application of [77, Theorem 1], which requires a C^4 metric to apply Allard's regularity theorem. This assumption can be substantially relaxed by using the extension of Allard's theorem to Alexandrov spaces established in [1] and then applies in our setting; see also the discussion therein.

9.2 THE EQUALITY CASE.

Since $g_i \to g_Z$ in the $C^{1,\alpha}$-topology, for i large ∂E is the normal graph over $\partial B_R^{g_i}(p)$ of a function u_i with small C^1-norm. Apply Lemma 7.0.1 to $\tilde{g}_i = ((\exp_p^{g_i})^{-1})_* g_i$ and $\tilde{E}_i := (\exp_p^{g_i})^{-1}(E)$, all constants are independent of i, to get

$$\begin{aligned} cR^{n-1} \int_{\mathbb{S}^{n-1}} (q_i(R + u_i(\theta), \theta) - 1)\, d\sigma &\leq P_{\tilde{g}_i}(\tilde{E}_i) - I_{(\mathbb{M}^n_{\bar{k}}, \delta^{\bar{k}})}(\mathrm{vol}_{\tilde{g}_i}(\tilde{E}_i)) \\ &= P_{g_i}(E) - I_{(\mathbb{M}^n_{\bar{k}}, \delta^{\bar{k}})}(\mathrm{vol}_{g_i}(E)) \\ &\longrightarrow P_{g_Z}(E) - I_{(\mathbb{M}^n_{\bar{k}}, \delta^{\bar{k}})}(\mathrm{vol}_{g_Z}(E)) = 0, \end{aligned}$$

where q_i is given by (16) for $\tilde{g}_i$. By the monotonicity of q_i and the argument in the proof of the rigidity statement of Lemma 7.0.1, the maps $\exp_p^{g_i} : (\tilde{E}_i, dr^2 + s_{\bar{k}}(r)^2\sigma) \to (E, g_i)$ are almost isometries. Since $g_i \to g_Z$ in the $C^{1,\alpha}$-topology, they converge in C^1 to $\exp_p^{g_Z}$, which is therefore an isometry from $(\tilde{E}, dr^2 + s_{\bar{k}}(r)^2\sigma)$ onto (E, g_Z). As in the proof of Lemma 7.0.1, $\tilde{E}$ is then an isoperimetric region in $(\mathbb{M}^n_{\bar{k}}, \delta^{\bar{k}})$, hence a geodesic ball, and $g_Z \simeq \delta^{\bar{k}}$ on E. This concludes the proof of Proposition 8.0.3.

10 REDUCTION TO THE CASE WITH LOWER RICCI CURVATURE BOUND

In this chapter, we show that it suffices to prove Theorem 1.2.4 under the additional assumption $\mathrm{Ric}_g \geq (n-1)\underline{k}$. Namely, we show how Proposition 10.0.1 implies Theorem 1.2.4: the inequality in Section 10.1 and the moreover part in Section 10.3. We then prove Proposition 10.0.1 in Chapter 11.

Proposition 10.0.1. *Let (M^n, g) be a Cartan-Hadamard manifold with $\mathrm{Sec}_g \leq \bar{k} \leq 0$ and $\mathrm{Ric}_g \geq (n-1)\underline{k}$. Assume in addition that Assumption 1 holds. Then we have*

$$I_{(M^n,g)}(v) \geq I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(v) \text{ for any } v \geq 0.$$

Moreover, if there exists a finite perimeter set $E \subset M$ with $\mathrm{vol}_g(E) = v > 0$ and $P_g(E) = I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(v)$, then E is isometric to a geodesic ball in $\mathbb{M}^n_{\bar{k}}$.

10.1 THE INEQUALITY

Assume the validity of Proposition 10.0.1. Let (M^n, g) be as in Theorem 1.2.4, Ω be a bounded set in M^n, and choose a big enough $R > 0$ and $p \in \Omega$ such that $\Omega \Subset B^g_R(p)$. We will show below that there exists a metric $\hat{g}$ on M^n such that $g = \hat{g}$ on $B^g_R(p)$ and $(M^n, \hat{g})$ is under the assumptions of Proposition 10.0.1. Applying it and using $g = \hat{g}$ on $B^g_R(p) \ni \Omega$ yield

$$P_g(\Omega) = P_{\hat{g}}(\Omega) \geq I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(\mathrm{vol}_{\hat{g}}(\Omega)) = I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(\mathrm{vol}_g(\Omega)). \tag{24}$$

Recalling that the isoperimetric profile defined in (5) can be equivalently defined by taking the infimum over bounded smooth sets of the fixed volume by [72, Theorem 1] and applying (24) to a minimizing sequence Ω_j of volume $v > 0$, we obtain

$$I_{(M^n,g)}(v) = \lim_{j\to+\infty} P_g(\Omega_j) \geq I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(v).$$

We now prove the existence of such a metric $\hat{g}$ to conclude the reduction by basically adapting our argument in the proof of Proposition 5.1.2(iv). Indeed, we fix a function $f : [0, R+1] \to [0, +\infty)$ such that

$$f \in C^\infty([0, R+1]),\ f \equiv 0 \text{ on } [0, R],\ f(R+1) = 1, \text{ and } f'' \geq \sqrt{-\bar{k}}f' > 0 \text{ on } (R, R+1).$$

Fix $r(\cdot) := d_g(p,\cdot)$ and define $\tilde{g} := (1-f\circ r)^{-2}g$. Notice that, by the Hessian comparison theorem, we have $\mathrm{Hess}_g r \geq \sqrt{-\bar{k}}(g - dr\otimes dr)$ and therefore

$$\mathrm{Hess}_g(f\circ r) = f^{''}dr\otimes dr + f^{'}\mathrm{Hess}_g r \geq \sqrt{-\bar{k}}f^{'}g. \tag{25}$$

By a direct computation, for any $q \in B^g_{R+1}(p)$ and any plane $\Pi := \mathrm{span}\{v,w\} \subset T_qM$, we obtain that

$$\begin{aligned}\mathrm{Sec}_{\tilde{g}}(\Pi) = (1-f\circ r)^2\mathrm{Sec}_g(\Pi) - (1-f\circ r)(\mathrm{Hess}_g(f\circ r)(v,v)\\ +\mathrm{Hess}_g(f\circ r)(w,w)) - |\nabla^g(f\circ r)|^2_g.\end{aligned} \tag{26}$$

Using (25), the fact that $|Dr|_g = 1$, and $\mathrm{Sec}_g \leq \bar{k}$, we obtain

$$\mathrm{Sec}_{\tilde{g}} \leq \bar{k}(1-f\circ r)^2 - 2\sqrt{-\bar{k}}(1-f\circ r)f^{'}\circ r - (f^{'}\circ r)^2 = -(\sqrt{-\bar{k}}(1-f\circ r) + f^{'}\circ r)^2 \leq \bar{k},$$

where the last inequality follows from $f^{'} \geq \sqrt{-\bar{k}}f$ on $[0, R+1]$. The latter is obtained by integrating $f^{''} \geq \sqrt{-\bar{k}}f^{'}$ from R and using $f(R) = f^{'}(R) = 0$. The lower bound is a simple consequence of (26). Indeed, set

$$C := \sup_{\overline{B^g_{R+1}(p)}}\left(|\mathrm{Sec}_g| + |\nabla^g(f\circ r)|^2_g, \|\mathrm{Hess}_g(f\circ r)\|_g\right),$$

by (26), we readily obtain $\mathrm{Sec}_{\tilde{g}} \geq -C$. We also notice that, since $(1-f\circ r(x))^{-2}$ explodes to $+\infty$ as x approaches $\partial B^g_{R+1}(p)$, $(B^g_{R+1}(p), \tilde{g})$ is complete and therefore it's a Cartan-Hadamard manifold.

Define $s : [0, R+1) \to [0, +\infty)$ and $s(t) := \int_0^t \frac{1}{1-f(u)}du$, then $s^{'}(t) > 0$ for any $t \in [0, R+1)$. Finally, set $\rho := s^{-1}$ and

$$\Phi : M \to B^g_{R+1}(p) \text{ by } \Phi(\exp^g_p(tv)) = \exp^g_p(s^{-1}(t)v),$$

for any $v \in \{T_pM : |v|_g = 1\}$ and $t \geq 0$. It's easy to see that Φ is a diffeomorphism and $\Phi(x) = x$ for any $x \in B^g_R(p)$. Setting $\hat{g} := \Phi^*\tilde{g}$, we have that $(M^n, \hat{g})$ is now under the assumptions of Proposition 10.0.1. Applying it we conclude the proof of the inequality.

10.2 PROOF OF COROLLARY 1.2.7

The corollary states that isoperimetric regions are bounded. Since we have already established the inequality (3) of Theorem 1.2.4 (assuming Proposition 10.0.1 that we prove later), we can replicate the very same proof of [75, Theorem 3] by the third-named author. Indeed, the author only uses the Ricci lower bound to apply the isoperimetric inequality for small volumes, namely, the results [75, Theorems 2.3 and 2.4]. This can now be replaced by our (3) and the argument proceeds verbatim.

10.3 THE EQUALITY CASE

Assuming the validity of Proposition 10.0.1, we prove the moreover part of Theorem 1.2.4. Let E be a finite perimeter set in M^n with $\mathrm{vol}_g(E) = v > 0$ and $P_g(E) = I_{(\mathbb{M}^n_{\bar{k}}, \delta^{\bar{k}})}(v)$. By Corollary 1.2.7, E is bounded. Then we apply the same argument above with $\Omega = E$ to obtain $\hat{g}$ under the assumptions of Proposition 10.0.1 and with $g = \hat{g}$ on $B^g_R(p) \supset E$. By Assumption 1 applied to E in $(M^n, \hat{g})$, we obtain that $(E, \hat{g})$ is isometric to a geodesic ball in $\mathbb{M}^n_{\bar{k}}$. Since $\hat{g} = g$ on $B^g_R(p)$, we conclude the proof.

11 ON THE ARBITRARY VOLUME REGIME

We now prove Proposition 10.0.1, which implies Theorem 1.2.4 as explained in Chapter 10, in the more general setting of Alexandrov spaces.

Proposition 11.0.1. *Let (Z, d_Z) be an Alexandrov space with no Alexandrov boundary, curvature bounded from above by $\bar{k} \leq 0$ and below by $\underline{k}$, and noncollapsing constant $v_{nc} > 0$ as in Definition 5.1.1. Suppose in addition that Assumption 1 holds. Then*

$$I_{(Z,d_Z)}(v) \geq I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(v), \text{ for any } v \geq 0.$$

Moreover, if there exists a finite perimeter set $E \subset Z$ with $\mathcal{H}^n_{d_Z}(E) = v > 0$ and $P_{d_Z}(E) = I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(v)$, then E is isometric to a geodesic ball in $\mathbb{M}^n_{\bar{k}}$.

Remark 11.0.2. We use Assumption 1 only to apply Lemma 7.0.1. Our argument produces functions that have small C^1-norms and are defined on the boundary of an isoperimetric region F_∞. The volume of such isoperimetric region F_∞ is as in Assumption 1. Lemma 7.0.1 requires the domain of the functions to be a geodesic sphere and this is what we ask in Assumption 1.

Proof. Let g_Z be the $C^{1,\alpha}$ Riemannian metric on Z and $v_0 > 0$ be the small volume threshold from Proposition 8.0.3, and define $S \subseteq \mathbb{R}_+ := [0, +\infty)$ as the set of volumes v for which the conjecture holds true. By Proposition 8.0.3, we have $[0, v_0] \subseteq S$. We will show that S is both closed and open, which implies, by connectedness, $S = \mathbb{R}_+$.

The fact that S is closed follows from the continuity of $I_{(Z,g_Z)}$ and $I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}$, see [72, Theorem 2] for the continuity of $I_{(Z,g_Z)}$.

We now prove that S is open by means of contradiction. Let $v_* \geq v_0$ be the supremum among all $s > 0$ such that $[0, s] \subseteq S$, and assume that $v_* < +\infty$. Take $v_i \searrow v_*$ with $v_i \notin S$, i.e., $I_{(Z,g_Z)}(v_i) < I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(v_i)$ for every i. By [72, Definition 1.1 and Theorem 1], $I_{(Z,g_Z)}$ is infimized over smooth bounded sets and then there are smooth bounded sets $E_i \subset Z$ of volume v_i such that

$$P_{g_Z}(E_i) < I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(v_i) \text{ for each } i. \tag{27}$$

By the definition of v_* and the continuity of both isoperimetric profiles, we must have

$$I_{(Z,g_Z)}(v_*) = I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(v_*). \tag{28}$$

Since $\sup_i P_{g_Z}(E_i) < +\infty$, $\sup_i \mathcal{H}^n_{d_Z}(E_i) = \sup_i v_i < +\infty$, and, by Remark 5.2.2, (Z, g_Z) is of $C^{1,\alpha}$-locally asymptotic bounded geometry, we may apply [16, Theorem 1.2] (see also [73, Theorem 1,

pg 64]). This yields a nondecreasing, possibly unbounded, sequence $\{N_i\} \subset \mathbb{N}_{\geq 1}$, points $p_{i,j} \in Z$, with $1 \leq j \leq N_i$ for any i, and pairwise disjoint subsets $E_{i,j} \subset E_i$ such that

- $\lim_i d_Z(p_{i,j}, p_{i,l}) = +\infty$, for any $j \neq l < \bar{N} + 1$, where $\bar{N} := \lim_i N_i \in \mathbb{N} \cup \{+\infty\}$;
- for every $1 \leq j < \bar{N} + 1$, the sequence $(Z, g_Z, \mathcal{H}^n_{d_Z}, p_{i,j})$ converges in the $C^{1,\alpha}$-topology to a $C^{1,\alpha}$ pointed Riemannian manifold $(Z_j, g_j, \mathcal{H}^n_{d_j}, p_j)$ as $i \to +\infty$;
- for every $1 \leq j < \bar{N} + 1$, there exist sets of finite perimeter $F_j \subset Z_j$ such that $(E_{i,j}, g_Z, p_{i,j}) \to (F_j, g_j, p_j)$ in the pointed $C^{1,\alpha}$-flat topology[1], cf. Definition 5.2.3, and there holds

$$v_* = \sum_{j=1}^{\bar{N}} \mathcal{H}^n_{d_j}(F_j) \tag{29}$$

and

$$P_{g_j}(F_j) \leq \lim_i P_{g_Z}(E_{i,j}). \tag{30}$$

By definition of pointed $C^{1,\alpha}$-flat convergence, for each $1 \leq j \leq \bar{N}$ and any $R_j > 0$, there exists a domain Ω_{R_j} satisfying $F_j \subset B_{R_j}(p_j) \subset \Omega_{R_j} \subset Z_j$, an integer $\nu_{R_j} \in \mathbb{N}$, and $C^{2,\alpha}$ embeddings $\Psi_{i,R_j} : \Omega_{R_j} \to Z$ such that, for all $i \geq \nu_{R_j}$, we have that $E_{i,j} \subset B_{R_j}(p_{i,j}) \subset \Psi_{i,R_j}(\Omega_{R_j})$.

We now claim that the decomposition $\{E_{i,j}\}_j$ of E_i only has one connected component at the limit. More precisely, the claim—which we prove at the end—is the following:

Claim 1: $\bar{N} = 1$.

Since, by claim 1, $\bar{N} = 1$, denote $(Z_\infty, g_\infty) := (Z_1, g_1)$, $F_\infty := F_1$, $E_i := E_{i,1}$, and $p_i := p_{i,1}$. By [16, Proposition 2.19, eq. (2.17)], we derive that

$$I_{(Z,g_Z)}(v_*) \leq I_{(Z_\infty,g_\infty)}(v_*) \leq P_{g_\infty}(F_\infty) \overset{(30)}{\leq} \lim_i P_{g_Z}(E_i) = I_{(Z,g_Z)}(v_*).$$

By the equality above above, F_∞ is an isoperimetric region of volume v_* and, by (28), we have

$$P_{g_\infty}(F_\infty) = I_{(\mathbb{M}^n_{\bar{k}}, \delta^{\bar{k}})}(v_*). \tag{31}$$

As explained in Chapter 8, (Z_∞, d_{g_∞}) is a non-collapsing Alexandrov space with no Alexandrov boundary and the same curvature bounds as (Z, d_Z) and [16, Proposition 2.19, eq. (2.17)] gives $I_{(Z_\infty,g_\infty)} \geq I_{(Z,g_Z)}$. We can then apply Assumption 1.

Therefore, by the definition of v_*, $\mathrm{vol}_{g_\infty}(F_\infty) = v_*$, (31), and Assumption 1, F_∞ is smooth and hence we can apply [77, Theorem 1][2] with $T_i = \Psi^{-1}_{i,R_1}(E_i)$, $(M, g) = (Z_\infty, \Psi^{-1}_{i,R_1 *} g_Z)$, and $B = F_\infty$, to obtain that $\partial \Psi^{-1}_{i,R_1}(E_i)$ are normal $C^{2,\alpha}$-graphs of a function u_i over ∂F_∞. So, defining $\tilde{u}_i := u_i \circ \Psi^{-1}_{i,R_1}$, we have that ∂E_i is a normal $C^{2,\alpha}$-graph of $\tilde{u}_i$ over $\partial \Psi_{i,R_1}(F_\infty)$.

1 Although the authors in [73] assumes C^0-locally asymptotic bounded geometry and therefore only obtains C^0-flat convergence, their proof readily gives $C^{1,\alpha}$-flat convergence if one assumes $C^{1,\alpha}$-locally asymptotic bounded geometry.

2 The assumption of C^4-regularity of the metric in [77, Theorem 1] arises solely from the use of Allard's regularity theorem. It can be substantially weakened using the extension of Allard's theorem to Alexandrov spaces established in [1] and then applies in our setting; see also the discussion therein.

Using that ∂E_i is a normal graph over $\partial\Psi_{i,R_1}(F_\infty)$ and, by Assumption 1, F_∞ is isometric to geodesic ball in $\mathbb{M}^n_{\bar{k}}$, we will show at the end the following claim.

Let g_j be the C^∞ approximants of Proposition 5.1.2(iv) that converge to g_Z in $C^{1,\alpha}$ and that satisfy $2\underline{k} \leq \mathrm{Sec}_{g_j} \leq \bar{k} \leq 0$.

Claim 2: ∂E_i is the normal graph of $w_{i,j} \in C^1(\partial B^{g_j}_{r_j}(p_i))$ with $\|w_{i,j}\|_{C^1} \to 0$ as $i \to \infty$ for $p_i \in Z$ and $r_j > 0$ with $\sup_j r_j < +\infty$.

Let $\epsilon = \epsilon(n, \underline{k}, \bar{k}, R_0)$ be given by Lemma 7.0.1 with $R_0 = \sup_j r_j < +\infty$; it does not depend on i nor j. By Claim 2, for i large enough, $w_{i,j}$ has small C^1-norm with $R = r_j$. Hence applying Lemma 7.0.1 with $\tilde{E}^j_i := (\exp^{g_j}_{p_i})^{-1}(E_i)$ and $\tilde{g}^j_i = (\exp^{g_j}_{p_i})^{-1}_* g_Z$ to obtain

$$\begin{aligned} P_{g_Z}(E_i) = P_{\tilde{g}_i}(\tilde{E}_i) = \lim_j P_{\tilde{g}^j_i}(\tilde{E}^j_i) &\geq \lim_j I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(\mathrm{vol}_{\tilde{g}^j_i}(\tilde{E}^j_i)) \\ &\geq I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(\mathrm{vol}_{\tilde{g}_i}(\tilde{E}_i)) = I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(\mathrm{vol}_{g_Z}(E_i)), \end{aligned}$$

where $\tilde{E}_i := (\exp^{g_Z}_{p_i})^{-1}(E_i)$ and $\tilde{g}_i = (\exp^{g_Z}_{p_i})^{-1}_* g_Z$ and we used the fact that perimeter and volume depend C^0 on the metric and $g_j \to g_Z$ in $C^{1,\alpha}$. This contradicts (27). This contradiction shows that S is open and then $S = \mathbb{R}_+$ which concludes the proof of the inequality.

For the rigidity part, since we have already established $I_{(Z,d_Z)} \geq I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}$ for *every* volume, we are under the conditions of Assumption 1 for any choice of volume $V \geq 0$ and therefore the rigidity statement follows. This completes the proof of the proposition up to Claims 1 and 2.

Proof of claim 1: Since the sets $E_{i,j} \subset E_i$ are pairwise disjoint and satisfy $\cup_{j=1}^{\bar{N}} E_{i,j} = E_i$ we have

$$\sum_{j=1}^{\bar{N}} P_{g_j}(F_j) \overset{(30)}{\leq} \sum_{j=1}^{\bar{N}} \lim_i P_{g_Z}(E_{i,j}) = \lim_i P_{g_Z}(E_i) = I_{(Z,g_Z)}(v_*) \overset{(28)}{=} I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(v_*). \tag{32}$$

Furthermore, if $\bar{N} > 1$, then, by (29), we have $v_j < v_*$ for every $1 \leq j \leq \bar{N}$. Hence, by the definition of v_* and [16, Proposition 2.19, eq. (2.17)],

$$P_{g_j}(F_j) \geq I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(v_j). \tag{33}$$

Moreover, by Lemma 6.0.3, we have

$$\begin{aligned} \sum_{j=1}^{\bar{N}} I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(v_j) \geq I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}\left(\sum_{j=1}^{\bar{N}} v_j\right) &\overset{(29)}{=} I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(v_*) \\ &\overset{(32)}{\geq} \sum_{j=1}^{\bar{N}} P_{g_j}(F_j) \overset{(33)}{\geq} \sum_{j=1}^{\bar{N}} I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(v_j). \end{aligned}$$

Therefore,

$$\sum_{j=1}^{\bar{N}} I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}(v_j) = I_{(\mathbb{M}^n_{\bar{k}},\delta^{\bar{k}})}\left(\sum_{j=1}^{\bar{N}} v_j\right).$$

Since $v_j > 0$ for at least two numbers j (otherwise $\bar{N} = 1$) and, by Lemma 6.0.3, the function $t \mapsto I_{(\mathbb{M}^n_{\bar{k}}, \delta^{\bar{k}})}(t)$ is *strictly* subadditive, the last displayed equality gives a contradiction and then $\bar{N} = 1$ and this proves the claim.

Proof of claim 2: To simplify notation, let $F_i := \Psi_{i,R_1}(F_\infty)$, $g := g_j$, $g_i := \Psi_{i,R_1 *} g_\infty$, $r = r_j$, and $\mathcal{I}_i : (B^{\delta_{\bar{k}}}_r(0), \delta_{\bar{k}}) \longrightarrow (F_i, g_i)$ be an isometry, i.e., $\mathcal{I}_i^* g_i = \delta_{\bar{k}}$. Set $p_i := \mathcal{I}_i(0)$ and notice that the fact that $\sup_j r_j < +\infty$ is clear from the set-up.

$\rightarrow$ *Step 1: ∂E_i is close to $\partial B^g_r(p_i)$ both in position and in tangent plane.*

Up to choosing R_1 big enough, we have that

$$\varepsilon_i := \sup_{B^g_{2r}(p_i)} \|g_i - g\|_{C^{1,\alpha}} \longrightarrow 0.$$

Let $G_i : \partial F_i \longrightarrow \partial E_i$ be the normal-graph parametrization of ∂E_i over ∂F_i, namely,

$$G_i(x) = \exp^{g_i}_x \left(\tilde{u}_i(x) \nu^{g_i}_{\partial F_i}(x) \right).$$

Since $\|\tilde{u}_i\|_{C^{2,\alpha}} \to 0$ as i grows (by [77]), it is easy to see that

$$\eta_i := \sup_{x \in \partial F_i} \left[\frac{1}{r} d_g(G_i(x), x) + \left| \nu^g_{\partial E_i}(G_i(x)) - P^g_{x, G_i(x)} \nu^g_{\partial F_i}(x) \right|_g \right] \longrightarrow 0,$$

where $P^g_{x,y} : T_x M \to T_y M$ denotes g-parallel transport along the unique g-geodesic joining x to y for any $x, y \in \partial F_i$.

For $v \in T_{p_i} M$ with $|v|_g = 1$ and $0 \le t \le r$, define

$$A_i(v) := \frac{v}{|v|_{g_i}}, \quad \gamma_{i,v}(t) := \exp^{g_i}_{p_i}(t A_i(v)), \quad \gamma_v(t) := \exp^g_{p_i}(tv).$$

Since $d\mathcal{I}_i(0)$ maps the $\delta_{\bar{k}}$-unit sphere onto $\{v \in T_{p_i} M, |v|_{g_i} = 1\}$, every vector $A_i(v)$ is of the form $A_i(v) = d\mathcal{I}_i(0)[\theta]$ for some $|\theta|_{\delta_{\bar{k}}} = 1$. The radial curve $t \longmapsto \mathcal{I}_i(\exp^{\delta_{\bar{k}}}_0(t\theta))$ is a g_i-geodesic, and hence $\gamma_{i,v}(t) = \mathcal{I}_i(\exp^{\delta_{\bar{k}}}_0(t\theta))$. This in turn implies that

$$\partial F_i = \left\{ \gamma_{i,v}(r) : v \in T_{p_i} M, |v|_g = 1 \right\}.$$

We now compare $\gamma_{i,v}$ with γ_v. By the formula for the difference of the Levi-Civita connections, we obtain

$$\sup_{B^g_{2r}(p_i)} |\nabla^{g_i} - \nabla^g|_g \le C\varepsilon_i. \tag{34}$$

Moreover,

$$|A_i(v) - v|_g \le C\varepsilon_i. \tag{35}$$

Because $\gamma_{i,v}$ is a g_i-geodesic, it satisfies

$$\nabla^{g}_{\gamma'_{i,v}}\gamma'_{i,v} = -(\nabla^{g_i} - \nabla^{g})(\gamma'_{i,v}, \gamma'_{i,v}). \tag{36}$$

Thus, when viewed in the metric g, the curve $\gamma_{i,v}$ has acceleration bounded by $C\varepsilon_i$ (by (34) and (36)), while its initial velocity differs from that of γ_v by at most $C\varepsilon_i$ (by (35)).

From $\underline{k} \le \mathrm{Sec}_g \le \bar{k} \le 0$, the continuous dependence for the geodesic equation, and the Jacobi field description of the differential of the geodesic flow, we can derive the following uniform estimate

$$\sup_{\substack{\{v\in T_{p_i}M:|v|_g=1\}\\ 0\le t\le r}} \left(d_g(\gamma_{i,v}(t), \gamma_v(t)) + \left|\gamma'_{i,v}(t) - P^{g}_{\gamma_v(t),\gamma_{i,v}(t)}\gamma'_v(t)\right|_g \right) \le C\varepsilon_i. \tag{37}$$

For the sake of readability and since the proof of (37) is very technical, we prove it at the end.

Here and below, vectors at nearby points are compared by g-parallel transport along the connecting geodesic. In particular, notice that $\gamma_{i,v}(r) \in \partial F_i$, $\gamma_v(r) \in \partial B^g_r(p_i)$, and $d_g(\gamma_{i,v}(r), \gamma_v(r)) \le C\varepsilon_i$.

Through the isometry I_i, the curve $\gamma_{i,v}$ corresponds to a radial geodesic in the constant-curvature space form $(\mathbb{R}^n, \delta_{\bar{k}})$. By the Gauss lemma, $\gamma'_{i,v}(r)$ is the outward g_i-unit normal to ∂F_i.

Consequently, for $W \in T_{\gamma_{i,v}(r)}\partial F_i$,

$$\left|g(\gamma'_{i,v}(r), W)\right| = \left|(g - g_i)(\gamma'_{i,v}(r), W)\right| \le C\varepsilon_i |W|_g.$$

This and (37) imply

$$\left|\nu^{g}_{\partial F_i}(\gamma_{i,v}(r)) - P^{g}_{\gamma_v(r),\gamma_{i,v}(r)}\nu^{g}_{\partial B^g_r(p_i)}(\gamma_v(r))\right|_g \le C\varepsilon_i. \tag{38}$$

Since $v \mapsto \gamma_{i,v}(r)$ and G_i are diffeomorphisms onto their images, $v \mapsto G_i(\gamma_{i,v}(r))$ parametrizes ∂E_i. The definition of η_i, the triangle inequality, and (37), ensure the position estimate, namely,

$$d_g(G_i(\gamma_{i,v}(r)), \gamma_v(r)) \le C(\varepsilon_i + \eta_i), \tag{39}$$

The triangle inequality, $\|\tilde{u}\|_{C^{2,\alpha}} \to 0$, (38), and the same Jacobi field estimates to control the change from parallel transport along the two short connecting geodesics, guarantee the tangent plane estimate, namely,

$$\left|\nu^{g}_{\partial E_i}(G_i(\gamma_{i,v}(r))) - P^{g}_{\gamma_v(r),G_i(\gamma_{i,v}(r))}\nu^{g}_{\partial B^g_r(p_i)}(\gamma_v(r))\right|_g \le C(\varepsilon_i + \eta_i). \tag{40}$$

→ *Step 2: A diffeomorphism between ∂E_i and $\partial B^g_r(p_i)$.*

Since (Z, g) is Cartan-Hadamard, $\exp^g_{p_i} : T_{p_i}M \to M$ is a global diffeomorphism and so is

$$\begin{aligned} \mathcal{T}_i : \partial B^g_r(p_i) \times (-r, \infty) &\longrightarrow M \setminus \{p_i\}, \\ \mathcal{T}_i(x, s) &:= \exp^g_x\left(s\nu^{g}_{\partial B^g_r(p_i)}(x)\right) \end{aligned}$$

In particular, $\partial B_r^g(p_i)$ admits a tubular neighborhood of uniform width, say r, and, by Step 1 (cf. (39)), ∂E_i is contained in this tubular neighborhood. Let

$$\pi_i : \mathcal{T}_i\left(\partial B_r^g(p_i) \times \left(-\frac{r}{2}, \frac{r}{2}\right)\right) \longrightarrow \partial B_r^g(p_i)$$

be the corresponding normal projection. In these coordinates, we have $\ker D\pi_i = \operatorname{span}\{\partial_s\}$, where ∂_s is the radial unit vector. By (40) in Step 1, we derive

$$g(\nu^g_{\partial E_i}, \partial_s) \geq 1 - C(\varepsilon_i + \eta_i). \tag{41}$$

Therefore, $\pi_i|_{\partial E_i}\colon \partial E_i \longrightarrow \partial B_r^g(p_i)$ is a local diffeomorphism. It remains to show that this local diffeomorphism is globally one-to-one. Consider $\Theta_i : \{T_{p_i}M : |v|_g = 1\} \longrightarrow \{T_{p_i}M : |v|_g = 1\}$ defined by

$$\Theta_i(v) := \frac{1}{r}\left(\exp^g_{p_i}\right)^{-1}\left(\pi_i(G_i(\gamma_{i,v}(r)))\right).$$

The position estimate (39) implies that $\sup_{v\in\{T_{p_i}M:|v|_g=1\}}|\Theta_i(v) - v|_g \longrightarrow 0$. Therefore, for large i, Θ_i is homotopic to the identity through $H_i(t, v) := |V|/|V|_g$ for $V := (1-t)v + t\Theta_i(v)$.

It follows that Θ_i, and hence $\pi_i|_{\partial E_i}$, has degree one. Since $\pi_i|_{\partial E_i}$ is a local diffeomorphism between compact connected spheres, it is a covering map. Its degree is one, so it has only one sheet and is therefore a diffeomorphism.

$\rightarrow$ *Step 3: Identifying the function whose normal graph is* ∂E_i.

We can finally define $w_i : \partial B_r^g(p_i) \longrightarrow \mathbb{R}$ by

$$\left(\pi_i|_{\partial E_i}\right)^{-1}(x) = \exp^g_x\left(w_i(x)\nu^g_{\partial B_r^g(p_i)}(x)\right).$$

It is then clear that ∂E_i is given by the normal graph of w_i over $\partial B_r^g(p_i)$. By the position estimate (39), we readily get

$$\|w_i\|_{C^0(\partial B_r^g(p_i))} \leq C(\varepsilon_i + \eta_i).$$

In the coordinates given by $\mathcal{T}_i$, the metric has the form $\mathcal{T}_i^* g = ds^2 + h_{i,s}$, where $h_{i,s}$ is the metric induced on the parallel hypersurface to $\partial B_r^g(p_i)$ at signed distance s. The unit normal to the graph of w_i is

$$\frac{\partial_s - \nabla^{h_{i,w_i}} w_i}{\sqrt{1 + |\nabla^{h_{i,w_i}} w_i|^2_{h_{i,w_i}}}}.$$

We notice that $\underline{k} \leq \operatorname{Sec}_g \leq \bar{k} \leq 0$, $|s| \leq r/2$, and classical Jacobi field estimates, imply that $h_{i,s}$ and $h_{i,0}$ are uniformly equivalent. With this and since, by (41), this normal differs from ∂_s by at most $C(\varepsilon_i + \eta_i)$, we conclude that

$$\|\nabla^{\partial B_r^g(p_i)} w_i\|_{C^0(\partial B_r^g(p_i))} \leq C(\varepsilon_i + \eta_i).$$

This finally finishes the proof of claim 2.

Proof of (37)*:* Fix i, $v \in T_{p_i}M$, $|v|_g = 1$, and we will work along the reference geodesic γ_v.

→ *Step 1: Everything in the same g-connection.*

Since (Z, g) is Cartan-Hadamard, $\exp_q^g$ is a global diffeomorphism for every $q \in M$, so γ_v has no conjugate points and the difference vector

$$y(t) := (\exp^g_{\gamma_v(t)})^{-1}(\gamma_{i,v}(t)) \ \in \ T_{\gamma_v(t)}M, \qquad 0 \le t \le r,$$

is well defined, with $|y(t)|_g = d_g(\gamma_v(t), \gamma_{i,v}(t))$. Fix a g-parallel orthonormal frame $E_1(t), \dots, E_n(t)$ along γ_v and identify $y(t)$ with its coordinate vector in $\mathbb{R}^n$; write $\dot{y}, \ddot{y}$ for the covariant derivative of y along γ_v, read in this frame, so that covariant differentiation along γ_v becomes ordinary differentiation of coordinates. By definition $y(0) = 0$, differentiating $\exp^g_{\gamma_v(t)}(y(t)) = \gamma_{i,v}(t)$ at $t = 0$,and (35), give

$$\dot{y}(0) = \gamma'_{i,v}(0) - \gamma'_v(0) = A_i(v) - v, \qquad |\dot{y}(0)|_g \le C\varepsilon_i.$$

→ *Step 2: The ODE for y: a forced Jacobi equation.*

Differentiate $\exp^g_{\gamma_v(t)}(y(t)) = \gamma_{i,v}(t)$ twice in t. For the sake of readability, for $a \in \{1, \dots, n\}$, set

$$H(t, z) := \exp^g_{\gamma_v(t)}(z^a E_a(t)), \qquad V_0 := H_*\partial_t = dH(\partial_t), \qquad V_a := H_*\partial_{z^a} = dH(\partial_{z^a}),$$

so that $\gamma_{i,v}(t) = H(t, y(t))$. Since $[\partial_t, \partial_{z^a}] = 0$ and ∇^g is torsion-free, for $z = y(t)$, we get

$$\nabla^g_{\gamma'_{i,v}} \gamma'_{i,v} = \nabla^g_{V_0} V_0 + 2\dot{y}^a \nabla^g_{V_0} V_a + \dot{y}^a \dot{y}^b \nabla^g_{V_a} V_b + \ddot{y}^a V_a, \tag{42}$$

At $z = 0$, since the differential of $\exp^g_{\gamma_v(t)}$ at the origin is the identity, we have $V_a(t, 0) = E_a(t)$. It also follows that $\nabla^g_{V_0} V_0(t, 0) = \nabla^g_{\gamma'_v} \gamma'_v = 0$. Because E_a is parallel, $\nabla^g_{V_0} V_a(t, 0) = \nabla_{\partial_t} E_a(t) = 0$. This gives that

$$\nabla^g_{\gamma'_{i,v}} \gamma'_{i,v} = \dot{y}^a \dot{y}^b \nabla^g_{V_a} V_b + \ddot{y}^a V_a \ \text{ at } \ z = 0.$$

To isolate the coefficient of $\ddot{y}$, define

$$A_t(z) : \mathbb{R}^n \longrightarrow T_{H(t,z)}M, \qquad A_t(z)e_a := V_a(t, z).$$

Since (Z, g) is Cartan-Hadamard, the exponential map is a global diffeomorphism and thus $A_t(z)$ is invertible.

Define $B(t, z, w) := \nabla^g_{V_0} V_0(t, z) + 2w^a \nabla^g_{V_0} V_a(t, z) + w^a w^b \nabla^g_{V_a} V_b(t, z)$ and $G(t, z, w) := A_t(z)^{-1} B(t, z, w)$. Since $\ddot{y}^a V_a = A_t(y)\ddot{y}$, equation (42) can be rewritten as

$$\nabla^g_{\gamma'_{i,v}} \gamma'_{i,v} = A_t(y)(\ddot{y} + G(t, y, \dot{y})) \ \text{ and } \ \ddot{y} + G(t, y, \dot{y}) = A_t(y)^{-1} \nabla^g_{\gamma'_{i,v}} \gamma'_{i,v}. \tag{43}$$

We now compute the linearization of G at $(z, w) = (0, 0)$. First, we have that $B(t, 0, 0) = \nabla^g_{V_0} V_0(t, 0) = 0$, and hence $G(t, 0, 0) = 0$.

The derivative with respect to w readily gives

$$\partial_{w^c} G(t,z,w) = A_t(z)^{-1}\left(2\nabla^g_{V_0} V_c(t,z) + 2w^b \nabla^g_{V_c} V_b(t,z)\right),$$

and, at $(z,w)=(0,0)$, since $V_c(t,0)=E_c(t)$ and E_c is parallel, we derive

$$D_w G(t,0,0)e_c = 2A_t(0)^{-1}\nabla^g_{V_0} V_c(t,0) = 0. \tag{44}$$

We must also compute the derivative with respect to z. We see that $D_z G[e_c] = D_z(A^{-1})[e_c]\,B + A^{-1}D_z B[e_c]$ and, at $(z,w)=(0,0)$, the first term vanishes because $B(t,0,0)=0$. Thus

$$D_z G(t,0,0)e_c = A_t(0)^{-1}[\nabla^g_{V_c}\nabla^g_{V_0} V_0]_{(t,0)}. \tag{45}$$

By the curvature commutation identity, we have $\nabla^g_{V_c}\nabla^g_{V_0}V_0 - \nabla^g_{V_0}\nabla^g_{V_c}V_0 = R^g(V_c,V_0)V_0$ and, since $\nabla^g_{V_c}V_0 = \nabla^g_{V_0}V_c$, we get

$$\nabla^g_{V_c}\nabla^g_{V_0}V_0 = \nabla^g_{V_0}\nabla^g_{V_0}V_c + R^g(V_c,V_0)V_0.$$

At $z=0$, $V_c(t,0)=E_c(t)$, $V_0(t,0)=\gamma'(t)$, and since E_c is parallel, we see that $\nabla^g_{V_0}\nabla^g_{V_0}V_c(t,0)=0$ and thus $\nabla^g_{V_c}\nabla^g_{V_0}V_0(t,0) = R^g(E_c(t),\gamma'(t))\gamma'(t)$. Plugging this into (45), we get

$$D_z G(t,0,0)e_c = A_t(0)^{-1}R^g(E_c(t),\gamma'(t))\gamma'(t).$$

Define $\mathcal{R}(t):\mathbb{R}^n \longrightarrow \mathbb{R}^n$ by $\mathcal{R}(t)u := A_t(0)^{-1}\left(R^g(A_t(0)u,\gamma'(t))\gamma'(t)\right)$. So, by the last displayed equation, we have

$$D_z G(t,0,0) = \mathcal{R}(t). \tag{46}$$

By Taylor expansion, (44), and (46), we then derive

$$G(t,z,w) = \mathcal{R}(t)z - \mathcal{N}(t,z,w),$$

where $\mathcal{N}(t,0,0)=0$ and $D_{(z,w)}\mathcal{N}(t,0,0)=0$. Define $\mathcal{F}_i(t) := A_t(y(t))^{-1}\nabla^g_{\gamma'_{i,v}}\gamma'_{i,v}$. By the last displayed equation and (43), we finally derive the ODE—the forced Jacobi equation—for y:

$$\ddot{y}(t) + \mathcal{R}(t)y(t) = \mathcal{N}(t,y(t),\dot{y}(t)) + \mathcal{F}_i(t), \qquad 0\le t\le r. \tag{47}$$

It remains to estimate the two terms on the right-hand side of (47). Since $\mathcal{N}$ and its first derivative vanish at $(z,w)=(0,0)$, Taylor's formula with integral remainder gives, with $\xi=(z,w)\in\overline{B_1}\times\overline{B_1}$ and $t\in[0,r]$,

$$|\mathcal{N}(t,z,w)| \le C(|z|+|w|).$$

Finally, since $A_t(y(t))^{-1}$ is uniformly bounded on $\overline{B_1}$, by (34), we get

$$\sup_{0\le t\le r}|\mathcal{F}_i(t)|_g = \sup_{0\le t\le r}\left|A_t(y(t))^{-1}\left(\nabla^{g}_{\gamma'_{i,v}}-\nabla^{g_i}_{\gamma'_{i,v}}\right)\gamma'_{i,v}\right|_g \le C\varepsilon_i. \tag{48}$$

→ *Step 3: A first order ODE.*

Set $Z(t) := (y(t), \dot y(t)) \in \mathbb{R}^{2n}$, so that (47) becomes the first-order linear system

$$\dot Z(t) = \mathcal{A}(t)Z(t) + (0,\ \mathcal{N}(t)+\mathcal{F}_i(t)),\quad \mathcal{A}(t) = \begin{pmatrix} 0 & I_n \\ -\mathcal{R}(t) & 0\end{pmatrix},\quad \|\mathcal{A}(t)\|\le 1+\rho. \tag{49}$$

This is the variational equation of [51, Chapter V, Theorem 3.1]. Let $\Phi(t,s)$ be the state-transition matrix of the homogeneous system $\dot Z = \mathcal{A}(t)Z$, i.e., $\partial_t\Phi(t,s) = \mathcal{A}(t)\Phi(t,s), \Phi(s,s) = I$; a column of $\Phi(t,0)$ is $(J(t), \dot J(t))$ for the Jacobi field J along γ_v with the corresponding initial data. So, bounding $\|\Phi\|$ amounts to bounding Jacobi fields on $[0,r]$. Since $\mathrm{Sec}_g \le 0$, for any Jacobi field J along the unit-speed geodesic γ_v,

$$\frac{d^2}{dt^2}|J|^2 = 2|\dot J|^2 - 2\mathrm{Sec}_g(J,\gamma'_v)(|J|^2 - \langle J,\gamma'_v\rangle^2)\ \ge\ 2|\dot J|^2 \ge\ 0;$$

hence $|J|^2$ is convex along γ_v, γ_v has no conjugate points, and Φ cannot blow up on $[0,r]$. On the other hand, by Rauch comparison, we get

$$|J(t)| \le \cosh(\sqrt{-\underline{k}}t)|J(0)| + \frac{\sinh(\sqrt{-\underline{k}}t)}{\sqrt{-\underline{k}}}|\dot J(0)|,$$
$$|\dot J(t)| \le \sqrt{-\underline{k}}\sinh(\sqrt{-\underline{k}}t)|J(0)| + \cosh(\sqrt{-\underline{k}}t)|\dot J(0)|.$$

Together, these two bounds give a constant $K = K(n,r,\underline{k},\bar{k}) < \infty$, depending only on n, r, $\underline{k}$, $\bar{k}$, in particular not on i or on v, such that

$$\|\Phi(t,s)\| \le K \qquad \text{for all } 0\le s\le t\le r.$$

→ *Step 4: Duhamel's Principle and Grönwall's Inequality.*

Variation of constants on (49), with $Z(0) = (0,\ A_i(v) - v)$, gives

$$Z(t) = \Phi(t,0)Z(0) + \int_0^t \Phi(t,s)(0,\ \mathcal{N}(s)+\mathcal{F}_i(s))\,ds.$$

As long as $|Z(s)| \le 1$ for $0\le s\le t$, taking norms and using $\|\Phi\|\le K$, $|Z(0)|\le C\varepsilon_i$ (by (35)), $|\mathcal{F}_i(s)|\le C\varepsilon_i$ (by (48)), and $|\mathcal{N}(s)|\le C|Z(s)|$ gives

$$|Z(t)| \le KC(1+r)\varepsilon_i + KC\int_0^t |Z(s)|\,ds,$$

and Grönwall's inequality yields

$$|Z(t)| \le KC(1+r)e^{KCr}\varepsilon_i =: C'\varepsilon_i, \qquad 0 \le t \le r, \tag{50}$$

on the interval where $|Z| \le 1$ holds. Since C' does not depend on i, once i is large enough that $C'\varepsilon_i < 1$ we have $|Z| \le 1$ on $[0, r]$; so the bound above holds on all of $[0, r]$ for every large i.

It remains to translate the bound on $Z = (y, \dot{y})$ into the two terms of (37). By definition $|y(t)|_g = d_g(\gamma_{i,v}(t), \gamma_v(t))$, which is the first term.

For the second term, fix $t \in [0, r]$ and introduce the connecting variation

$$\sigma(s, t) = \exp^g_{\gamma_v(t)}(s\, y(t)), \qquad 0 \le s \le 1,$$

so that $\sigma(0, t) = \gamma_v(t)$, $\sigma(1, t) = \gamma_{i,v}(t)$, and, for each fixed t, the curve $s \mapsto \sigma(s, t)$ is the unique minimizing g-geodesic joining $\gamma_v(t)$ to $\gamma_{i,v}(t)$ of length $|y(t)|_g$. Put $S := \partial_s\sigma$ and $J := \partial_t\sigma$. Since every s-curve is a geodesic, S is parallel along it, so $|S|_g \equiv |y(t)|_g$, and J is a Jacobi field along it with $J(0) = \gamma_v'(t)$, $J(1) = \gamma_{i,v}'(t)$, and $\nabla_s J(0) = \dot{y}(t)$.

Let P_s be g-parallel transport along $s \mapsto \sigma(s, t)$ from 0 to s, so $P_1 = P^g_{\gamma_v(t), \gamma_{i,v}(t)}$ is the operator in (37), and set $\widetilde{J} := P_s^{-1} J \in T_{\gamma_v(t)}M$. The Jacobi equation becomes

$$\widetilde{J}''(s) = -P_s^{-1} R(J, S)S,$$

and $\underline{k} \le \mathrm{Sec}_g \le \bar{k} \le 0$, together with $|S|_g = |y(t)|_g$ and the Rauch bound $|J(s)|_g \le C$, gives $|\widetilde{J}''(s)|_g \le C\rho\, |y(t)|_g^2$. Taylor's formula with integral remainder across $s \in [0, 1]$,

$$\widetilde{J}(1) = \widetilde{J}(0) + \widetilde{J}'(0) + E, \qquad |E|_g \le C\rho\, |y(t)|_g^2,$$

reads $P_1^{-1}\gamma_{i,v}'(t) = \gamma_v'(t) + \dot{y}(t) + E$; applying the isometry P_1, we get

$$\gamma_{i,v}'(t) - P^g_{\gamma_v(t), \gamma_{i,v}(t)}\gamma_v'(t) = P^g_{\gamma_v(t), \gamma_{i,v}(t)}(\dot{y}(t) + E).$$

Since $P^g_{\gamma_v(t), \gamma_{i,v}(t)}$ is a linear g-isometry the right-hand side has g-norm $|\dot{y}(t) + E|_g$. By the triangle inequality and the remainder bound $|E|_g \le C\rho\, |y(t)|_g^2$,

$$|\gamma_{i,v}'(t) - P^g_{\gamma_v(t), \gamma_{i,v}(t)}\gamma_v'(t)|_g \;=\; |\dot{y}(t) + E|_g \;\le\; |\dot{y}(t)|_g + C\rho\, |y(t)|_g^2.$$

The Grönwall bound (cf. (50)) gives $|y(t)|_g \le |Z(t)| \le C\varepsilon_i$ and $|\dot{y}(t)|_g \le |Z(t)| \le C\varepsilon_i$; in particular $|y(t)|_g \le 1$ for i large, so $|y(t)|_g^2 \le |y(t)|_g \le C\varepsilon_i$. Hence

$$|\gamma_{i,v}'(t) - P^g_{\gamma_v(t), \gamma_{i,v}(t)}\gamma_v'(t)|_g \;\le\; C\varepsilon_i,$$

and, adding the first term $d_g(\gamma_{i,v}(t), \gamma_v(t)) = |y(t)|_g \le C\varepsilon_i$, we obtain

$$d_g(\gamma_{i,v}(t), \gamma_v(t)) + |\gamma_{i,v}'(t) - P^g_{\gamma_v(t), \gamma_{i,v}(t)}\gamma_v'(t)|_g \;\le\; C\varepsilon_i, \qquad 0 \le t \le r.$$

$\rightarrow$ *Step 5: Uniformity in v and i.*

Every constant occurring above ρ, a, K, C, C' depends only on n, r, and the curvature bounds $\underline{k} \leq \mathrm{Sec}_g \leq \bar{k} \leq 0$ on the fixed ball $B^g_{2r}(p_i)$. None of them depends on the direction v, which enters only through the initial datum $A_i(v) - v$, controlled uniformly on the compact sphere $\{v \in T_{p_i}M : |v|_g = 1\}$ by (35); and, once i is large enough for the bootstrap step above to apply, none of them depends on i either. Thus the same constant C works for every unit vector $v \in T_{p_i}M$ and every $t \in [0, r]$, and taking the supremum gives (37). This finishes the proof. □

Part III

ALLARD'S INTERIOR ε-REGULARITY THEOREM IN ALEXANDROV SPACES

12 THEORY OF GENERAL VARIFOLDS IN ABSTRACT RIEMANNIAN MANIFOLDS

12.1 INTRINSIC DEFINITION OF VARIFOLDS IN RIEMANNIAN MANIFOLDS

Let $m, n \in \mathbb{N}$ such that $1 \leq m < n$. We denote by (M^n, g) an n-dimensional complete connected Riemannian manifold with differentiable structure of class $\mathcal{C}^3$ and abstract metric tensor g of class $\mathcal{C}^2$. We also denote by $\mathrm{Gr}_m(TM^n)$ the Grassmannian m-plane bundle of M^n and

$$\mathrm{Gr}_m(M^n) := \left\{(x, S) \in M^n \times \mathrm{Gr}_m(TM^n) : S \in \tilde{\pi}^{-1}(x)\right\},$$

where $\tilde{\pi} : \mathrm{Gr}_m(TM^n) \to M^n$ is the natural fiber bundle projection.

Definition 12.1.1. We say that V is an m-dimensional varifold in M^n, if V is a nonnegative real-extended valued Radon measure on $\mathrm{Gr}_m(M^n)$. We denote by $\mathbf{V}_m(M^n)$ the space of all m-dimensional varifolds in M^n.

Definition 12.1.2. Let $V \in \mathbf{V}_m(M^n)$ and $\pi : \mathrm{Gr}_m(M^n) \to M^n$ the projection onto the first factor. The weight of V, denoted by $\|V\|$, is the measure given by the push-forward of V by π.

Definition 12.1.3. Let (M^n, g) be a Riemannian manifold, μ a Radon measure on M^n, and $m \in (0, n]$. We define the m-dimensional upper (resp., lower) density of μ as

$$\Theta^{*m}(\mu, x) := \limsup_{r \downarrow 0} \frac{\mu(B_{d_g}(x, r))}{\omega_m r^m} \left(\text{resp., } \Theta^m_*(\mu, x) := \liminf_{r \downarrow 0} \frac{\mu(B_{d_g}(x, r))}{\omega_m r^m}\right), \quad x \in M^n,$$

where $B_{d_g}(x, r)$ denotes the geodesic ball of radius r centered at x. When the two coincide, their common value is the density $\Theta^m(\mu, x)$.

We fix $r_0 \in \mathbb{R}$ such that

$$0 < r_0 < \frac{\mathrm{inj}(M^n, g)}{4}, \tag{r_0}$$

where $\mathrm{inj}(M^n, g)$ is the injectivity radius of (M^n, g).

Let $\Gamma \subset M^n$ be a countably m-rectifiable set and $\theta \in L^1_{loc}(\Gamma,]0, \infty[)$. For every $B \in \mathcal{B}(\mathrm{Gr}_m(M^n))$, we define

$$V(\Gamma, \theta, g)(B) := \int_{\{x \in \Gamma : (x, T_x\Gamma) \in B\}} \theta d\mathcal{H}^m_g,$$

where $\mathcal{H}_g$ is the Hausdorff measure in M^n with respect to the metric g.

Definition 12.1.4. We say that $V \in \mathbf{V}_m(M^n)$ is a rectifiable m-varifold in M^n when there exist a countably m-rectifiable set $\Gamma \subset M^n$ and $\theta \in L^1_{loc}(\Gamma,]0, \infty[)$ such that $V = V(\Gamma, \theta, g)$. Furthermore, when $\theta : \Gamma \to \mathbb{N}$ we say that V is an integer m-varifold in M^n. We denote by $\mathbf{RV}_m(M^n)$ the space of m-rectifiable varifolds in M^n and by $\mathbf{IV}_m(M^n)$ the set of all integer m-varifolds in M^n.

12.2 FIRST VARIATION OF A VARIFOLD

Let $k \in \mathbb{N}$. We denote by $\mathfrak{X}^k_c(M^n)$ the set of k-differentiable vector fields on M^n with compact support, by $\mathfrak{X}^0_c(M^n)$ the set of continuous vector fields on M^n with compact support, and by $\mathfrak{X}^\infty_c(M^n)$ the set of smooth vector fields on M^n with compact support.

Furthermore, we denote by ∇^g the Levi-Civita connection on M^n, and, for $x \in M^n$ and $S \in \mathrm{Gr}(m, T_xM^n)$, we denote by $P_S : T_xM^n \to S$ the orthogonal projection of T_xM^n onto S.

Definition 12.2.1. Let $V \in \mathbf{V}_m(M^n)$. We define the first variation of the varifold V as the linear functional $\delta_g V : \mathfrak{X}^1_c(M^n) \to \mathbb{R}$ given by

$$\delta_g V(X) := \int_{\mathrm{Gr}_m(M^n)} \mathrm{div}_S(X)(x) dV(x, S),$$

where $\mathrm{div}_S(X)$ denotes the tangential divergence of X along the m-plane S.

Definition 12.2.2. Let $d, d_1, d_2 \in \mathbb{N}$ such that $d < \min\{d_1, d_2\}$, (N_1, g_1) and (N_2, g_2) be d_1 and d_2-dimensional Riemannian manifolds, respectively, $V \in \mathbf{V}_d(N_1)$, and $f : N_1 \to N_2$ a proper $\mathcal{C}^1$ map. We define the push-forward of V by f, for every $B \in \mathcal{B}(\mathrm{Gr}_d(N_2))$, by

$$f^\# V(B) := \int_{\{(x,S) \in \mathrm{Gr}_d(N_1) : (f(x), df_x(S)) \in B\}} \mathrm{Jac}_S(f)(x) dV(x, S),$$

where $\mathrm{Jac}_S(f)$ denotes the tangential Jacobian of f along the m-plane S.

Given $X \in \mathfrak{X}^1_c(M^n)$, the one-parameter family of diffeomorphisms generated by X is the map $\Phi : \mathbb{R} \times M^n \to M^n$, which is, for all $(t, x) \in \mathbb{R} \times M^n$, the unique solution of

$$\begin{cases} \Phi(0, x) & = x \\ \left.\dfrac{\partial \Phi}{\partial t}\right|_{t=0} & = X(x). \end{cases} \tag{51}$$

Note that the flow $\Phi(\cdot, x)$ is defined on all $\mathbb{R}$ and any $x \in M^n$ because X is assumed to have compact support in M^n.

Proposition 12.2.3. *Let $V \in \mathbf{V}_m(M^n)$ and $X \in \mathfrak{X}_c^1(M^n)$. Then*

$$\delta_g V(X) = \left.\frac{d}{dt}\right|_{t=0} \left(\|\Phi_t{}^{\#} V\|(M^n) \right), \tag{52}$$

where $\Phi_t(\cdot) := \Phi(t, \cdot)$ and Φ is the one-parameter family of diffeomorphisms generated by X.

Proof. Let $x \in M^n$ and (U, φ) be a normal coordinate chart centered at x. By [65, Proposition 5.24, p. 123] we have that

1. $\phi(x) = (\phi^1(x), \dots, \phi^n(x)) = (0, \dots, 0)$, where $\phi^i : M^n \to \mathbb{R}$ for all $i = 1, \dots, n$;
2. The components of the metric at x are $g_{ij}(x) = \delta_{ij}$ for all $i, j = 1, \dots, n$;
3. $\Gamma_{ij}^k(x) = 0$ for all $i, j, k = 1, \dots, n$;
4. $\frac{\partial g_{ij}}{\partial \phi^k}(x) = 0$ for all $i, j, k = 1, \dots, n$.

Let $\Psi : \mathbb{R} \times M^n \to M^n$ be given by $\Psi(t, x) := \exp_x^g(tX(x))$ and note that

$$\Psi(0, x) = \exp_x(0X(x)) = x$$

and

$$\left.\frac{\partial \Psi}{\partial t}\right|_{t=0} = \left.\frac{\partial}{\partial t}\right|_{t=0} \exp_x(tX(x)) = X(x).$$

So, Ψ is a solution to Equation (51), and by the uniqueness of solutions for ODEs, it is the only one. In this case, we denote it by Φ.

Since g is of class $\mathcal{C}^2$ we have, by a standard argument of mollification of the Riemannian metric g (see [43]) for all $t > 0$ sufficiently close to 0, for all $(x, S) \in \mathrm{Gr}_m(M^n)$, and for all $i, j = 1, \dots, m$, that

$$[\mathrm{Jac}_S(\Phi_t)(x)]_{ij} = \delta_{ij} + t \left(\left\langle \nabla_{e_i} X, e_j \right\rangle_{g_x} + \left\langle e_i, \nabla_{e_j} X \right\rangle_{g_x} + E_{ij}(\Phi_t(x)) \right), \tag{53}$$

where $\{e_1, \dots, e_m\}$ is an orthonormal basis for S and $\sup_{x \in \mathrm{supp}(X)} E_{ij}(\Phi_t(x)) \to 0$ when $t \downarrow 0$.

By [41, Section 1.4.5, p. 21] and Equation (53) we have, for all $t > 0$ sufficiently close to 0, and for all $(x, S) \in \mathrm{Gr}_m(M^n)$, that

$$\mathrm{Jac}_S(\Phi_t)(x) = \sqrt{1 + t\,(2\mathrm{div}_S(X)(x) + E_{ii}(\Phi_t(x))) + O(t^2)}.$$

Which implies, for all $(x, S) \in \mathrm{Gr}_m(M^n)$, that

$$\left.\frac{d}{dt}\right|_{t=0} \mathrm{Jac}_S(\Phi_t)(x) = \mathrm{div}_S(X)(x). \tag{54}$$

So, by Equation (54) and [42, Theorem 2.27, p. 56], we have that Equation (52) holds. □

Remark 12.2.4. We initially defined the first variation of a varifold as a linear functional on $\mathfrak{X}_c^1(M^n)$. We now extend this definition to the space $\mathfrak{X}_c^0(M^n)$ for varifolds that has $L^p_{\mathrm{loc}}(\|V\|)$-bounded first variation in M^n (as a linear functional), for every $1 \leq p \leq \infty$.

We define $\widetilde{\delta_g V} : \mathfrak{X}^0_c(M^n) \to \mathbb{R}$ by

$$\widetilde{\delta_g V}(X) := \lim_{\varepsilon \downarrow 0} \delta_g V(X_\varepsilon),$$

where X_ε is a smoothing mollifier of the vector field $X \in \mathfrak{X}^0_c(M^n)$.

Note that $\widetilde{\delta_g V}$ is well-defined and is equal to $\delta_g V$ on $\mathfrak{X}^1_c(M^n)$. From now on we identify $\widetilde{\delta_g V}$ with $\delta_g V$, although technically they may differ.

Definition 12.2.5. We say that $V \in \mathbf{V}_m(M^n)$ is stationary when $\delta_g V(X) = 0$ for any $X \in \mathfrak{X}^0_c(M^n)$.

Definition 12.2.6. We say that $V \in \mathbf{V}_m(M^n)$ has locally bounded first variation when, for every open subset $W \subset\subset M^n$, there exists a real constant $C_W > 0$ such that

$$|\delta_g V(X)| \leq C_W \|X\|_{L^\infty(W, \|V\|)},$$

holds for every $X \in \mathfrak{X}^0_c(W)$.

Proposition 12.2.7. *Let $V \in \mathbf{V}_m(M^n)$ with locally bounded first variation. Then the total variation of $\delta_g V$, denoted by $\|\delta_g V\|_{\mathrm{TV}}$, is a Radon measure. Furthermore, there exists a $\|V\|$-measurable function $\overrightarrow{H_g} : M^n \to TM^n$, a real constant $C_W > 0$ and $Z \subset M^n$ with $\|V\|(Z) = 0$, such that, for every open subset $W \subset\subset M^n$ and $X \in \mathfrak{X}^0_c(M^n)$, we have that*

$$\delta_g V(X) = -\int_{M^n} \left\langle \overrightarrow{H_g}, X \right\rangle_g d\|V\| + \int_{M^n} \langle \nu, X \rangle_g \, d\|\delta_g V\|_{\mathrm{sing}} \tag{55}$$

and $\|\overrightarrow{H_g}\|_{L^1(W, \|V\|)} \leq C_W$, where ν is a $\|\delta_g V\|_{\mathrm{TV}}$-measurable function with $\|\nu(x)\|_g = 1$, for all $x \in M^n$, and $\|\delta_g V\|_{\mathrm{sing}} := \|\delta_g V\|_{\mathrm{TV}} \llcorner Z$.

Proof. Since V has locally bounded first variation, for all open set $W \subset\subset M^n$, there exists a real constant $C_W > 0$ such that $|\delta_g V(X)| \leq C_W \|X\|_{L^\infty(W, \|V\|)}$. Then, the total variation of $\delta_g V$,

$$\|\delta_g V\|_{\mathrm{TV}}(W) := \sup\{|\delta_g V(X)| : X \in \mathfrak{X}^0_c(W), \|X\|_{L^\infty(W, \|V\|)} \leq 1\} \leq C_W < \infty.$$

By Riesz representation theorem [41, Theorem 2.5.13, p. 106] $\|\delta_g V\|_{\mathrm{TV}}$ is a Radon measure on M^n and there exists a $\|\delta_g V\|_{\mathrm{TV}}$-mensurable function $\nu : M^n \to TM^n$ with $\|\nu(x)\|_{g_x} = 1$, for $\|\delta_g V\|_{\mathrm{TV}}$-a.e. $x \in M^n$. Furthermore, for all $X \in \mathfrak{X}^0_c(M^n)$,

$$\delta_g V(X) = \int_{M^n} \langle \nu, X \rangle_g d\|\delta_g V\|_{\mathrm{TV}}. \tag{56}$$

By Radon-Nikodym Theorem ([41, Theorem 2.9.2, p. 153] and [42, Theorem 2.9.7, p. 155]) we have, for all $B \in \mathcal{B}(M^n)$,

$$\|\delta_g V\|_{\mathrm{TV}}(B) = \int_B \Theta^{\|V\|}(\|\delta_g V\|_{\mathrm{TV}}, x) d\|V\|(x) + \left(\|\delta_g V\|_{\mathrm{TV}} \llcorner Z\right)(B) \tag{57}$$

with $(\Theta^{\|V\|}\|\delta_g V\|_{\mathrm{TV}}, x) \in L^1(B, \|V\|)$ and $Z = \{x \in M^n : \Theta^{\|V\|}\left(\|\delta_g V\|_{\mathrm{TV}}, x\right) = \infty\}$.

Defining $-\overrightarrow{H_g}(x) := \Theta^{\|V\|}(\|\delta_g V\|_{\mathrm{TV}}, x)\nu(x)$, for all $x \in M^n$, and $\|\delta_g V\|_{\mathrm{sing}} := \|\delta_g V\|_{\mathrm{TV}} \llcorner Z$, by Equations (56) and (57) we conclude that Equation (55) holds.

Furthermore, for $W \subset\subset M^n$ open, we have that by Hölder inequality that

$$\|\overrightarrow{H_g}\|_{L^1(W,\|V\|)} = \left\| -\Theta^{\|V\|}\left(\|\delta_g V\|_{\mathrm{TV}}, x\right)\nu \right\|_{L^1(W,\|V\|)} \leq C_W,$$

where $C_W \geq 0$ exists because $\Theta^{\|V\|}(\|\delta_g V\|_{\mathrm{TV}}, x) \in L^1(W, \|V\|)$ and $\|\nu(x)\|_{g_x} = 1$, for $\|\delta_g V\|_{\mathrm{TV}}$-a.e. $x \in M^n$. □

Let $1 < p < \infty$. We say that $q \in \mathbb{R}$ is the conjugate exponent of p when q is such that

$$\frac{1}{p} + \frac{1}{q} = 1.$$

When $p = 1$ we say that $q = \infty$ is its conjugate exponent and when $p = \infty$ we say that $q = 1$ is its conjugate exponent. By symmetry of this definition we have that, for $1 \leq p, q \leq \infty$, p is the conjugate exponent of q if, and only if, q is the conjugate exponent of p.

Proposition 12.2.8. *Let $1 \leq p < \infty$, q the conjugate exponent of p, and $V \in \mathbf{V}_m(M^n)$. Then the following statements are equivalent:*

1. *For every open subset $W \subset\subset M^n$, there exists a real constant $C_W > 0$ (which depends of W) such that*
$$|\delta_g V(X)| \leq C_W \|X\|_{L^p(W,\|V\|)}$$
holds for every $X \in \mathfrak{X}_c^0(W)$.

2. *The total variation of $\delta_g V$, denoted by $\|\delta_g V\|_{\mathrm{TV}}$, is a Radon measure. Furthermore, there exists a $\|V\|$-measurable function $\overrightarrow{H_g} : M^n \to TM^n$ such that*
$$\delta_g V(X) = -\int_{M^n} \left\langle \overrightarrow{H_g}, X \right\rangle_g d\|V\| \tag{58}$$
and $\|\overrightarrow{H_g}\|_{L^q(W,\|V\|)} \leq C_W$, for every open subset $W \subset\subset M^n$ and $X \in \mathfrak{X}_c^0(W)$.

Proof. (*Item 1* $\implies$ *Item 2*): Suppose that, for every open subset $W \subset\subset M^n$, there exists a real constant $C_W > 0$ such that $|\delta_g V(X)| \leq C_W \|X\|_{L^p(W,\|V\|)}$ holds for every $X \in \mathfrak{X}_c^0(W)$. Then, the total variation of $\delta_g V$,

$$\|\delta_g V\|_{\mathrm{TV}}(W) := \sup\{|\delta_g V(X)| : X \in \mathfrak{X}_c^0(W), \|X\|_{L^\infty(W,\|V\|)} \leq 1\} < \infty.$$

By Riesz representation theorem [41, Theorem 2.5.13, p. 106] $\|\delta_g V\|_{\mathrm{TV}}$ is a Radon measure on M^n and there exists a $\|\delta_g V\|_{\mathrm{TV}}$-mensurable function $\nu : M^n \to TM^n$ with $\|\nu(x)\|_{g_x} = 1$, for $\|\delta_g V\|_{\mathrm{TV}}$-a.e. $x \in M^n$. Furthermore, for all $X \in \mathfrak{X}_c^0(M^n)$,

$$\delta_g V(X) = \int_{M^n} \langle \nu, X \rangle_g d\|\delta_g V\|_{\mathrm{TV}}. \tag{59}$$

Furthermore, note that $\|\delta_g V\|_{\mathrm{TV}}$ is absolutely continuous with respect to $\|V\|$. In fact, let $B \in \mathcal{B}(M^n)$ such that $\|V\|(B) = 0$ then

$$\begin{aligned}\|\delta_g V\|_{\mathrm{TV}}(B) &\leq \sup\left\{C_B\|X\|_{L^p(B,\|V\|)}\colon X \in \mathfrak{X}^0_c(B), \|X\|_{L^\infty(B,\|V\|)} \leq 1\right\}\\ &\leq \sup\left\{C_B\|X\|_{L^\infty(B,\|V\|)}(\|V\|(B))^{\frac{1}{p}} : X \in \mathfrak{X}^0_c(B), \|X\|_{L^\infty(B,\|V\|)} \leq 1\right\} = 0.\end{aligned}$$

By Radon-Nikodym Theorem ([41, Theorem 2.9.2, p. 153] and [42, Theorem 2.9.7, p. 155]) we have, for all $B \in \mathcal{B}(M^n)$,

$$\|\delta_g V\|_{\mathrm{TV}}(B) = \int_B \Theta^{\|V\|}(\|\delta_g V\|_{\mathrm{TV}}, x)d\|V\|(x) \tag{60}$$

with $\Theta^{\|V\|}\|(\delta_g V\|_{\mathrm{TV}}, x) \in L^1(B, \|V\|)$.

Defining $-\overrightarrow{H_g}(x) := \Theta^{\|V\|}(\|\delta_g V\|_{\mathrm{TV}}, x)\nu(x)$, for all $x \in M^n$, by Equations (59) and (60) we conclude that Equation (58) holds.

Furthermore, since for every open subset $W \subset\subset M^n$ and for every $X \in \mathfrak{X}^0_c(W)$ there exists a real constant $C_W > 0$ such that $|\delta_g V(X)| \leq C_W\|X\|_{L^p(W,\|V\|)}$, we have that $\delta_g V$ is a bounded linear functional on $L^p(W, \|V\|)$. By [89, Theorem 6.16, p. 127] there exists a unique $F \in L^q(W, \|V\|)$ such that

$$\delta_g V(X) = \int_{M^n} \langle F, X\rangle_g \, d\|V\|. \tag{61}$$

Combining Equation (58) and Equation (61) we obtain that $\overrightarrow{H_g} = -F$ and, therefore, there exists a real constant $C_W > 0$ such that $\|\overrightarrow{H_g}\|_{L^q(W,\|V\|)} \leq C_W$, for every open subset $W \subset\subset M^n$.

$(Item\ 2 \implies Item\ 1)$: Now, suppose that $\|\delta_g V\|_{\mathrm{TV}}$ is a Radon measure, there exists a $\|V\|$-measurable function $\overrightarrow{H_g} : M^n \to TM^n$ such that Equation (58) holds for all $X \in \mathfrak{X}^0_c(M^n)$, and $\|\overrightarrow{H_g}\|_{L^q(W,\|V\|)} \leq C_W$, for every open subset $W \subset\subset M^n$. Let $W \subset\subset M^n$ and $X \in \mathfrak{X}^0_c(W)$ be arbitrary. Thus by Hölder inequality

$$|\delta_g V(X)| = \left|-\int_M \left\langle \overrightarrow{H_g}, X\right\rangle_g d\|V\|\right| \leq \int_M \|\overrightarrow{H_g}\|_g\|X\|_g d\|V\| \leq C_W\|X\|_{L^p(W,\|V\|)}.$$

□

Definition 12.2.9. Using the same notation as in Proposition 12.2.7 and Proposition 12.2.8, we denote by $\overrightarrow{H_g}$ the generalized mean curvature vector of V, by Z the generalized boundary of V, by $\|\delta_g V\|_{\mathrm{sing}}$ the generalized boundary measure of V, and by $\nu\restriction_Z$ the generalized unit conormal of V.

Definition 12.2.10. Let $1 \leq p < \infty$, q the conjugate exponent of p, and $V \in \mathbf{V}_m(M^n)$. We say that V has L^q_{loc}-bounded generalized mean curvature vector in M^n when V satisfies Item 1 or Item 2 in the Proposition 12.2.8.

12.3 MONOTONICITY FORMULA

Let $0 < \varepsilon < 1$, and $\varphi_\varepsilon \in C_c^1(]-1,1[)$ an even function such that

$$\varphi_\varepsilon(x) := \begin{cases} 1 \text{ if } x \leq \varepsilon, \\ 0 \text{ if } x \geq 1, \end{cases} \qquad \text{and,} \qquad \varphi_\varepsilon' < 0, \text{ if } 0 < \varepsilon < x < 1.$$

Definition 12.3.1. Let $\xi \in M^n$, $0 < \delta < 1$, $0 < \rho < r_0$, $y \in B_{d_g}(\xi, \delta\rho)$, and $\sigma \in]0, (1-\delta)\rho]$. We define the ε, σ-radial perturbation of the gradient of u_y by

$$\begin{aligned} X_{\varepsilon,\sigma} : M^n &\to TM^n \\ x &\mapsto X_{\varepsilon,\sigma}(x) := \varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) u_y(x) \nabla_{g_x}(u_y)(x), \end{aligned}$$

where $u_y(x) := d_g(y,x)$, for all $x \in M^n$, and $\nabla_{g_x}(u_y)(x)$ is the gradient of u_y at x, for all $x \in M^n$.

Remark 12.3.2. Note that the ε, σ-radial perturbation of the gradient of u_y is a vector field with compact support. Moreover, $\operatorname{supp}(X_{\varepsilon,\sigma}) \subset B_{d_g}(y, \sigma)$.

Lemma 12.3.3. *Let $\xi \in M^n$, $0 < \delta < 1$, $0 < \rho < r_0$. Then, for all $y \in B_{d_g}(\xi, \delta\rho)$, for all $\sigma \in]0, (1-\delta)\rho]$, for all $x \in B_{d_g}(y, \sigma)$, and for all $S(x) \in Gr(m, T_xM)$, we have that*

$$\begin{aligned} \operatorname{div}_{S(x)}(X_{\varepsilon,\sigma})(x) = {} & \varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) \operatorname{div}_{S(x)}\left(u_y \nabla_g u_y\right)(x) + \varphi_\varepsilon'\left(\frac{u_y(x)}{\sigma}\right) \frac{u_y(x)}{\sigma} \\ & - \varphi_\varepsilon'\left(\frac{u_y(x)}{\sigma}\right) \frac{u_y(x)}{\sigma} \left\| P_{S(x)^\perp}(\nabla_{g_x}(u_y)(x)) \right\|_{g_x}^2, \end{aligned} \tag{62}$$

where $P_{S(x)^\perp} : T_xM^n \to S(x)^\perp$ is the orthogonal projection onto $S(x)^\perp$ with respect to the metric g_x.

Proof. Let $y \in B_{d_g}(\xi, \delta\rho)$, $\sigma \in]0, (1-\delta)\rho]$, $x \in B_{d_g}(y, \sigma)$, and $S(x) \in Gr(m, T_xM)$ arbitrary. By a standard computation we have that

$$\operatorname{div}_{S(x)}(X_{\varepsilon,\sigma})(x) = \varphi_\varepsilon'\left(\frac{u_y(x)}{\sigma}\right) \frac{u_y(x)}{\sigma} \left\| P_{S(x)}\left(\nabla_{g_x}(u_y)(x)\right) \right\|_{g_x}^2 + \varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) \operatorname{div}_S\left(u_y \nabla_g u_y\right)(x). \tag{63}$$

where $P_{S(x)} : T_xM \to S(x)$ is the orthogonal projection onto $S(x)$ with respect to the metric g_x and $P_{S(x)^\perp} : T_xM \to S(x)^\perp$ is the orthogonal projection onto $S(x)^\perp$ with respect to the metric g_x. Since u_y is the distance function we have that

$$\left\| P_{S(x)}(\nabla_{g_x}(u_y)(x)) \right\|_{g_x}^2 + \left\| P_{S(x)^\perp}(\nabla_{g_x}(u_y)(x)) \right\|_{g_x}^2 = \left\| \nabla_{g_x}(u_y)(x) \right\|_{g_x}^2 = 1. \tag{64}$$

By Equation (63) and Equation (64), we obtain Equation (62). □

Although this choice is enough to get many useful information, let us consider a general case, which will be used in the sequel when we will prove a fundamental height estimates.

Definition 12.3.4. Let $h \in \mathcal{C}^1(M^n, [0, +\infty[)$, $\xi \in M^n$, $0 < \delta < 1$, $0 < \rho < r_0$, $y \in B_{d_g}(\xi, \delta\rho)$, and $\sigma \in]0, (1-\delta)\rho]$. We define the h, ε, σ-radial perturbation of the gradient of u_y by

$$\begin{aligned} X_{h,\varepsilon,\sigma} : M^n &\to TM^n \\ x &\mapsto X_{h,\varepsilon,\sigma}(x) := h(x)X_{\varepsilon,\sigma}(x) = h(x)\varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) u_y(x)\nabla_{g_x}(u_y)(x), \end{aligned}$$

where $u_y(x) := d_g(y, x)$, for all $x \in M^n$, and $\nabla_{g_x}(u_y)(x)$ is the gradient of u_y at x, for all $x \in M^n$.

Lemma 12.3.5. *Let $h \in \mathcal{C}^1(M, [0, +\infty[)$, $\xi \in M$, $0 < \delta < 1$, $0 < \rho < r_0$, and $V \in \mathbf{V}_m(M^n)$. Then, for all $y \in B_{d_g}(\xi, \delta\rho)$ and for all $\sigma \in]0, (1-\delta)\rho]$, we have that*

$$\begin{aligned} \delta_g V\left(X_{h,\varepsilon,\sigma}\right) = &\int_{\mathrm{Gr}_m(M^n)} \varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) \left\langle u_y(x)\nabla_{g_x}(u_y)(x), P_{S(x)}(\nabla_{g_x}(h)(x))\right\rangle_{g_x} dV(x, S(x)) \\ &+ \int_{\mathrm{Gr}_m(M^n)} h(x)\varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) \mathrm{div}_S\left(u_y\nabla_g u_y\right)(x) dV(x, S(x)) \\ &+ \int_{\mathrm{Gr}_m(M^n)} h(x)\varphi'_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) \frac{u_y(x)}{\sigma} dV(x, S(x)) \\ &- \int_{\mathrm{Gr}_m(M^n)} h(x)\varphi'_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) \frac{u_y(x)}{\sigma} \left\|P_{S(x)^\perp}(\nabla_g(u_y)(x))\right\|^2_{g_x} dV(x, S(x)). \end{aligned} \tag{65}$$

Proof. Let $y \in B_{d_g}(\xi, \delta\rho)$, $\sigma \in]0, (1-\delta)\rho]$, $x \in B_{d_g}(y, \sigma)$, and $S(x) \in Gr(m, T_xM)$ arbitrary. By Lemma 12.3.3 we have that

$$\begin{aligned} \mathrm{div}_S\left(X_{h,\varepsilon,\sigma}\right)(x) = &\varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) \left\langle u_y(x)\nabla_{g_x}(u_y)(x), P_{S(x)}\left(\nabla_{g_x}(h)(x)\right)\right\rangle_{g_x} \\ &+ h(x)\varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) \mathrm{div}_S\left(u_y\nabla_g u_y\right)(x) \\ &+ h(x)\varphi'_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) \frac{u_y(x)}{\sigma} \\ &- h(x)\varphi'_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) \frac{u_y(x)}{\sigma} \left\|P_{S(x)^\perp}(\nabla_{g_x}(u_y)(x))\right\|^2_{g_x}. \end{aligned} \tag{66}$$

Then, by Equation (66) we have Equation (65). □

Remark 12.3.6. When comparing Lemma 12.3.5 to the Euclidean case [91, Chapter 8, Section 3, p. 240], we find a term that depends of the metric in addition to the dimensional Euclidean term. Therefore, in order to obtain our desired result, we need to make a comparison with the Euclidean case and we do it by introducing a curvature term.

Definition 12.3.7. We define the function $c : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ by $c(t, b) := t\cot_b(t)$, where $\cot_b(t) := \frac{s'_b(t)}{s_b(t)}$ and

$$s_b(t) = \begin{cases} \dfrac{\sinh(\sqrt{-b}t)}{\sqrt{-b}}, & \text{if } b < 0, \\ t, & \text{if } b = 0, \\ \dfrac{\sin\left(\sqrt{b}t\right)}{\sqrt{b}}, & \text{if } b > 0. \end{cases}$$

From now on, we assume that there exists $K > 0$ such that $|\mathrm{sec}_g| \leq K$, where sec_g is the sectional curvature of (M^n, g).

Lemma 12.3.8. *Let $\xi \in M^n$ and $0 < \delta < 1$. There exists $\rho_0 := \rho_0(r_0, K) > 0$ such that for all $0 < \rho < \rho_0$, for all $y \in B_{d_g}(\xi, \delta\rho)$, for all $\sigma \in]0, (1-\delta)\rho]$, for all $x \in B_{d_g}(y, \sigma) \setminus \{y\}$, and for all $S(x) \in Gr(m, T_xM^n)$, we have that*

$$mc(u_y(x), K) \leq \operatorname{div}_{S(x)} \left(u_y \nabla_g u_y\right)(x) \leq mc(u_y(x), -K). \tag{67}$$

Proof. Let us fix $\rho_0 := \rho_0(r_0) < r_0$. Consider $0 < \rho < \rho_0$, $y \in B_{d_g}(\xi, \delta\rho)$, $\sigma \in]0, (1-\delta)\rho]$, $x \in B_{d_g}(y, \sigma) \setminus \{y\}$, and $S(x) \in Gr(m, T_xM)$ arbitrary. By a standard computation we have that

$$\operatorname{div}_{S(x)}(u_y \nabla_g u_y)(x) = \left\| P_{S(x)} \left(\nabla_{g_x}(u_y)(x)\right)\right\|^2_{g_x} + u_y(x) \sum_{i=1}^{m} \operatorname{Hess}_{g_x}(u_y)(x) \left(e_{j_i}(x), e_{j_i}(x)\right). \tag{68}$$

The existence of ρ_0 (with dependence only on r_0 and K) follows by applying the Hessian comparison theorem [65, Theorem 11.7, p. 327] to the distance function u_y in Equation (68). Indeed, for ρ_0 sufficiently small, the sectional curvature bounds imply corresponding upper and lower estimates for $\operatorname{Hess}_{g_x}(u_y)$, which substituted into Equation (68) yield Equation (67). □

Lemma 12.3.9. *Let $h \in \mathcal{C}^1(M^n, [0, +\infty[)$, $\xi \in M^n$, $0 < \delta < 1$, and $V \in \mathbf{V}_m(M^n)$. There exists $\rho_0 := \rho_0(r_0, K) > 0$ such that for all $0 < \rho < \rho_0$, for all $y \in B_{d_g}(\xi, \delta\rho)$, and for all $\sigma \in]0, (1-\delta)\rho]$, we have that*

$$\begin{aligned}
&\frac{d}{d\sigma}\left(\frac{1}{\sigma^m}\int_{B_{d_g}(y,\sigma)} h(x)\varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) d\|V\|(x)\right) \geq \\
&\geq \frac{1}{\sigma^{m+1}}\int_{\mathrm{Gr}_m(B_{d_g}(y,\sigma))} \varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) \left\langle u_y(x)\nabla_{g_x}(u_y)(x), P_{S(x)}\left(\nabla_{g_x}(h)(x)\right)\right\rangle_{g_x} dV(x, S(x)) \\
&+ \frac{c(\sigma, K) - 1}{\sigma}\frac{m}{\sigma^m}\int_{B_{d_g}(y,\sigma)} h(x)\varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) d\|V\|(x) \\
&+ \int_{\mathrm{Gr}_m(B_{d_g}(y,\sigma))} \frac{h(x)}{\sigma^m}\frac{d}{d\sigma}\left(\varphi_\varepsilon\left(\frac{u_y}{\sigma}\right)\right)(x) \left\| P_{S(x)^\perp}(\nabla_{g_x}(u_y)(x))\right\|^2_{g_x} dV(x, S(x)) \\
&- \frac{\delta_g V\left(X_{h,\varepsilon,\sigma}\right)}{\sigma^{m+1}},
\end{aligned} \tag{69}$$

and

$$\begin{aligned}
&\frac{d}{d\sigma}\left(\frac{1}{\sigma^m}\int_{B_{d_g}(y,\sigma)} h(x)\varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) d\|V\|(x)\right) \leq \\
&\leq \frac{1}{\sigma^{m+1}}\int_{\mathrm{Gr}_m(B_{d_g}(y,\sigma))} \varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) \left\langle u_y(x)\nabla_{g_x}(u_y)(x), P_{S(x)}\left(\nabla_{g_x}(h)(x)\right)\right\rangle_{g_x} dV(x, S(x)) \\
&+ \frac{c(\sigma, -K) - 1}{\sigma}\frac{m}{\sigma^m}\int_{B_{d_g}(y,\sigma)} h(x)\varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) d\|V\|(x) \\
&+ \int_{\mathrm{Gr}_m(B_{d_g}(y,\sigma))} \frac{h(x)}{\sigma^m}\frac{d}{d\sigma}\left(\varphi_\varepsilon\left(\frac{u_y}{\sigma}\right)\right)(x) \left\| P_{S(x)^\perp}(\nabla_{g_x}(u_y)(x))\right\|^2_{g_x} dV(x, S(x)) \\
&- \frac{\delta_g V\left(X_{h,\varepsilon,\sigma}\right)}{\sigma^{m+1}},
\end{aligned} \tag{70}$$

Proof. Let us fix $\rho_0 := \rho_0(r_0, K) > 0$ given by Lemma 12.3.8. Consider $0 < \rho < \rho_0$, $y \in B_{d_g}(\xi, \delta\rho)$, and $\sigma \in]0, (1-\delta)\rho]$ arbitrary.

Note that if $x \in M^n \setminus B_{d_g}(y, \sigma)$ then $u_y(x) \geq \sigma$, that is, $\frac{u_y(x)}{\sigma} \geq 1$. Thus,

$$x \in M \setminus B_{d_g}(y,\sigma) \implies \varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) = 0 \text{ and } \varphi'_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) = 0. \tag{71}$$

Since $M = B_{d_g}(y,\sigma) \sqcup \left(M \setminus B_{d_g}(x,\sigma)\right)$, by Equation (71) and Lemma 12.3.5, we have that

$$\begin{aligned}
\delta_g V\left(X_{h,\varepsilon,\sigma}\right) = &\int_{\mathrm{Gr}_m\left(B_{d_g}(y,\sigma)\right)} \varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) \left\langle u_y(x)\nabla_{g_x}(u_y)(x), P_{S(x)}(\nabla_{g_x}(h)(x))\right\rangle_{g_x} dV(x,S(x)) \\
&+ \int_{\mathrm{Gr}_m\left(B_{d_g}(y,\sigma)\right)} h(x)\varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) \mathrm{div}_S\left(u_y\nabla_g u_y\right)(x) dV(x,S(x)) \\
&+ \int_{\mathrm{Gr}_m\left(B_{d_g}(y,\sigma)\right)} h(x)\varphi'_\varepsilon\left(\frac{u_y(x)}{\sigma}\right)\frac{u_y(x)}{\sigma} dV(x,S(x)) \\
&- \int_{\mathrm{Gr}_m\left(B_{d_g}(y,\sigma)\right)} h(x)\varphi'_\varepsilon\left(\frac{u_y(x)}{\sigma}\right)\frac{u_y(x)}{\sigma} \left\|P_{S(x)^\perp}(\nabla_g(u_y)(x))\right\|^2_{g_x} dV(x,S(x)).
\end{aligned} \tag{72}$$

We will divide the proof in the following two cases. Firstly, the case when $\sec_g \leq K$. By Equation (72) and Lemma 12.3.8, we obtain

$$\begin{aligned}
\delta_g V\left(X_{h,\varepsilon,\sigma}\right) \geq &\int_{\mathrm{Gr}_m\left(B_{d_g}(y,\sigma)\right)} \varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) \left\langle u_y(x)\nabla_{g_x}(u_y)(x), P_{S(x)}(\nabla_{g_x}(h)(x))\right\rangle_{g_x} dV(x,S(x)) \\
&+ \int_{\mathrm{Gr}_m\left(B_{d_g}(y,\sigma)\right)} h(x)\varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) mc(u_y(x),K) dV(x,S(x)) \\
&+ \int_{\mathrm{Gr}_m\left(B_{d_g}(y,\sigma)\right)} h(x)\varphi'_\varepsilon\left(\frac{u_y(x)}{\sigma}\right)\frac{u_y(x)}{\sigma} dV(x,S(x)) \\
&- \int_{\mathrm{Gr}_m\left(B_{d_g}(y,\sigma)\right)} h(x)\varphi'_\varepsilon\left(\frac{u_y(x)}{\sigma}\right)\frac{u_y(x)}{\sigma} \left\|P_{S(x)^\perp}(\nabla_g(u_y)(x))\right\|^2_{g_x} dV(x,S(x)).
\end{aligned} \tag{73}$$

Note that $c(t,b)$ is non-increasing for $t > 0$ and $b > 0$. Thus, since $u_y(x) < \sigma$, for every $x \in B_{d_g}(y,\sigma)$, we have that $c(u_y(x),K) \geq c(\sigma,K)$. Therefore, by Equation (73), we have

$$\begin{aligned}
\frac{\delta_g V\left(X_{h,\varepsilon,\sigma}\right)}{\sigma^{m+1}} \geq &\frac{1}{\sigma^{m+1}} \int_{\mathrm{Gr}_m\left(B_{d_g}(y,\sigma)\right)} \varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) \left\langle u_y(x)\nabla_{g_x}(u_y)(x), P_{S(x)}(\nabla_{g_x}(h)(x))\right\rangle_{g_x} dV(x,S(x)) \\
&+ \frac{c(\sigma,K)-1}{\sigma}\frac{m}{\sigma^m} \int_{\mathrm{Gr}_m\left(B_{d_g}(y,\sigma)\right)} h(x)\varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) dV(x,S(x)) \\
&+ \int_{\mathrm{Gr}_m\left(B_{d_g}(y,\sigma)\right)} \left(h(x)\varphi'_\varepsilon\left(\frac{u_y(x)}{\sigma}\right)\frac{u_y(x)}{\sigma^{m+2}} + \frac{m}{\sigma^{m+1}}h(x)\varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right)\right) dV(x,S(x)) \\
&- \int_{\mathrm{Gr}_m\left(B_{d_g}(y,\sigma)\right)} \frac{h(x)}{\sigma^{m+1}}\varphi'_\varepsilon\left(\frac{u_y(x)}{\sigma}\right)\frac{u_y(x)}{\sigma} \left\|P_{S(x)^\perp}(\nabla_g(u_y)(x))\right\|^2_{g_x} dV(x,S(x)).
\end{aligned} \tag{74}$$

By [42, Theorem 2.27, p. 56], we have that

$$\frac{d}{d\sigma}\left(\frac{1}{\sigma^m}\int_{\mathrm{Gr}_m\left(B_{d_g}(y,\sigma)\right)} h(x)\varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) dV(x,S(x))\right) =$$
$$= -\int_{\mathrm{Gr}_m\left(B_{d_g}(y,\sigma)\right)}\left(h(x)\varphi_\varepsilon'\left(\frac{u_y(x)}{\sigma}\right)\frac{u_y(x)}{\sigma^{m+2}} + \frac{m}{\sigma^{m+1}}h(x)\varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right)\right) dV(x,S(x)). \tag{75}$$

By Equations (74) and (75), we obtain

$$\begin{aligned}\frac{\delta_g V\left(X_{h,\varepsilon,\sigma}\right)}{\sigma^{m+1}} \geq{}& \frac{1}{\sigma^{m+1}}\int_{\mathrm{Gr}_m\left(B_{d_g}(y,\sigma)\right)}\varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right)\left\langle u_y(x)\nabla_{g_x}(u_y)(x), P_{S(x)}(\nabla_{g_x}(h)(x))\right\rangle_{g_x} dV(x,S(x))\\
&+\frac{c(\sigma,K)-1}{\sigma}\frac{m}{\sigma^m}\int_{\mathrm{Gr}_m\left(B_{d_g}(y,\sigma)\right)} h(x)\varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) dV(x,S(x))\\
&-\frac{d}{d\sigma}\left(\frac{1}{\sigma^m}\int_{\mathrm{Gr}_m\left(B_{d_g}(y,\sigma)\right)} h(x)\varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) dV(x,S(x))\right)\\
&-\int_{\mathrm{Gr}_m\left(B_{d_g}(y,\sigma)\right)}\frac{h(x)}{\sigma^{m+1}}\varphi_\varepsilon'\left(\frac{u_y(x)}{\sigma}\right)\frac{u_y(x)}{\sigma}\left\|P_{S(x)^\perp}(\nabla_g(u_y)(x))\right\|^2_{g_x} dV(x,S(x)).\end{aligned} \tag{76}$$

By Disintegration Lemma there exists a Radon measure $V^{(x)}$ on $\mathrm{Gr}(m, T_xM^n)$ for $\|V\|$-a.e. $x \in M^n$ such that $V^{(x)}(\mathrm{Gr}(m, T_xM^n)) = 1$ and

$$\int_{\mathrm{Gr}_m\left(B_{d_g}(y,\sigma)\right)} h(x)\varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) dV(x,S(x)) = \int_{B_{d_g}(y,\sigma)} h(x)\varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) d\|V\|(x). \tag{77}$$

By Equations (76) and (77), we obtain

$$\begin{aligned}&\frac{d}{d\sigma}\left(\frac{1}{\sigma^m}\int_{B_{d_g}(y,\sigma)} h(x)\varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) d\|V\|(x)\right) \geq\\
&\geq \frac{1}{\sigma^{m+1}}\int_{\mathrm{Gr}_m\left(B_{d_g}(y,\sigma)\right)}\varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right)\left\langle u_y(x)\nabla_{g_x}(u_y)(x), P_{S(x)}(\nabla_{g_x}(h)(x))\right\rangle_{g_x} dV(x,S(x))\\
&+\frac{c(\sigma,K)-1}{\sigma}\frac{m}{\sigma^m}\int_{B_{d_g}(y,\sigma)} h(x)\varphi_\varepsilon\left(\frac{u_y(x)}{\sigma}\right) d\|V\|(x)\\
&+\int_{\mathrm{Gr}_m\left(B_{d_g}(y,\sigma)\right)}\frac{h(x)}{\sigma^m}\frac{d}{d\sigma}\left(\varphi_\varepsilon\left(\frac{u_y}{\sigma}\right)\right)(x)\left\|P_{S(x)^\perp}(\nabla_g(u_y)(x))\right\|^2_{g_x} dV(x,S(x))\\
&-\frac{\delta_g V\left(X_{h,\varepsilon,\sigma}\right)}{\sigma^{m+1}}.\end{aligned}$$

The case $\sec_g \geq -K$ is handled analogously to $\sec_g \leq K$, noting that $c(t,b)$ is non-decreasing for $t > 0$ and $b < 0$. □

Now we apply Lemma 12.3.9 to varifolds that satisfy certain hypotheses on their first variation.

Theorem 12.3.10. *Let $1 < p \leq \infty$, $h \in \mathcal{C}^1(M^n, [0, +\infty[)$, $\xi \in M^n$, $0 < \delta < 1$, and $V \in \mathbf{V}_m(M^n)$. There exists $\rho_0 := \rho_0(r_0, K) > 0$ such that if V has L^p_{loc}-bounded generalized mean curvature vector in $B_{d_g}(\xi, \rho)$, for some $0 < \rho < \rho_0$, then, for all $y \in B_{d_g}(\xi, \delta\rho)$ and for all $\sigma \in]0, (1-\delta)\rho]$, we have that*

$$\begin{aligned}
&\frac{d}{d\sigma}\left(\frac{1}{\sigma^m}\int_{B_{d_g}(y,\sigma)} h(x)d\|V\|(x)\right) \geq \\
&\geq \frac{1}{\sigma^{m+1}}\int_{\mathrm{Gr}_m(B_{d_g}(y,\sigma))} \left\langle u_y(x)\nabla_{g_x}(u_y)(x), P_{S(x)}\left(\nabla_{g_x}(h)(x)\right)\right\rangle_{g_x} dV(x,S(x)) \\
&+ \frac{c(\sigma,K)-1}{\sigma}\frac{m}{\sigma^m}\int_{B_{d_g}(y,\sigma)} h(x)d\|V\|(x) \\
&+ \frac{d}{d\sigma}\left(\int_{\mathrm{Gr}_m(B_{d_g}(y,\sigma))} \frac{h(x)}{u_y(x)^m}\left\|P_{S(x)^\perp}(\nabla_{g_x}(u_y)(x))\right\|^2_{g_x} dV(x,S(x))\right) \\
&+ \frac{1}{\sigma^{m+1}}\int_{B_{d_g}(y,\sigma)} \left\langle \overrightarrow{H_g}(x), h(x)u_y(x)\nabla_{g_x}(u_y)(x)\right\rangle_{g_x} d\|V\|(x),
\end{aligned} \tag{78}$$

and

$$\begin{aligned}
&\frac{d}{d\sigma}\left(\frac{1}{\sigma^m}\int_{B_{d_g}(y,\sigma)} h(x)d\|V\|(x)\right) \leq \\
&\leq \frac{1}{\sigma^{m+1}}\int_{\mathrm{Gr}_m(B_{d_g}(y,\sigma))} \left\langle u_y(x)\nabla_{g_x}(u_y)(x), P_{S(x)}\left(\nabla_{g_x}(h)(x)\right)\right\rangle_{g_x} dV(x,S(x)) \\
&+ \frac{c(\sigma,-K)-1}{\sigma}\frac{m}{\sigma^m}\int_{B_{d_g}(y,\sigma)} h(x)d\|V\|(x) \\
&+ \frac{d}{d\sigma}\left(\int_{\mathrm{Gr}(B_{d_g}(y,\sigma))} \frac{h(x)}{u_y(x)^m}\left\|P_{S(x)^\perp}(\nabla_{g_x}(u_y)(x))\right\|^2_{g_x} dV(x,S(x))\right) \\
&+ \frac{1}{\sigma^{m+1}}\int_{B_{d_g}(y,\sigma)} \left\langle \overrightarrow{H_g}(x), h(x)u_y(x)\nabla_{g_x}(u_y)(x)\right\rangle_{g_x} d\|V\|(x).
\end{aligned} \tag{79}$$

Proof. Let us fix $\rho_0 := \rho_0(r_0, K) > 0$ given by Lemma 12.3.9. Suppose that V has L^p_{loc}-bounded generalized mean curvature vector in $B_{d_g}(\xi, \rho)$, for some $0 < \rho < \rho_0$. Consider $y \in B_{d_g}(\xi, \delta\rho)$ and $\sigma \in]0, (1-\delta)\rho]$ arbitrary.

Let $\{\varepsilon_j\}_{j\in\mathbb{N}}$ a increasing sequence such that $\varepsilon_j > 0$, for all $j \in \mathbb{N}$, and $\varepsilon_j \uparrow 1$, when $j \uparrow \infty$. Note that, for all $t \in \mathbb{R}$, $\varphi_{\varepsilon_j}(t) \leq \mathcal{X}_{]-1,1[}(t)$, for all $j \in \mathbb{N}$, and $\lim_{j\uparrow\infty}\varphi_{\varepsilon_j}(t) = \mathcal{X}_{]-1,1[}(t)$. Thus, for all $x \in B_{d_g}(y, \sigma)$, we have that

$$\lim_{j\uparrow\infty}\varphi_{\varepsilon_j}\left(\frac{u_y(x)}{\sigma}\right) = \mathcal{X}_{]-1,1[}\left(\frac{u_y(x)}{\sigma}\right).$$

However, note that $\mathcal{X}_{]-1,1[}\left(\frac{u_y(x)}{\sigma}\right) = 1$ if, and only if, $-1 < \frac{u_y(x)}{\sigma} < 1$ which occurs if, and only if, $|u_y(x)| < \sigma$, that is, $x \in B_{d_g}(y,\sigma)$. Therefore,

$$\lim_{j\uparrow\infty} \varphi_{\varepsilon_j}\left(\frac{u_y(x)}{\sigma}\right) = \mathcal{X}_{]-1,1[}\left(\frac{u_y(x)}{\sigma}\right) = \mathcal{X}_{B_{d_g}(y,\sigma)}(x). \tag{80}$$

Applying this to Equations (69) and (70) and taking the limit as $j \uparrow \infty$, we obtain, respectively,

$$\begin{aligned}
&\lim_{j\uparrow\infty}\left(\frac{d}{d\sigma}\left(\frac{1}{\sigma^m}\int_{B_{d_g}(y,\sigma)} h(x)\varphi_{\varepsilon_j}\left(\frac{u_y(x)}{\sigma}\right) d\|V\|(x)\right)\right) \geq \\
&\geq \lim_{j\uparrow\infty}\left(\frac{1}{\sigma^{m+1}}\int_{\mathrm{Gr}_m(B_{d_g}(y,\sigma))} \varphi_{\varepsilon_j}\left(\frac{u_y(x)}{\sigma}\right)\left\langle u_y(x)\nabla_{g_x}(u_y)(x), P_{S(x)}\left(\nabla_{g_x}(h)(x)\right)\right\rangle_{g_x} dV(x,S(x))\right) \\
&+ \lim_{j\uparrow\infty}\left(\frac{c(\sigma,K)-1}{\sigma}\frac{m}{\sigma^m}\int_{B_{d_g}(y,\sigma)} h(x)\varphi_{\varepsilon_j}\left(\frac{u_y(x)}{\sigma}\right) d\|V\|(x)\right) \\
&+ \lim_{j\uparrow\infty}\int_{\mathrm{Gr}_m(B_{d_g}(y,\sigma))} \frac{h(x)}{\sigma^m}\frac{d}{d\sigma}\left(\varphi_{\varepsilon_j}\left(\frac{u_y}{\sigma}\right)\right)(x)\left\|P_{S(x)^\perp}(\nabla_{g_x}(u_y)(x))\right\|^2_{g_x} dV(x,S(x)) \\
&- \lim_{j\uparrow\infty}\frac{\delta_g V\left(X_{h,\varepsilon_j,\sigma}\right)}{\sigma^{m+1}},
\end{aligned} \tag{81}$$

and

$$\begin{aligned}
&\lim_{j\uparrow\infty}\left(\frac{d}{d\sigma}\left(\frac{1}{\sigma^m}\int_{B_{d_g}(y,\sigma)} h(x)\varphi_{\varepsilon_j}\left(\frac{u_y(x)}{\sigma}\right) d\|V\|(x)\right)\right) \leq \\
&\leq \lim_{j\uparrow\infty}\left(\frac{1}{\sigma^{m+1}}\int_{\mathrm{Gr}_m(B_{d_g}(y,\sigma))} \varphi_{\varepsilon_j}\left(\frac{u_y(x)}{\sigma}\right)\left\langle u_y(x)\nabla_{g_x}(u_y)(x), P_{S(x)}\left(\nabla_{g_x}(h)(x)\right)\right\rangle_{g_x} dV(x,S(x))\right) \\
&+ \lim_{j\uparrow\infty}\left(\frac{c(\sigma,-K)-1}{\sigma}\frac{m}{\sigma^m}\int_{B_{d_g}(y,\sigma)} h(x)\varphi_{\varepsilon_j}\left(\frac{u_y(x)}{\sigma}\right) d\|V\|(x)\right) \\
&+ \lim_{j\uparrow\infty}\left(\int_{\mathrm{Gr}_m(B_{d_g}(y,\sigma))} \frac{h(x)}{\sigma^m}\frac{d}{d\sigma}\left(\varphi_{\varepsilon_j}\left(\frac{u_y}{\sigma}\right)\right)(x)\left\|P_{S(x)^\perp}(\nabla_{g_x}(u_y)(x))\right\|^2_{g_x} dV(x,S(x))\right) \\
&- \lim_{j\uparrow\infty}\frac{\delta_g V\left(X_{h,\varepsilon_j,\sigma}\right)}{\sigma^{m+1}},
\end{aligned} \tag{82}$$

By a standard application of [42, Theorem 2.24, p. 54] and Equation (80) in each term of Equations (81) and (82), we obtain the desired result □

Remark 12.3.11. In the context of Theorem 12.3.10. Firstly note that

$$\lim_{K\downarrow 0}\frac{c(\sigma,K)-1}{\sigma} = 0 \qquad \text{and} \qquad \lim_{K\downarrow 0}\frac{c(\sigma,-K)-1}{\sigma} = 0.$$

Thus taking the limit $K \downarrow 0$ in Equations (78) and (79) we obtain (compare with [91, Equation 4.1, Chapter 4, Section 4, p. 93]) that

$$\begin{aligned}\frac{d}{d\sigma}\left(\frac{1}{\sigma^m}\int_{B_{d_g}(y,\sigma)} h(x)d\|V\|(x)\right) &= \frac{1}{\sigma^{m+1}}\int_{\mathrm{Gr}_m(B_{d_g}(y,\sigma))} \left\langle u_y(x)\nabla_{g_x}(u_y)(x), P_{S(x)}\left(\nabla_{g_x}(h)(x)\right)\right\rangle_{g_x} dV(x,S(x))\\ &+ \frac{d}{d\sigma}\left(\int_{\mathrm{Gr}_m(B_{d_g}(y,\sigma))} \frac{h(x)}{u_y(x)^m}\left\|P_{S(x)^\perp}(\nabla_{g_x}(u_y)(x))\right\|^2_{g_x} dV(x,S(x))\right)\\ &+ \frac{1}{\sigma^{m+1}}\int_{B_{d_g}(y,\sigma)} \left\langle \overrightarrow{H_g}(x), h(x)u_y(x)\nabla_{g_x}(u_y)(x)\right\rangle_{g_x} d\|V\|(x),\end{aligned}$$

Theorem 12.3.12. *Let $\xi \in M$, $0 < \delta < 1$, and $V \in \mathbf{V}_m(M^n)$. There exists $\rho_0 := \rho_0(r_0, K) > 0$ such that if V has L^∞_{loc}-bounded generalized mean curvature vector in $B_{d_g}(\xi,\rho)$, for some $0 < \rho < \rho_0$, then, for all $y \in B_{d_g}(\xi,\delta\rho)$ and for all $\sigma \in]0,(1-\delta)\rho]$, we have that the function defined by*

$$\begin{aligned} f :]0,\sigma[&\to [0,+\infty[\\ t &\mapsto e^{\mathbf{C_1}t}\frac{\|V\|(B_{d_g}(y,t))}{t^m},\end{aligned} \tag{83}$$

is non-decreasing, where $\mathbf{C_1} = \mathbf{C_1}(C(\rho), m, \rho, K) > 0$ and $C(\rho) > 0$ is given by $\|\overrightarrow{H_g}\|_{L^\infty(B_{d_g}(\xi,\rho),\|V\|)} \leq C(\rho)$. Furthermore, for $0 < \sigma_1 < \sigma_2 < \sigma$ we have that

$$\begin{aligned} &e^{\mathbf{C_1}\sigma_2}\frac{\|V\|(B_{d_g}(y,\sigma_2))}{\sigma_2^m} - e^{\mathbf{C_1}\sigma_1}\frac{\|V\|(B_{d_g}(y,\sigma_1))}{\sigma_1^m} \geq\\ &\geq \int_{\mathrm{Gr}_m\left(B_{d_g}(y,\sigma_2)\setminus B_{d_g}(y,\sigma_1)\right)} \frac{\left\|P_{S(x)^\perp}(\nabla_{g_x}(u_y)(x))\right\|^2_{g_x}}{u_y(x)^m} dV(x,S(x)),\end{aligned} \tag{84}$$

and

$$\begin{aligned} &e^{-\mathbf{C_2}\sigma_2}\frac{\|V\|(B_{d_g}(y,\sigma_2))}{\sigma_2^m} - e^{-\mathbf{C_2}\sigma_1}\frac{\|V\|(B_{d_g}(y,\sigma_1))}{\sigma_1^m} \leq\\ &\leq \int_{\mathrm{Gr}_m\left(B_{d_g}(y,\sigma_2)\setminus B_{d_g}(y,\sigma_1)\right)} \frac{\left\|P_{S(x)^\perp}(\nabla_{g_x}(u_y)(x))\right\|^2_{g_x}}{u_y(x)^m} dV(x,S(x)),\end{aligned} \tag{85}$$

where $\mathbf{C_2} = \mathbf{C_2}(C(\rho), m, \rho, K) > 0$.

Proof. Let us fix $\rho_0 := \rho_0(r_0, K) > 0$ given by Theorem 12.3.10. Suppose that V has L^∞_{loc}-bounded generalized mean curvature vector in $B_{d_g}(\xi,\rho)$, for some $0 < \rho < \rho_0$. Consider $y \in B_{d_g}(\xi,\delta\rho)$ and $\sigma \in]0,(1-\delta)\rho]$ arbitrary.

Let $t \in]0,\sigma[$ and $h : \mathrm{supp}(\|V\|) \to \mathbb{R}$ given by $h \equiv 1$. Applying Equation (78) for t we have that

$$\begin{aligned}\frac{d}{dt}\left(\frac{\|V\|\left(B_{d_g}(y,t)\right)}{t^m}\right) &\geq \frac{c(t,K)-1}{t}m\frac{\|V\|\left(B_{d_g}(y,t)\right)}{t^m} + \frac{d}{dt}\left(\int_{\mathrm{Gr}_m(B_{d_g}(y,t))}\frac{\left\|P_{S(x)^\perp}(\nabla_{g_x}(u_y)(x))\right\|^2_{g_x}}{u_y(x)^m}dV(x,S(x))\right)\\ &+ \frac{1}{t^{m+1}}\int_{B_{d_g}(y,t)}\left\langle \overrightarrow{H_g}(x), u_y(x)\nabla_{g_x}(u_y)(x)\right\rangle_{g_x} d\|V\|(x),\end{aligned} \tag{86}$$

By Cauchy-Schwarz inequality and standard computations, we obtain that

$$\frac{1}{t^{m+1}}\int_{B_{d_g}(y,t)}\left\langle \overrightarrow{H_g}(x), u_y(x)\nabla_{g_x}(u_y)(x)\right\rangle_{g_x} d\|V\|(x) \geq -\frac{1}{t^m}\int_{B_{d_g}(y,t)}\left\|\overrightarrow{H_g}(x)\right\|_{g_x} d\|V\|(x). \quad (87)$$

By Equations (86) and (87), and Hölder inequality we obtain that

$$\frac{d}{dt}\left(\frac{\|V\|\left(B_{d_g}(y,t)\right)}{t^m}\right) \geq -\left(\left\|\overrightarrow{H_g}\right\|_{L^\infty(B_{d_g}(y,t),\|V\|)} - m\frac{c(t,K)-1}{t}\right)\frac{\|V\|\left(B_{d_g}(y,t)\right)}{t^m} + \frac{d}{dt}\left(\int_{\mathrm{Gr}_m(B_{d_g}(y,t))}\frac{\left\|P_{S(x)^\perp}(\nabla_{g_x}(u_y)(x))\right\|^2_{g_x}}{u_y(x)^m}dV(x,S(x))\right). \quad (88)$$

Note that for $t > 0$ and $b > 0$ the function $\frac{c(t,b)-1}{t}$ is non-increasing and negative. Thus, since $0 < t < \sigma < \rho$, we have that $\frac{c(t,K)-1}{t} \geq \frac{c(\rho,K)-1}{\rho}$. Moreover, since we are assuming that $V \in \mathbf{V}_m(M)$ has L^∞_{loc}-bounded generalized mean curvature vector in $B_{d_g}(\xi,\rho)$, there exists $C(\rho) > 0$ such that $\left\|\overrightarrow{H_g}\right\|_{L^\infty(B_{d_g}(\xi,\rho),\|V\|)} \leq C(\rho)$. Thus, by Equation (88), we obtain

$$\frac{d}{dt}\left(\frac{\|V\|\left(B_{d_g}(y,t)\right)}{t^m}\right) \geq -\mathbf{C_1}\frac{\|V\|\left(B_{d_g}(y,t)\right)}{t^m} + \frac{d}{dt}\left(\int_{\mathrm{Gr}_m(B_{d_g}(y,t))}\frac{\left\|P_{S(x)^\perp}(\nabla_{g_x}(u_y)(x))\right\|^2_{g_x}}{u_y(x)^m}dV(x,S(x))\right). \quad (89)$$

Since $e^{\mathbf{C_1}t} > 1$, for all $t > 0$, Equation (89) implies

$$f'(t) \geq \frac{d}{dt}\left(\int_{\mathrm{Gr}_m(B_{d_g}(y,t))}\frac{\left\|P_{S(x)^\perp}(\nabla_{g_x}(u_y)(x))\right\|^2_{g_x}}{u_y(x)^m}dV(x,S(x))\right) \geq 0. \quad (90)$$

By Equation (90), f is non-decreasing. Furthermore, for $0 < \sigma_1 < \sigma_2 < \sigma$, integrating Equation (90) over $[\sigma_1,\sigma_2]$ with respect to dt, by Fundamental Theorem of Calculus, we obtain Equation (84).

By a similar argument, we obtain Equation (85). □

Theorem 12.3.13. *Let $m < p < \infty$, $\xi \in M^n$, $0 < \delta < 1$, and $V \in \mathbf{V}_m(M^n)$. There exists $\rho_0 := \rho_0(r_0,K) > 0$ such that if V has L^p_{loc}-bounded generalized mean curvature vector in $B_{d_g}(\xi,\rho)$, for some $0 < \rho < \rho_0$, then for all $y \in B_{d_g}(\xi,\delta\rho)$ and for all $\sigma \in]0,(1-\delta)\rho]$, we have that the function defined by*

$$\begin{aligned} f :\,]0,\sigma[&\to [0,+\infty[\\ t &\mapsto \left(\frac{\|V\|\left(B_{d_g}(y,t)\right)}{t^m}\right)^{\frac{1}{p}} + \mathbf{C}t^{1-\frac{m}{p}}, \end{aligned} \quad (91)$$

is non-decreasing, where $\mathbf{C} = \mathbf{C}(C(\rho), m, \rho, K) > 0$ *and* $C(\rho) > 0$ *is given by* $\|\overrightarrow{H_g}\|_{L^p(B_{d_g}(\xi,\rho),\|V\|)} \leq C(\rho)$. *Furthermore, for* $0 < \sigma_1 < \sigma_2 < \sigma$ *we have that*

$$\left(\frac{\|V\|\left(B_{d_g}(y,\sigma_2)\right)}{\sigma_2^m}\right)^{\frac{1}{p}} - \left(\frac{\|V\|\left(B_{d_g}(y,\sigma_1)\right)}{\sigma_1^m}\right)^{\frac{1}{p}} \geq \mathbf{C}\left(\sigma_1^{1-\frac{m}{p}} - \sigma_2^{1-\frac{m}{p}}\right). \tag{92}$$

Proof. Let us fix $\rho_0 := \rho_0(r_0, K) > 0$ given by Theorem 12.3.10. Suppose that V has L^p_{loc}-bounded generalized mean curvature vector in $B_{d_g}(\xi,\rho)$, for some $0 < \rho < \rho_0$. Consider $y \in B_{d_g}(\xi, \delta\rho)$ and $\sigma \in]0, (1-\delta)\rho]$ arbitrary.

Let $t \in]0, \sigma[$ and $h : \text{supp}(\|V\|) \to \mathbb{R}$ given by $h \equiv 1$. Applying Equation (78) for t we have that

$$\begin{aligned}\frac{d}{dt}\left(\frac{\|V\|\left(B_{d_g}(y,t)\right)}{t^m}\right) \geq{}& \frac{c(t,K)-1}{t} m \frac{\|V\|\left(B_{d_g}(y,t)\right)}{t^m} \\ &+ \frac{d}{dt}\left(\int_{\text{Gr}_m(B_{d_g}(y,t))} \frac{\left\|P_{S(x)^\perp}(\nabla_{g_x}(u_y)(x))\right\|^2_{g_x}}{u_y(x)^m} dV(x,S(x))\right) \\ &+ \frac{1}{t^{m+1}} \int_{B_{d_g}(y,t)} \left\langle \overrightarrow{H_g}(x), u_y(x)\nabla_{g_x}(u_y)(x)\right\rangle_{g_x} d\|V\|(x),\end{aligned} \tag{93}$$

By Cauchy-Schwarz inequality and standard computations, we obtain that

$$\frac{1}{t^{m+1}} \int_{B_{d_g}(y,t)} \left\langle \overrightarrow{H_g}(x), u_y(x)\nabla_{g_x}(u_y)(x)\right\rangle_{g_x} d\|V\|(x) \geq -\frac{1}{t^m} \int_{B_{d_g}(y,t)} \left\|\overrightarrow{H_g}(x)\right\|_{g_x} d\|V\|(x). \tag{94}$$

By Equations (93) and (94), and Hölder inequality we obtain that

$$\begin{aligned}\frac{d}{dt}\left(\frac{\|V\|\left(B_{d_g}(y,t)\right)}{t^m}\right) \geq{}& -\left(\left\|\overrightarrow{H_g}\right\|_{L^p(B_{d_g}(y,t),\|V\|)} - \frac{c(t,K)-1}{t} m \left(\|V\|\left(B_{d_g}(y,t)\right)\right)^{\frac{1}{p}}\right) \frac{\left(\|V\|\left(B_{d_g}(y,t)\right)\right)^{\frac{p-1}{p}}}{t^m} \\ &+ \frac{d}{dt}\left(\int_{\text{Gr}_m(B_{d_g}(y,t))} \frac{\left\|P_{S(x)^\perp}(\nabla_{g_x}(u_y)(x))\right\|^2_{g_x}}{u_y(x)^m} dV(x,S(x))\right).\end{aligned} \tag{95}$$

Note that for $t > 0$ and $b > 0$ the function $\frac{c(t,b)-1}{t}$ is non-increasing and negative. Moreover, since we are assuming that $V \in \mathbf{V}_m(M)$ has L^p_{loc}-bounded generalized mean curvature vector in $B_{d_g}(\xi,\rho)$, there exists $C(\rho) > 0$ such that $\left\|\overrightarrow{H_g}\right\|_{L^p(B_{d_g}(\xi,\rho),\|V\|)} \leq C(\rho)$. Thus, by Equation (95), we obtain

$$f'(t) = \frac{d}{dt}\left(\left(\frac{\|V\|(B_{d_g}(y,t))}{t^m}\right)^{\frac{1}{p}}\right) + \mathbf{C}\frac{d}{dt}\left(t^{1-\frac{m}{p}}\right) \geq 0. \tag{96}$$

By Equation (96) we have that $f'(t) \geq 0$ for all $0 < t < \sigma$, that is, f is non-decreasing. Furthermore, for $0 < \sigma_1 < \sigma_2 < \sigma$, integrating Equation (96) over $[\sigma_1, \sigma_2]$ with respect to dt, by Fundamental Theorem of Calculus, we obtain Equation (92). □

Corollary 12.3.14. *Let $m < p \leq \infty$, $\xi \in M^n$, $0 < \delta < 1$, and $V \in \mathbf{V}_m(M^n)$. There exists $\rho_0 := \rho_0(r_0, K) > 0$ such that if V has L^p_{loc}-bounded generalized mean curvature vector in $B_{d_g}(\xi, \rho)$, for some $0 < \rho < \rho_0$, then, for all $y \in B_{d_g}(\xi, \delta\rho)$, there exists the density*

$$\Theta^m(\|V\|, y) = \lim_{t \downarrow 0} \frac{\|V\|\left(B_{d_g}(y,t)\right)}{\omega_m t^m}. \tag{97}$$

Furthermore, $\Theta^m(\|V\|, \cdot)$ is an upper-semicontinuous function in $B_{d_g}(\xi, \delta\rho)$, that is, for every $y \in B_{d_g}(\xi, \delta\rho)$, we have that

$$\limsup_{x \to y} \Theta^m(\|V\|, x) \leq \Theta^m(\|V\|, y). \tag{98}$$

Proof. The proof we be divide in the cases $p = \infty$ and $m < p < \infty$.

- $p = \infty$

  Let us fix $\rho_0 := \rho_0(r_0, K) > 0$ given by Theorem 12.3.12. Suppose that V has L^∞_{loc}-bounded generalized mean curvature vector in $B_{d_g}(\xi, \rho)$, for some $0 < \rho < \rho_0$. Consider $y \in B_{d_g}(\xi, \delta\rho)$ and $\sigma \in]0, (1-\delta)\rho]$ arbitrary.

  Define the function $f :]0, \sigma[\to [0, +\infty[$ by

  $$f(t) = e^{\mathbf{C_1} t} \frac{\|V\|(B_{d_g}(y,t))}{t^m},$$

  where $\mathbf{C_1} = \mathbf{C_1}(C(\rho), m, \rho, K) > 0$ and $C(\rho) > 0$ is given by $\|\overrightarrow{H_g}\|_{L^\infty(B_{d_g}(\xi,\rho),\|V\|)} \leq C(\rho)$.

  Note that by Theorem 12.3.12 we have that f is non-decreasing. Furthermore, f is bounded from below by 0. Thus, the limit $\lim_{t \downarrow 0} f(t)$ exists. Let $M \geq 0$ be such limit.

  Let $\{t_i\}_{i \in \mathbb{N}}$ be a sequence such that $0 < t_i < \sigma$, for all $i \in \mathbb{N}$, and $t_i \downarrow 0$ when $i \uparrow \infty$. Then,

  $$\Theta^m(\|V\|, y) = \lim_{i \uparrow \infty} \frac{1}{\omega_m} \frac{f(t_i)}{e^{\mathbf{C_1} t_i}} < \infty.$$

  Which implies that the density $\Theta^m(\|V\|, y)$ exists.

  Let $\varepsilon \in]0, 1[$ arbitrary and $\{y_i\}_{i \in \mathbb{N}} \subset B_{d_g}(\xi, \delta\rho)$ any sequence such that $y_i \to y$ when $i \uparrow \infty$.

  Note that for i sufficiently large we have that $B_{d_g}(y_i, (1-\varepsilon)\sigma) \subset B_{d_g}(y, \sigma)$.

  Take $\sigma_1 < (1-\varepsilon)\sigma$. Note that $y_i \in B_{d_g}(\xi, \delta\rho)$, for all $i \in \mathbb{N}$, and $0 < \sigma_1 < (1-\varepsilon)\sigma < \sigma$. By Theorem 12.3.12 we have, for each $i \in \mathbb{N}$, that

  $$e^{\mathbf{C_1}(1-\varepsilon)\sigma} \frac{\|V\|(B_{d_g}(y_i, (1-\varepsilon)\sigma))}{((1-\varepsilon)\sigma)^m} - e^{\mathbf{C_1}\sigma_1} \frac{\|V\|(B_{d_g}(y_i, \sigma_1))}{\sigma_1^m} \geq 0.$$

Thus, by standard computations, we obtain that $\Theta^m(\|V\|,\cdot)$ is an upper-semicontinuous function in $B_{d_g}(\xi,\delta\rho)$.

- $m < p < \infty$

  Let us fix $\rho_0 := \rho_0(r_0, K) > 0$ given by Theorem 12.3.13. Suppose that V has L^p_{loc}-bounded generalized mean curvature vector in $B_{d_g}(\xi,\rho)$, for some $0 < \rho < \rho_0$. Consider $y \in B_{d_g}(\xi,\delta\rho)$ and $\sigma \in]0,(1-\delta)\rho]$ arbitrary.

  Define the function $f :]0, r_0[\to [0, +\infty[$ by

$$f(t) = \left(\frac{\|V\|(B_{d_g}(y,t))}{t^m}\right)^{\frac{1}{p}} + \mathbf{C_2} t^{1-\frac{m}{p}}$$

  Note that by Theorem 12.3.13 we have that f is non-decreasing. Furthermore, f is bounded from below by 0. Thus, the limit $\lim_{t\downarrow 0} f(t)$ exists. Let $M \geq 0$ be such limit.

  Let $\{t_i\}_{i\in\mathbb{N}}$ be a sequence such that $0 < t_i < \sigma$, for all $i \in \mathbb{N}$, and $t_i \downarrow 0$ when $i \uparrow \infty$. Then,

$$\Theta^m(\|V\|,y) = \lim_{i\uparrow\infty} \frac{1}{\omega_m}\left(f(t_i) - \mathbf{C_2} t_i^{1-\frac{m}{p}}\right)^p < \infty,$$

  where $\mathbf{C_2} = \mathbf{C_2}(C(\rho), m, \rho, K) > 0$ and $C(\rho) > 0$ is given by $\|\overrightarrow{H_g}\|_{L^p(B_{d_g}(\xi,\rho),\|V\|)} \leq C(\rho)$, which implies that the density $\Theta^m(\|V\|,y)$ exists.

  Let $\varepsilon \in]0,1[$ arbitrary and $\{y_i\}_{i\in\mathbb{N}} \subset B_{d_g}(\xi,\delta\rho)$ any sequence such that $y_i \to y$ when $i \uparrow \infty$.

  Note that for i sufficiently large we have that $B_{d_g}(y_i,(1-\varepsilon)\sigma) \subset B_{d_g}(y,\rho)$.

  Take $\sigma_1 < (1-\varepsilon)\sigma$. Note that $y_i \in B_{d_g}(\xi,\delta\rho)$, for all $i \in \mathbb{N}$, and $0 < \sigma_1 < (1-\varepsilon)\sigma < \sigma$. By Theorem 12.3.13 we have, for each $i \in \mathbb{N}$, that

$$\left(\frac{\|V\|\left(B_{d_g}(y_i,(1-\varepsilon)\sigma)\right)}{((1-\varepsilon)\sigma)^m}\right)^{\frac{1}{p}} - \left(\frac{\|V\|\left(B_{d_g}(y_i,\sigma_1)\right)}{\sigma_1^m}\right)^{\frac{1}{p}} \geq \mathbf{C_2}\left(\sigma_1^{1-\frac{m}{p}} - ((1-\varepsilon)\sigma)^{1-\frac{m}{p}}\right).$$

  Thus, by standard computations, we obtain that $\Theta^m(\|V\|,\cdot)$ is an upper-semicontinuous function in $B_{d_g}(\xi,\delta\rho)$.

To conclude the proof, we choose $\rho_0 := \rho_0(r_0, K) > 0$ as the minimum value ensuring that both cases above hold. □

Remark 12.3.15. In the context of Corollary 12.3.14, since $\Theta^m(\|V\|,\cdot)$ is upper-semicontinuous in $B_{d_g}(\xi,\delta\rho)$ then we have that $\Theta^m(\|V\|,\cdot)$ is a $\|V\|$-measurable function in $B_{d_g}(\xi,\delta\rho)$.

Theorem 12.3.16. *Let $1 < p \leq \infty$, $\xi \in M^n$, $0 < \delta < 1$, and $V \in \mathbf{V}_m(M^n)$. There exists $\rho_0 := \rho_0(r_0, K) > 0$ such that if V has L^p_{loc}-bounded generalized mean curvature vector in $B_{d_g}(\xi, \rho)$, for some $0 < \rho < \rho_0$, and*

$$\left\| \overrightarrow{H_g} \right\|_{L^1(B_{d_g}(y,t), \|V\|)} \leq \alpha \Lambda \left(\frac{t}{\rho} \right)^{\alpha - 1} \|V\|(B_{d_g}(y,t)), \tag{99}$$

for all $y \in B_{d_g}(\xi, \rho)$, for all $t > 0$ such that $B_{d_g}(y,t) \subset B_{d_g}(\xi, \rho)$, for some $\alpha > 0$, and for some $\Lambda > 0$, then, for all $y \in B_{d_g}(\xi, \delta\rho)$ and for all $\sigma \in]0, (1-\delta)\rho]$, we have that the function defined by

$$\begin{aligned} f :]0, \sigma[&\to [0, +\infty[\\ t &\mapsto e^{\lambda(t)} \frac{\|V\|(B_{d_g}(y,t))}{t^m} \end{aligned} \tag{100}$$

is non-decreasing, where

$$\begin{aligned} \lambda :]0, \sigma[&\to]0, +\infty[\\ t &\mapsto \left(\Lambda \left(\frac{t}{\rho} \right)^{\alpha - 1} - \mathbf{C_1} m \right) t, \end{aligned} \tag{101}$$

and $\mathbf{C_1} = \mathbf{C_1}(\rho, K) > 0$.
Furthermore, for $0 < \sigma_1 < \sigma_2 < \sigma$, we have that

$$\begin{aligned} & e^{\lambda(\sigma_2)} \frac{\|V\|(B_{d_g}(y,\sigma_2))}{\sigma_2^m} - e^{\lambda(\sigma_1)} \frac{\|V\|(B_{d_g}(y,\sigma_1))}{\sigma_1^m} \geq \\ & \geq \int_{\mathrm{Gr}_m\left(B_{d_g}(y,\sigma_2) \setminus B_{d_g}(y,\sigma_1)\right)} \frac{\left\| P_{S(x)^\perp}(\nabla_{g_x}(u_y)(x)) \right\|^2_{g_x}}{u_y(x)^m} dV(x, S(x)), \end{aligned} \tag{102}$$

and

$$\begin{aligned} & e^{-\ell(\sigma_2)} \frac{\|V\|(B_{d_g}(y,\sigma_2))}{\sigma_2^m} - e^{-\ell(\sigma_1)} \frac{\|V\|(B_{d_g}(y,\sigma_1))}{\sigma_1^m} \leq \\ & \leq \int_{\mathrm{Gr}_m\left(B_{d_g}(y,\sigma_2) \setminus B_{d_g}(y,\sigma_1)\right)} \frac{\left\| P_{S(x)^\perp}(\nabla_{g_x}(u_y)(x)) \right\|^2_{g_x}}{u_y(x)^m} dV(x, S(x)), \end{aligned} \tag{103}$$

where

$$\begin{aligned} \ell :]0, \sigma[&\to]0, +\infty[\\ t &\mapsto \left(\Lambda \left(\frac{t}{\rho} \right)^{\alpha - 1} + \mathbf{C_2} m \right) t, \end{aligned} \tag{104}$$

and $\mathbf{C_2} = \mathbf{C_2}(\rho, K) > 0$.

Proof. Let us fix $\rho_0 := \rho_0(r_0, K) > 0$ given by Theorem 12.3.10. Suppose that V has L^p_{loc}-bounded generalized mean curvature vector in $B_{d_g}(\xi, \rho)$, for some $0 < \rho < \rho_0$, and Equation (99) holds, for all $0 < t < \rho$, for some $\alpha > 0$, and for some $\Lambda > 0$. Consider $y \in B_{d_g}(\xi, \delta\rho)$ and $\sigma \in]0, (1-\delta)\rho]$ arbitrary.

Let $t \in]0, \sigma[$ and $h : \text{supp}(\|V\|) \to \mathbb{R}$ given by $h \equiv 1$. Applying Equation (78) for t, we have that

$$\begin{aligned}\frac{d}{dt}\left(\frac{\|V\|\left(B_{d_g}(y,t)\right)}{t^m}\right) \geq \frac{c(t,K)-1}{t} m \frac{\|V\|\left(B_{d_g}(y,t)\right)}{t^m} \\ + \frac{d}{dt}\left(\int_{\text{Gr}_m(B_{d_g}(y,t))} \frac{\left\|P_{S(x)^\perp}(\nabla_{g_x}(u_y)(x))\right\|^2_{g_x}}{u_y(x)^m} dV(x,S(x))\right) \\ + \frac{1}{t^{m+1}} \int_{B_{d_g}(y,t)} \left\langle \overrightarrow{H_g}(x), u_y(x)\nabla_{g_x}(u_y)(x)\right\rangle_{g_x} d\|V\|(x). \end{aligned} \tag{105}$$

By Cauchy-Schwarz inequality and standard computations, we obtain that

$$\frac{1}{t^{m+1}} \int_{B_{d_g}(y,t)} \left\langle \overrightarrow{H_g}(x), u_y(x)\nabla_{g_x}(u_y)(x)\right\rangle_{g_x} d\|V\|(x) \geq -\frac{1}{t^m} \int_{B_{d_g}(y,t)} \left\|\overrightarrow{H_g}(x)\right\|_{g_x} d\|V\|(x). \tag{106}$$

Since we are assuming that Equation (99) holds, for all $y \in B_{d_g}(\xi, \rho)$, for all $t > 0$ such that $B_{d_g}(y,t) \subset B_{d_g}(\xi, \rho)$, for some $\alpha > 0$, and for some $\Lambda > 0$. By Equations (105) and (106), and Hölder inequality we obtain that

$$\begin{aligned}\frac{d}{dt}\left(\frac{\|V\|\left(B_{d_g}(y,t)\right)}{t^m}\right) \geq -\left(\alpha\Lambda\left(\frac{t}{\rho}\right)^{\alpha-1} - \frac{c(t,K)-1}{t} m\right) \frac{\|V\|\left(B_{d_g}(y,t)\right)}{t^m} \\ + \frac{d}{dt}\left(\int_{\text{Gr}_m(B_{d_g}(y,t))} \frac{\left\|P_{S(x)^\perp}(\nabla_{g_x}(u_y)(x))\right\|^2_{g_x}}{u_y(x)^m} dV(x,S(x))\right). \end{aligned} \tag{107}$$

Note that for $t > 0$ and $b > 0$ the function $\frac{c(t,b)-1}{t}$ is non-increasing and negative. Thus, since $0 < t < \sigma < \rho$, we have that $\frac{c(t,K)-1}{t} \geq \frac{c(\rho,K)-1}{\rho}$. Moreover, $e^{\lambda(t)} > 0$, for all $t \in \mathbb{R}$, and $\lambda(t) > 0$, for all $t \in]0, \sigma[$. Thus, by Equation (107), we obtain that

$$f'(t) \geq \frac{d}{dt}\left(\int_{\text{Gr}_m(B_{d_g}(y,t))} \frac{\left\|P_{S(x)^\perp}(\nabla_{g_x}(u_y)(x))\right\|^2_{g_x}}{u_y(x)^m} dV(x,S(x))\right) \geq 0. \tag{108}$$

By Equation (108), f is non-decreasing. Furthermore, for $0 < \sigma_1 < \sigma_2 < \sigma$, integrating Equation (108) over $[\sigma_1, \sigma_2]$ with respect to dt, by Fundamental Theorem of Calculus, we obtain Equation (102).

By a similar argument, we obtain Equation (103). □

Theorem 12.3.17. *Let $\xi \in M^n$, $0 < \delta < 1$, and $V \in \mathbf{V}_m(M^n)$. There exists $\rho_0 := \rho_0(r_0, K) > 0$ such that if V has locally bounded first variation in $B_{d_g}(\xi, \rho)$, for some $0 < \rho < \rho_0$, then, for $\|V\|$-a.e. $y \in B_{d_g}(\xi, \delta\rho)$, there exists the density*

$$\Theta^m(\|V\|, y) = \lim_{t \downarrow 0} \frac{\|V\|(B_{d_g}(y,t))}{\omega_m t^m}.$$

Furthermore, $\Theta^m(\|V\|, y)$ is an upper-semicontinuous function for $\|V\|$-a.e. $y \in B_{d_g}(\xi, \delta\rho)$, that is, for $\|V\|$-a.e. $y \in B_{d_g}(\xi, \delta\rho)$, we have that

$$\limsup_{x \to y} \Theta^m(\|V\|, x) \leq \Theta^m(\|V\|, y).$$

Proof. Let us fix $\rho_0 := \rho_0(r_0, K) > 0$ given by Lemma 12.3.9. Suppose that V has locally bounded first variation in $B_{d_g}(\xi, \rho)$, for some $0 < \rho < \rho_0$. Consider $y \in B_{d_g}(\xi, \delta\rho)$ and $\sigma \in]0, (1-\delta)\rho]$ arbitrary.

The first part of the proof is to establish a monotonicity formula for varifolds with locally bounded first variation, following exactly the same lines as Theorem 12.3.10, except for the differences addressed below.

Since we are assuming that V has locally bounded first variation in $B_{d_g}(\xi, \rho)$, by Proposition 12.2.7, [42, Theorem 2.24, p. 54] and Equation (80), we have that

$$-\lim_{j \uparrow \infty} \frac{\delta_g V\left(X_{h,\varepsilon_j,\sigma}\right)}{\sigma^{m+1}} = -\frac{\delta_g V(\mathbf{X}^{h,\sigma})}{\sigma^m}, \tag{109}$$

where $\mathbf{X}^{h,\sigma} := h(x)\frac{u_y(x)}{\sigma}\nabla_{g_x}(u_y)(x)$. Therefore,

$$\begin{aligned}
&\frac{d}{d\sigma}\left(\frac{1}{\sigma^m}\int_{B_{d_g}(y,\sigma)} h(x)d\|V\|(x)\right) \geq \\
&\geq \frac{1}{\sigma^{m+1}}\int_{\mathrm{Gr}_m(B_{d_g}(y,\sigma))} \left\langle u_y(x)\nabla_{g_x}(u_y)(x), P_{S(x)}\left(\nabla_{g_x}(h)(x)\right)\right\rangle_{g_x} dV(x,S(x)) \\
&+ \frac{c(\sigma,K)-1}{\sigma}\frac{m}{\sigma^m}\int_{B_{d_g}(y,\sigma)} h(x)d\|V\|(x) \\
&+ \frac{d}{d\sigma}\left(\int_{\mathrm{Gr}_m(B_{d_g}(y,\sigma))} \frac{h(x)}{u_y(x)^m}\left\|P_{S(x)^\perp}(\nabla_{g_x}(u_y)(x))\right\|^2_{g_x} dV(x,S(x))\right) \\
&- \frac{\delta_g V(\mathbf{X}^{h,\sigma})}{\sigma^m},
\end{aligned} \tag{110}$$

Let $t \in]0, \sigma[$ and $h : \mathrm{supp}(\|V\|) \to \mathbb{R}$ given by a mollification of $\mathcal{X}_{B_{d_g}(y,t)}$. Applying Equation (110) for t and taking the appropriates limits for the function h, we have that

$$\frac{d}{dt}\left(\frac{\|V\|(B_{d_g}(y,t))}{t^m}\right) \geq \frac{c(t,K)-1}{t} m \frac{\|V\|(B_{d_g}(y,t))}{t^m}$$

$$+\frac{d}{dt}\left(\int_{\mathrm{Gr}_m(B_{d_g}(y,t))}\frac{\left\|P_{S(x)^\perp}(\nabla_{g_x}(u_y)(x))\right\|^2_{g_x}}{u_y(x)^m}dV(x,S(x))\right)$$
$$-\frac{\delta_g V(\mathbf{X}^t)}{t^m}, \tag{111}$$

where $\mathbf{X}^t := \mathcal{X}_{B_{d_g}(y,t)}(x)\frac{u_y(x)}{t}\nabla_{g_x}(u_y)(x)$.

Since we are assuming that V has locally bounded first variation in $B_{d_g}(\xi,\rho)$, by Proposition 12.2.7 we have that $\|\delta_g V\|_{\mathrm{TV}}$ is a Radon measure. Thus, by [41, Theorem 295, p. 154], we have that $\Theta^{\|V\|}(\|\delta_g V\|_{TV},x)$ there exists for $\|V\|$-a.e. $x\in B_{d_g}(\xi,\rho)$. Therefore, by definition of density, we have that there exists $0<C(x)<\infty$ such that

$$\|\delta_g V\|_{\mathrm{TV}}(B_{d_g}(x,t)) \leq C(x)\|V\|(B_{d_g}(x,t)), \tag{112}$$

for $\|V\|$-a.e. $x\in B_{d_g}(\xi,\rho)$ and for all $t>0$ such that $B_{d_g}(x,t)\subset B_{d_g}(\xi,\rho)$. On the other hand, note that $\mathbf{X}^t\in\mathfrak{X}^0_c(B_{d_g}(y,t))$ and $\left\|\mathbf{X}^t\right\|_{L^\infty(B_{d_g}(y,t),\|V\|)}=1$. Let $y\in B_{d_g}(\xi,\delta\rho)$, which was chosen arbitrarily at the start of this proof, now be fixed such that Equation (112) holds for it. By Equation (112), we have that

$$-\frac{\delta_g V(\mathbf{X}^t)}{t^m}\geq -C(y)\frac{\|V\|(B_{d_g}(y,t))}{t^m}. \tag{113}$$

By Equations (111) and (113) we obtain that

$$\frac{d}{dt}\left(\frac{\|V\|(B_{d_g}(y,t))}{t^m}\right)-\left(C(y)-\frac{c(t,K)-1}{t}m\right)\frac{\|V\|(B_{d_g}(y,t))}{t^m}\geq$$
$$\geq\frac{d}{dt}\left(\int_{\mathrm{Gr}_m(B_{d_g}(y,t))}\frac{\left\|P_{S(x)^\perp}(\nabla_{g_x}(u_y)(x))\right\|^2_{g_x}}{u_y(x)^m}dV(x,S(x))\right). \tag{114}$$

Since we have now established a monotonicity formula for varifolds with locally bounded first variation, the proof follows the same lines as the previous cases. □

Remark 12.3.18. In the context of Theorem 12.3.17, since $\Theta^m(\|V\|,\cdot)$ is upper-semicontinuous for $\|V\|$-a.e. $y\in B_{d_g}(\xi,\delta\rho)$, we have that $\Theta^m(\|V\|,\cdot)$ is a $\|V\|$-measurable function in $B_{d_g}(\xi,\delta\rho)$.

12.4 RECTIFIABILITY THEOREM

Before proving the rectifiability theorem, we note that the argument will be carried out using local charts on the manifold. The reason for this is that rectifiability is fundamentally a local measure-theoretic and geometric property, rather than a genuinely metric one. In particular, the proof of rectifiability relies on Euclidean tools such as density estimates, tangent planes, and covering arguments, and is stable under

bi-Lipschitz changes of coordinates. Consequently, once we work in sufficiently small coordinate neighborhoods, the problem reduces to the classical Euclidean setting. Moreover, the validity of the argument is independent of the specific geometric constants of the ambient Riemannian manifold, provided the coordinate maps are uniformly bi-Lipschitz on small balls.

Theorem 12.4.1. *Let $\xi \in M^n$, $0 < \delta \leq 1$, and $V \in \mathbf{V}_m(M^n)$. There exists $\rho_0 := \rho_0(r_0) > 0$ such that if V has locally bounded first variation in $B_{d_g}(\xi, \rho)$, for some $0 < \rho < \rho_0$, and $\Theta^m(\|V\|, x) > 0$ for $\|V\|$-a.e. $x \in B_{d_g}(\xi, \rho)$, then $V \in \mathbf{RV}_m(B_{d_g}(\xi, \delta\rho))$.*

Proof. Let us fix $\rho_0 := \rho_0(r_0) < r_0$. There exists a smooth coordinate chart

$$\varphi : B_{d_g}(\xi, \rho) \to \mathbb{R}^n,$$

such that φ is bi-Lipschitz on $B_{d_g}(\xi, \rho)$.

Define the pushforward varifold $\widetilde{V} := \varphi^{\#}V \in \mathbf{V}_m(\mathbb{R}^n)$. Since φ is a smooth bi-Lipschitz diffeomorphism, the hypotheses of locally bounded first variation and positivity of the density are preserved under pushforward.

Therefore, by the Euclidean Rectifiability Theorem [91, Theorem 5.5, p. 246], it follows that

$$\widetilde{V} \in \mathbf{RV}_m(\varphi(B_{d_g}(\xi, \rho))).$$

Finally, since rectifiability is preserved under bi-Lipschitz changes of coordinates, we conclude that

$$V \in \mathbf{RV}_m(B_{d_g}(\xi, \delta\rho)),$$

which completes the proof. □

13 RIEMANNIAN INTRINSIC ALLARD'S INTERIOR ε-REGULARITY THEOREM

In the subsequent sections, we will require a more extensive background in Riemannian geometry than was needed in Chapter 12. For a comprehensive overview, we recommend the following references: [65, Chapter 8], [27, Chapter IX], and particularly [48, Chapters 2, 3, 8, and 9].

13.1 PRELIMINARIES ON SUBMANIFOLDS OF RIEMANNIAN MANIFOLDS

Before proceeding, let us establish some notation and review key aspects of the theory of submanifolds in abstract Riemannian manifolds.

Definition 13.1.1. Let $\Sigma \subset M^n$ be an m-dimensional submanifold of M^n of class $\mathcal{C}^2$, $x \in \Sigma$ and $(U, \{x_i\}_{i=1}^n)$ a Fermi coordinate chart at x. For $u_\Sigma > 0$ we define

$$u_\Sigma^2 := \sum_{i=m+1}^{n} x_i^2 \qquad \textit{and} \qquad N := \sum_{i=m+1}^{n} \frac{x_i}{u_\Sigma} \frac{\partial}{\partial x_i}. \tag{115}$$

The following lemma characterizes the function u_Σ and the vector field N. A proof can be found in [48, Lemma 2.7, p. 23].

Lemma 13.1.2. *Let $\Sigma \subset M^n$ be an m-dimensional submanifold of M^n of class $\mathcal{C}^2$ and $x \in M^n$. Suppose that there exists a unique geodesic γ from x to Σ meeting Σ orthogonally. Then*

$$u_\Sigma(x) = \operatorname{dist}_g(x, \Sigma) \qquad \textit{and} \qquad N(\gamma(s)) = \gamma'(s).$$

Definition 13.1.3. Let $\Sigma \subset M^n$ be an m-dimensional submanifold of M^n of class $\mathcal{C}^2$ and $r \geq 0$. We define the tube of radius r around Σ, denoted by $\operatorname{Tube}(\Sigma, r)$, as the set of points in M^n such that there exists a geodesic parting through the points, meeting Σ orthogonally and of length less than or equal to r.

Note that we can visualize a tubular neighborhood around an embedded submanifold using the exponential map.

$$\operatorname{Tube}(\Sigma, r) = \bigcup_{x \in \Sigma} \{\exp_x(v) : v \in T_x\Sigma^\perp \text{ and } \|v\|_{g_x} \leq r\}.$$

Definition 13.1.4. Let $\Sigma \subset M^n$ be an m-dimensional submanifold of M^n of class $\mathcal{C}^2$ and $0 < t \leq r$. We define the tubular hypersurface at a distance t from Σ by

$$\Sigma_t := \{x \in \text{Tube}(\Sigma, r) : \text{dist}_g(p, \Sigma) = t\}.$$

The tubular hypersurfaces $\{\Sigma_t\}_{0<t\leq r}$ form a natural foliation of the set $\text{Tube}(\Sigma, r) \setminus \Sigma$.

Definition 13.1.5. Let $\Sigma \subset M^n$ be an m-dimensional submanifold of M^n of class $\mathcal{C}^2$ and $x \in \Sigma$. We define the shape operator of Σ at x as the map

$$\begin{aligned} S_x : T_x\Sigma \times T_x\Sigma^\perp &\to T_x\Sigma \\ (X, \nu) &\mapsto S_x(X, \nu) := P_{T_x\Sigma}\left(\nabla^g_X \nu\right). \end{aligned}$$

Definition 13.1.6. Let $\Sigma \subset M^n$ be an m-dimensional submanifold of M^n of class $\mathcal{C}^2$, $x \in \Sigma$ and $\nu \in T_x\Sigma^\perp$. We define the second fundamental form of Σ with respect to ν at x as the map

$$\begin{aligned} \text{II}_{\nu,x} : T_x\Sigma \times T_x\Sigma &\to \mathbb{R} \\ (X, Y) &\mapsto \text{II}_{\nu,x}(X, Y) := \langle S_x(X, \nu), Y\rangle_{g_x}. \end{aligned}$$

Note that the second fundamental form of Σ with respect to ν at x is symmetric. Moreover, for a fixed $\nu \in T_x\Sigma^\perp$, we denote by $S_{\nu,x}$ the map

$$\begin{aligned} S_{\nu,x} : T_x\Sigma &\to T_x\Sigma \\ X &\mapsto S_{\nu,x}(X) := S_x(X, \nu). \end{aligned}$$

Therefore, $S_{\nu,x}$ is self-adjoint. Now, assume that $\nu \in T_x\Sigma^\perp$ has $\|\nu\|_g = 1$. The m eigenvalues of $S_{\nu,x}$, all of which are real due to its self-adjointness, are called the principal curvatures of Σ in the direction ν. Correspondingly, the associated eigenvectors are referred to as the principal curvature vectors of Σ in the direction ν.

Consider the scenario where Σ has co-dimension 1, meaning $n = m + 1$. In this case, for each $x \in \Sigma$, there exist exactly two normal vectors $\nu \in T_x\Sigma^\perp$ satisfying $\|\nu\|_g = 1$. It's important to observe that the second fundamental form and the shape operator change sign when we switch the normal vector.

Furthermore, note that for $X \in T_x\Sigma$ arbitrary, we have

$$S_{\nu,x}(X) = \nabla^g_X \nu.$$

Let $\Sigma \subset M^n$ be an m-dimensional submanifold of M^n of class $\mathcal{C}^2$, $0 < t \leq r_0$, Σ_t the tubular hypersurface at a distance t from Σ.

Assume that every point at Σ_t has only one nearest point projection onto Σ and fix $x \in \Sigma_t$ arbitrarily. We denote by $\mathbf{p}^\Sigma(x) \in \Sigma$ the nearest point projection of x onto Σ and $\gamma : [0, r_0[\to M^n$ the geodesic

that minimizes the distance between $\mathbf{p}^{\Sigma}(x)$ and x, with the constraint that $\gamma(0) = \mathbf{p}^{\Sigma}(x)$ and $\gamma(t) = x$. By Lemma 13.1.2 we have that

$$u_{\Sigma}(\gamma(s)) = \operatorname{dist}_g^{\Sigma}(\gamma(s)) = s \qquad \text{and} \qquad N(\gamma(l)) = \gamma'(l).$$

for $0 \leq s \leq t$ and $0 < l \leq t$. Consider $f_1(\mathbf{p}^{\Sigma}(x)), \ldots, f_m(\mathbf{p}^{\Sigma}(x))$ the principal curvature vectors of Σ at $\mathbf{p}^{\Sigma}(x)$ in the direction $\gamma'(0)$ and $\kappa_1(\mathbf{p}^{\Sigma}(x)), \ldots, \kappa_m(\mathbf{p}^{\Sigma}(x))$ their respective principal curvatures. Now, we extend these tangent vectors to unit vector fields $F_1(\mathbf{p}^{\Sigma}(x))(s), \ldots, F_m(\mathbf{p}^{\Sigma}(x))(s)$ along γ in such a way that, for each $s \in]0, t]$ and $j = 1, \ldots, m$,

$$S(s)(F_j(\mathbf{p}^{\Sigma}(x))(s)) = \kappa_j(\mathbf{p}^{\Sigma}(x))(s) F_j(\mathbf{p}^{\Sigma}(x))(s),$$

where $S(s)$ is given by Definition 13.1.5. That is, for $j = 1, \ldots, m$, $F_j(\mathbf{p}^{\Sigma}(x))(s)$ is the principal curvature vector of Σ_s at $\gamma(s)$ in the direction of $N(\gamma(s)) = \gamma'(s)$ and $\kappa_j(\mathbf{p}^{\Sigma}(x))(s)$ its respective principal curvature, for all $0 < s \leq t$.

For all $0 < s \leq t$, let $F_{m+2}(\mathbf{p}^{\Sigma}(x))(s), \ldots, F_n(\mathbf{p}^{\Sigma}(x))(s)$ be the remaining principal curvature vectors of Σ_s at $\gamma(s)$ in the direction $N(\gamma(s)) = \gamma'(s)$ with principal curvatures $\kappa_{m+2}(\mathbf{p}^{\Sigma}(x))(s), \ldots, \kappa_n(\mathbf{p}^{\Sigma}(x))(s)$, respectively. That is, for each $s \in]0, t]$ and $j = 1, \ldots, m$,

$$S(s)(F_j(\mathbf{p}^{\Sigma}(x))(s)) = \kappa_j(s) F_j(\mathbf{p}^{\Sigma}(x))(s).$$

Let $F_{m+1}(\mathbf{p}^{\Sigma}(x))(s) := \gamma'(s)$, for all $0 < s \leq t$. Then $s \mapsto \{F_1(\mathbf{p}^{\Sigma}(x))(s), \ldots, F_n(\mathbf{p}^{\Sigma}(x))(s)\}$ is an orthonormal frame field along γ, for all $0 < s \leq t$. We will also denote, for all $0 < s \leq t$, by

$$R^g(s)(X) := R^g(X, N)N$$

for all $X \in T_{\gamma(s)}\Sigma_s$, and

$$\mathrm{II}(s)(X, Y) := \mathrm{II}_{N,x}(X, Y)$$

for all $X, Y \in T_{\gamma(s)}\Sigma_s$.

The following lemma is a standard consequence of [48, Lemma 2.8, p. 23], Generalized Gauss Lemma [48, Lemma 2.11, p. 26], and results from Elliptic Regularity Theory [39, Chapter 5, p. 251]. The main difficulty arises from the fact that Σ is only of class $\mathcal{C}^2$, meaning that u_{Σ} is of class $\mathcal{C}^1$, and $\nabla_g(u_{\Sigma})$ is of class $\mathcal{C}^0$. Consequently, $S(t)$ is also of class $\mathcal{C}^0$, requiring us to consider distributional derivatives. However, by Elliptic Regularity Theory, we can conclude *a posteriori* that the derivative exists in the usual sense.

Lemma 13.1.7. *Let $\Sigma \subset M^n$ be an m-dimensional submanifold of M^n of class $\mathcal{C}^2$, $0 < t \leq r_0$, Σ_t a tubular hypersurface at a distance t, and $x \in \Sigma_t$. Then, for $X, Y \in T_x\Sigma_t$, we have*

1. $\mathrm{Hess}_g(u_{\Sigma})(X, X) = \mathrm{II}(t)(X, X)$.
2. $S'(t)(X) = -S^2(t)(X) - R^g(t)(X)$.
3. $\kappa_j(\mathbf{p}^{\Sigma}(x))'(t) = -\kappa_j(\mathbf{p}^{\Sigma}(x))^2(s) - \sec_g(F_j(t), N)$.

Definition 13.1.8. Let $\kappa, K \in \mathbb{R}$ with $K > 0$. Define, for $t \in \left[0, \frac{1}{\sqrt{K}}\left(\pi - \cot^{-1}\left(\frac{\kappa}{\sqrt{K}}\right)\right)\right[$:

$$\cot_{K,\kappa}(t) := \sqrt{K}\cot\left(\sqrt{K}t + \cot^{-1}\left(\frac{\kappa}{\sqrt{K}}\right)\right).$$

Remark 13.1.9. By straightforward calculation, we can deduce that

$$\begin{cases} \cot'_{K,\kappa}(t) = -\cot^2_{K,\kappa}(t) - K \\ \cot_{K,\kappa}(0) = \kappa, \end{cases} \tag{116}$$

and

$$\lim_{t\uparrow\infty} \cot^{-1}(t) = 0. \tag{117}$$

Moreover, we have

$$\cot_{K,\kappa}(t) = \kappa - \left(\kappa^2 + K\right)t + O\left(t^2\right).$$

Therefore, there exists $t_0 := t_0(K,\kappa) > 0$ such that, for all $0 < t < t_0$, we have

$$\kappa - 2(\kappa^2 + K)t \leq \cot_{K,\kappa}(t) \leq \kappa - \frac{\kappa^2 + K}{2}t \tag{118}$$

On the other hand, in the notation of Definition 12.3.7, we have

$$t\cot_K(t) = 1 - \frac{K}{3}t^2 + O\left(t^4\right).$$

Therefore, there exists $t_0 := t_0(K) > 0$ such that, for all $0 < t < t_0$, we have

$$1 - \frac{2K}{3}t^2 \leq t\cot_K(t) \leq 1 - \frac{K}{6}t^2. \tag{119}$$

Let $\Sigma \subset M^n$ be an m-dimensional submanifold of M^n of class $\mathcal{C}^2$, $0 < t \leq r_0$, Σ_t the tubular hypersurface at a distance t from Σ.

Assume that every point at Σ_t has only one nearest point projection onto Σ and fix $x \in \Sigma_t$ arbitrarily.

- $j = 1, \ldots, m$:

  By the item Item 3 of Lemma 13.1.7, we have that

  $$\begin{cases} \kappa_j(\mathbf{p}^\Sigma(x))'(t) \geq -\kappa_j(\mathbf{p}^\Sigma(x))^2(t) - K, \\ \kappa_j(\mathbf{p}^\Sigma(x))(0) = \kappa_j(\mathbf{p}^\Sigma(x)). \end{cases}$$

  Which implies that $\kappa_j(\mathbf{p}^\Sigma(x))(t)$ is a super solution for $x'(t) = -x^2(t) - K$. By Equation (116), $\cot_{K,\kappa_j(\mathbf{p}^\Sigma(x))}(t)$ is a solution of $x'(t) = -x^2(t) - K$ and

  $$\cot_{K,\kappa_j(\mathbf{p}^\Sigma(x))}(0) = \kappa_j(\mathbf{p}^\Sigma(x)) = \cot_{K,\kappa_j(\mathbf{p}^\Sigma(x))}(0).$$

By [94, Lemma 1.2, p. 24] we have that

$$\kappa_j(\mathbf{p}^\Sigma(x))(t) > \cot_{K,\kappa_j(\mathbf{p}^\Sigma(x))}(t), \tag{120}$$

for $t \in \left]0, \min\left\{\frac{1}{\sqrt{K}}\left(\pi - \cot^{-1}\left(\frac{\kappa_j(\mathbf{p}^\Sigma(x))}{\sqrt{K}}\right)\right), r_0\right\}\right[$.

- $j = m+2, \ldots, n$:

  By Item 3 of Lemma 13.1.7, we have that

$$\begin{cases} \kappa_j(\mathbf{p}^\Sigma(x))'(t) \geq -\kappa_j(\mathbf{p}^\Sigma(x))^2(t) - K \\ \kappa_j(\mathbf{p}^\Sigma(x))(t) \xrightarrow{t\downarrow 0} +\infty. \end{cases}$$

  Let $\varepsilon > 0$ arbitrary and define $\overline{\kappa_j(\mathbf{p}^\Sigma(x))}(t) := \kappa_j(\mathbf{p}^\Sigma(x))(t+\varepsilon)$. Note that

$$\begin{cases} \overline{\kappa_j(\mathbf{p}^\Sigma(x))}'(t) \geq -\overline{\kappa_j(\mathbf{p}^\Sigma(x))}^2(t) - K \\ \overline{\kappa_j(\mathbf{p}^\Sigma(x))}(0) = \kappa_j(\mathbf{p}^\Sigma(x))(\varepsilon). \end{cases}$$

  Which implies that $\overline{\kappa_j(\mathbf{p}^\Sigma(x))}(t)$ is a super solution for $x'(t) = -x^2(t) - K$. By Equation (116), $\cot_{K,\kappa_j(\mathbf{p}^\Sigma(x))(\varepsilon)}(t)$ is a solution of $x'(t) = -x^2(t) - K$ and

$$\cot_{K,\kappa_j(\mathbf{p}^\Sigma(x))(\varepsilon)}(0) = \kappa_j(\mathbf{p}^\Sigma(x))(\varepsilon) = \overline{\kappa_j(\mathbf{p}^\Sigma(x))}(0).$$

  By [94, Lemma 1.2, p. 24] we have that

$$\overline{\kappa_j(\mathbf{p}^\Sigma(x))}(t) > \cot_{K,\kappa_j(\mathbf{p}^\Sigma(x))(\varepsilon)}(t),$$

  for $t \in \left]0, \min\left\{\frac{1}{\sqrt{K}}\left(\pi - \cot^{-1}\left(\frac{\kappa_j(\mathbf{p}^\Sigma(x))(\varepsilon)}{\sqrt{K}}\right)\right), r_0\right\}\right[$. Therefore, by Equation (117) and continuity of cot in $\left]0, \frac{1}{\sqrt{K}}\left(\pi - \cot^{-1}\left(\frac{\kappa_j(\mathbf{p}^\Sigma(x))(\varepsilon)}{\sqrt{K}}\right)\right)\right[$, we have that

$$\kappa_j(\mathbf{p}^\Sigma(x))(t) > \cot_K(t) \tag{121}$$

  for $t \in]0, r_0[$.

For the remainder of this section, we fix $r_H \in \mathbb{R}$ satisfying

$$0 < r_H < r_H(M^n, g), \tag{r_H}$$

where $r_H(M^n, g)$ denotes the harmonic radius of (M^n, g) as defined in [53, Definition 7, p. 579]. Such a choice of r_H is possible because we assume that $|\sec_g| \leq K$, which implies the existence of

$K_1 > 0$ such that $|\mathrm{Ric}_g| \leq K_1$. By [53, Theorem 6, p. 577], it follows that $r_H(M^n, g) > 0$. We now present some fundamental results, using the notation of this paper, that will be repeatedly employed throughout this section. For a detailed presentation, see [55], particularly Lemma 2.8.4 on page 69, which provides estimates on the principal curvatures .

Lemma 13.1.10. *Let $\xi \in M^n$. Then, for each $y \in B_{d_g}(\xi, r_H)$ and $S \in Gr(m, T_yM^n)$, there exists a unique m-dimensional submanifold of M^n of class C^2, denoted by $\Sigma := \Sigma(y, S)$ and refereed as the* **harmonic m-submanifold of M^n at** *(y, S), such that*

1. $y \in \Sigma$,
2. $T_y\Sigma = S$,
3. $\mathrm{cut}(\Sigma) \cap B_{d_g}(\xi, r_H) = \emptyset$, *and*
4. $\kappa_j(\mathbf{p}^\Sigma(x)) = o(1)$ *for all* $j = 1, \ldots, m$ *and all* $x \in B_{d_g}(\xi, r_H)$,

where $\mathbf{p}^\Sigma(x)$ is the nearest point projection of x onto Σ and $\kappa_j\left(\mathbf{p}^\Sigma(x)\right)$ are the principal curvatures of Σ at $\mathbf{p}^\Sigma(x)$.

Lemma 13.1.11. *Let $\xi \in M^n$, $S \in Gr(m, T_\xi M^n)$, $\Sigma \subset M^n$ the harmonic m-submanifold of M^n at (ξ, S). Then, for each $y \in B_{d_g}(\xi, r_H)$, there exists a unique m-dimensional submanifold of M^n of class C^2, denoted by Σ_y and refereed as the* **Fermi transport of Σ from ξ to** *y, such that*

1. $y \in \Sigma_y$,
2. $T_x\Sigma_y = \mathrm{span}\{F_1(\mathbf{p}^\Sigma(x))(t), \ldots, F_m(\mathbf{p}^\Sigma(x))(t)\}$,
3. $\mathrm{dist}_g^{\Sigma(y, T_y\Sigma_y)}(x) = o(1)$, *for all* $x \in \Sigma_y \cap B_{d_g}(\xi, r_H)$, *where* $\Sigma(y, T_y\Sigma_y) \subset M^n$ *is the harmonic m-submanifold of M at* $(y, T_y\Sigma_y)$,
4. $\mathrm{cut}(\Sigma_y) \cap B_{d_g}(\xi, r_H) = \emptyset$,
5. $\kappa_j(\mathbf{p}^{\Sigma_y}(x)) = o(1)$, *for all* $j = 1, \ldots, m$ *and all* $x \in B_{d_g}(\xi, r_H)$.

The following lemmas are standard consequences of the fact that if g is of class C^2, then the distance induced by g is bi-Lipschitz equivalent to the Euclidean distance. Moreover, the fact that the first term of the expansion of the Riemannian metric is the Euclidean metric, and the second term depends on the sectional curvature, which we assume to be bounded. With this assumption, we can then make the estimates below.

Lemma 13.1.12. *Let $\xi \in M^n$, $S \in Gr(m, T_\xi M^n)$, $\Sigma \subset M^n$ the harmonic m-submanifold of M^n at (ξ, S). Then, for each $y \in B_{d_g}(\xi, r_H)$, we have that*

$$\frac{1}{2}\left\|P_{S^\perp}\left(u_y(\xi)\nabla_g(u_y)(\xi)\right)\right\|_{g_\xi}^2 \leq \left(\mathrm{dist}_g^\Sigma(y)\right)^2 \leq 2\left\|P_{S^\perp}\left(u_y(\xi)\nabla_g(u_y)(\xi)\right)\right\|_{g_\xi}^2,$$

where $u_y(\xi) := d_g(y, \xi)$.

Lemma 13.1.13. *Let $\xi \in M^n$, $y_i \in B_{d_g}(\xi, r_H)$, $S_i \in Gr(m, T_{y_i}M^n)$, $\Sigma_i \subset M^n$ the harmonic m-submanifold of M^n at (y_i, S_i), for $i = 1, 2$. Then, for each $x \in B_{d_g}(\xi, r_H)$, we have that*

$$\frac{1}{2}\left\|P_{T_x\Sigma_{1x}} - P_{T_x\Sigma_{2x}}\right\|^2_{\mathrm{HS}_{g_x}} \leq \left\|P_{T_\xi\Sigma_{1\xi}} - P_{T_\xi\Sigma_{2\xi}}\right\|^2_{\mathrm{HS}_{g_\xi}} \leq 2\left\|P_{T_x\Sigma_{1x}} - P_{T_x\Sigma_{2x}}\right\|^2_{\mathrm{HS}_{g_x}}.$$

Lemma 13.1.14. *Let $\xi \in M^n$, $S_i \in Gr(m, T_\xi M^n)$, $\Sigma_i \subset M^n$ the harmonic m-submanifold of M at (ξ, S_i), for $i = 1, 2$. Then, for each $y \in B_{d_g}(\xi, r_H)$, we have that*

$$\begin{aligned}\frac{1}{2}\left\|P_{S_1}\left(\exp_\xi^{-1}(y)\right) - P_{S_2}\left(\exp_\xi^{-1}(y)\right)\right\|_{g_\xi} &\leq d_g\left(\mathbf{p}^{\Sigma_1}(y), \mathbf{p}^{\Sigma_2}(y)\right)\\ &\leq 2\left\|P_{S_1}\left(\exp_\xi^{-1}(y)\right) - P_{S_2}\left(\exp_\xi^{-1}(y)\right)\right\|_{g_\xi},\end{aligned}$$

where $\mathbf{p}^{\Sigma_i}(y)$ is the nearest point projection of y onto Σ_i, for $i = 1, 2$.

Definition 13.1.15. Let $\xi \in M^n$, $S \in \mathrm{Gr}(m, T_\xi M^n)$, $0 < \sigma < r_H$, and $V \in \mathbf{RV}_m(M^n)$. We define the ***tilt-excess of V over the ball $B_{d_g}(\xi, \sigma)$ relative to the m-dimensional space*** S by

$$E(\xi, \sigma, S, V) := \frac{1}{\sigma^m}\int_{B_{d_g}(\xi,\sigma)} \left\|P_{T_xV} - P_{T_x\Sigma(\xi,S)_x}\right\|^2_{\mathrm{HS}_{g_x}} d\|V\|(x),$$

where $\Sigma(\xi, S) \subset M^n$ is the harmonic m-submanifold of M^n at (ξ, S), and $\Sigma(\xi, S)_x$ the Fermi transport of $\Sigma(\xi, S)$ from ξ to x, for all $x \in \mathrm{supp}(\|V\|)$.

In the following Remark, we explain why the point ξ in the tilt-excess definition may not be in the support of $\|V\|$ and why the tilt-excess is interesting (in our approach) only for points in the support of $\|V\|$.

Remark 13.1.16. Note that in the definition of the tilt-excess, ξ is not necessarily in $\mathrm{supp}(\|V\|)$. Since one of our objectives is to find a small σ such that the excess $E(\xi, \sigma, S, V)$ is also small, the only points of interest are those where $\xi \in \mathrm{supp}(\|V\|)$. That is because if $\xi \notin \mathrm{supp}(\|V\|)$, we can simply choose σ_0 small enough to ensure that the ball $B_{d_g}(\xi, \sigma_0)$ does not intersect $\mathrm{supp}(\|V\|)$. In this case, it is immediate that $E(\xi, \sigma_0, S, V) = 0$.

13.2 CACCIOPPOLI INEQUALITY

Note that because of the adaptations we had to make in the definition of the tilt-excess, we also obtained a different Caccioppoli inequality, which depends on the principal curvatures of the harmonic submanifold that we considered. Additionally, note that in the Euclidean sense, the harmonic submanifold that we can take is the linear subspace, which has all zero principal curvatures. Thus, our version of the Caccioppoli inequality extends the Euclidean one shown in [91, Lemma 2.5, Chapter 5, Section 2, p. 131].

Lemma 13.2.1 (Intrinsic Riemannian Caccioppoli Inequality)**.** *Let $m < p < \infty$, $0 < \delta < 1$, $V \in \mathbf{RV}_m(M^n)$, and $\xi \in \operatorname{supp}(\|V\|)$. There exists $\rho_0 := \rho_0(m, K, r_H) > 0$ such that if V has L^p_{loc}-bounded generalized mean curvature vector in $B_{d_g}(\xi, \rho)$, for some $\rho < \rho_0$, then, for all $y \in B_{d_g}(\xi, \delta\rho)$, for all $S \in Gr(m, T_yM^n)$, for all $\sigma \in]0, (1-\delta)\rho]$, and for all $\alpha \in]0, 1[$, we have that*

$$
\begin{aligned}
E(y, \alpha\sigma, S, V) \leq \frac{\mathbf{C}}{\sigma^m} \int_{B_{d_g}(y,\sigma)} \left(\frac{\operatorname{dist}^{\Sigma}_g(x)}{\sigma} \right)^2 d\|V\|(x) + \frac{2\alpha^{-m}}{\sigma^{m-2}} \int_{B_{d_g}(y,\sigma)} \left\| \overrightarrow{H_g}(x) \right\|^2_{g_x} d\|V\|(x) \\
+ \frac{4m\rho\alpha^{-m}}{\sigma^m} \int_{B_{d_g}(y,\sigma)} \frac{\operatorname{dist}^{\Sigma}_g(x)}{\sigma} \sum_{j=1}^{m} \left| \kappa_j(\mathbf{p}^{\Sigma}(x)) \right| d\|V\|(x) \\
+ \frac{8m\rho^2\alpha^{-m}}{\sigma^m} \int_{B_{d_g}(y,\sigma)} \left(\frac{\operatorname{dist}^{\Sigma}_g(x)}{\sigma} \right)^2 \sum_{j=1}^{m} \kappa_j(\mathbf{p}^{\Sigma}(x))^2 d\|V\|(x). \qquad (122)
\end{aligned}
$$

where $\Sigma \subset M$ is the harmonic m-submanifold of M^n at (y, S) and $\mathbf{C} = \mathbf{C}(m, n, \alpha, K, \rho) > 0$.

Remark 13.2.2. The constant obtained in Lemma 13.2.1 is given explicitly by

$$
\mathbf{C} = \mathbf{C}(m, n, \alpha, K, \rho) = \alpha^{-m} \left(2 + \frac{1}{(1-\alpha)^2} + 8K \left(\frac{m(n-m-1)}{3} + m^2 \right) \rho^2 \right).
$$

Observe that the part of the constant which does not depend on the sectional curvature bound K, namely

$$
\alpha^{-m} \left(2 + \frac{1}{(1-\alpha)^2} \right),
$$

is exactly the Euclidean constant. In particular, if the ambient manifold has zero sectional curvature, then our formula recovers the Euclidean one. Indeed, in this case the K-dependent term vanishes, and moreover the last two lines in Equation (122) are also equal to zero, since the principal curvatures $\kappa_j(\mathbf{p}^{\Sigma}(x))$ vanish for totally geodesic submanifolds. This is precisely the situation in the Euclidean ambient space, where Σ is an affine plane.

Proof of Lemma 13.2.1. Let us fix $\rho_0 > 0$ arbitrarily. Suppose that V has L^p_{loc}-bounded generalized mean curvature vector in $B_{d_g}(\xi, \rho)$, for some $0 < \rho < \rho_0$. Consider $y \in B_{d_g}(\xi, \delta\rho)$, $S \in Gr(m, T_yM^n)$, $\sigma \in]0, (1-\delta)\rho]$, and $\alpha \in]0, 1[$ arbitrary.

Let $x \in \operatorname{supp}(\|V\|) \cap B_{d_g}(y, \sigma)$ fixed arbitrarily. Note that

$$
\frac{1}{2} \|P_{T_xV} - P_{T_x\Sigma_x}\|^2_{\text{HS}_{g_x}} = m - \operatorname{tr}_{g_x}(P_{T_xV} \circ P_{T_x\Sigma_x}). \qquad (123)
$$

If $x \in \Sigma$, as $V \in \mathbf{RV}_m(M^n)$, there exists a countable m-rectifiable set $\Gamma \subset M^n$ and a Borel map $f : \Gamma \to]0, +\infty[$ such that, for all $B \in \mathcal{B}(\text{Gr}_m(M^n))$, we have that

$$
V(B) = V(\Gamma, f, g)(B) = \int_{\{x \in \Gamma : (x, T_x\Gamma) \in B\}} f(x) d\mathcal{H}^m_{d_g}(x). \qquad (124)
$$

Thus, by Equations (123) and (124), we have that

$$\int_{B_{d_g}(y,\sigma)\cap\Sigma} \|P_{T_xV} - P_{T_x\Sigma_x}\|^2_{\mathrm{HS}_{g_x}} d\|V\|(x) = 0. \tag{125}$$

Now, if $x \notin \Sigma$, let $\left(\mathrm{Tube}(\Sigma, r_H), \{x_i\}_{i=1}^n\right)$ be a system of Fermi coordinates for Σ at y. We define

$$u_\Sigma^2 := \sum_{i=m+1}^n x_i^2,$$

and, for $u_\Sigma > 0$,

$$N := \sum_{i=m+1}^n \frac{x_i}{u_\Sigma}\frac{\partial}{\partial x_i}.$$

By Generalized Gauss Lemma [48, Lemma 2.11, p. 26], $N = \nabla_g(u_\Sigma)$ on $\mathrm{Tube}(\Sigma, r_H) \setminus \Sigma$.

Let $\zeta \in \mathcal{C}^1_c(\mathrm{Tube}(\Sigma, r_H))$ be non-negative, $X := \zeta^2 u_\Sigma \nabla_g(u_\Sigma)$, $\{e_1, \ldots, e_m\}$ an orthonormal basis for T_xV, $t_x := \mathrm{dist}_g^\Sigma(x)$, and Σ_{t_x} the tubular hypersurface at a distance t_x around Σ. Since $T_x\Sigma_x \subset T_x\Sigma_{t_x}$, we have that $\nabla_g(u_\Sigma) \perp v$, for all $v \in T_x\Sigma_x$. Thus

$$\mathrm{div}_{T_xV}(X) = 2\zeta u_\Sigma \left\langle \nabla_g(u_\Sigma), (P_{T_xV} - P_{T_x\Sigma_x})\left(\nabla_g(\zeta)\right)\right\rangle_g + \zeta^2 \mathrm{div}_{T_xV}\left(u_\Sigma \nabla_g(u_\Sigma)\right). \tag{126}$$

Since we are assuming that V has L^p_{loc}-bounded generalized mean curvature vector in $B_{d_g}(\xi,\rho)$, by Proposition 12.2.8, we have that

$$\int_{B_{d_g}(y,\sigma)} \mathrm{div}_{T_xV}(X)(x) d\|V\|(x) = -\int_{B_{d_g}(y,\sigma)} \left\langle \overrightarrow{H_g}(x), X(x)\right\rangle_{g_x} d\|V\|(x).$$

By Equation (126), we have that

$$\begin{aligned}\int_{B_{d_g}(y,\sigma)} \zeta^2(x)\frac{1}{2}\|P_{T_xV} - P_{T_x\Sigma_x}\|^2_{\mathrm{HS}_{g_x}} d\|V\|(x) &= -\int_{B_{d_g}(y,\sigma)} \left\langle \overrightarrow{H_g}(x), \zeta^2(x)u_\Sigma(x)\nabla_g(u_\Sigma)(x)\right\rangle_{g_x} d\|V\|(x)\\ &- \int_{B_{d_g}(y,\sigma)} 2\zeta(x)u_\Sigma(x)\left\langle \nabla_g(u_\Sigma)(x), (P_{T_xV} - P_{T_x\Sigma_x})\left(\nabla_g(\zeta)(x)\right)\right\rangle_{g_x} d\|V\|(x)\\ &+ \int_{B_{d_g}(y,\sigma)} \zeta^2(x)\left(\frac{1}{2}\|P_{T_xV} - P_{T_x\Sigma_x}\|^2_{\mathrm{HS}_{g_x}} - \mathrm{div}_{T_xV}\left(u_\Sigma\nabla_g(u_\Sigma)\right)(x)\right) d\|V\|(x)\end{aligned} \tag{127}$$

Next, we will proceed to estimate the quantity $\frac{1}{2}\|P_{T_xV} - P_{T_x\Sigma_x}\|^2_{\mathrm{HS}_g} - \mathrm{div}_{T_xV}\left(u_\Sigma\nabla_g(u_\Sigma)\right)$. Firstly, we note that

$$\mathrm{div}_{T_xV}\left(u_\Sigma\nabla_g(u_\Sigma)\right) = \sum_{i=1}^m \left|\left\langle \nabla_g(u_\Sigma), e_i\right\rangle_g\right|^2 + u_\Sigma \sum_{i=1}^m \left\langle \nabla^g_{e_i}\left(\nabla_g(u_\Sigma)\right), e_i\right\rangle_g. \tag{128}$$

Let $\gamma : [0,\sigma] \to M^n$ be the minimizing geodesic between x and $\mathbf{p}^\Sigma(x)$ such that $\gamma(0) = \mathbf{p}^\Sigma(x)$ and $\gamma(t_x) = x$, and $t \mapsto \{F_1(\mathbf{p}^\Sigma(x))(t), \ldots, F_n(\mathbf{p}^\Sigma(x))(t)\}$ the orthonormal frame field along γ of principal curvature vectors. Thus

$$e_i = \sum_{j=1}^n \left\langle e_i, F_j(\mathbf{p}^\Sigma(x))(t_x)\right\rangle_g F_j(\mathbf{p}^\Sigma(x))(t_x).$$

By Equation (128), we have that

$$\operatorname{div}_{T_xV}\left(u_\Sigma \nabla_g(u_\Sigma)\right) = \sum_{i=1}^{m}\left|\left\langle F_{m+1}(\mathbf{p}^\Sigma(x))(t_x), e_i\right\rangle_g\right|^2 + u_\Sigma \sum_{i=1}^{m}\sum_{\substack{j=1\\ j\neq m+1}}^{n}\left|\left\langle e_i, F_j(\mathbf{p}^\Sigma(x))(t_x)\right\rangle_g\right|^2 \kappa_j(\mathbf{p}^\Sigma(x))(t_x). \tag{129}$$

By Lemma 13.1.11, we note that $\left\{F_1(\mathbf{p}^\Sigma(x))(t_x), \ldots, F_n(\mathbf{p}^\Sigma(x))(t_x)\right\}$ is an orthonormal basis for T_xM such that $\left\{F_1(\mathbf{p}^\Sigma(x))(t_x), \ldots, F_m(\mathbf{p}^\Sigma(x))(t_x)\right\}$ is an orthonormal basis for $T_x\Sigma_x$. By using the basis $\left\{F_1(\mathbf{p}^\Sigma(x))(t_x), \ldots, F_n(\mathbf{p}^\Sigma(x))(t_x)\right\}$ in Equation (123), we obtain

$$\frac{1}{2}\left\|P_{T_xV} - P_{T_x\Sigma_x}\right\|^2_{\mathrm{HS}_{g_x}} = \sum_{i=1}^{m}\sum_{j=m+1}^{n}\left|\left\langle e_i, F_j(\mathbf{p}^\Sigma(x))(t_x)\right\rangle_{g_x}\right|^2. \tag{130}$$

Note that $x \in B_{d_g}(y,\sigma)$, then $\mathbf{p}^\Sigma(x) \in B_{d_g}(y,\sigma)$. Therefore, since $0 < \delta < 1$ we have that $0 < (1-\delta) < 1$, thus

$$u_\Sigma(x) = d_g\left(x, \mathbf{p}^\Sigma(x)\right) \leq d_g(x,y) + d_g\left(y, \mathbf{p}^\Sigma(x)\right) < 2\sigma < 2(1-\delta)\rho < 2\rho. \tag{131}$$

Then there exists $\rho_0 := \rho_0(r_H) > 0$ such that $u_\Sigma(x) < r_H$. By Equations (121), (129) and (130) we have that

$$\begin{aligned}\frac{1}{2}\left\|P_{T_xV} - P_{T_x\Sigma_x}\right\|^2_{\mathrm{HS}_g} - \operatorname{div}_{T_xV}\left(u_\Sigma\nabla_g(u_\Sigma)\right) \leq{}& (1 - u_\Sigma \cot_K(u_\Sigma))\sum_{i=1}^{m}\sum_{j=m+1}^{n}\left|\left\langle e_i, F_j(\mathbf{p}^\Sigma(x))(t_x)\right\rangle_g\right|^2 \\ &- u_\Sigma\sum_{i=1}^{m}\sum_{j=1}^{m}\left|\left\langle e_i, F_j(\mathbf{p}^\Sigma(x))(t_x)\right\rangle_g\right|^2 \kappa_j(\mathbf{p}^\Sigma(x))(u_\Sigma(x)).\end{aligned} \tag{132}$$

By Lemma 13.1.10 we have that $\kappa_j(\mathbf{p}^\Sigma(x)) = o(1)$ for all $j = 1, \ldots, m$ and for all $x \in B_{d_g}(\xi, r_H)$, thus, by Equation (131) there exists $\rho_0 := \rho_0(K, m, r_H) > 0$ such that

$$u_\Sigma(x) < \min\left\{\min_{j=1,\ldots,m}\frac{1}{\sqrt{K}}\left(\pi - \cot^{-1}\left(\frac{\kappa_j(\mathbf{p}^\Sigma(x))}{\sqrt{K}}\right)\right), r_H\right\} \tag{133}$$

Thus, by Equations (118), (120) and (133), there exists $\rho_0 := \rho_0(K, m, r_H) > 0$ such that

$$\begin{aligned}\kappa_j(\mathbf{p}^\Sigma(x))(u_\Sigma(x)) &\geq \cot_{K,\kappa_j(\mathbf{p}^\Sigma(x))}(u_\Sigma(x)) \\ &\geq \kappa_j(\mathbf{p}^\Sigma(x)) - 2\left(\kappa_j(\mathbf{p}^\Sigma(x))^2 + K\right)u_\Sigma(x),\end{aligned} \tag{134}$$

for all $j = 1, \ldots, m$. On the other hand, by Equations (119) and (133) there exists $\rho_0 := \rho_0(K, m, r_H) > 0$ such that

$$1 - u_\Sigma(x)\cot_K(u_\Sigma(x)) \leq \frac{2K}{3}u_\Sigma(x)^2. \tag{135}$$

Thus, by Equations (132), (134) and (135), we have that

$$\frac{1}{2}\left\|P_{T_xV} - P_{T_x\Sigma_x}\right\|^2_{\mathrm{HS}_g} - \operatorname{div}_{T_xV}\left(u_\Sigma\nabla_g(u_\Sigma)\right) \leq \mathbf{C_1}u_\Sigma^2 + mu_\Sigma\sum_{j=1}^{m}\left|\kappa_j(\mathbf{p}^\Sigma(x))\right| + 2mu_\Sigma^2\sum_{j=1}^{m}\kappa_j(\mathbf{p}^\Sigma(x))^2. \tag{136}$$

where $\mathbf{C_1} = \mathbf{C_1}(m, n, K) > 0$. By Equations (127) and (136), Cauchy-Schwarz inequality, and the fact that $\left\|\nabla_g(u_\Sigma)(x)\right\|_{g_x} = 1$ for all $x \in \mathrm{Tube}(\Sigma, r_H) \setminus \Sigma$, we have

$$\begin{aligned}\int_{B_{d_g}(y,\sigma)} \zeta^2(x)\frac{1}{2}\left\|P_{T_xV} - P_{T_x\Sigma_x}\right\|^2_{\mathrm{HS}_{g_x}} d\|V\|(x) \leq & \int_{B_{d_g}(y,\sigma)} \zeta^2(x)u_\Sigma(x)\left\|\overrightarrow{H_g}(x)\right\|_{g_x} d\|V\|(x)\\ &+ \int_{B_{d_g}(\xi,\rho)} 2\zeta(x)u_\Sigma(x)\left\|\left(P_{T_xV} - P_{T_x\Sigma_x}\right)\left(\nabla_g\left(\zeta\right)(x)\right)\right\|_{g_x} d\|V\|(x)\\ &+ \int_{B_{d_g}(y,\sigma)} \zeta^2(x)\mathbf{C_1}u^2_\Sigma(x) d\|V\|(x)\\ &+ \int_{B_{d_g}(y,\sigma)} \zeta^2(x)mu_\Sigma(x)\sum_{j=1}^m\left|\kappa_j(\mathbf{p}^\Sigma(x))\right| d\|V\|(x)\\ &+ \int_{B_{d_g}(y,\sigma)} \zeta^2(x)2mu^2_\Sigma(x)\sum_{j=1}^m\kappa_j(\mathbf{p}^\Sigma(x))^2 d\|V\|(x). \end{aligned} \tag{137}$$

By a standard calculation using Hölder's inequality, we obtain $\left\|\left(P_{T_xV} - P_{T_x\Sigma_x}\right)\left(\nabla_g\left(\zeta\right)(x)\right)\right\|_{g_x} \leq \left\|P_{T_xV} - P_{T_x\Sigma_x}\right\|_{\mathrm{HS}_{g_x}} \left\|\nabla_g\left(\zeta\right)(x)\right\|_{g_x}$ which, together with Equation (137) and some computations, implies that

$$\begin{aligned}\int_{B_{d_g}(y,\sigma)} \zeta^2(x)\left\|P_{T_xV} - P_{T_x\Sigma_x}\right\|^2_{\mathrm{HS}_{g_x}} d\|V\|(x) \leq & \int_{B_{d_g}(y,\sigma)} 2\zeta^2(x)\frac{u^2_\Sigma(x)}{\sigma^2} d\|V\|(x) + \int_{B_{d_g}(y,\sigma)} 2\zeta^2(x)\sigma^2\left\|\overrightarrow{H_g}(x)\right\|^2_{g_x} d\|V\|(x)\\ &+ \int_{B_{d_g}(y,\sigma)} 16u^2_\Sigma(x)\left\|\nabla_g\left(\zeta\right)(x)\right\|^2_{g_x} d\|V\|(x) + \int_{B_{d_g}(y,\sigma)} 4\zeta^2(x)\mathbf{C_1}u^2_\Sigma(x) d\|V\|(x)\\ &+ \int_{B_{d_g}(y,\sigma)} 4\zeta^2(x)mu_\Sigma(x)\sum_{j=1}^m\left|\kappa_j(\mathbf{p}^\Sigma(x))\right| d\|V\|(x)\\ &+ \int_{B_{d_g}(y,\sigma)} 4\zeta^2(x)2mu^2_\Sigma(x)\sum_{j=1}^m\kappa_j(\mathbf{p}^\Sigma(x))^2 d\|V\|(x). \end{aligned} \tag{138}$$

Let $\zeta \equiv 1$ on $B_{d_g}(y, \alpha\sigma)$, $\zeta \equiv 0$ outside of $B_{d_g}(y, \sigma)$ and $\left\|\nabla_g\left(\zeta\right)(x)\right\|_g \leq \frac{1}{\sqrt{16(1-\alpha)}\sigma}$ on $B_{d_g}(y, \sigma)$. Since $\sigma < (1-\delta)\rho < \rho$, we have that $1 < \frac{\rho}{\sigma}$ and $1 < \frac{\rho^2}{\sigma^2}$, which implies that $u_\Sigma(x) < \rho\frac{u_\Sigma(x)}{\sigma}$ and $u^2_\Sigma(x) < \rho^2\left(\frac{u_\Sigma(x)}{\sigma}\right)^2$. Moreover, since $\alpha \in]0,1[$ we have that $B_{d_g}(y, \alpha\sigma) \subset B_{d_g}(y, \sigma)$. Thus, also by Equation (138), we obtain

$$\begin{aligned}\frac{1}{(\alpha\sigma)^m}\int_{B_{d_g}(y,\alpha\sigma)} \left\|P_{T_xV} - P_{T_x\Sigma_x}\right\|^2_{\mathrm{HS}_{g_x}} d\|V\|(x) \leq & \frac{\alpha^{-m}\left(2 + \frac{1}{(1-\alpha)^2} + 4\mathbf{C_1}\rho^2\right)}{\sigma^m}\int_{B_{d_g}(y,\sigma)}\left(\frac{u_\Sigma(x)}{\sigma}\right)^2 d\|V\|(x)\\ &+ \frac{2\alpha^{-m}}{\sigma^{m-2}}\int_{B_{d_g}(y,\sigma)}\left\|\overrightarrow{H_g}(x)\right\|^2_{g_x} d\|V\|(x)\\ &+ \frac{4m\rho\alpha^{-m}}{\sigma^m}\int_{B_{d_g}(y,\sigma)}\frac{u_\Sigma(x)}{\sigma}\sum_{j=1}^m\left|\kappa_j(\mathbf{p}^\Sigma(x))\right| d\|V\|(x)\\ &+ \frac{8m\rho^2\alpha^{-m}}{\sigma^m}\int_{B_{d_g}(y,\sigma)}\left(\frac{u_\Sigma(x)}{\sigma}\right)^2\sum_{j=1}^m\kappa_j(\mathbf{p}^\Sigma(x))^2 d\|V\|(x). \end{aligned}$$

To conclude the proof, we choose $\rho_0 := \rho_0(m, K, r_H) > 0$ as the minimum value ensuring that all the above estimates hold. Then, Equation (122) holds. □

13.3 DENSITY ESTIMATES

Definition 13.3.1. Let $m < p < \infty$; in case $m = 1$, we require that $p \geq 2$, $\xi \in M^n$, $0 < \delta < 1$, $0 < \rho < r_H$, and $V \in \mathbf{V}_m(M^n)$. We say that V satisfies the regularity condition in $B_{d_g}(\xi, \rho)$ with constants δ and p when $\xi \in \operatorname{supp}(\|V\|)$,

$$\Theta^m(\|V\|, x) \geq 1, \tag{139}$$

for $\|V\|$-a.e. $x \in B_{d_g}(\xi, \rho)$,

$$\frac{\|V\|(B_{d_g}(\xi, \rho))}{\omega_m \rho^m} \leq 1 + \delta. \tag{140}$$

and

$$\left|\delta_g V(X)\right| \leq \frac{\delta}{\rho^{1-\frac{m}{p}}} \|X\|_{L^{\frac{p}{p-1}}(M^n, \|V\|)}, \tag{141}$$

for all $X \in \mathfrak{X}_c^0(M^n)$ such that $\operatorname{supp}(X) \subset B_{d_g}(\xi, \rho)$. Furthermore, $\mathcal{AC}(\xi, \rho, \delta, p)$ denotes the set al all varifolds that satisfies the regularity condition in $B_{d_g}(\xi, \rho)$ with constants δ and p.

The conditions presented in Theorem 13.6.8 also depend on a parameter d, as given by equations Equations (293) to (295). However, since we can always modify the density of the varifold without changing its support, we will restrict ourselves to varifolds in $\mathcal{AC}(\xi, \rho, \delta, p)$, that is, when $d = 1$.

Remark 13.3.2. Let $m < p < \infty$, $\xi \in M^n$, $0 < \delta < 1$, and $0 < \rho < r_H$.

If $V \in \mathcal{AC}(\xi, \rho, \delta, p)$ then, for every open subset $W \subset\subset B_{d_g}(\xi, \rho)$, there exists a real constant $C = C(\delta, \rho, m) := \frac{\delta}{\rho^{1-\frac{m}{p}}} > 0$ such that

$$\left|\delta_g V(X)\right| \leq C \|X\|_{L^{\frac{p}{p-1}}(W, \|V\|)}$$

holds for every $X \in \mathfrak{X}_c^0(W)$.

Therefore, by Proposition 12.2.8, we have that the total variation of $\delta_g V$ is a Radon measure. Furthermore, there exists a $\|V\|$-measurable function $\overrightarrow{H_g} : B_{d_g}(\xi, \rho) \to TB_{d_g}(\xi, \rho)$ such that

$$\delta_g V(X) = -\int_{B_{d_g}(\xi,\rho)} \left\langle \overrightarrow{H_g}, X \right\rangle_g d\|V\|$$

and $\|\overrightarrow{H_g}\|_{L^p(W, \|V\|)} \leq C$, for every open subset $W \subset\subset B_{d_g}(\xi, \rho)$ and $X \in \mathfrak{X}_c^0(W)$.

Remark 13.3.3. Let $m < p < \infty$, $\xi \in M^n$, $0 < \delta < 1$, and $0 < \rho < r_H$.

If $V \in \mathcal{AC}(\xi, \rho, \delta, p)$ then $V \in \mathbf{RV}_m\left(B_{d_g}(\xi, \delta\rho)\right)$. In fact, let $W \subset\subset B_{d_g}(\xi, \rho)$ and $X \in \mathfrak{X}_c^0(W)$

$$|\delta_g V(X)| \leq \frac{\delta}{\rho^{1-\frac{m}{p}}} (\omega_m \rho^m)^{\frac{p}{p-1}} (1 + \delta)^{\frac{p}{p-1}} \|X\|_{L^\infty(W, \|V\|)}.$$

Which implies that V has locally bounded first variation in $B_{d_g}(\xi, \rho)$. Therefore, by Theorem 12.4.1, we conclude that $V \in \mathbf{RV}_m\left(B_{d_g}(\xi, \delta\rho)\right)$.

Lemma 13.3.4. *Let $m < p < \infty$ and $\xi \in M^n$. There exist $\delta_0 := \delta_0(m, p, K, r_H) > 0$ and $\rho_0 := \rho_0(m, p, K, r_H) > 0$ such that if $V \in \mathcal{AC}(\xi, \rho, \delta, p)$, for some $0 < \delta < \delta_0$ and for some $0 < \rho < \rho_0$, then, for all $y \in \operatorname{supp}(\|V\|) \cap B_{d_g}(\xi, 2\delta\rho)$ and for all $\sigma \in]0, (1-2\delta)\rho]$, we have that*

$$\frac{1}{2} \leq 1 - \mathbf{C_1}\delta - \mathbf{C_2}\rho^2 \leq \frac{\|V\|\left(B_{d_g}(y,\sigma)\right)}{\omega_m \sigma^m} \leq 1 + \mathbf{C_1}\delta + \mathbf{C_2}\rho^2 \leq 2, \tag{142}$$

where $\mathbf{C_1} = \mathbf{C_1}(m, p) > 0$ and $\mathbf{C_2} = \mathbf{C_2}(K, m) > 0$. Furthermore, if $\rho < \sqrt{\delta}$ then

$$1 - \mathbf{C_3}\delta \leq \frac{\|V\|\left(B_{d_g}(y,\sigma)\right)}{\omega_m \sigma^m} \leq 1 + \mathbf{C_3}\delta, \tag{143}$$

where $\mathbf{C_3} = \mathbf{C_3}(m, p, K) > 0$.

Proof. Let us fix $\delta_0 > 0$ and $\rho_0 > 0$ arbitrarily. Suppose that $V \in \mathcal{AC}(\xi, \rho, \delta, p)$, for some $0 < \delta < \delta_0$ and for some $0 < \rho < \rho_0$. Consider $y \in \operatorname{supp}(\|V\|) \cap B_{d_g}(\xi, 2\delta\rho)$ and $\sigma \in]0, (1-2\delta)\rho]$ arbitrary.

By Remark 13.3.2, we have, for all $t \in]0, (1-2\delta)\rho]$, that

$$\left\| \overrightarrow{H_g} \right\|_{L^p\left(B_{d_g}(y,t), \|V\|\right)} \leq \left(1 - \frac{m}{p}\right) G,$$

where

$$G := \frac{\delta}{\left(1 - \frac{m}{p}\right) \rho^{1 - \frac{m}{p}}}.$$

By Hölder inequality, for all $t \in]0, (1-2\delta)\rho]$, we have that

$$\left\| \overrightarrow{H_g} \right\|_{L^1\left(B_{d_g}(y,t), \|V\|\right)} \leq \left(1 - \frac{m}{p}\right) G \left(\|V\|\left(B_{d_g}(y,t)\right)\right)^{-\frac{1}{p}} \left(\|V\|\left(B_{d_g}(y,t)\right)\right). \tag{144}$$

Also note that by Remark 13.3.2 we have, in particular, that V has L^p_{loc}-bounded generalized mean curvature vector in $B_{d_g}(\xi, \rho)$. Let $0 < \delta_0 < \frac{1}{2}$, thus, by Theorem 12.3.13, there exists $\rho_0 := \rho_0(r_H, K) > 0$ such that, for all $0 < \gamma < t < (1-2\delta)\rho$, we have that

$$\left(\frac{\|V\|(B_{d_g}(y,t))}{t^m}\right)^{\frac{1}{p}} \geq (\omega_m)^{\frac{1}{p}} - \mathbf{C_4} t^{1 - \frac{m}{p}}, \tag{145}$$

where $\mathbf{C_4} = \mathbf{C_4}(\delta, m, p, \rho, K) > 0$. By Equations (119) and (145) and Taylor expansion of $(1+\delta)^{\frac{1}{p}}$ in δ around 0 we have that there exist $\delta_0 := \delta_0(m, K, r_H, p) > 0$ and $\rho_0 := \rho_0(m, K, r_H, p) > 0$ such that

$$\left(\|V\|(B_{d_g}(y,t))\right)^{-\frac{1}{p}} \leq \left(\frac{2}{(\omega_m)^{\frac{1}{p}}}\right) t^{-\frac{m}{p}}. \tag{146}$$

Finally, by Equations (144) and (146), we have that

$$\left\|\overrightarrow{H_g}\right\|_{L^1\left(B_{d_g}(y,t),\|V\|\right)} \leq \alpha\Lambda\left(\frac{t}{(1-\delta)\rho}\right)^{\alpha-1}\|V\|\left(B_{d_g}(y,t)\right), \tag{147}$$

where

$$\begin{cases}\Lambda := \dfrac{2\delta}{\left(1-\frac{m}{p}\right)\rho^{1-\frac{m}{p}}\left(\omega_m\right)^{\frac{1}{p}}\left((1-2\delta)\rho\right)^{\frac{m}{p}}},\\ \alpha := 1-\dfrac{m}{p}.\end{cases}$$

By Remark 13.3.2 we have, in particular, that V has L^p_{loc}-bounded generalized mean curvature vector in $B_{d_g}(\xi,\rho)$ which with Equation (147) permits us to apply the Theorem 12.3.16. Therefore, there exists $\rho_0 := \rho_0(r_H, K) > 0$, such that the function

$$\begin{aligned} f :]0,(1-2\delta)\rho] &\to [0,+\infty[\\ t &\mapsto e^{\lambda(t)}\frac{\|V\|\left(B_{d_g}(y,t)\right)}{t^m}\end{aligned}$$

is non-decreasing, where

$$\begin{aligned}\lambda :]0,(1-2\delta)\rho] &\to]0,+\infty[\\ t &\mapsto \left(\frac{2\delta}{\left(1-\frac{m}{p}\right)\rho^{1-\frac{m}{p}}\left(\omega_m\right)^{\frac{1}{p}}\left((1-2\delta)\rho\right)^{\frac{m}{p}}}\left(\frac{t}{(1-2\delta)\rho}\right)^{-\frac{m}{p}} - \frac{c((1-2\delta)\rho,K)-1}{(1-2\delta)\rho}m\right)t.\end{aligned}$$

Since f is non-decreasing on $]0,(1-2\delta)\rho]$ we have that

$$\frac{\|V\|\left(B_{d_g}(y,\sigma)\right)}{\omega_m\sigma^m} \leq e^{\lambda((1-2\delta)\rho)-\lambda(\sigma)}\frac{1+\delta}{(1-2\delta)^m}. \tag{148}$$

Since $\lambda(\sigma) > 0$, Equation (119) and Taylor expansion of $\delta(1-2\delta)^{1-\frac{m}{p}}$ in δ around 0 we have that there exist $\rho_0 := \rho_0(K) > 0$ and $\delta_0 := \delta_0(m,p) > 0$ such that

$$\lambda\left((1-2\delta)\rho\right) - \lambda\left(\sigma\right) \leq \mathbf{C_5}\delta + \frac{2K}{3}m\rho^2, \tag{149}$$

where $\mathbf{C_5} = \mathbf{C_5}(m,p) > 0$. Therefore, by Equations (148) and (149) and Taylor expansion of $e^{\mathbf{C_5}\delta}\frac{1+\delta}{(1-2\delta)^m}$ in δ around 0 we have that there exists $\delta_0 := \delta_0(m,p) > 0$ such that

$$\frac{\|V\|\left(B_{d_g}(y,\sigma)\right)}{\omega_m\sigma^m} \leq e^{\frac{2K}{3}m\rho^2}\left(1+2\left(\frac{4}{\left(1-\frac{m}{p}\right)\left(\omega_m\right)^{\frac{1}{p}}}+2m+1\right)\delta\right). \tag{150}$$

Now, let $\sigma_1 < \sigma$ and note that

$$\frac{1}{e^{\lambda(\sigma)}} \leq \frac{\|V\|\left(B_{d_g}(y,\sigma)\right)}{\omega_m \sigma^m}. \tag{151}$$

Since $\sigma \leq (1-2\delta)\rho$ we have that

$$\lambda(\sigma) \leq \frac{2}{\left(1-\frac{m}{p}\right)(\omega_m)^{\frac{1}{p}}}\delta(1-2\delta)^{1-\frac{m}{p}} - ((1-2\delta)\rho\cot_K((1-2\delta)\rho)-1)\,m. \tag{152}$$

Since $\lambda(\sigma) > 0$, Equations (119) and (152) and Taylor expansion of $\delta(1-2\delta)^{1-\frac{m}{p}}$ we have that there exist $\rho_0 := \rho_0(K) > 0$ and $\delta_0 := \delta_0(m,p) > 0$ such that

$$\frac{1}{e^{\lambda(\sigma)}} \geq \frac{1}{e^{\mathbf{C_5}\delta + \frac{2K}{3}m\rho^2}}. \tag{153}$$

Therefore, by Equations (150), (151) and (153) and Taylor expansion of $e^{-\mathbf{C_5}\delta}$ in δ at 0, we obtain that there exists $\delta_0 := \delta_0(m,p) > 0$ such that

$$e^{-\frac{2K}{3}m\rho^2}(1-\mathbf{C_1}\delta) \leq \frac{\|V\|\left(B_{d_g}(y,\sigma)\right)}{\omega_m\sigma^m} \leq e^{\frac{2K}{3}m\rho^2}(1+\mathbf{C_1}\delta), \tag{154}$$

By Equation (154), and Taylor expansion of $e^{\frac{2K}{3}m\rho^2}$ and $e^{-\frac{2K}{3}m\rho^2}$ in ρ around 0 we have that there exist $\delta_0 := \delta_0(m,p) > 0$ and $\rho_0 := \rho_0(K,m) > 0$ such that

$$1-\mathbf{C_1}\delta-\mathbf{C_2}\rho^2 \leq \frac{\|V\|\left(B_{d_g}(y,\sigma)\right)}{\omega_m\sigma^m} \leq 1+\mathbf{C_1}\delta+\mathbf{C_2}\rho^2. \tag{155}$$

To conclude the proof, we choose $\delta_0 := \delta_0(m,K,r_H,p) > 0$ and $\rho_0 := \rho_0(m,K,r_H,p) > 0$ as the minimum value ensuring that all the above estimates hold. Then, we have that Equation (142) holds. Now, supposing that $\rho < \sqrt{\delta}$. By Equation (155) we conclude that Equation (143) holds. □

Corollary 13.3.5. *Let $m < p < \infty$ and $\xi \in M^n$. There exist $\delta_0 := \delta_0(m,p,K,r_H) > 0$ and $\rho_0 := \rho_0(m,p,K,r_H) > 0$ such that if $V \in \mathcal{AC}(\xi,\rho,\delta,p)$, for some $0 < \delta < \delta_0$ and for some $0 < \rho < \rho_0$, then, for all $y \in \operatorname{supp}(\|V\|) \cap B_{d_g}(\xi, 2\delta\rho)$, we have that*

$$\int_{B_{d_g}(y,(1-2\delta)\rho)} \frac{\left(\operatorname{dist}_g^{\Sigma(x,T_xV)}(y)\right)^2}{u_y(x)^{m+2}} d\|V\|(x) \leq 4\left(\mathbf{C_1}\delta+\mathbf{C_2}\rho^2\right)\omega_m, \tag{156}$$

where $\Sigma(x, T_xV)$ *is the harmonic* m*-submanifold of* M^n *at* (x, T_xV)*, for all* $x \in \operatorname{supp}(\|V\|)$ *where it exists and* $u_y(x) := d_g(x, y)$ *for all* $x \in M^n$*,* $\mathbf{C_1} = \mathbf{C_1}(m, p) > 0$*, and* $\mathbf{C_2} = \mathbf{C_2}(K, m) > 0$*. Furthermore, if* $\rho < \sqrt{\delta}$ *then*

$$\int_{B_{d_g}(y,(1-2\delta)\rho)} \frac{\left(\operatorname{dist}_g^{\Sigma(x,T_xV)}(y)\right)^2}{u_y(x)^{m+2}} d\|V\|(x) \leq 4\omega_m \mathbf{C_3}\delta, \tag{157}$$

where $\mathbf{C_3} = \mathbf{C_3}(m, p, K) > 0$.

Proof. Let us fix $\delta_0 := \delta_0(m, p, K, r_H) > 0$ and $\rho_0 := \rho_0(m, p, K, r_H) > 0$ given by Lemma 13.3.4. Suppose that $V \in \mathcal{AC}(\xi, \rho, \delta, p)$, for some $0 < \delta < \delta_0$ and for some $0 < \rho < \rho_0$. Consider $y \in \operatorname{supp}(\|V\|) \cap B_{d_g}(\xi, 2\delta\rho)$ arbitrary.

Letting $\sigma \downarrow 0$ in Equation (142), we have that

$$1 - \mathbf{C_4}\delta - \mathbf{C_5}\rho^2 \leq \Theta^m(\|V\|, y) \leq 1 + \mathbf{C_4}\delta + \mathbf{C_5}\rho^2, \tag{158}$$

where $\mathbf{C_4} = \mathbf{C_4}(m, p) > 0$ and $\mathbf{C_5} = \mathbf{C_5}(K, m) > 0$. By Theorem 12.3.16 there exists $\rho_0 := \rho_0(r_H, K) > 0$ such that, for $0 < \sigma < (1 - 2\delta)\rho$, we have that

$$\begin{aligned} &e^{\lambda((1-2\delta)\rho)} \frac{\|V\|(B_{d_g}(y, (1-2\delta)\rho))}{(1-2\delta)^m \rho^m} - e^{\lambda(\sigma)} \frac{\|V\|(B_{d_g}(y, \sigma))}{\sigma^m} \geq \\ &\geq \int_{\operatorname{Gr}_m\left(B_{d_g}(y,(1-2\delta)\rho) \setminus B_{d_g}(y,\sigma)\right)} \frac{\left\|P_{S(x)^\perp}(\nabla_g(u_y)(x))\right\|_{g_x}^2}{u_y(x)^m} dV(x, S(x)), \end{aligned} \tag{159}$$

Taking $\sigma \downarrow 0$ in Equation (159) and using the same approach adopted in the proof of Lemma 13.3.4, we obtain that

$$\int_{\operatorname{Gr}_m(B_{d_g}(y,(1-2\delta)\rho))} \frac{\left\|P_{S(x)^\perp}(\nabla_g(u_y)(x))\right\|_{g_x}^2}{u_y(x)^m} dV(x, S) \leq 2\left(\mathbf{C_4}\delta + \mathbf{C_5}\rho^2\right)\omega_m \tag{160}$$

By Remark 13.3.3 we have that $V \in \mathbf{RV}_m(B_{d_g}(\xi, \delta\rho))$, which implies that T_xV exists for $\|V\|$-a.e. $x \in \operatorname{supp}(\|V\|) \cap B_{d_g}(\xi, \delta\rho)$. Thus, by Equation (160) and Disintegration Lemma, we have that

$$\int_{B_{d_g}(y,(1-2\delta)\rho)} \frac{\left\|P_{(T_xV)^\perp}(\nabla_g(u_y)(x))\right\|_{g_x}^2}{u_y(x)^m} d\|V\|(x) \leq 2\left(\mathbf{C_4}\delta + \mathbf{C_5}\rho^2\right)\omega_m. \tag{161}$$

Let $\Sigma(x, T_xV) \subset M^n$ be the harmonic m-submanifold of M^n at (x, T_xV) for all $x \in \operatorname{supp}(\|V\|)$ where T_xV exists, by Lemma 13.1.12 there exists $\rho_0 := \rho_0(K, r_H) > 0$ such that

$$\frac{1}{2}\left\|P_{(T_xV)^\perp}(u_y(x)\nabla_g(u_y)(x))\right\|_{g_x}^2 \leq \left(\operatorname{dist}_g^{\Sigma(x,T_xV)}(y)\right)^2 \leq 2\left\|P_{(T_xV)^\perp}(u_y(x)\nabla_g(u_y)(x))\right\|_{g_x}^2, \tag{162}$$

for all $x \in B_{d_g}(y, (1-2\delta)\rho)$.

Therefore, by Equations (161) and (162), we obtain that Equation (156) holds. To conclude the proof, we choose $\delta_0 := \delta_0(m, K, r_H, p) > 0$ and $\rho_0 := \rho_0(m, K, r_H, p) > 0$ as the minimum value ensuring that all the above estimates hold. Now suppose that $\rho < \sqrt{\delta}$. By Equations (161) and (162) we conclude that Equation (157) holds. □

13.4 AFFINE APPROXIMATION

Remark 13.4.1. Let $m < p < \infty$ and $\xi \in M^n$. There exist $\delta_0 := \delta_0(m, p, K, r_H) > 0$ and $\rho_0 := \rho_0(m, p, K, r_H) > 0$ such that if $V \in \mathcal{AC}(\xi, \rho, \delta, p)$, for some $0 < \delta < \delta_0$ and for some $0 < \rho < \min\{\rho_0, \sqrt{\delta}\}$, then, for all $y \in \operatorname{supp}(\|V\|) \cap B_{d_g}(\xi, \delta\rho)$ and for all $\sigma \in]0, 4\delta\rho]$, we have that

$$\frac{1}{2} \leq \frac{\|V\|\left(B_{d_g}(y, \sigma)\right)}{\omega_m \sigma^m} \leq 2, \tag{163}$$

and

$$\int_{B_{d_g}(y,2\sigma)} \left(\operatorname{dist}_g^{\Sigma(x,T_xV)}(y)\right)^2 d\|V\|(x) \leq 2^{m+2}\sigma^{m+2} 4\omega_m \mathbf{C}\delta, \tag{164}$$

where $\mathbf{C} = \mathbf{C}(m, p, K) > 0$. In fact, let us fix $\delta_0 := \delta_0(m, p, K, r_H) > 0$ and $\rho_0 := \rho_0(m, p, K, r_H) > 0$ given by Lemma 13.3.4. Suppose that $V \in \mathcal{AC}(\xi, \rho, \delta, p)$ for some $0 < \delta < \delta_0$ and for some $0 < \rho < \min\{\rho_0, \sqrt{\delta}\}$. Consider $y \in \operatorname{supp}(\|V\|) \cap B_{d_g}(\xi, \delta\rho)$ and $\sigma \in]0, 4\delta\rho]$ arbitrary.

Taking $0 < \delta_0 \frac{1}{6}$ we have that $4\delta\rho \leq (1 - 2\delta)\rho$. Thus, by Lemma 13.3.4, we have that Equation (163) holds.

On the other hand, note that for every $x \in B_{d_g}(y, 2\sigma)$ we have

$$\left(\operatorname{dist}_g^{\Sigma(x,T_xV)}(y)\right)^2 \leq \frac{\left(\operatorname{dist}_g^{\Sigma(x,T_xV)}(y)\right)^2}{u_y(x)^{m+2}} 2^{m+1}\sigma^{m+2}.$$

By integrating the above inequality over $B_{d_g}(y, 2\sigma)$ with respect to $\|V\|$, we obtain

$$\int_{B_{d_g}(y,2\sigma)} \left(\operatorname{dist}_g^{\Sigma(x,T_xV)}(y)\right)^2 d\|V\|(x) \leq \int_{B_{d_g}(y,2\sigma)} \frac{\left(\operatorname{dist}_g^{\Sigma(x,T_xV)}(y)\right)^2}{u_y(x)^{m+2}} 2^{m+1}\sigma^{m+2} d\|V\|(x). \tag{165}$$

Taking $0 < \delta_0 < \frac{1}{10}$ we have that $2(4\delta\rho) < (1 - 2\delta)\rho$. By Equation (165) and Corollary 13.3.5 there exist $\delta_0 := \delta_0(m, p, K, r_H) > 0$ and $\rho_0 := \rho_0(m, p, K, r_H) > 0$ such that Equation (164) holds.

To conclude the proof, we choose $\delta_0 := \delta_0(m, K, r_H, p) > 0$ and $\rho_0 := \rho_0(m, K, r_H, p) > 0$ as the minimum value ensuring that all the above estimates hold.

Lemma 13.4.2. *Let $m < p < \infty$ and $\xi \in M^n$. There exist $\delta_0 := \delta_0(m, p, K, r_H) > 0$ and $\rho_0 := \rho_0(m, p, K, r_H) > 0$ such that if $V \in \mathcal{AC}(\xi, \rho, \delta, p)$, for some $0 < \delta < \delta_0$ and for some*

$0 < \rho < \min\{\rho_0, \sqrt{\delta}\}$, then, for all $y \in \operatorname{supp}(\|V\|) \cap B_{d_g}(\xi, \delta\rho)$ and for all $\sigma \in]0, 4\delta\rho]$, we have that

$$\sup_{x \in \operatorname{supp}(\|V\|) \cap B_{d_g}(y,\sigma)} \operatorname{dist}_g^{\Sigma(y,S(y))}(x) \leq \mathbf{C}\delta^{\frac{1}{2(m+1)}}\sigma, \tag{166}$$

for some $S(y) \in \operatorname{Gr}(m, T_y M^n)$, where $\Sigma(y, S(y)) \subset M^n$ is the harmonic m-submanifold of M^n at $(y, S(y))$ and $\mathbf{C} = \mathbf{C}(m, p, K) > 0$.

Proof. Let us fix $\delta_0 := \delta_0(m, p, K, r_H) > 0$ and $\rho_0 := \rho_0(m, p, K, r_H) > 0$ given by Remark 13.4.1. Suppose that $V \in \mathcal{AC}(\xi, \rho, \delta, p)$ for some $0 < \delta < \delta_0$ and for some $0 < \rho < \min\{\rho_0, \sqrt{\delta}\}$. Consider $y \in \operatorname{supp}(\|V\|) \cap B_{d_g}(\xi, \delta\rho)$ and $\sigma \in]0, 4\delta\rho]$ arbitrary.

By Remark 13.4.1 we have that

$$\frac{1}{2} \leq \frac{\|V\|\left(B_{d_g}(y, \sigma)\right)}{\omega_m \sigma^m} \leq 2. \tag{167}$$

Recall the general principle that if $W \subset M^n$ is compact and $\eta > 0$, then any maximal pairwise disjoint collection of closed balls $\left\{\overline{B_{d_g}\left(w_i, \frac{\eta}{2}\right)}\right\}_{i=1,\ldots,N}$ with $w_i \in W$, for all $i = 1, \ldots, N$, will automatically have the property that $W \subset \cup_{i=1}^N \overline{B_{d_g}(w_i, \eta)}$.

Let $\alpha \in]0, 1[$ arbitrary. Using the general principle with $\eta = \delta^\alpha\sigma$ we have pairwise disjoint closed balls $\left\{\overline{B_{d_g}\left(y_i, \frac{\delta^\alpha\sigma}{2}\right)}\right\}_{i=1,\ldots,N}$ with $y_i \in \operatorname{supp}(\|V\|) \cap \overline{B_{d_g}\left(y, \frac{\sigma}{4}\right)}$, for all $i = 1, \ldots, N$, for some $N \in \mathbb{N}$, such that

$$\operatorname{supp}(\|V\|) \cap \overline{B_{d_g}\left(y, \frac{\sigma}{4}\right)} \subset \bigcup_{i=1}^N \overline{B_{d_g}(y_i, \delta^\alpha\sigma)}. \tag{168}$$

Let $i = 1, \ldots, N$ be arbitrary. We claim that $y_i \in \operatorname{supp}(\|V\|) \cap B_{d_g}(\xi, 2\delta\rho)$ and $\frac{\delta^\alpha\sigma}{2} \in]0, (1 - 2\delta)\rho]$.

In fact, since $y_i \in \operatorname{supp}(\|V\|) \cap \overline{B_{d_g}\left(y, \frac{\sigma}{4}\right)}$, $\sigma \in]0, 4\delta\rho]$, and $y \in \operatorname{supp}(\|V\|) \cap B_{d_g}(\xi, \delta\rho)$ we have that

$$d_g(y_i, \xi) \leq d_g(y_i, y) + d_g(y, \xi) \leq \frac{\sigma}{4} + \delta\rho < \frac{4\delta\rho}{4} + \delta\rho = 2\delta\rho.$$

Furthermore, since $\sigma \in]0, 4\delta\rho]$, there exists $\delta_0 > 0$ such that $\frac{\delta^\alpha\sigma}{2} < (1 - 2\delta)\rho$. Thus, by Lemma 13.3.4, we have that

$$\frac{\omega_m \delta^{m\alpha}\sigma^m}{2^{m+1}} \leq \|V\|\left(B_{d_g}\left(y_i, \frac{\delta^\alpha\sigma}{2}\right)\right) \leq \|V\|\left(\overline{B_{d_g}\left(y_i, \frac{\delta^\alpha\sigma}{2}\right)}\right). \tag{169}$$

On the other hand, since $\overline{B_{d_g}\left(y_i, \frac{\delta^\alpha\sigma}{2}\right)} \cap \overline{B_{d_g}\left(y_j, \frac{\delta^\alpha\sigma}{2}\right)} = \emptyset$ for all $i, j = 1, \ldots, N$ with $i \neq j$, we have, by Equation (169), that

$$\frac{N\omega_m \delta^{m\alpha}\sigma^m}{2^{m+1}} \leq \sum_{i=1}^N \|V\|\left(\overline{B_{d_g}\left(y_i, \frac{\delta^\alpha\sigma}{2}\right)}\right) = \|V\|\left(\bigcup_{i=1}^N \overline{B_{d_g}\left(y_i, \frac{\delta^\alpha\sigma}{2}\right)}\right). \tag{170}$$

Let $i = 1, \dots, N$ be arbitrary. We claim that $\overline{B_{d_g}\left(y_i, \frac{\delta^\alpha \sigma}{2}\right)} \subset B_{d_g}(y, \sigma)$. Which implies that $\cup_{i=1}^N \overline{B_{d_g}\left(y_i, \frac{\delta^\alpha \sigma}{2}\right)} \subset B_{d_g}(y, \sigma)$.

In fact, let $x \in \overline{B_{d_g}\left(y_i, \frac{\delta^\alpha \sigma}{2}\right)}$. Since $y_i \in \operatorname{supp}(\|V\|) \cap \overline{B_{d_g}\left(y, \frac{\sigma}{4}\right)}$ we have that

$$d_g(x, y) \leq d_g(x, y_i) + d_g(y_i, y) \leq \frac{\delta^\alpha \sigma}{2} + \frac{\sigma}{4} = \sigma\left(\frac{\delta^\alpha}{2} + \frac{1}{4}\right) < \sigma.$$

Therefore, by Equations (167) and (170), we have that

$$N \leq \frac{2^{m+2}}{\delta^{m\alpha}}. \tag{171}$$

Since $y_i \in \operatorname{supp}(\|V\|) \cap B_{d_g}(\xi, 2\delta\rho)$ and $\sigma \in]0, 4\delta\rho]$ by Remark 13.4.1 (applied to y_i and σ) we have that

$$\int_{B_{d_g}(y_i, 2\sigma)} \left(\operatorname{dist}_g^{\Sigma(x, T_x V)}(y_i)\right)^2 d\|V\|(x) \leq 2^{m+2} \sigma^{m+2} 4\omega_m \mathbf{C_1} \delta \tag{172}$$

holds for all $i = 1, \dots, N$, where $\mathbf{C_1} = \mathbf{C_1}(m, p, K) > 0$. We claim that $B_{d_g}(y, \sigma) \subset B_{d_g}(y_i, 2\sigma)$ for all $i = 1, \dots, N$.

In fact, let $i = 1, \dots, N$ arbitrary, and $x \in B_{d_g}(y, \sigma)$, since $y_i \in \operatorname{supp}(\|V\|) \cap \overline{B_{d_g}\left(y, \frac{\sigma}{4}\right)}$, we have that

$$d_g(x, y_i) \leq d_g(x, y) + d_g(y, y_i) \leq \sigma + \frac{\sigma}{4} < 2\sigma.$$

Therefore, by Equations (171) and (172), we have that

$$\int_{B_{d_g}(y, \sigma)} \sum_{i=1}^N \left(\operatorname{dist}_g^{\Sigma(x, T_x V)}(y_i)\right)^2 d\|V\|(x) \leq 2^{2m+6} \sigma^{m+2} \omega_m \frac{\mathbf{C_1} \delta}{\delta^{m\alpha}}.$$

Thus, for any given $\lambda \geq 1$, we have that

$$\sum_{i=1}^N \left(\operatorname{dist}_g^{\Sigma(x, T_x V)}(y_i)\right)^2 \leq 2^{2m+6} \sigma^{m+2} \omega_m \frac{\mathbf{C_1} \delta}{\delta^{m\alpha}} \lambda \tag{173}$$

holds for $\|V\|$-a.e. $y \in \operatorname{supp}(\|V\|) \cap B_{d_g}(y, \sigma) \cap A$ such that $\|V\|(A) \geq \frac{1}{\lambda}$.

Since $\delta^\alpha \sigma < \sigma$ we have by Equation (167) that $\|V\|\left(B_{d_g}(y, \delta^\alpha \sigma)\right) \geq \frac{\omega_m \delta^{m\alpha} \sigma^m}{2}$, and we can select $\lambda = \frac{2}{\omega_m \delta^{m\alpha} \sigma^m}$ in such a way that Equation (173) holds for some $x_0 \in \operatorname{supp}(\|V\|) \cap B_{d_g}(y, \delta^\alpha \sigma)$.

So we have shown that there is $x_0 \in \operatorname{supp}(\|V\|) \cap B_{d_g}(y, \delta^\alpha \sigma)$ such that

$$\operatorname{dist}_g^{\Sigma(x_0, T_{x_0} V)}(y_i) \leq 2^{m+\frac{7}{2}} \sigma \sqrt{\mathbf{C_1}} \delta^{\frac{1}{2} - m\alpha}$$

holds for all $i = 1, \dots, N$. Now let $\Sigma(x_0, T_{x_0} V)_y$ be the Fermi transport of $\Sigma(x_0, T_{x_0} V)$ from x_0 to y and note that for all $i = 1, \dots, N$, we have that

$$\operatorname{dist}_g^{\Sigma(x_0, T_{x_0} V)_y}(y_i) \leq \operatorname{dist}_g^{\Sigma(x_0, T_{x_0} V)}(y_i) + d_g(x_0, y). \tag{174}$$

Define $S(y) := T_y\Sigma(x_0, T_{x_0}V)_y$ and $\Sigma(y, S)$ the harmonic m-submanifold of M at $(y, S(y))$. By Lemma 13.1.11, for all $i = 1, \dots, N$, we have

$$\operatorname{dist}_g^{\Sigma(y,S(y))}(y_i) \leq \operatorname{dist}_g^{\Sigma(x_0,T_{x_0}V)_y}(y_i) + o(1). \tag{175}$$

Therefore, by Equations (174) and (175), we have for all $i = 1, \dots, N$

$$\operatorname{dist}_g^{\Sigma(y,S(y))}(y_i) \leq 2^{m+\frac{7}{2}}\sigma\sqrt{\mathbf{C_1}}\delta^{\frac{1}{2}-m\alpha} + \delta^\alpha\sigma + \delta^\alpha\sigma$$

Furthermore, for all $i = 1, \dots, N$ and all $x \in B_{d_g}(y_i, \delta^\alpha\sigma)$ we have

$$\operatorname{dist}_g^{\Sigma(y,S(y))}(x) \leq \delta^\alpha\sigma + 2^{m+\frac{7}{2}}\sigma\sqrt{\mathbf{C_1}}\delta^{\frac{1}{2}-m\alpha} + 2\delta^\alpha\sigma.$$

Then, selecting α such that $\frac{1}{2} - m\alpha = \alpha$, that is, $\alpha = \frac{1}{2(m+1)}$, we have, for all $i = 1, \dots, N$ and all $x \in B_{d_g}(y_i, \delta^\alpha\sigma)$, that

$$\operatorname{dist}_g^{\Sigma(y,S(y))}(x) \leq \left(3 + 2^{m+\frac{7}{2}}\sqrt{\mathbf{C_1}}\right)\delta^{\frac{1}{2(m+1)}}\sigma$$

Therefore, by Equation (168) and the fact that $\operatorname{supp}(\|V\|) \cap B_{d_g}(y, \sigma) \subset \operatorname{supp}(\|V\|) \cap \overline{B_{d_g}(y, \sigma)}$, we have that Equation (166) holds. To conclude the proof, we choose $\delta_0 := \delta_0(m, K, r_H, p) > 0$ and $\rho_0 := \rho_0(m, K, r_H, p) > 0$ as the minimum value ensuring that all the above estimates hold. □

Corollary 13.4.3. *Let $m < p < \infty$ and $\xi \in M^n$. There exist $\delta_0 := \delta_0(m, p, K, r_H) > 0$ and $\rho_0 := \rho_0(m, p, K, r_H) > 0$ such that if $V \in \mathcal{AC}(\xi, \rho, \delta, p)$, for some $0 < \delta < \delta_0$ and for some $0 < \rho < \min\{\rho_0, \sqrt{\delta}\}$, then, for all $y \in \operatorname{supp}(\|V\|) \cap B_{d_g}(\xi, \delta\rho)$, for all $\sigma \in]0, 4\delta\rho]$, and for all $\alpha \in]0, 1[$, we have that*

$$\sup_{x \in \operatorname{supp}(\|V\|) \cap B_{d_g}(y,\sigma)} \operatorname{dist}_g^{\Sigma(y,S(y))}(x) \leq \mathbf{C_1}\delta^{\frac{1}{2(m+1)}}\sigma, \tag{176}$$

where $\mathbf{C_1} = \mathbf{C_1}(m, p, K) > 0$ and

$$E(y, \alpha\sigma, S(y), V) \leq \mathbf{C_2}\delta^{\frac{1}{m+1}} \tag{177}$$

for some $S(y) \in \operatorname{Gr}(m, T_yM^n)$, where $\Sigma(y, S(y)) \subset M^n$ is the harmonic m-submanifold of M^n at $(y, S(y))$ and $\mathbf{C_2} = \mathbf{C_2}(m, n, p, \alpha, \rho, K) > 0$.

Proof. Let us fix $\delta_0 := \delta_0(m, p, K, r_H) > 0$ and $\rho_0 := \rho_0(m, p, K, r_H) > 0$ given by Lemma 13.4.2. Suppose that $V \in \mathcal{AC}(\xi, \rho, \delta, p)$, for some $0 < \delta < \delta_0$ and for some $0 < \rho < \min\{\rho_0, \sqrt{\delta}\}$. Consider $y \in \operatorname{supp}(\|V\|) \cap B_{d_g}(\xi, \delta\rho)$, $\sigma \in]0, 4\delta\rho]$, and $\alpha \in]0, 1[$ arbitrary.

By Lemma 13.4.2, we have that

$$\sup_{x \in \operatorname{supp}(\|V\|) \cap B_{d_g}(y,\sigma)} \operatorname{dist}_g^{\Sigma(y,S(y))}(x) \leq \mathbf{C_3}\delta^{\frac{1}{2(m+1)}}\sigma, \tag{178}$$

for some $S(y) \in \mathrm{Gr}(m, T_yM^n)$, where $\Sigma(y, S(y)) \subset M^n$ is the harmonic m-submanifold of M^n at $(y, S(y))$ and $\mathbf{C_3} = \mathbf{C_3}(m, p, K) > 0$.

By Remark 13.3.2 V has L^p_{loc}-bounded generalized mean curvature vector in $B_{d_g}(\xi, \rho)$. Furthermore, taking $0 < \delta_0 < \frac{1}{5}$ we have that $4\delta\rho < (1-\delta)\rho$. Thus, by Lemma 13.2.1, we have that

$$\begin{aligned} E(y, \alpha\sigma, S(y), V) \leq \frac{\mathbf{C_4}}{\sigma^m} \int_{B_{d_g}(y,\sigma)} \left(\frac{\mathrm{dist}_g^{\Sigma(y,S(y))}(x)}{\sigma} \right)^2 d\|V\|(x) + \frac{2\alpha^{-m}}{\sigma^{m-2}} \int_{B_{d_g}(y,\sigma)} \left\| \overrightarrow{H_g}(x) \right\|_{g_x}^2 d\|V\|(x) \\ + \frac{4m\rho\alpha^{-m}}{\sigma^m} \int_{B_{d_g}(y,\sigma)} \frac{\mathrm{dist}_g^{\Sigma(y,S(y))}(x)}{\sigma} \sum_{j=1}^m \left| \kappa_j(\mathbf{p}^\Sigma(x)) \right| d\|V\|(x) \\ + \frac{8m\rho^2\alpha^{-m}}{\sigma^m} \int_{B_{d_g}(y,\sigma)} \left(\frac{\mathrm{dist}_g^{\Sigma(y,S(y))}(x)}{\sigma} \right)^2 \sum_{j=1}^m \kappa_j(\mathbf{p}^\Sigma(x))^2 d\|V\|(x), \end{aligned} \tag{179}$$

where $\mathbf{C_4} = \mathbf{C_4}(m, n, \alpha, K, \rho) > 0$. Therefore, by Equations (178) and (179) and Lemma 13.1.10, we conclude that

$$\begin{aligned} E(y, \alpha\sigma, S(y), V) \leq \left(\mathbf{C_4}\mathbf{C_3}^2 + 4m\rho\alpha^{-m}\mathbf{C_3} + 8m^2\rho^2\alpha^{-m}\mathbf{C_3}^2 \right) \delta^{\frac{1}{m+1}} \frac{\|V\|(B_{d_g}(y,\sigma))}{\sigma^m} \\ + \frac{2\alpha^{-m}}{\sigma^{m-2}} \int_{B_{d_g}(y,\sigma)} \left\| \overrightarrow{H_g}(x) \right\|_{g_x}^2 d\|V\|(x). \end{aligned} \tag{180}$$

Since $\sigma < 4\delta\rho$ taking $0 < \delta_0 < \frac{1}{6}$ we have that $\sigma < (1-2\delta)\rho$. On the other hand, $y \in B_{d_g}(\xi, \delta\rho) \subset B_{d_g}(\xi, 2\delta\rho)$. Thus, by Lemma 13.3.4, there exists $\delta_0 := \delta_0(m, p, K, r_H) > 0$ and $\rho_0 := \rho_0(m, p, K, r_H) > 0$ such that

$$\frac{\|V\|\left(B_{d_g}(y,\sigma)\right)}{\sigma^m} \leq 2\omega_m. \tag{181}$$

Denote by $h(x) := \left\| \overrightarrow{H_g}(x) \right\|_{g_x}^2$. By Hölder inequality, the fact that $B_{d_g}(y,\sigma) \subset B_{d_g}(\xi,\rho)$ and $1 \leq m < p$, we have that

$$\frac{2\alpha^{-m}}{\sigma^{m-2}} \int_{B_{d_g}(y,\sigma)} \left\| \overrightarrow{H_g}(x) \right\|_{g_x}^2 d\|V\|(x) \leq 2^{\frac{6p-4m-2}{p}} \alpha^{-m} (\omega_m)^{\frac{p-2}{p}} \delta^{\frac{1}{m+1}}. \tag{182}$$

Therefore, by Equations (180) to (182), we have that Equation (177) holds. To conclude the proof, we choose $\delta_0 := \delta_0(m, K, r_H, p) > 0$ and $\rho_0 := \rho_0(m, K, r_H, p) > 0$ as the minimum value ensuring that all the above estimates hold. □

13.5 LIPSCHITZ APPROXIMATION

Definition 13.5.1. Let $\Sigma \subset M^n$ be an m-dimensional of class $\mathcal{C}^2$ submanifold of M^n, $\xi \in \Sigma$, $\left(\text{Tube}(\Sigma, r_H), \{x_i\}_{i=1}^n\right)$ a Fermi coordinate chart for Σ at ξ, and

$$\begin{aligned} f : \Sigma &\to \mathbb{R}^{n-m} \\ x &\mapsto \left(f^{m+1}(x), \dots, f^n(x)\right), \end{aligned}$$

where $f^j : \Sigma \to \mathbb{R}$ for all $j = m+1, \dots, n$. We define the normal graph of f over Σ by

$$\text{Graph}^{\Sigma}(f) := \left\{ \exp_x \left(\sum_{j=m+1}^{n} f^j(x) \frac{\partial}{\partial x_j}(x) \right) : x \in \Sigma \right\}.$$

Lemma 13.5.2. *Let $m < p < \infty$, $\xi \in M^n$ and $0 < L \leq 1$. There exists $\delta_0 := \delta_0(m, n, p, K, r_H) > 0$ and $\rho_0 := \rho_0(m, K, r_H, p) > 0$ such that if $V \in \mathcal{AC}(\xi, \rho, \delta, p)$, for some $0 < \delta < (\delta_0 L)^{2(m+1)}$ and for some $0 < \rho < \min\left\{\rho_0, \sqrt{\delta}\right\}$, then, for all $y \in \text{supp}(\|V\|) \cap B_{d_g}\left(\xi, \frac{\delta\rho}{2}\right)$, for all $\sigma \in]0, 2\delta\rho]$ and for all $S(y) \in \text{Gr}(m, T_y M^n)$ such that Equations (176) and (177) hold, we have that there exists a Lipschitz function $\mathbf{f}_{S(y)} : B_{d_g}(y, \sigma) \cap \Sigma(y, S(y)) \to \mathbb{R}^{n-m}$ such that*

$$\text{Lip}(\mathbf{f}_{S(y)}) \leq L, \tag{183}$$

$$\sup_{x \in \Sigma(y,S(y)) \cap B_{d_g}(y, 2\delta\sigma)} \left\| \mathbf{f}_{S(y)}(x) \right\|_{g_e} \leq \mathbf{C_1} \delta^{\frac{1}{2(m+1)}} \sigma, \tag{184}$$

where $\mathbf{C_1} = \mathbf{C_1}(m, p, K) > 0$,

$$\frac{\|V\| \llcorner B_{d_g}(y, \sigma) \left(\text{supp}(\|V\|) \setminus \text{Graph}^{\Sigma(y,S(y))}\left(\mathbf{f}_{S(y)}\right) \right)}{\sigma^m} \leq \frac{\mathbf{C_2}}{L^2} E(y, \sigma, S(y), V) \leq \frac{\mathbf{C_3}}{L^2} \delta^{\frac{1}{m+1}}, \tag{185}$$

where $\mathbf{C_2} = \mathbf{C_2}(m, p, K, r_H) > 0$ and $\mathbf{C_3} = \mathbf{C_3}(m, n, p, K, r_H) > 0$, and

$$\frac{\mathcal{H}^m_{d_g} \llcorner B_{d_g}(y, \sigma) \left(\text{Graph}^{\Sigma(y,S(y))}\left(\mathbf{f}_{S(y)}\right) \setminus \text{supp}(\|V\|) \right)}{\sigma^m} \leq \frac{\mathbf{C_4}}{L^2} E(y, \sigma, S(y), V) \leq \frac{\mathbf{C_5}}{L^2} \delta^{\frac{1}{m+1}}, \tag{186}$$

where where $\mathbf{C_4} = \mathbf{C_4}(m, n, p, K, r_H) > 0$ and $\mathbf{C_5} = \mathbf{C_5}(m, n, p, K, r_H) > 0$.

Proof. Let $\delta_0 := \delta_0(m, p, K, r_H) > 0$ and $\rho_0 := \rho_0(m, p, K, r_H) > 0$ given by Lemma 13.3.4. Suppose that $V \in \mathcal{AC}(\xi, \rho, \delta, p)$, for some $0 < \delta < (\delta_0 L)^{2(m+1)}$ and for some $0 < \rho < \min\left\{\rho_0, \sqrt{\delta}\right\}$. Consider $y \in \text{supp}(\|V\|) \cap B_{d_g}\left(\xi, \frac{\delta\rho}{2}\right)$ and $\sigma \in]0, 2\delta\rho]$ arbitrary, and $S(y) \in \text{Gr}(m, T_y M^n)$ such that Equations (176) and (177) hold. Define

$$G := \left\{ z \in \text{supp}(\|V\|) \cap B_{d_g}(y, 2\delta\sigma) : \sup_{t \in]0, (1-2\delta)\sigma]} \frac{1}{t^m} \int_{B_{d_g}(z,t)} \left\| P_{T_x V} - P_{T_x \Sigma(y,S(y))_x} \right\|^2_{\text{HS}_{g_x}} d\|V\|(x) \leq \delta_0^2 L^2 \right\}. \tag{187}$$

Thus, if $z \in \operatorname{supp}(\|V\|) \cap B_{d_g}(y, 2\delta\sigma) \setminus G$ then there exists $t \in]0, (1-2\delta)\sigma]$ such that

$$\delta_0^2 L^2 < \frac{1}{t^m} \int_{B_{d_g}(z,t)} \left\| P_{T_x V} - P_{T_x \Sigma(y,S(y))_x} \right\|^2_{\mathrm{HS}_{g_x}} d\|V\|(x). \tag{188}$$

By the five-times covering lemma we can pick pairwise disjoint balls $B_{d_g}(z_j, t_j)$ such that Equation (188) holds for $t = t_j \in]0, (1-2\delta)\sigma]$ and $z = z_j \in \operatorname{supp}(\|V\|) \cap B_{d_g}(y, 2\delta\sigma) \setminus G$, and such that

$$\operatorname{supp}(\|V\|) \cap B_{d_g}(y, 2\delta\sigma) \setminus G \subset \bigcup_j B_{d_g}(z_j, 5t_j). \tag{189}$$

By Equations (188) and (189) and since $\delta_0 := \delta_0(m, p, K, r_H) > 0$ and $\rho_0 := \rho_0(m, p, K, r_H) > 0$ were given by Lemma 13.3.4, we have that

$$\|V\|\Big(\operatorname{supp}(\|V\|) \cap B_{d_g}(y, 2\delta\sigma) \setminus G\Big) < \frac{2\omega_m 5^m}{\delta_0^2 L^2} \int_{B_{d_g}(y,\sigma)} \left\| P_{T_x V} - P_{T_x \Sigma(y,S(y))_x} \right\|^2_{\mathrm{HS}_{g_x}} d\|V\|(x). \tag{190}$$

We now claim that G is the normal graph Lipschitz function over $\Sigma(y, S(y))$. To check this, let z_1, z_2 be distinct points of G. Let $\gamma : [0, r_H[\to M^n$ be the unique minimizing geodesic between z_1 and z_2 such that $\gamma(0) = z_1$ and $\gamma(t) = z_2$, where $t := d_g(z_1, z_2)$.

Taking $0 < \delta_0 < \frac{1}{6}$ we have that $4\delta\sigma < (1-2\delta)\sigma$. Since $z_1, z_2 \in G \subset B_{d_g}(y, 2\delta\sigma)$, we obtain that

$$t = d_g(z_1, z_2) \leq d_g(z_1, y) + d_g(y, z_2) < 2\delta\sigma + 2\delta\sigma = 4\delta\sigma < (1-2\delta)\sigma. \tag{191}$$

Since $z_1 \in G$ we have that

$$\frac{1}{t^m} \int_{B_{d_g}\left(z_1, \frac{t}{2}\right)} \left\| P_{T_x V} - P_{T_x \Sigma(y,S)_x} \right\|^2_{\mathrm{HS}_{g_x}} d\|V\|(x) \leq \frac{1}{2^m} \delta_0^2 L^2. \tag{192}$$

Since $\sigma \in]0, 2\delta\rho]$ there exists $\delta_0 > 0$ such that $\sigma < 2\delta\rho < \frac{\rho}{5}$. On the other hand, since $z_1 \in B_{d_g}(y, 2\delta\rho)$ and $y \in B_{d_g}\left(\xi, \frac{\delta\rho}{2}\right)$ we have that

$$d_g(z_1, \xi) \leq d_g(z_1, y) + d_g(y, \xi) < 2\delta\sigma + \frac{\delta\rho}{2} < \frac{2\delta\rho}{5} + \frac{\delta\rho}{2} = \frac{9}{10}\delta\rho < \delta\rho.$$

Therefore, $z_1 \in B_{d_g}(\xi, \delta\rho)$, and $t < 4\delta\sigma < 4\delta\rho$. By Corollary 13.4.3 there exists $\delta_0 := \delta_0(m, p, K, r_H) > 0$ and $\rho_0 := \rho_0(m, p, K, r_H) > 0$ such that

$$\frac{1}{t^m} \int_{B_{d_g}\left(z_1, \frac{t}{2}\right)} \left\| P_{T_x V} - P_{T_x \Sigma(z_1,S(z_1))_x} \right\|^2_{\mathrm{HS}_{g_x}} d\|V\|(x) \leq \frac{1}{2^m} \mathbf{C_6} \delta_0^2 L^2, \tag{193}$$

for some $S(z_1) \in \mathrm{Gr}(m, T_{z_1} M^n)$, where $\mathbf{C_6} = \mathbf{C_6}(m, n, p, K, r_H) > 0$. On the other hand, fix $s \in [0, t]$ arbitrarily. By Lemma 13.1.13 we have, for all $x \in \operatorname{supp}(\|V\|) \cap B_{d_g}\left(z_1, \frac{t}{2}\right)$, that

$$\begin{aligned} \frac{1}{2} \left\| P_{T_{\gamma(s)}\Sigma(y,S(y))_{\gamma(s)}} - P_{T_{\gamma(s)}\Sigma(z_1,S(z_1))_{\gamma(s)}} \right\|^2_{\mathrm{HS}_{g_{\gamma(s)}}} &\leq \left\| P_{T_x \Sigma(y,S(y))_x} - P_{T_x \Sigma(z_1,S(z_1))_x} \right\|^2_{\mathrm{HS}_{g_x}} \\ &\leq 2 \left\| P_{T_{\gamma(s)}\Sigma(y,S(y))_{\gamma(s)}} - P_{T_{\gamma(s)}\Sigma(z_1,S(z_1))_{\gamma(s)}} \right\|^2_{\mathrm{HS}_{g_{\gamma(s)}}}. \end{aligned} \tag{194}$$

Then, by Equation (194) and Lemma 13.3.4 there exists $\delta_0 := \delta_0(m, p, K, r_H) > 0$ and $\rho_0 := \rho_0(m, p, K, r_H) > 0$ such that

$$\begin{aligned}&\frac{1}{t^m}\int_{B_{d_g}\left(z_1,\frac{t}{2}\right)} \left\|P_{T_x\Sigma(y,S(y))_x} - P_{T_x\Sigma(z_1,S(z_1))_x}\right\|^2_{\mathrm{HS}_{g_x}} d\|V\|(x)\\ &\geq \frac{\omega_m}{2^{m+2}}\left\|P_{T_{\gamma(s)}\Sigma(y,S(y))_{\gamma(s)}} - P_{T_{\gamma(s)}\Sigma(z_1,S(z_1))_{\gamma(s)}}\right\|^2_{\mathrm{HS}_{g_{\gamma(s)}}}.\end{aligned} \tag{195}$$

For all $x \in B_{d_g}(y,\sigma)$ where T_xV exists we have, by the squared triangle inequality, that

$$\begin{aligned}\left\|P_{T_x\Sigma(y,S(y))_x} - P_{T_x\Sigma(z_1,S(z_1))_x}\right\|^2_{\mathrm{HS}_{g_x}} &\leq 2\left\|P_{T_x\Sigma(y,S(y))_x} - P_{T_xV}\right\|^2_{\mathrm{HS}_{g_x}}\\ &\quad + 2\left\|P_{T_xV} - P_{T_x\Sigma(z_1,S(z_1))_x}\right\|^2_{\mathrm{HS}_{g_x}}.\end{aligned} \tag{196}$$

By Equation (191) we have that $\frac{t}{2} < t < (1-2\delta)\sigma$, which implies, since $z_1 \in G$, with Equations (187), (195) and (196), that

$$\left\|P_{\left(T_{\gamma(s)}\Sigma(y,S(y))_{\gamma(s)}\right)^\perp}\left(\gamma'(s)\right)\right\|_{g_{\gamma(s)}} \leq \left\|P_{\left(T_{\gamma(s)}\Sigma(z_1,S(z_1))_{\gamma(s)}\right)^\perp}\left(\gamma'(s)\right)\right\|_{g_{\gamma(s)}} + \sqrt{\frac{8\left(\mathbf{C_7}+1\right)}{\omega_m}}\delta_0 L\left\|\gamma'(s)\right\|_{g_{\gamma(s)}}, \tag{197}$$

where $\mathbf{C_7} = \mathbf{C_7}(m, n, p, K, r_H) > 0$. Integrating Equation (197) in $[0,t]$ with respect to ds we obtain that

$$\int_0^t \left\|P_{\left(T_{\gamma(s)}\Sigma(y,S(y))_{\gamma(s)}\right)^\perp}\left(\gamma'(s)\right)\right\|_{g_{\gamma(s)}} ds \leq \left(\mathbf{C_8} + \sqrt{\frac{8\left(\mathbf{C_7}+1\right)}{\omega_m}}\right)\delta_0 L d_g(z_1,z_2), \tag{198}$$

where $\mathbf{C_8} = \mathbf{C_8}(m, p, K) > 0$. On the other hand

$$d_g(z_1,z_2) \leq \int_0^t \left\|P_{\left(T_{\gamma(s)}\Sigma(y,S(y))_{\gamma(s)}\right)^\perp}\left(\gamma'(s)\right)\right\|_{g_{\gamma(s)}} ds + d_g\left(\mathbf{p}^{\Sigma(y,S(y))}(z_1), \mathbf{p}^{\Sigma(y,S(y))}(z_2)\right). \tag{199}$$

By Equations (198) and (199), we have that

$$\begin{aligned}&\int_0^t \left\|P_{\left(T_{\gamma(s)}\Sigma(y,S(y))_{\gamma(s)}\right)^\perp}\left(\gamma'(s)\right)\right\|_{g_{\gamma(s)}} ds\\ &\leq \left(\mathbf{C_8} + \sqrt{\frac{8\left(\mathbf{C_7}+1\right)}{\omega_m}}\right)\delta_0 L\int_0^t \left\|P_{\left(T_{\gamma(s)}\Sigma(y,S(y))_{\gamma(s)}\right)^\perp}\left(\gamma'(s)\right)\right\|_{g_{\gamma(s)}} ds\\ &+ \left(\mathbf{C_8} + \sqrt{\frac{8\left(\mathbf{C_7}+1\right)}{\omega_m}}\right)\delta_0 L d_g\left(\mathbf{p}^{\Sigma(y,S(y))}(z_1), \mathbf{p}^{\Sigma(y,S(y))}(z_2)\right).\end{aligned} \tag{200}$$

Note that there exists $\delta_0 := \delta_0(m, n, p, K, r_H) > 0$ such that

$$\left(\mathbf{C_8} + \sqrt{\frac{8\left(\mathbf{C_7}+1\right)}{\omega_m}}\right)\delta_0 \leq \frac{1}{2}. \tag{201}$$

Since $0 < L \leq 1$, by Equations (200) and (201), we conclude that

$$\begin{aligned}&\int_0^t \left\| P_{\left(T_{\gamma(s)}\Sigma(y,S(y))_{\gamma(s)}\right)^{\perp}}\left(\gamma'(s)\right)\right\|_{g_{\gamma(s)}} ds\\ &\leq 2\left(\mathbf{C_8}+\sqrt{\frac{8\left(\mathbf{C_7}+1\right)}{\omega_m}}\right)\delta_0 L d_g\left(\mathbf{p}^{\Sigma(y,S(y))}(z_1),\mathbf{p}^{\Sigma(y,S(y))}(z_2)\right).\end{aligned} \tag{202}$$

By Equations (199), (201) and (202), we conclude that

$$d_g\left(\mathbf{p}^{\Sigma(y,S(y))}(z_1),\mathbf{p}^{\Sigma(y,S(y))}(z_2)\right) \geq \frac{d_g(z_1,z_2)}{1+L} > 0 \tag{203}$$

Therefore, $\mathbf{p}^{\Sigma(y,S(y))} : G \to \Sigma(y,S)$ is an injective map. Thus,

$$\mathbf{p}^{\Sigma(y,S)} : G \to \mathbf{p}^{\Sigma(y,S)}(G)$$

is bijective. Let $\left(\mathrm{Tube}(\Sigma(y,S(y)), r_H), \{x^i\}_{i=1}^n\right)$ be a Fermi coordinate chart for $\Sigma(y,S(y))$ at y and define

$$\begin{aligned}f_{S(y)} : \mathbf{p}^{\Sigma(y,S(y))}(G) \subset \Sigma(y,S(y)) &\to \mathbb{R}^{n-m}\\ \mathbf{p}^{\Sigma(y,S(y))}(z) &\mapsto \left(f_{S(y)}^{m+1}\left(\mathbf{p}^{\Sigma(y,S(y))}(z)\right),\ldots,f_{S(y)}^{n}\left(\mathbf{p}^{\Sigma(y,S)}(z)\right)\right),\end{aligned}$$

where $f_{S(y)}^i\left(\mathbf{p}^{\Sigma(y,S)}(z)\right) := x^i(z)$, for all $i = m+1,\ldots,n$. By Lemma 13.1.2, for all $z \in G$, we have that

$$\left\|f_{S(y)}\left(\mathbf{p}^{\Sigma(y,S(y))}(z)\right)\right\|_{g_e} = \mathrm{dist}_g^{\Sigma(y,S(y))}(z), \tag{204}$$

We also have $\mathrm{Graph}^{\Sigma(y,S(y))}\left(f_{S(y)}\right) = G$. Therefore, by Equations (176) and (204), we have that

$$\sup_{z\in G}\left\|f_{S(y)}\left(\mathbf{p}^{\Sigma(y,S(y))}(z)\right)\right\|_{g_e} \leq \mathbf{C_6}\delta^{\frac{1}{2(m+1)}}2\delta\sigma, \tag{205}$$

where $\mathbf{C_6} = \mathbf{C_6}(m,p,K) > 0$. By Equation (202), for all $z_1, z_2 \in G$, we have

$$\begin{aligned}&\left\|f_{S(y)}\left(\mathbf{p}^{\Sigma(y,S)}(z_1)\right) - f_{S(y)}\left(\mathbf{p}^{\Sigma(y,S)}(z_2)\right)\right\|_{g_e}\\ &\leq 2\left(\mathbf{C_8}+\sqrt{\frac{8\left(\mathbf{C_7}+1\right)}{\omega_m}}\right)\delta_0 L d_g\left(\mathbf{p}^{\Sigma(y,S)}(z_1),\mathbf{p}^{\Sigma(y,S)}(z_2)\right).\end{aligned} \tag{206}$$

Then, by Lipschitz Extension Theorem, there exists $\mathbf{f}_{S(y)} : \Sigma(y,S) \cap B_{d_g}(y,2\delta\sigma) \to \mathbb{R}^{n-m}$ such that, by Equations (201) and (206), we have that

$$\mathrm{Lip}\left(\mathbf{f}_{S(y)}\right) = 2\left(\mathbf{C_8}+\sqrt{\frac{8\left(\mathbf{C_7}+1\right)}{\omega_m}}\right)\delta_0 L \leq L, \tag{207}$$

which implies Equation (183). Furthermore,

$$\mathbf{f}_{S(y)}{}_{\restriction_{\mathbf{p}^{\Sigma(y,S(y))}(G)\cap B_{d_g}(y,2\delta\sigma)}} \equiv f_{S(y)}.$$

Note that there exists $\delta_0 > 0$ such that $2\delta < 1$, thus, by Equation (205), we also have

$$\sup_{x\in\Sigma(y,S(y))\cap B_{d_g}(y,2\delta\sigma)} \left\|\mathbf{f}_{S(y)}(x)\right\|_{g_e} \leq \mathbf{C_6}\delta^{\frac{1}{2(m+1)}}\sigma,$$

which implies Equation (184).
Since $G \subset \mathrm{Graph}^{\Sigma(y,S(y))}\left(\mathbf{f}_{S(y)}\right)$, by Equation (190), we have that

$$\frac{\|V\|\llcorner B_{d_g}(y,2\delta\sigma)\left(\mathrm{supp}(\|V\|)\setminus \mathrm{Graph}^{\Sigma(y,S(y))}\left(\mathbf{f}_{S(y)}\right)\right)}{\sigma^m} \leq \frac{\mathbf{C_3}}{L^2}\delta^{\frac{1}{m+1}},$$

which implies Equation (185). It thus remains only to prove Equation (186).
To check this, take $z_0 \in B_{d_g}(y,2\delta\sigma)\cap \mathrm{Graph}^{\Sigma(y,S)}\left(\mathbf{f}_{S(y)}\right)\setminus \mathrm{supp}(\|V\|)$ and let

$$t := \frac{3}{2}\mathrm{dist}_g^{\mathrm{supp}(\|V\|)}(z_0) \leq \frac{3}{2}d_g(z_0,y) \leq \frac{3}{2}2\delta\sigma = 3\delta\sigma, \tag{208}$$

because $y \in \mathrm{supp}(\|V\|)$. Note that

$$B_{d_g}\left(z_0,\frac{2}{3}t\right)\cap \mathrm{supp}(\|V\|) = \emptyset, \tag{209}$$

and

$$\overline{B_{d_g}\left(z_0,\frac{2}{3}t\right)}\cap \mathrm{supp}(\|V\|) \neq \emptyset.$$

By an analogous argument to that presented in Lemma 13.3.4, we obtain that

$$t^m \leq \frac{3^m e^{\ell(\rho)}}{2^{m-3}\omega_m}\left(\|V\|\left(B_{d_g}(z_0,t)\setminus \mathrm{Graph}^{\Sigma(y,S(y))}\left(\mathbf{f}_{S(y)}\right)\right) + \sigma^m E(y,\sigma,S(y),V)\right). \tag{210}$$

Now we observe that the collection of such balls $B_{d_g}(z_0,t)$ by definition cover all of $B_{d_g}(y,2\delta\sigma)\cap \mathrm{Graph}^{\Sigma(y,S)}\left(\mathbf{f}_{S(y)}\right)\setminus \mathrm{supp}(\|V\|)$, so by the 5-times covering lemma and Equation (210) we can find a pairwise disjoint collection $\left\{B_{d_g}\left(z_j,t_j\right)\right\}_{j\in\mathbb{N}}$ of such balls with

$$t_j^m \leq \frac{3^m e^{\ell(\rho)}}{2^{m-3}\omega_m}\left(\|V\|\left(B_{d_g}(z_j,t_j)\setminus \mathrm{Graph}^{\Sigma(y,S(y))}\left(\mathbf{f}_{S(y)}\right)\right) + \sigma^m E(y,\sigma,S(y),V)\right) \tag{211}$$

for each $j \in \mathbb{N}$ and $B_{d_g}(y,2\delta\sigma)\cap \mathrm{Graph}^{\Sigma(y,S)}\left(\mathbf{f}_{S(y)}\right)\setminus \mathrm{supp}(\|V\|) \subset \cup_{j\in\mathbb{N}} B_{d_g}\left(z_j,5t_j\right)$.

By [48, Theorem 9.12, p. 196] and Equation (208) we have that $t_j \leq 3\delta\sigma$ and then there exists $\rho_0 := \rho_0(m, K, r_H) > 0$ such that

$$\mathcal{H}^m_{d_g}\left(B_{d_g}\left(z_j, 5t_j\right) \cap \mathrm{Graph}^{\Sigma(y,S(y))}\left(\mathbf{f}_{S(y)}\right)\right) \leq 5^m \omega_m \left(2 + \frac{n^2K5^2}{6(m+2)}\right) t_i^m. \tag{212}$$

Since the collection $\left\{B_{d_g}\left(z_j, t_j\right)\right\}_{j\in\mathbb{N}}$ is disjoint, $B_{d_g}(y, 2\delta\sigma) \cap \mathrm{Graph}^{\Sigma(y,S)}\left(\mathbf{f}_{S(y)}\right) \setminus \mathrm{supp}(\|V\|) \subset \cup_{j\in\mathbb{N}} B_{d_g}\left(z_j, 5t_j\right)$, by Equations (185), (211) and (212), we have that

$$\begin{aligned}&\frac{\mathcal{H}^m_{d_g} \llcorner B_{d_g}(y, 2\delta\sigma)\left(\mathrm{Graph}^{\Sigma(y,S(y))}\left(\mathbf{f}_{S(y)}\right) \setminus \mathrm{supp}(\|V\|)\right)}{\sigma^m} \leq \\ &\leq \frac{5^m 3^m e^{\ell(\rho)}}{2^{m-3}}\left(2 + \frac{n^2K5^2}{6(m+2)}\right)\left(\frac{\mathbf{C_2}}{L^2}E(y,\sigma,S,V) + E(y,\sigma,S,V)\right).\end{aligned} \tag{213}$$

Since $0 < L \leq 1$ and we can choose $\delta_0 := \delta_0(m) > 0$ such that $0 < \delta_0 \leq \sqrt{2\omega_m 5^m}$, then $1 \leq \frac{\mathbf{C_2}}{L^2}$. By Equation (213) we have that

$$\frac{\mathcal{H}^m_{d_g} \llcorner B_{d_g}(y, 2\delta\sigma)\left(\mathrm{Graph}^{\Sigma(y,S(y))}\left(\mathbf{f}_{S(y)}\right) \setminus \mathrm{supp}(\|V\|)\right)}{\sigma^m} \leq \frac{\mathbf{C_5}}{L^2}\delta^{\frac{1}{m+1}},$$

which implies Equation (186). To conclude the proof, we choose $\delta_0 := \delta_0(m, n, p, K, r_H) > 0$ and $\rho_0 := \rho_0(m, K, r_H, p) > 0$ as the minimum value ensuring that all the above estimates hold. □

Corollary 13.5.3. *Let $m < p < \infty$, $\xi \in M^n$ and $0 < L \leq 1$. There exists $\delta_0 := \delta_0(m, n, p, K, r_H) > 0$ and $\rho_0 := \rho_0(m, K, r_H, p) > 0$ such that if $V \in \mathcal{AC}(\xi, \rho, \delta, p)$ for some $0 < \delta < (\delta_0 L)^{2(m+1)}$ and for some $0 < \rho < \min\left\{\rho_0, \sqrt{\delta}\right\}$, then, for all $y \in \mathrm{supp}(\|V\|) \cap B_{d_g}\left(\xi, \frac{\delta\rho}{2}\right)$, for all $\sigma \in]0, 2\delta\rho]$ and for all $S(y) \in \mathrm{Gr}(m, T_yM^n)$ such that Equation (176), Equation (177), and*

$$\sup_{t\in]0,(1-2\delta)\sigma]} \frac{1}{t^m}\int_{B_{d_g}(z,t)} \left\|P_{T_xV} - P_{T_x\Sigma(y,S(y))_x}\right\|^2_{\mathrm{HS}_{gx}} d\|V\|(x) \leq \delta_0^2 L^2 \tag{214}$$

hold for all $z \in \mathrm{supp}(\|V\|) \cap B_{d_g}(y, 2\delta\sigma)$, then there exists a Lipschitz function $\mathbf{f}_{S(y)} : B_{d_g}(y, \sigma) \cap \Sigma(y, S(y)) \to \mathbb{R}^{n-m}$ such that Equations (183) and (184) hold and

$$\mathrm{supp}(\|V\|) \cap B_{d_g}(y, \sigma) = \mathrm{Graph}^{\Sigma(y,S(y))}\left(\mathbf{f}_{S(y)}\right) \cap B_{d_g}(y, \sigma). \tag{215}$$

Proof. Let $\delta_0 := \delta_0(m, n, p, K, r_H) > 0$ and $\rho_0 := \rho_0(m, p, K, r_H) > 0$ given by Lemma 13.5.2. Suppose that $V \in \mathcal{AC}(\xi, \rho, \delta, p)$, for some $0 < \delta < (\delta_0 L)^{2(m+1)}$ and for some $0 < \rho < \min\left\{\rho_0, \sqrt{\delta}\right\}$. Consider $y \in \mathrm{supp}(\|V\|) \cap B_{d_g}\left(\xi, \frac{\delta\rho}{2}\right)$ and $\sigma \in]0, 2\delta\rho]$ arbitrary, and $S(y) \in \mathrm{Gr}(m, T_yM^n)$ such that Equations (176), (177) and (214) hold for all $z \in \mathrm{supp}(\|V\|) \cap B_{d_g}(y, 2\delta\sigma)$.

By Lemma 13.5.2, there a Lipschitz function $\mathbf{f}_{S(y)} : B_{d_g}(y, \sigma) \cap \Sigma(y, S(y)) \to \mathbb{R}^{n-m}$ such that Equations (183) and (184) hold.

Thus we have that the set G, defined in Equation (187), contains the set $\operatorname{supp}(\|V\|) \cap B_{d_g}(y, 2\delta\sigma)$, Which implies that

$$\operatorname{supp}(\|V\|) \cap B_{d_g}(y, 2\delta\sigma) \subset G \subset \operatorname{Graph}^{\Sigma(y,S(y))}\left(\mathbf{f}_{S(y)}\right). \tag{216}$$

Suppose, by contradiction, that there exists $z_0 \in B_{d_g}(y, 2\delta\sigma) \cap \operatorname{Graph}^{\Sigma(y,S)}(\mathbf{F}(y,S)) \setminus \operatorname{supp}(\|V\|)$ and let $t := \frac{3}{2}\operatorname{dist}_g^{\operatorname{supp}(\|V\|)}(z_0) > 0$. By Equation (216) we have that

$$B_{d_g}(z_0, t) \setminus \operatorname{Graph}^{\Sigma(y,S)}(\mathbf{F}(y,S)) \subset B_{d_g}(z_0, t) \setminus \Big(\operatorname{supp}(\|V\|) \cap B_{d_g}(y, 2\delta\sigma)\Big) \tag{217}$$

By Equations (210) and (217), we have that

$$t^m \leq \frac{3^m e^{\ell(\rho)}}{2^{m-3}\omega_m}\sigma^m E(y, \sigma, S(y), V). \tag{218}$$

By Equations (177) and (218), we have that

$$t^m \leq \left(\frac{3^m e^{\ell(\rho)}\mathbf{C}\delta_0^2 L^2}{2^{m-3}\omega_m}\right)\sigma^m,$$

where $\mathbf{C} = \mathbf{C}(m, n, p, K, r_H) > 0$. Since $\sigma \in]0, 2\delta\rho]$ is arbitrary, we conclude that for all $\sigma^m < \frac{2^{m-3}\omega_m}{3^m e^{\ell(\rho)}\mathbf{C}\delta_0^2 L^2}\frac{t^m}{2}$ we have a contradiction, because it implies that $t^m < \frac{t^m}{2}$ and $t > 0$.

Therefore, there exists $\sigma \in]0, 2\delta\rho]$ such that $B_{d_g}(y, 2\delta\sigma) \cap \operatorname{Graph}^{\Sigma(y,S)}(\mathbf{F}(y,S)) \setminus \operatorname{supp}(\|V\|) = \emptyset$ which implies that

$$B_{d_g}(y, 2\delta\sigma) \cap \operatorname{Graph}^{\Sigma(y,S)}(\mathbf{F}(y,S)) \subset \operatorname{supp}(\|V\|). \tag{219}$$

Thus Equations (216) and (219) implies Equation (215). □

13.6 HARMONIC APPROXIMATION, TILT-EXCESS DECAY AND MAIN REGULARITY THEOREM

The proof of Allard's Interior ε-Regularity Theorem (Theorem 13.6.8) rests on three main ingredients: the Lipschitz approximation, the monotonicity formula, and the Caccioppoli inequality, all of which were established in the previous sections. Having these in hand, the proof follows the structure of [91, Theorem 5.2, p. 146].

The previous sections required more significant adaptations from the Euclidean proofs due to the curvature of the ambient space. Here, this is no longer the case, and we only point out the main differences with respect to [91, Theorem 5.2, p. 146].

In the Harmonic Approximation, the main difference is that the Mean-Value property used in the Euclidean setting must be replaced by the Schauder Interior Estimates for elliptic operators.

Definition 13.6.1. Let $m < p < \infty$, $\xi \in M^n$, $0 < \delta < 1$, $0 < \rho < r_H$, and $V \in \mathcal{AC}(\xi, \rho, \delta, p)$. Consider $S(\xi) \in \mathrm{Gr}(m, T_\xi M^n)$ and $0 < \sigma < \rho$. We define

$$\mathbf{E}(\xi, \sigma, S(\xi), V) := \max\left\{ E(\xi, \sigma, S(\xi), V), \frac{1}{\delta}\left(\sigma^{p-m} \int_{B_{d_g}(\xi,\sigma)} \left\| \overrightarrow{H_g}(x) \right\|_{g_x}^p d\|V\|(x) \right)^{\frac{2}{p}} \right\}.$$

Remark 13.6.2. Let $m < p < \infty$, $\xi \in M^n$, $0 < \delta < 1$, $0 < \rho < r_H$, and $V \in \mathcal{AC}(\xi, \rho, \delta, p)$. Consider $S(\xi) \in \mathrm{Gr}(m, T_\xi M^n)$ and $0 < \sigma < \rho$. By Remark 13.3.2, we have that

$$\frac{1}{\delta}\left(\sigma^{p-m} \int_{B_{d_g}(\xi,\sigma)} \left\| \overrightarrow{H_g}(x) \right\|_{g_x}^p d\|V\|(x) \right)^{\frac{2}{p}} \leq \delta.$$

Theorem 13.6.3 (Tilt-Excess Decay Theorem)**.** *Let $m < p < \infty$ and $\xi \in M^n$. There exist $\eta := \eta(m, n, p, K, r_H) > 0$, $\delta_0 := \delta_0(m, n, p, K, r_H) > 0$ and $\rho_0 := \rho_0(m, p, K, r_H) > 0$ such that if $V \in \mathcal{AC}(\xi, \rho, \delta, p)$, for some $0 < \delta < \delta_0$ and for some $0 < \rho < \min\left\{\rho_0, \sqrt{\delta}\right\}$, then, for all $y \in \mathrm{supp}(\|V\|) \cap B_{d_g}\left(\xi, \frac{\delta\rho}{2}\right)$, for all $T(y) \in \mathrm{Gr}(m, T_y M^n)$, and for all $\sigma \in]0, 2\delta\rho]$, we have that there exists $S(y) \in \mathrm{Gr}(m, T_y M^n)$ such that*

$$\mathbf{E}(y, \eta\sigma, S(y), V) \leq \eta^{2\left(1-\frac{m}{p}\right)} \mathbf{E}(y, \sigma, T(y), V). \tag{220}$$

Proof. Let $\delta_0 := \delta_0(m, p, K, r_H) > 0$ and $\rho_0 := \rho_0(m, p, K, r_H) > 0$ given by Corollary 13.4.3. Suppose that $V \in \mathcal{AC}(\xi, \rho, \delta, p)$, for some $0 < \delta < \delta_0$ and for some $0 < \rho < \min\left\{\rho_0, \sqrt{\delta}\right\}$. Consider $y \in \mathrm{supp}(\|V\|) \cap B_{d_g}\left(\xi, \frac{\delta\rho}{2}\right)$, $T(y) \in \mathrm{Gr}(m, T_y M^n)$ and $\sigma \in]0, 2\delta\rho]$ arbitrary.

By Corollary 13.4.3 there exists $\mathbf{S}(y) \in \mathrm{Gr}(m, T_y M^n)$ such that Equations (176) and (177) holds.

Note that we have the following two possibilities

$$E(y, \sigma, T(y), V) \leq \mathbf{C_1} \delta^{\frac{1}{m+1}} \tag{221}$$

$$E(y, \sigma, T(y), V) > \mathbf{C_1} \delta^{\frac{1}{m+1}}, \tag{222}$$

where $\mathbf{C_1} > 0$ is the constant given by Equation (177). If Equation (222) holds then, by Equation (177), we have that

$$\mathbf{E}(y, \sigma, \mathbf{S}(y), V) \leq \mathbf{E}(y, \sigma, T(y), V), \tag{223}$$

which implies Equation (220) with $\eta = 1$. From now on assume that Equation (221) holds. We claim that

$$\sup_{x \in \mathrm{supp}(\|V\|) \cap B_{d_g}(y,\sigma)} \mathrm{dist}_g^{\Sigma(y,T(y))}(x) \leq \mathbf{C_2} \delta^{\frac{1}{2(m+1)}} \sigma, \tag{224}$$

where $\mathbf{C_2} = \mathbf{C_2}(m, n, p, r_H, K) > 0$. In fact, note that

$$\sup_{x \in \mathrm{supp}(\|V\|) \cap B_{d_g}(y,\sigma)} \mathrm{dist}_g^{\Sigma(y,T(y))}(x)$$

$$\leq \sup_{x\in \operatorname{supp}(\|V\|)\cap B_{d_g}(y,\sigma)} \operatorname{dist}_g^{\Sigma(y,\mathbf{S}(y))}(x) + \sup_{x\in \operatorname{supp}(\|V\|)\cap B_{d_g}(y,\sigma)} d_g\left(\mathbf{p}^{\Sigma(y,\mathbf{S}(y))}(x), \mathbf{p}^{\Sigma(y,T(y))}(x)\right). \tag{225}$$

By Lemma 13.1.14 we have, for all $x \in \operatorname{supp}(\|V\|) \cap B_{d_g}(y,\sigma)$, that

$$\begin{aligned}\frac{1}{2}\left\|P_{T(y)}\left(\exp_y^{-1}(x)\right) - P_{\mathbf{S}(y)}\left(\exp_y^{-1}(x)\right)\right\|_{g_y} &\leq d_g\left(\mathbf{p}^{\Sigma(y,\mathbf{S}(y))}(x), \mathbf{p}^{\Sigma(y,T(y))}(x)\right)\\ &\leq 2\left\|P_{T(y)}\left(\exp_y^{-1}(x)\right) - P_{\mathbf{S}(y)}\left(\exp_y^{-1}(x)\right)\right\|_{g_y}\end{aligned} \tag{226}$$

Thus, by Equations (176), (225) and (226), we have that

$$\sup_{x\in \operatorname{supp}(\|V\|)\cap B_{d_g}(y,\sigma)} \operatorname{dist}_g^{\Sigma(y,T(y))}(x) \leq \mathbf{C_3}\delta^{\frac{1}{2(m+1)}}\sigma + 2\left\|P_{T(y)} - P_{\mathbf{S}(y)}\right\|_{\mathrm{HS}_{g_y}}\sigma, \tag{227}$$

where $\mathbf{C_3} = \mathbf{C_3}(m,p,K) > 0$. By Lemma 13.1.13 and the squared triangle inequality we have, for all $x \in \operatorname{supp}(\|V\|) \cap B_{d_g}(y,\sigma)$,

$$\left\|P_{T(y)} - P_{\mathbf{S}(y)}\right\|^2_{\mathrm{HS}_{g_y}} \leq 4\left\|P_{T_x\Sigma(y,\mathbf{S}(y))_x} - P_{T_xV}\right\|^2_{\mathrm{HS}_{g_x}} + 4\left\|P_{T_xV} - P_{T_x\Sigma(y,T(y))_x}\right\|^2_{\mathrm{HS}_{g_x}} \tag{228}$$

By Lemma 13.3.4 there exists $\delta_0 := \delta_0(m,p,K,r_H) > 0$ and $\rho_0 := \rho_0(m,p,K,r_H) > 0$ such that

$$\frac{1}{\sigma^m}\int_{B_{d_g}(y,\sigma)}\left\|P_{\mathbf{S}(y)} - P_{T(y)}\right\|^2_{\mathrm{HS}_{g_y}} d\|V\|(x) \geq \left\|P_{\mathbf{S}(y)} - P_{T(y)}\right\|^2_{\mathrm{HS}_{g_y}}\frac{\omega_m}{2}. \tag{229}$$

Note that there exists $\delta_0 > 0$ such that $\delta^{\frac{1}{m+1}} \leq \delta^{\frac{1}{2(m+1)}}$, therefore, by Equations (177), (221) and (227) to (229), we conclude the claim.

Since $T(y) \in \mathrm{Gr}(m, T_yM^n)$ is such that Equations (221) and (224) hold we have by the Lemma 13.5.2 that there exists $\delta_0 := \delta_0(m,n,K,r_H,p) > 0$, $\rho_0 := \rho_0(m,K,r_H,p) > 0$, and a Lipschitz function $\mathbf{f}_{T(y)} : B_{d_g}(y,\sigma) \cap \Sigma(y,T(y)) \to \mathbb{R}^{n-m}$ such that

$$\operatorname{Lip}(\mathbf{f}_{T(y)}) \leq 1,$$

$$\sup_{x\in\Sigma(y,T(y))\cap B_{d_g}(y,2\delta\sigma)}\left\|\mathbf{f}_{T(y)}(x)\right\|_{g_e} \leq \mathbf{C_1}\delta^{\frac{1}{2(m+1)}}\sigma, \tag{230}$$

$$\|V\|\llcorner B_{d_g}(y,\sigma)\left(\operatorname{supp}(\|V\|)\setminus \operatorname{Graph}^{\Sigma(y,T(y))}\left(\mathbf{f}_{T(y)}\right)\right) \leq \mathbf{C_3}\mathbf{E}(y,\sigma,T(y),V)\sigma^m, \tag{231}$$

where $\mathbf{C_3} > 0$ is the constant given by Equation (185) and

$$\mathcal{H}^m_{d_g}\llcorner B_{d_g}(y,\sigma)\left(\operatorname{Graph}^{\Sigma(y,T(y))}\left(\mathbf{f}_{T(y)}\right)\setminus \operatorname{supp}(\|V\|)\right) \leq \mathbf{C_4}\mathbf{E}(y,\sigma,T(y),V)\sigma^m,$$

where $\mathbf{C_4} > 0$ is given by Equation (186). Furthermore, there exists $\delta_0 := \delta_0(m,n,p,r_H,K) > 0$ such that $\mathbf{C_2}\delta^{\frac{1}{2(m+1)}} < \frac{1}{4}$ and, by Equation (224), we conclude that

$$\operatorname{supp}(\|V\|)\cap B_{d_g}(y,\sigma)\cap \operatorname{Tube}\left(\Sigma(y,T(y)^\perp),\frac{\sigma}{4}\right) \subset$$

$$\subset \text{Tube}\left(\Sigma(y,T(y)),\frac{\sigma}{4}\right)\cap \text{Tube}\left(\Sigma(y,T(y)^{\perp}),\frac{\sigma}{4}\right)\subset$$
$$\subset B_{d_g}\left(y,\frac{\sigma}{3}\right). \tag{232}$$

Our aim now is to prove that each component of the Lipschitz function $\mathbf{f}_{T(y)}$ is well-approximated by a harmonic function.

Let $\left(B_{d_g}\left(y,\frac{\sigma}{3}\right),\{x_i\}_{i=1}^n\right)$ a Fermi coordinate chart at y for $\Sigma(y,T(y))$ such that $\frac{\partial}{\partial x_i}(x)=F_i(\mathbf{p}^{\Sigma(y,T(y))}(x))\left(\text{dist}^{\Sigma(y,T(y))}(x)\right)$ in the notation of Chapter 13 for all $x\in B_{d_g}\left(y,\frac{\sigma}{3}\right)$ and for all $i=1,\dots,n$. Let $\zeta\in\mathcal{C}^1_c\left(B_{d_g}\left(y,\frac{\sigma}{3}\right)\right)$ and define, for each $j=m+1,\dots,n$, the vector field $X(x):=\zeta(x)\frac{\partial}{\partial x_j}(x)$, for all $x\in B_{d_g}\left(y,\frac{\sigma}{3}\right)$. Let $Z(x)\in \text{Gr}(m,T_xM^n)$ such that $\text{span}\{e_1(Z(x)),\dots,e_m(Z(x))\}=Z(x)$, thus for all $j=m+1,\dots,n$, we have that

$$\text{div}_{Z(x)}(X)=\left\langle P_{Z(x)}\left(\frac{\partial}{\partial x_j}\right),P_{Z(x)}\left(\nabla_g(\zeta)\right)\right\rangle_g+\zeta\text{div}_{Z(x)}\left(\frac{\partial}{\partial x_j}\right)$$

By Remark 13.3.2 and Remark 13.3.3 we have that there exists a countably m-rectifiable set $\Gamma\subset M^n$ and $\theta\in L^1_{loc}(\Gamma,]0,\infty[)$ such that $V=V(\Gamma,\theta,g)$. Thus, by Disintegration Lemma, we conclude that

$$\int_{B_{d_g}(y,\frac{\sigma}{3})\cap\Gamma}\left\langle P_{T_x\Gamma}\left(\frac{\partial}{\partial x_j}(x)\right),P_{T_x\Gamma}\left(\nabla_g(\zeta)(x)\right)\right\rangle_{g_x}d\|V\|(x)=-\int_{B_{d_g}(y,\frac{\sigma}{3})\cap\Gamma}\left\langle\overrightarrow{H_g}(x),\zeta(x)\frac{\partial}{\partial x_j}(x)\right\rangle_{g_x}d\|V\|(x)$$
$$-\int_{B_{d_g}(y,\frac{\sigma}{3})\cap\Gamma}\zeta(x)\text{div}_{T_x\Gamma}\left(\frac{\partial}{\partial x_j}\right)(x)d\|V\|(x). \tag{233}$$

We denote by $\nabla^{\Gamma}(f)(x):=P_{T_x\Gamma}(\nabla_g(f)(x))$, for every $f\in\mathcal{C}^1(M^n)$ and for all $x\in M^n$. Also note that by the construction of $\mathbf{f}_{T(y)}$ in Lemma 13.5.2 we have that $x_j(x)=\tilde{\mathbf{f}}^j_{T(y)}(x)$ for all $x\in\Gamma\cap\text{Graph}^{\Sigma(y,T(y))}\left(\mathbf{f}_{T(y)}\right)$ and for all $j=m+1,\dots,n$, where $\tilde{\mathbf{f}}^j_{T(y)}$ is defined by $\tilde{\mathbf{f}}^j_{T(y)}(x):=\mathbf{f}^j_{T(y)}\left(\mathbf{p}^{\Sigma(y,T(y))}(x)\right)$, for all $x\in B_{d_g}(y,\sigma)$ and for all $j=m+1,\dots,n$. Therefore,

$$\nabla^{\Gamma}(x_j)(x)=\nabla^{\Gamma}\left(\tilde{\mathbf{f}}^j_{T(y)}\right)(x), \tag{234}$$

for $\|V\|$-a.e. $x\in\Gamma\cap\text{Graph}^{\Sigma(y,T(y))}\left(\mathbf{f}_{T(y)}\right)$ and for all $j=m+1,\dots,n$. Hence Equations (233) and (234) implies that

$$\left|\int_{B_{d_g}(y,\frac{\sigma}{3})\cap\Gamma\cap\text{Graph}^{\Sigma(y,T(y))}(\mathbf{f}_{T(y)})}\left\langle\nabla^{\Gamma}\left(\tilde{\mathbf{f}}^j_{T(y)}\right)(x),\nabla^{\Gamma}(\zeta)(x)\right\rangle_{g_x}d\|V\|(x)\right|\leq$$
$$\leq\sup_{x\in B_{d_g}(y,\frac{\sigma}{3})\cap\Gamma}(|\zeta(x)|)\int_{B_{d_g}(y,\frac{\sigma}{3})\cap\Gamma}\left|\left\langle\overrightarrow{H_g}(x),\frac{\partial}{\partial x_j}(x)\right\rangle_{g_x}\right|d\|V\|(x)$$
$$+\sup_{x\in B_{d_g}(y,\frac{\sigma}{3})\cap\Gamma}(|\zeta(x)|)\int_{B_{d_g}(y,\frac{\sigma}{3})\cap\Gamma}\left|\text{div}_{T_x\Gamma}\left(\frac{\partial}{\partial x_j}\right)(x)\right|d\|V\|(x)$$
$$+\int_{\left(B_{d_g}(y,\frac{\sigma}{3})\cap\Gamma\right)\setminus\text{Graph}^{\Sigma(y,T(y))}(\mathbf{f}_{T(y)})}\left|\left\langle\nabla^{\Gamma}\left(x_j\right)(x),\nabla^{\Gamma}(\zeta)(x)\right\rangle_{g_x}\right|d\|V\|(x). \tag{235}$$

Now we analyze each term on the right side of Equation (235). Firstly note that there exists $\rho_0 := \rho_0(g) > 0$ such that $g_{jj}(x) = O(1)$, for all $x \in B_{d_g}\left(y, \frac{\sigma}{3}\right)$. Furthermore, by Lemma 13.3.4 there exists $\delta_0 := \delta_0(m, p, K, r_H) > 0$ and $\rho_0 := \rho_0(m, p, K, r_H) > 0$ such that $\|V\|\left(B_{d_g}(y, \sigma) \cap \Gamma\right) \leq 2\omega_m \sigma^m$, thus by Cauchy-Schwarz and Hölder inequalities we have that

$$\int_{B_{d_g}\left(y, \frac{\sigma}{3}\right)\cap\Gamma} \left| \left\langle \overrightarrow{H_g}(x), \frac{\partial}{\partial x_j}(x) \right\rangle_{g_x} \right| d\|V\|(x) \leq 2\left(2\omega_m\right)^{\frac{p-1}{p}} \delta^{\frac{1}{2}} \mathbf{E}(y, \sigma, T(y), V)^{\frac{1}{2}} \sigma^{m-1}. \tag{236}$$

For all $x \in B_{d_g}\left(y, \frac{\sigma}{3}\right)$ consider $\{e_1, \ldots, e_m\}$ an orthonormal basis for $T_x\Gamma$ and note that

$$\operatorname{div}_{T_x\Gamma}\left(\frac{\partial}{\partial x_j}\right) = \sum_{i=1}^{m}\sum_{k=1}^{n}\sum_{l=1}^{n}\sum_{\alpha=1}^{n} \left\langle e_i, \frac{\partial}{\partial x_k} \right\rangle \left\langle e_i, \frac{\partial}{\partial x_l} \right\rangle g_{\alpha l} \Gamma^{\alpha}_{kj}. \tag{237}$$

Note that there exists $\rho_0 := \rho_0(m, n, K) > 0$ such that $\Gamma^{\alpha}_{kj}(x) = o(1)$, for all $x \in B_{d_g}\left(y, \frac{\sigma}{3}\right)$ and for all $k, j, \alpha = 1, \ldots, n$, and $g_{\alpha l}(x) = O(1)$, for all $x \in B_{d_g}\left(y, \frac{\sigma}{3}\right)$ and for all $\alpha, l = 1, \ldots, n$. Therefore, by Cauchy–Schwarz inequality and Equation (237), we have that

$$\int_{B_{d_g}\left(y, \frac{\sigma}{3}\right)\cap\Gamma} \left| \operatorname{div}_{T_x\Gamma}\left(\frac{\partial}{\partial x_j}\right)(x) \right| d\|V\|(x) \leq 2mn^3 \mathbf{E}(y, \sigma, T(y), V)\sigma^{-1} \|V\|\left(B_{d_g}\left(y, \frac{\sigma}{3}\right) \cap \Gamma\right).$$

By Lemma 13.3.4 there exists $\delta_0 := \delta_0(m, p, K, r_H) > 0$ and $\rho_0 := \rho_0(m, p, K, r_H) > 0$ such that $\|V\|\left(B_{d_g}(y, \sigma) \cap \Gamma\right) \leq 2\omega_m \sigma^m$, thus

$$\int_{B_{d_g}\left(y, \frac{\sigma}{3}\right)\cap\Gamma} \left| \operatorname{div}_{T_x\Gamma}\left(\frac{\partial}{\partial x_j}\right)(x) \right| d\|V\|(x) \leq 4\omega_m mn^3 \mathbf{E}(y, \sigma, T(y), V)\sigma^{m-1} \tag{238}$$

Finally, by Cauchy–Schwarz inequality and Equation (231), we have that

$$\begin{aligned} &\int_{\left(B_{d_g}\left(y, \frac{\sigma}{3}\right)\cap\Gamma\right)\setminus \operatorname{Graph}^{\Sigma(y,T(y))}\left(\mathbf{f}_{T(y)}\right)} \left| \left\langle \nabla^{\Gamma}\left(x_j\right)(x), \nabla^{\Gamma}(\zeta)(x) \right\rangle_{g_x} \right| d\|V\|(x) \leq \\ &\leq \sup_{x \in B_{d_g}\left(y, \frac{\sigma}{3}\right)\cap\Gamma} \left\| \nabla^{\Gamma}(\zeta)(x) \right\|_{g_x} \mathbf{C_3} \mathbf{E}(y, \sigma, T(y), V)\sigma^m. \end{aligned} \tag{239}$$

Thus, by Equations (235), (236), (238) and (239), we obtain that

$$\begin{aligned} &\left| \frac{1}{\sigma^m} \int_{B_{d_g}\left(y, \frac{\sigma}{3}\right)\cap\Gamma\cap\operatorname{Graph}^{\Sigma(y,T(y))}\left(\mathbf{f}_{T(y)}\right)} \left\langle \nabla^{\Gamma}\left(\tilde{\mathbf{f}}^j_{T(y)}\right)(x), \nabla^{\Gamma}(\zeta)(x) \right\rangle_{g_x} d\|V\|(x) \right| \leq \\ &\leq \mathbf{C_5} \left(\delta^{\frac{1}{2}} \mathbf{E}(y, \sigma, T(y), V)^{\frac{1}{2}} + \mathbf{E}(y, \sigma, T(y), V) \right) \sup_{x \in B_{d_g}\left(y, \frac{\sigma}{3}\right)\cap\Gamma} \left\| \nabla^{\Gamma}(\zeta)(x) \right\|_{g_x}. \end{aligned} \tag{240}$$

where $\mathbf{C_5} = \mathbf{C_5}(m, p, K, r_H) > 0$. Furthermore, by Equations (130) and (234) we have that

$$\frac{1}{\sigma^m} \int_{B_{d_g}\left(y, \frac{\sigma}{3}\right)\cap\Gamma\cap\operatorname{Graph}^{\Sigma(y,T(y))}\left(\mathbf{f}_{T(y)}\right)} \left\| \nabla^{\Gamma}\left(\tilde{\mathbf{f}}^j_{T(y)}\right)(x) \right\|^2_{g_x} d\|V\|(x) \leq \frac{1}{2} \mathbf{E}(y, \sigma, T(y), V). \tag{241}$$

Now let $\gamma \in \mathcal{C}^1_c\left(B_{d_g}\left(y,\frac{\sigma}{3}\right)\cap\Sigma(y,T(y))\right)$ and $\tilde{\gamma}(x) := \gamma\left(\mathbf{p}^{\Sigma(y,T(y))}(x)\right)$. Note that $\operatorname{supp}(\tilde{\gamma}) = \operatorname{supp}(\gamma)\cap \operatorname{Tube}\left(\Sigma\left(y,T(y)^{\perp}\right),\frac{\sigma}{3}\right) \subset \left(B_{d_g}\left(y,\frac{\sigma}{3}\right)\cap\Sigma(y,T(y))\right)\cap \operatorname{Tube}\left(\Sigma\left(y,T(y)^{\perp}\right),\frac{\sigma}{3}\right)$, which implies, by Equation (232), that there is a function $\tilde{\zeta} \in \mathcal{C}^1_c\left(B_{d_g}\left(y,\frac{\sigma}{3}\right)\right)$ which is equal $\tilde{\gamma}$ in a neighborhood of $\operatorname{supp}(\|V\|)\cap\operatorname{supp}(\tilde{\gamma})$, and hence we can use $\tilde{\gamma}$ in plane of ζ in the above calculations.

Moreover, for all $B_{d_g}\left(y,\frac{\sigma}{3}\right)$, by Equation (130) and Cauchy–Schwarz inequality, we have that

$$\left|\left\langle\nabla^{\Gamma}\left(\tilde{\mathbf{f}}^j_{T(y)}\right)(x),\nabla^{\Gamma}(\tilde{\gamma})(x)\right\rangle_{g_x} - \left\langle\nabla_g\left(\tilde{\mathbf{f}}^j_{T(y)}\right)(x),\nabla_g(\tilde{\gamma})(x)\right\rangle_{g_x}\right| \leq \\ \leq \frac{1}{2}\left\|P_{T_x\Gamma} - P_{T_x\Sigma(y,T(y))_x}\right\|^2_{\mathrm{HS}_{g_x}}\left\|\nabla_g(\tilde{\gamma})(x)\right\|_{g_x}. \tag{242}$$

Note that by Equations (240) and (242) we have that

$$\left|\frac{1}{\sigma^m}\int_{B_{d_g}\left(y,\frac{\sigma}{3}\right)\cap\Gamma\cap\mathrm{Graph}^{\Sigma(y,T(y))}\left(\mathbf{f}_{T(y)}\right)}\left\langle\nabla_g\left(\tilde{\mathbf{f}}^j_{T(y)}\right)(x),\nabla_g(\tilde{\gamma})(x)\right\rangle_{g_x} d\|V\|(x)\right| \leq \\ \leq \left(\mathbf{C_5}\delta^{\frac{1}{2}}\mathbf{E}(y,\sigma,T(y),V)^{\frac{1}{2}} + \left(\mathbf{C_5}+\frac{1}{2}\right)\mathbf{E}(y,\sigma,T(y),V)\right)\sup_{x\in B_{d_g}\left(y,\frac{\sigma}{3}\right)\cap\Gamma}\left\|\nabla_g(\tilde{\gamma})(x)\right\|_{g_x}. \tag{243}$$

Also note that by Equations (241) and (242) (for $\tilde{\mathbf{f}}^j_{T(y)}$ in place of $\tilde{\gamma}$), we have that

$$\frac{1}{\sigma^m}\int_{B_{d_g}\left(y,\frac{\sigma}{3}\right)\cap\Gamma\cap\mathrm{Graph}^{\Sigma(y,T(y))}\left(\mathbf{f}_{T(y)}\right)}\left\|\nabla_g\left(\tilde{\mathbf{f}}^j_{T(y)}\right)(x)\right\|^2_{g_x} d\|V\|(x) \leq \mathbf{E}(y,\sigma,T(y),V). \tag{244}$$

By an adaptation of the Area formula [41, Theorem 3.2.3, p. 243] for the context of normal graphs in Riemannian manifolds, Equations (243) and (244) we have that

$$\left|\frac{1}{\sigma^m}\int_{B_{d_g}\left(y,\frac{\sigma}{3}\right)\cap\Sigma(y,T(y))}\left\langle\nabla_g\left(\mathbf{f}^j_{T(y)}\right)(x),\nabla_g(\gamma)(x)\right\rangle_{g_x}\theta(\mathbf{F}(x))\operatorname{Jac}(\mathbf{F})(x)d\mathcal{L}^m_{d_g}(x)\right| \leq \\ \leq \left(\mathbf{C_5}\delta^{\frac{1}{2}}\mathbf{E}(y,\sigma,T(y),V)^{\frac{1}{2}} + \left(\mathbf{C_5}+\frac{1}{2}\right)\mathbf{E}(y,\sigma,T(y),V)\right)\sup_{x\in B_{d_g}\left(y,\frac{\sigma}{3}\right)\cap\Gamma}\left\|\nabla_g(\tilde{\gamma})(x)\right\|_{g_x} \tag{245}$$

and

$$\frac{1}{\sigma^m}\int_{B_{d_g}\left(y,\frac{\sigma}{3}\right)\cap\Sigma(y,T(y))}\left\|\nabla_g\left(\mathbf{f}^j_{T(y)}\right)(x)\right\|^2_{g_x}\theta(\mathbf{F}(x))\operatorname{Jac}(\mathbf{F})(x)d\mathcal{L}^m_{d_g}(x) \leq \mathbf{E}(y,\sigma,T(y),V), \tag{246}$$

where $\mathbf{F} : B_{d_g}\left(y,\frac{\sigma}{3}\right)\cap\Sigma(y,T(y)) \to M^n$ is defined by

$$\mathbf{F}(x) := \exp_x\left(\sum_{j=m+1}^{n}\mathbf{f}^j_{T(y)}(x)\frac{\partial}{\partial x_j}(x)\right)$$

and $\mathrm{Jac}(\mathbf{F})(x)$ is the Jacobian of $\mathbf{F}$ at x, for all $x \in B_{d_g}\left(y, \frac{\sigma}{3}\right) \cap \Sigma(y, T(y))$. By Lemma 13.3.4 we have that there exists $\delta_0 := \delta_0(m, p, K, r_H) > 0$ and $\rho_0 := \rho_0(m, p, K, r_H) > 0$ such that $1 \leq \theta(x) \leq 1 + \mathbf{C_6}\delta \leq 2$, where $\mathbf{C_6}$ is given by Equation (143), for all $x \in B_{d_g}\left(y, \frac{\sigma}{3}\right)$, and by standard estimates on $\mathrm{Jac}(\mathbf{F})$, that is, $1 \leq \mathrm{Jac}(\mathbf{F})(x) \leq 1 + \mathbf{C_7}\left\|D\mathbf{f}_{T(y)}(x)\right\|^2$, for all $x \in B_{d_g}\left(y, \frac{\sigma}{3}\right)$ and for some constant $\mathbf{C_7} := \mathbf{C_7}(m, n, K) > 0$, we conclude from Equations (245) and (246) that

$$\left|\frac{1}{\sigma^m}\int_{B_{d_g}\left(y,\frac{\sigma}{3}\right)\cap\Sigma(y,T(y))}\left\langle\nabla_g\left(\mathbf{f}^j_{T(y)}\right)(x), \nabla_g(\gamma)(x)\right\rangle_{g_x} d\mathcal{L}^m_{d_g}(x)\right| \leq$$
$$\leq \mathbf{C_8}\left(\delta^{\frac{1}{2}}\mathbf{E}(y,\sigma,T(y),V)^{\frac{1}{2}} + \mathbf{E}(y,\sigma,T(y),V)\right)\sup_{x\in B_{d_g}\left(y,\frac{\sigma}{3}\right)\cap\Gamma\cap\Sigma(y,T(y))}\left\|\nabla_g(\gamma)(x)\right\|_{g_x}$$

and

$$\frac{1}{\sigma^m}\int_{B_{d_g}\left(y,\frac{\sigma}{3}\right)\cap\Sigma(y,T(y))}\left\|\nabla_g\left(\mathbf{f}^j_{T(y)}\right)(x)\right\|^2_{g_x} d\mathcal{L}^m_{d_g}(x) \leq \mathbf{C_8}\mathbf{E}(y,\sigma,T(y),V),$$

for some constant $\mathbf{C_8} := \mathbf{C_8}(m, n, p, K, r_H) > 0$. Define $\overline{\mathbf{f}}^j_{T(y)} \equiv (\mathbf{C_8}\mathbf{E}(y,\sigma,T(y),V))^{-\frac{1}{2}}\mathbf{f}^j_{T(y)}$, select $\delta_0 := \delta_0(m, n, p, K, r_H) > 0$ such that $\delta_0 \leq \frac{1}{\mathbf{C_8}}$, and note that

$$\left|\frac{1}{\sigma^m}\int_{B_{d_g}\left(y,\frac{\sigma}{3}\right)\cap\Sigma(y,T(y))}\left\langle\nabla_g\left(\overline{\mathbf{f}}^j_{T(y)}\right)(x), \nabla_g(\gamma)(x)\right\rangle_{g_x} d\mathcal{L}^m_{d_g}(x)\right| \leq$$
$$\leq (\mathbf{C_8}\mathbf{E}(y,\sigma,T(y),V))^{\frac{1}{2}}\sup_{x\in B_{d_g}\left(y,\frac{\sigma}{3}\right)\cap\Gamma\cap\Sigma(y,T(y))}\left\|\nabla_g(\gamma)(x)\right\|_{g_x}$$

and

$$\frac{1}{\sigma^m}\int_{B_{d_g}\left(y,\frac{\sigma}{3}\right)\cap\Sigma(y,T(y))}\left\|\nabla_g\left(\overline{\mathbf{f}}^j_{T(y)}\right)(x)\right\|^2_{g_x} d\mathcal{L}^m_{d_g}(x) \leq 1.$$

Therefore, by the Harmonic Estimates, we have that there exists a harmonic function $\overline{u}^j$ on $B_{d_g}\left(y, \frac{\sigma}{3}\right) \cap \Gamma \cap \mathrm{Graph}^{\Sigma(y,T(y))}\left(\mathbf{f}_{T(y)}\right)$ such that

$$\frac{1}{\sigma^{m+2}}\int_{B_{d_g}\left(y,\frac{\sigma}{3}\right)\cap\Sigma(y,T(y))}\left|\overline{u}^j(x) - \overline{\mathbf{f}}^j_{T(y)}(x)\right|^2 d\mathcal{L}^m_{d_g}(x) \leq \varepsilon,$$

and

$$\frac{1}{\sigma^m}\int_{B_{d_g}\left(y,\frac{\sigma}{3}\right)\cap\Sigma(y,T(y))}\left\|\nabla_g\left(\overline{u}^j\right)(x)\right\|^2_{g_x} d\mathcal{L}^m_{d_g}(x) \leq 1.$$

And finally, by taking $u^j \equiv (\mathbf{C_8}\mathbf{E}(y,\sigma,T(y),V))^{-\frac{1}{2}}\overline{u}^j$ we obtain that

$$\frac{1}{\sigma^{m+2}}\int_{B_{d_g}\left(y,\frac{\sigma}{3}\right)\cap\Sigma(y,T(y))}\left|u^j(x) - \mathbf{f}^j_{T(y)}(x)\right|^2 d\mathcal{L}^m_{d_g}(x) \leq \varepsilon\mathbf{C_8}\mathbf{E}(y,\sigma,T(y),V), \tag{247}$$

and

$$\frac{1}{\sigma^m}\int_{B_{d_g}\left(y,\frac{\sigma}{3}\right)\cap\Sigma(y,T(y))}\left\|\nabla_g\left(u^j\right)(x)\right\|^2_{g_x} d\mathcal{L}^m_{d_g}(x) \leq \mathbf{C_8}\mathbf{E}(y,\sigma,T(y),V). \tag{248}$$

Now define $\lambda : B_{d_g}\left(y, \frac{\sigma}{3}\right) \cap \Gamma \cap \Sigma\left(y, T(y)\right) \to \mathbb{R}^{n-m}$ by $\lambda(x) := \left(\lambda^{m+1}(x), \ldots, \lambda^n(x)\right)$ where

$$\lambda^j(x) := u^j(y) + \left\langle \nabla_g(u^j)(y), \exp_y^{-1}(x) \right\rangle_{g_y},$$

for all $x \in B_{d_g}\left(y, \frac{\sigma}{3}\right) \cap \Gamma \cap \Sigma\left(y, T(y)\right)$ and for all $j = m+1, \ldots, n$. Therefore, by the squared triangle inequality,

$$\begin{aligned}
&\frac{1}{(\eta\sigma)^{m+2}} \int_{B_{d_g}(y,\eta\sigma)\cap\Sigma(y,T(y))} \left\|\lambda(x) - \mathbf{f}_{T(y)}(x)\right\|_{g_e}^2 d\mathcal{L}_{d_g}^m(x) \leq \\
&\leq \frac{2}{(\eta\sigma)^{m+2}} \int_{B_{d_g}(y,\eta\sigma)\cap\Sigma(y,T(y))} \left\|\lambda(x) - u(x)\right\|_{g_e}^2 d\mathcal{L}_{d_g}^m(x) \\
&+ \frac{2}{(\eta\sigma)^{m+2}} \int_{B_{d_g}(y,\eta\sigma)\cap\Sigma(y,T(y))} \left\|u(x) - \mathbf{f}_{T(y)}(x)\right\|_{g_e}^2 d\mathcal{L}_{d_g}^m(x). \qquad (249)
\end{aligned}$$

Now we analyze each term on the right side of Equation (249). By an analogous argument to the one from the proof of the Lipschitz Approximation Lemma 13.5.2 we have that there exists $\rho_0 := \rho_0(K, r_H) > 0$ such that $\mathcal{L}_{d_g}^m\left(B_{d_g}\left(y, \eta\sigma\right) \cap \Sigma(y, T(y))\right) \leq \omega_m \eta^m \sigma^m \left(2 + \frac{n^2 K}{6(m+2)}\right)$. Moreover, by the Harmonic Estimates and Equation (248), we have that

$$\begin{aligned}
&\frac{2}{(\eta\sigma)^{m+2}} \int_{B_{d_g}(y,\eta\sigma)\cap\Sigma(y,T(y))} \left\|\lambda(x) - u(x)\right\|_{g_e}^2 d\mathcal{L}_{d_g}^m(x) \\
&\leq \mathbf{C_9} 2\omega_m (n-m-1) \left(2 + \frac{n^2 K}{6(m+2)}\right) \eta^2 \mathbf{E}(y, \sigma, T(y), V). \qquad (250)
\end{aligned}$$

For some constant $\mathbf{C_9} := \mathbf{C_9}(m, n, p, K, r_H) > 0$. On the other hand, by Equation (247), we have that

$$\frac{2}{(\eta\sigma)^{m+2}} \int_{B_{d_g}(y,\eta\sigma)\cap\Sigma(y,T(y))} \left\|u(x) - \mathbf{f}_{T(y)}(x)\right\|_{g_e}^2 d\mathcal{L}_{d_g}^m(x) \leq \mathbf{C_9} 2(n-m-1) \frac{\varepsilon}{\eta^{m+2}} \mathbf{E}(y, \sigma, T(y), V). \qquad (251)$$

Therefore, by Equations (249) to (251), we conclude that there is a constant $\mathbf{C_{10}} := \mathbf{C_{10}}(m, n, p, K, r_H) > 0$ such that

$$\frac{1}{(\eta\sigma)^{m+2}} \int_{B_{d_g}(y,\eta\sigma)\cap\Sigma(y,T(y))} \left\|\lambda(x) - \mathbf{f}_{T(y)}(x)\right\|_{g_e}^2 d\mathcal{L}_{d_g}^m(x) \leq \mathbf{C_{10}} \left(\frac{\varepsilon}{\eta^{m+2}} \mathbf{E}(y, \sigma, T(y), V) + \eta^2 \mathbf{E}(y, \sigma, T(y), V)\right) \qquad (252)$$

Now let $S(y) := \operatorname{span}\left\{\sum_{j=m+1}^n \frac{\partial u^j}{\partial x_i}(y) \frac{\partial}{\partial x_i}(y)\right\}_{i=1}^m$ and $M^n \ni \tau(y) := \exp_y\left(\sum_{j=m+1}^n u^j(y) \frac{\partial}{\partial x_j}(y)\right)$. By an analogous argument to the one from the proof of the Lipschitz Approximation Lemma 13.5.2, we have that

$$\operatorname{dist}^{\Sigma(y,S(y))_{\tau(y)}}(x) \leq \left\|\mathbf{f}_{T(y)}\left(\mathbf{p}^{\Sigma(y,T(y))}(x)\right) - \lambda\left(\mathbf{p}^{\Sigma(y,T(y))}(x)\right)\right\|_{g_e}, \qquad (253)$$

for all $x \in B_{d_g}\left(\tau(y), \eta\sigma\right) \cap \operatorname{Graph}^{\Sigma(y,T(y))}\left(\mathbf{f}_{T(y)}\right)$.

On the other hand, we have that

$$\begin{aligned}&\frac{2^{m+2}}{(\eta\sigma)^{m+2}}\int_{B_{d_g}\left(\tau(y),\frac{\eta\sigma}{2}\right)}\left(\mathrm{dist}^{\Sigma(y,S(y))_{\tau(y)}}(x)\right)^2 d\|V\|(x)\leq\\&\leq\frac{2^{m+2}}{(\eta\sigma)^{m+2}}\int_{B_{d_g}\left(\tau(y),\frac{\eta\sigma}{2}\right)\cap\mathrm{Graph}^{\Sigma(y,T(y))}(\mathbf{f}_T(y))}\left(\mathrm{dist}^{\Sigma(y,S(y))_{\tau(y)}}(x)\right)^2 d\|V\|(x)\\&+\frac{2^{m+2}}{(\eta\sigma)^{m+2}}\int_{B_{d_g}\left(\tau(y),\frac{\eta\sigma}{2}\right)\setminus\mathrm{Graph}^{\Sigma(y,T(y))}(\mathbf{f}_T(y))}\left(\mathrm{dist}^{\Sigma(y,S(y))_{\tau(y)}}(x)\right)^2 d\|V\|(x).\end{aligned}\tag{254}$$

Now we will analyze each term on the right side of Equation (254). By Lemma 13.3.4 there exist $\delta_0:=\delta_0(m,p,K,r_H)>0$ and $\rho_0:=\rho_0(m,p,K,r_H)>0$ such that $\theta(x)\leq 2$. Also by Equations (252) and (253), we have that

$$\begin{aligned}&\frac{2^{m+2}}{(\eta\sigma)^{m+2}}\int_{B_{d_g}\left(\tau(y),\frac{\eta\sigma}{2}\right)\cap\mathrm{Graph}^{\Sigma(y,T(y))}(\mathbf{f}_T(y))}\left(\mathrm{dist}^{\Sigma(y,S(y))_{\tau(y)}}(x)\right)^2 d\|V\|(x)\leq\\&\leq\frac{2^{m+2}}{(\eta\sigma)^{m+2}}\int_{B_{d_g}\left(\tau(y),\frac{\eta\sigma}{2}\right)}\left\|\mathbf{f}_{T(y)}\left(\mathbf{p}^{\Sigma(y,T(y))}(x)\right)-\lambda\left(\mathbf{p}^{\Sigma(y,T(y))}(x)\right)\right\|_{g_e}^2\theta(x)d\mathcal{H}^m_{d_g}(x).\end{aligned}\tag{255}$$

Note that by the Harmonic Estimates we have that $|u^j(y)|\leq \mathbf{C_{11}}\sigma^{-\frac{m}{2}}\|u^j\|_{L^2\left(B_{d_g}\left(y,\frac{\sigma}{3}\right)\cap\Sigma(y,T(y)),\mathcal{L}^m_{d_g}\right)}$, for some constant $\mathbf{C_{11}}:=\mathbf{C_{11}}(m,n,K)>0$. By an analogous argument to the one from the proof of the Lipschitz Approximation Lemma 13.5.2 we have that there exists $\rho_0:=\rho_0(K,r_H)>0$ such that $\mathcal{L}^m_{d_g}\left(B_{d_g}\left(y,\frac{\sigma}{3}\right)\cap\Sigma(y,T(y))\right)\leq\frac{\omega_m\sigma^m}{3^m}\left(2+\frac{n^2K3^2}{6(m+2)}\right)$. Thus, by Equations (221), (230) and (247), we have that

$$d_g\left(y,\tau(y)\right)\leq\mathbf{C_{12}}\delta^{\frac{1}{2(m+1)}}\sigma,\tag{256}$$

where $\mathbf{C_{12}}:=\mathbf{C_{12}}(m,n,p,K,r_H)>0$. Note that Equation (256) implies that we can choose $\delta_0:=\delta_0(m,n,p,K,r_H)>0$ such that

$$\mathbf{C_{12}}\delta^{\frac{1}{2(m+1)}}\sigma<\frac{\eta\sigma}{4}\implies B_{d_g}\left(y,\frac{\eta\sigma}{4}\right)\subset B_{d_g}\left(\tau(y),\frac{\eta\sigma}{2}\right)\subset B_{d_g}\left(y,\frac{3\eta\sigma}{4}\right)\subset B_{d_g}\left(y,\eta\sigma\right).\tag{257}$$

Thus, by Equations (247) and (255)

$$\begin{aligned}&\frac{2^{m+2}}{(\eta\sigma)^{m+2}}\int_{B_{d_g}\left(\tau(y),\frac{\eta\sigma}{2}\right)\cap\mathrm{Graph}^{\Sigma(y,T(y))}(\mathbf{f}_T(y))}\left(\mathrm{dist}^{\Sigma(y,S(y))_{\tau(y)}}(x)\right)^2 d\|V\|(x)\leq\\&\leq 2^{m+3}\mathbf{C_8}\left(\frac{\varepsilon}{\eta^{m+2}}\mathbf{E}(y,\sigma,T(y),V)+\eta^2\mathbf{E}(y,\sigma,T(y),V)\right)\end{aligned}\tag{258}$$

On the other hand, by Equations (224) and (231), we have that

$$\frac{2^{m+2}}{(\eta\sigma)^{m+2}}\int_{B_{d_g}\left(\tau(y),\frac{\eta\sigma}{2}\right)\setminus\mathrm{Graph}^{\Sigma(y,T(y))}(\mathbf{f}_T(y))}\left(\mathrm{dist}^{\Sigma(y,S(y))_{\tau(y)}}(x)\right)^2 d\|V\|(x)\leq\frac{2^{m+2}\mathbf{C_2}^2\mathbf{C_3}}{\eta^{m+2}}\delta^{\frac{1}{m+1}}\mathbf{E}(y,\sigma,T(y),V)\tag{259}$$

Thus, by Equations (254), (258) and (259), we conclude

$$\frac{2^{m+2}}{(\eta\sigma)^{m+2}}\int_{B_{d_g}\left(\tau(y),\frac{\eta\sigma}{2}\right)}\left(\mathrm{dist}^{\Sigma(y,S(y))_{\tau(y)}}(x)\right)^2 d\|V\|(x)\leq$$
$$\leq \mathbf{C_{13}}\left(\eta^{-m-2}\left(\varepsilon+\delta^{\frac{1}{m+1}}\right)\mathbf{E}(y,\sigma,T(y),V)+\eta^2\mathbf{E}(y,\sigma,T(y),V)\right), \tag{260}$$

where $\mathbf{C_{13}} := \mathbf{C_{13}}(m,n,p,K,r_H) > 0$. Therefore, by Lemma 13.2.1 and by Equations (257) and (260), we obtain that

$$E\left(y,\frac{\eta\sigma}{4},S(y),V\right)\leq \mathbf{C_{13}}\left(\eta^{-m-2}\left(\varepsilon+\delta^{\frac{1}{m+1}}\right)\mathbf{E}(y,\sigma,T(y),V)+\eta^2\mathbf{E}(y,\sigma,T(y),V)\right). \tag{261}$$

Select $\eta := \eta(m,n,p,K,r_H) > 0$ such that $\mathbf{C_{13}}\eta^2 \leq \frac{1}{3}\left(\frac{\eta}{4}\right)^{2\left(1-\frac{m}{p}\right)}$, $\varepsilon := \varepsilon(m,n,p,K,r_H) > 0$ such that $\mathbf{C_{13}}\eta^{-m-2}\varepsilon \leq \frac{1}{3}\left(\frac{\eta}{4}\right)^{2\left(1-\frac{m}{p}\right)}$, $\delta_0 := \delta_0(m,n,p,K,r_H) > 0$ such that Equation (257) holds (and the minimum value ensuring that all the above estimates hold), and $\rho_0 := \rho_0(m,n,p,K,r_H) > 0$ as the minimum value ensuring that all the above estimates hold. Therefore, by Equation (261), we have that

$$E\left(y,\tilde{\eta}\sigma,S(y),V\right)\leq \tilde{\eta}^{2\left(1-\frac{m}{p}\right)}\mathbf{E}(y,\sigma,T(y),V),$$

where $\tilde{\eta} = \frac{\eta}{4}$. Also note that

$$\frac{1}{\delta}\left((\tilde{\eta}\sigma)^{p-m}\int_{B_{d_g}(y,\tilde{\eta}\sigma)}\left\|\overrightarrow{H_g}(x)\right\|_{g_x}^p d\|V\|(x)\right)^{\frac{2}{p}}\leq \tilde{\eta}^{2\left(1-\frac{m}{p}\right)}\frac{1}{\delta}\left(\sigma^{p-m}\int_{B_{d_g}(y,\sigma)}\left\|\overrightarrow{H_g}(x)\right\|_{g_x}^p d\|V\|(x)\right)^{\frac{2}{p}}$$

implies that

$$\mathbf{E}\left(y,\tilde{\eta}\sigma,S(y),V\right)\leq \tilde{\eta}^{2\left(1-\frac{m}{p}\right)}\mathbf{E}(y,\sigma,T(y),V),$$

which concludes the proof with $\tilde{\eta}$ in place of η. □

Corollary 13.6.4. *Let $m < p < \infty$ and $\xi \in M^n$. There exist $\eta := \eta(m,n,p,K,r_H) > 0$, $\delta_0 := \delta_0(m,n,p,K,r_H) > 0$ and $\rho_0 := \rho_0(m,n,p,K,r_H) > 0$ such that if $V \in \mathcal{AC}(\xi,\rho,\delta,p)$, for some $0 < \delta < \delta_0$ and for some $0 < \rho < \min\left\{\rho_0,\sqrt{\delta}\right\}$, then, for all $y \in \mathrm{supp}(\|V\|)\cap B_{d_g}\left(\xi,\frac{\delta\rho}{2}\right)$, for all $T(y) \in \mathrm{Gr}(m,T_yM^n)$, for all $\sigma \in]0,2\delta\rho]$, and for all $S(y) \in \mathrm{Gr}(m,T_yM^n)$ such that Equation (220) holds, we have that*

$$\left\|P_{S(y)}-P_{T(y)}\right\|_{\mathrm{HS}_{g_y}}^2 \leq \mathbf{C}E(y,\sigma,T(y),V), \tag{262}$$

where $\mathbf{C} = \mathbf{C}(m,p) > 0$.

Proof. Let $\eta := \eta(m,n,p,K,r_H) > 0$, $\delta_0 := \delta_0(m,n,p,K,r_H) > 0$ and $\rho_0 := \rho_0(m,n,p,K,r_H) > 0$ given by Theorem 13.6.3. Suppose that $V \in \mathcal{AC}(\xi,\rho,\delta,p)$, for some $0 < \delta < \delta_0$ and for some $0 < \rho < \min\left\{\rho_0,\sqrt{\delta}\right\}$. Consider $y \in \mathrm{supp}(\|V\|)\cap B_{d_g}\left(\xi,\frac{\delta\rho}{2}\right)$, $T(y) \in \mathrm{Gr}(m,T_yM^n)$, $\sigma \in]0,2\delta\rho]$, and $S(y) \in \mathrm{Gr}(m,T_yM^n)$ such that Equation (220) holds.

Note that we have

$$E(y,\eta\sigma,T(y),V)\leq\frac{1}{\eta^m}\mathbf{E}(y,\sigma,T(y),V), \tag{263}$$

For all $x\in\operatorname{supp}(\|V\|)\cap B_{d_g}(y,\sigma)$, by Lemma 13.1.13 the squared triangle inequality, we have that

$$\left\|P_{S(y)}-P_{T(y)}\right\|^2_{\mathrm{HS}_{g_y}}\leq 4\left\|P_{T_x\Sigma(y,S(y))_x}-P_{T_xV}\right\|^2_{\mathrm{HS}_{g_x}}+4\left\|P_{T_xV}-P_{T_x\Sigma(y,T(y))_x}\right\|^2_{\mathrm{HS}_{g_x}} \tag{264}$$

By Lemma 13.3.4 there exists $\delta_0:=\delta_0(m,p,K,r_H)>0$ and $\rho_0:=\rho_0(m,p,K,r_H)>0$ such that $\frac{\|V\|\left(B_{d_g}(y,\eta\sigma)\right)}{\eta^m\sigma^m}\geq\frac{\omega_m}{2}$, thus

$$\frac{1}{\eta^m\sigma^m}\int_{B_{d_g}(y,\eta\sigma)}\left\|P_{S(y)}-P_{T(y)}\right\|^2_{\mathrm{HS}_{g_y}}d\|V\|(x)\geq\left\|P_{S(y)}-P_{T(y)}\right\|^2_{\mathrm{HS}_{g_y}}\frac{\omega_m}{2}. \tag{265}$$

Therefore, by Equations (263) to (265), we have that

$$\left\|P_{S(y)}-P_{T(y)}\right\|^2_{\mathrm{HS}_{g_y}}\leq\mathbf{C}\mathbf{E}(y,\sigma,T(y),V),$$

where $\mathbf{C}=\mathbf{C}(m,p)>0$. For $\eta<1$, since $0<2\left(1-\frac{m}{p}\right)<2$, we have that $\eta^{2\left(1-\frac{m}{p}\right)}<\eta^{-m}$. Thus, we also have that

$$\left\|P_{S(y)}-P_{T(y)}\right\|^2_{\mathrm{HS}_{g_y}}\leq\frac{16}{\omega_m}\eta^{-m}\mathbf{E}(y,\sigma,T(y),V).$$

To conclude the proof, we choose $\eta:=\eta(m,n,p,K,r_H)>0$, $\delta_0:=\delta_0(m,n,p,K,r_H)>0$ and $\rho_0:=\rho_0(m,n,p,K,r_H)>0$ as the minimum value ensuring that all the above estimates hold. □

Theorem 13.6.5. *Let $m<p<\infty$ and $\xi\in M^n$. There exist $\varepsilon:=\varepsilon(m,n,p,K,r_H)>0$, $\delta_0:=\delta_0(m,n,p,K,r_H)>0$, and $\rho_0:=\rho_0(m,n,p,K,r_H)>0$ such that if $V\in\mathcal{AC}(\xi,\rho,\delta,p)$, for some $0<\delta<\delta_0$ and for some $0<\rho<\min\left\{\rho_0,\sqrt{\delta}\right\}$, then there exist $T(\xi)\in\mathrm{Gr}(m,T_\xi M^n)$ and a function $\mathbf{f}_{T(\xi)}:B_{d_g}(\xi,\varepsilon\rho)\cap\Sigma(\xi,T(\xi))\to\mathbb{R}^{n-m}$ such that $\mathbf{f}_{T(\xi)}$ is of class $\mathcal{C}^{1-\frac{m}{p}}$, $\mathbf{f}_{T(\xi)}(\xi)=0$, $d\mathbf{f}_{T(\xi)_\xi}\equiv 0$,*

$$\operatorname{supp}(\|V\|)\cap B_{d_g}(\xi,\varepsilon\rho)=\mathrm{Graph}^{\Sigma(\xi,T(\xi))}\left(\mathbf{f}_{T(\xi)}\right)\cap B_{d_g}(\xi,\varepsilon\rho),$$

and the estimate of the scaling invariant $C^{1,1-\frac{m}{p}}$-norm of $\mathbf{f}_{T(\xi)}$ depend only on m,n,p,K,r_H and ε.

Proof. Let $\delta_0:=\delta_0(m,p,K,r_H)>0$ and $\rho_0:=\rho_0(m,p,K,r_H)>0$ given by Corollary 13.4.3. Suppose that $V\in\mathcal{AC}(\xi,\rho,\delta,p)$, for some $0<\delta<\delta_0$ and for some $0<\rho<\min\left\{\rho_0,\sqrt{\delta}\right\}$. Also by Corollary 13.4.3 we have that there exists $\mathbf{S}(\xi)\in\mathrm{Gr}(m,T_\xi M^n)$ such that

$$\sup_{x\in\operatorname{supp}(\|V\|)\cap B_{d_g}(\xi,2\sigma_0)}\mathrm{dist}_g^{\Sigma(\xi,\mathbf{S}(\xi))}(x)\leq\mathbf{C_1}\delta^{\frac{1}{2(m+1)}}2\sigma_0, \tag{266}$$

where $\mathbf{C_1}>0$ is given by Equation (176), and, for all $\alpha\in]0,1[$, we have that

$$E(\xi,\alpha 2\sigma_0,\mathbf{S}(\xi),V)\leq\mathbf{C_2}\delta^{\frac{1}{m+1}}, \tag{267}$$

where $\mathbf{C_2} > 0$ is given by Equation (177), and $\sigma_0 := \frac{\delta\rho}{2}$. Let $y \in \operatorname{supp}(\|V\|) \cap B_{d_g}(y, \sigma_0)$ and $S_0(y) \in \mathrm{Gr}(m, T_yM^n)$ be arbitrary. By Theorem 13.6.3 we have that there exist $\eta := \eta(m, n, p, K, r_H) > 0$, $\delta_0 := \delta_0(m, n, p, K, r_H) > 0$ and $\rho_0 := \rho_0(m, n, p, K, r_H) > 0$, such that there exists $S_1(y) \in \mathrm{Gr}(m, T_yM^n)$ with the property that

$$\mathbf{E}(y, \eta\sigma_0, S_1(y), V) \leq \eta^{2\left(1-\frac{m}{p}\right)}\mathbf{E}(y, \sigma_0, S_0(y), V). \tag{268}$$

Note that this can be repeated; by induction we have that there is a sequence $\{S_i(y)\}_{i\in\mathbb{N}} \subset \mathrm{Gr}(m, T_yM^n)$ such that

$$\mathbf{E}(y, \eta^i\sigma_0, S_i(y), V) \leq \eta^{2\left(1-\frac{m}{p}\right)}\mathbf{E}(y, \eta^{i-1}\sigma_0, S_{i-1}(y), V) \leq \eta^{2\left(1-\frac{m}{p}\right)i}\mathbf{E}(y, \sigma_0, S_0(y), V), \tag{269}$$

for each $i \in \mathbb{N}$.

Take $S_0(y) := T_y\Sigma(\xi, \mathbf{S}(\xi))_y \in \mathrm{Gr}(m, T_yM^n)$ and note that by the squared triangle inequality, we have, for all $x \in B_{d_g}(y, \sigma_0)$ and for some $\beta > 0$, that

$$E\left(y, \sigma_0, S_0(y), V\right) \leq 2((2-\beta)^m)E\left(\xi, (2-\beta)\sigma_0, S(\xi), V\right). \tag{270}$$

Therefore, by Equation (267) with $\alpha = \frac{2-\beta}{2} < 1$ and Equation (270), we have that

$$E\left(y, \sigma_0, S_0(y), V\right) \leq 2^{m+1}\mathbf{C_2}\delta^{\frac{1}{m+1}}. \tag{271}$$

On the other hand, by Corollary 13.6.4 we have that there exist $\eta := \eta(m, n, p, K, r_H) > 0$, $\delta_0 := \delta_0(m, n, p, K, r_H) > 0$ and $\rho_0 := \rho_0(m, n, p, K, r_H) > 0$ such that, with Equation (269), we have that

$$\left\|P_{S_i(y)} - P_{S_{i-1}(y)}\right\|^2_{\mathrm{HS}_{g_y}} \leq \mathbf{C_3}\mathbf{E}(y, \eta^{i-1}\sigma_0, S_{i-1}(y), V) \leq \mathbf{C_3}\eta^{2\left(1-\frac{m}{p}\right)(i-1)}\mathbf{E}(y, \sigma_0, S_0(y), V), \tag{272}$$

where $\mathbf{C_3} > 0$ is given by Equation (262) for each $i \in \mathbb{N}$. Let $0 \leq j < l$, by summation from $j+1$ to l, we have that

$$\left\|P_{S_l(y)} - P_{S_j(y)}\right\|_{\mathrm{HS}_{g_y}} \leq \sqrt{\mathbf{C_3}\mathbf{E}(y, \sigma_0, S_0(y), V)}\sum_{i=j+1}^{l}\eta^{\left(1-\frac{m}{p}\right)(i-1)}. \tag{273}$$

Taking $l \uparrow \infty$ we have that there is $S(y) \in \mathrm{Gr}(m, T_yM^n)$ (since $\mathrm{Gr}(m, T_yM^n)$ is compact) such that $S(y) = \lim_{l\uparrow\infty} S_l(y)$ and

$$\left\|P_{S(y)} - P_{S_j(y)}\right\|^2_{\mathrm{HS}_{g_y}} \leq \mathbf{C_4}\eta^{2\left(1-\frac{m}{p}\right)j}\mathbf{E}(y, \sigma_0, S_0(y), V), \tag{274}$$

where $\mathbf{C_4} = \mathbf{C_4}(m, p) > 0$. In particular, setting $j = 0$, we have that

$$\left\|P_{S(y)} - P_{S_0(y)}\right\|^2_{\mathrm{HS}_{g_y}} \leq \mathbf{C_4}\mathbf{E}(y, \sigma_0, S_0(y), V). \tag{275}$$

Let $\sigma \in]0, \sigma_0]$ be arbitrary and choose $j_0 \geq 0$ such that $\eta^{j_0}\sigma_0 < \sigma < \eta^{j_0-1}\sigma_0$. By Lemma 13.1.13, we have that

$$E(y,\sigma,S(y),V) \leq \frac{1}{\sigma^m}\frac{\eta^{(j_0-1)m}(\sigma_0)^m}{\eta^{(j_0-1)m}(\sigma_0)^m}\int_{B_{d_g}(y,\eta^{j_0-1}\sigma_0)} 2\left\|P_{T_xV} - P_{T_x\Sigma(y,S_{j_0-1}(y))_x}\right\|^2_{\mathrm{HS}_{gx}} d\|V\|(x)$$
$$+ \frac{1}{\sigma^m}\frac{\eta^{(j_0-1)m}(\sigma_0)^m}{\eta^{(j_0-1)m}(\sigma_0)^m}\int_{B_{d_g}(y,\eta^{j_0-1}\sigma_0)} 4\left\|P_{T_y\Sigma(y,S_{j_0-1}(y))_y} - P_{T_y\Sigma(y,S(y))_y}\right\|^2_{\mathrm{HS}_{gy}} d\|V\|(x). \tag{276}$$

By Lemma 13.3.4 we have that there exist $\delta_0 := \delta_0(m,p,K,r_H) > 0$ and $\rho_0 := \rho_0(m,p,K,r_H) > 0$ such that $\frac{\|V\|\left(B_{d_g}(y,\eta^{j_0-1}\sigma_0)\right)}{\eta^{(j_0-1)m}(\sigma_0)^m} \leq 2\omega_m$. Thus, by Equation (276), we have that

$$E(y,\sigma,S(y),V) \leq \mathbf{C_5}\left(\frac{\sigma}{\sigma_0}\right)^{2\left(1-\frac{m}{p}\right)}\mathbf{E}(y,\sigma_0,S_0(y),V), \tag{277}$$

where $\mathbf{C_5} = \mathbf{C_5}(m,p) > 0$. By Lemma 13.3.4 we have that there exist $\delta_0 := \delta_0(m,p,K,r_H) > 0$ and $\rho_0 := \rho_0(m,p,K,r_H) > 0$ such that $\frac{\|V\|(B_{d_g}(y,\sigma))}{\sigma^m} \leq 2\omega_m$. Thus, by Lemma 13.1.13 and Equations (275) and (277), we have that

$$\frac{1}{\sigma^m}\int_{B_{d_g}(y,\sigma)}\left\|P_{T_xV} - P_{T_x\Sigma(\xi,\mathbf{S}(\xi))_x}\right\|^2_{\mathrm{HS}_{gx}} d\|V\|(x) \leq \left(\mathbf{C_5}\left(\frac{\sigma}{\sigma_0}\right)^{2\left(1-\frac{m}{p}\right)} + 8\omega_m\mathbf{C_4}\right)\mathbf{E}(y,\sigma_0,S_0(y),V). \tag{278}$$

On the other hand, by Equation (271) and Remark 13.6.2 we have that

$$\mathbf{E}(y,\sigma_0,S_0(y),V) \leq 2^{m+1}\mathbf{C_2}\delta^{\frac{1}{m+1}}. \tag{279}$$

Note that there exists $0 < \delta_0 < 1$ such that, by Equations (278) and (279), we have that

$$\sup_{\sigma\in]0,\sigma_0]}\frac{1}{\sigma^m}\int_{B_{d_g}(y,\sigma)}\left\|P_{T_xV} - P_{T_x\Sigma(\xi,\mathbf{S}(\xi))_x}\right\|^2_{\mathrm{HS}_{gx}} d\|V\|(x) \leq \mathbf{C_6}\delta^{\frac{1}{m+1}}, \tag{280}$$

where $\mathbf{C_6} = \mathbf{C_6}(m,n,p,r_H,K) > 0$. Let $0 < L \leq 1$ be arbitrary. Note that there exists $0 < \delta_0 < 1$ such that $\delta^{\frac{1}{m+1}} < (\delta_0)^{\frac{1}{m+1}} < \delta_0^3$. Therefore, $\mathbf{C_6}\delta^{\frac{1}{m+1}} < \mathbf{C_6}\delta_0^3$. Also note that there exists $\delta_0 := \delta_0(m,n,p,K,r_H) > 0$ such that $\mathbf{C_6}\delta_0^3 < \delta_0^2L^2$. Thus

$$\sup_{\sigma\in]0,\sigma_0]}\frac{1}{\sigma^m}\int_{B_{d_g}(y,\sigma)}\left\|P_{T_xV} - P_{T_x\Sigma(\xi,\mathbf{S}(\xi))_x}\right\|^2_{\mathrm{HS}_{gx}} d\|V\|(x) \leq \delta_0^2L^2 \tag{281}$$

By Corollary 13.5.3 there exist $\delta_0 := \delta_0(m,n,p,K,r_H) > 0$ and $\rho_0 := \rho_0(m,K,r_H,p) > 0$ such that there exists a Lipschitz function $\mathbf{f}_{\mathbf{S}(\xi)} : B_{d_g}(\xi,\sigma_0) \cap \Sigma(\xi,\mathbf{S}(\xi)) \to \mathbb{R}^{n-m}$ such that

$$\mathrm{Lip}\left(\mathbf{f}_{\mathbf{S}(\xi)}\right) \leq L, \tag{282}$$

and

$$\mathrm{supp}(\|V\|) \cap B_{d_g}\left(\xi,\frac{\sigma_0}{2}\right) = \mathrm{Graph}^{\Sigma(\xi,\mathbf{S}(\xi))}\left(\mathbf{f}_{\mathbf{S}(\xi)}\right) \cap B_{d_g}\left(\xi,\frac{\sigma_0}{2}\right). \tag{283}$$

For $\mathcal{H}^m_{d_g}$-a.e. $y \in \mathrm{Graph}^{\Sigma(\xi,\mathbf{S}(\xi))}\left(\mathbf{f}_{\mathbf{S}(\xi)}\right) \cap B_{d_g}\left(\xi, \frac{\sigma_0}{2}\right)$, by Equations (277), (279) and (283), we have that

$$\lim_{\sigma\downarrow 0} \frac{1}{\sigma^m} \int_{\mathrm{Graph}^{\Sigma(\xi,\mathbf{S}(\xi))}\left(\mathbf{f}_{\mathbf{S}(\xi)}\right)\cap B_{d_g}(\xi,\sigma)} \left\| P_{T_x\mathrm{Graph}^{\Sigma(\xi,\mathbf{S}(\xi))}\left(\mathbf{f}_{\mathbf{S}(\xi)}\right)} - P_{T_x\Sigma(y,S(y))_x} \right\|^2_{\mathrm{HS}_{g_x}} d\mathcal{H}^m_{d_g}(x) \leq 0. \quad (284)$$

Which implies that at all such points $S(y) = T_y\mathrm{Graph}^{\Sigma(\xi,\mathbf{S}(\xi))}\left(\mathbf{f}_{\mathbf{S}(\xi)}\right)$. Therefore, by Equation (277), we have that

$$\frac{1}{\sigma^m} \int_{\mathrm{Graph}^{\Sigma(\xi,\mathbf{S}(\xi))}\left(\mathbf{f}_{\mathbf{S}(\xi)}\right)\cap B_{d_g}(y,\sigma)} \left\| P_{T_x\mathrm{Graph}^{\Sigma(\xi,\mathbf{S}(\xi))}\left(\mathbf{f}_{\mathbf{S}(\xi)}\right)} - P_{T_x\Sigma\left(y,T_y\mathrm{Graph}^{\Sigma(\xi,\mathbf{S}(\xi))}\left(\mathbf{f}_{\mathbf{S}(\xi)}\right)\right)_x} \right\|^2_{\mathrm{HS}_{g_x}} d\mathcal{H}^m_{d_g}(x) \leq$$

$$\leq \mathbf{C_5} \left(\frac{\sigma}{\sigma_0}\right)^{2\left(1-\frac{m}{p}\right)} \mathbf{E}(y, \sigma_0, S_0(y), V), \quad (285)$$

for all $0 < \sigma \leq \frac{\sigma_0}{2}$. Let $x \in \mathrm{Graph}^{\Sigma(\xi,\mathbf{S}(\xi))}\left(\mathbf{f}_{\mathbf{S}(\xi)}\right) \cap B_{d_g}(y,\sigma)$ and $\left(\mathrm{Tube}(\Sigma(\xi,\mathbf{S}(\xi)), r_H), \{x_i\}_{i=1}^n\right)$ a system of Fermi coordinates for $\Sigma(\xi,\mathbf{S}(\xi))$ at $\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi))}(x)$.

For $i = 1, \ldots, m$ and $\vartheta > 0$, define $\gamma_i :]-\vartheta, \vartheta[\to \Sigma(\xi,\mathbf{S}(\xi))$ such that $\gamma_i(0) = \mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi))}(x)$ and $\gamma_i'(0) = \frac{\partial}{\partial x_i}\left(\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi))}(x)\right)$. Note that

$$\alpha_i(t) := \exp_{\gamma_i(t)}\left(\sum_{j=m+1}^n \mathbf{f}^j_{\mathbf{S}(\xi)}(\gamma_i(t)) \frac{\partial}{\partial x_j}(\gamma_i(t))\right)$$

is a curve in $\mathrm{Graph}^{\Sigma(\xi,\mathbf{S}(\xi))}\left(\mathbf{f}_{\mathbf{S}(\xi)}\right)$ and $\left\{\alpha_i'(0)\right\}_{i=1}^m$ is a basis for $T_x\mathrm{Graph}^{\Sigma(\xi,\mathbf{S}(\xi))}\left(\mathbf{f}_{\mathbf{S}(\xi)}\right)$.

By the Gram-Schmidt orthogonalization process, starting with the basis $\left\{\alpha_i'(0)\right\}_{i=1}^m$ of $T_x\mathrm{Graph}^{\Sigma(\xi,\mathbf{S}(\xi))}\left(\mathbf{f}_{\mathbf{S}(\xi)}\right)$ as above, we have that the matrix of $P_{T_x\mathrm{Graph}^{\Sigma(\xi,\mathbf{S}(\xi))}\left(\mathbf{f}_{\mathbf{S}(\xi)}\right)}$ in the resultant basis is given by

$$\left[P_{T_x\mathrm{Graph}^{\Sigma(\xi,\mathbf{S}(\xi))}\left(\mathbf{f}_{\mathbf{S}(\xi)}\right)}\right]_{n\times n} = \begin{pmatrix} \mathrm{Id}_{m\times m} & \left[d\mathbf{f}_{\mathbf{S}(\xi)_{\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi))}(x)}}\right]_{m\times(n-m)} \\ \left[d\mathbf{f}_{\mathbf{S}(\xi)_{\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi))}(x))}}\right]^\top_{m\times(n-m)} & 0_{(n-m)\times(n-m)} \end{pmatrix}_{n\times n}$$
$$+ \mathcal{F}\left(\left[d\mathbf{f}_{\mathbf{S}(\xi)_{\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi))}(x)}}\right]_{m\times(n-m)}\right), \quad (286)$$

where $\mathcal{F}(v)$ is a real analytic function of $v = (v_{ij})_{i=1,\ldots,m;j=1,\ldots,n-m} \in \mathbb{R}^{n(n-m)}$ with $\mathcal{F}(0) = 0$, $d\mathcal{F}_v(0) = 0$ and hence

$$|\mathcal{F}(v_1) - \mathcal{F}(v_2)| \leq \mathbf{C_7}\left(\|v_1\|_{g_e} + \|v_2\|_{g_e}\right)\|v_1 - v_2\|_{g_e}, \quad (287)$$

where $\mathbf{C_7} = \mathbf{C_7}(m,n) > 0$, for $\|v_1\|_{g_e} \leq 1$, and $\|v_2\|_{g_e} \leq 1$. By Equation (282), we have $\mathrm{Lip}\left(\mathbf{f}_{\mathbf{S}(\xi)}\right) \leq L$. Since $0 < L \leq 1$ was chosen arbitrarily, we can further select L to be sufficiently small, depending only on n and m, such that, by Equations (286) and (287), it follows that

$$\begin{aligned}&\left\|d\mathbf{f}_{\mathbf{S}(\xi)\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi))}(x)} - \left(d\mathbf{f}_{\mathbf{S}(\xi)\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi))}(y)}\right)_{\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi))}(x)}\right\|^2_{\mathrm{HS}_{g_e}} \leq \\ &\leq \left\|P_{T_x\mathrm{Graph}^{\Sigma(\xi,\mathbf{S}(\xi))}\left(\mathbf{f}_{\mathbf{S}(\xi)}\right)} - P_{T_x\Sigma\left(y,T_y\mathrm{Graph}^{\Sigma(\xi,\mathbf{S}(\xi))}\left(\mathbf{f}_{\mathbf{S}(\xi)}\right)\right)_x}\right\|^2_{\mathrm{HS}_{g_x}},\end{aligned} \tag{288}$$

where $\left(d\mathbf{f}_{\mathbf{S}(\xi)\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi))}(y)}\right)_{\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi))}(x)}$ is defined as the composition of $d\mathbf{f}_{\mathbf{S}(\xi)\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi))}(y)}$ with the Fermi transport of a vector from $T_{\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi))}(x)}\Sigma(\xi,\mathbf{S}(\xi))$ to $T_{\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi))}(y)}\Sigma(\xi,\mathbf{S}(\xi))$. Therefore, by an adaptation of the Area formula [41, Theorem 3.2.3, p. 243] for the context of normal graphs in Riemannian manifolds, and by Equations (285) and (288), we have that

$$\frac{1}{\sigma^m}\int_{\Sigma(\xi,\mathbf{S}(\xi))\cap B_{d_g}\left(\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi))}(y),\sigma\right)}\left\|d\mathbf{f}_{\mathbf{S}(\xi)x} - \left(d\mathbf{f}_{\mathbf{S}(\xi)\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi))}(y)}\right)_x\right\|^2_{\mathrm{HS}_{g_e}} d\mathcal{L}^m_{d_g}(x) \leq \mathbf{C_5}\left(\frac{\sigma}{\sigma_0}\right)^{2\left(1-\frac{m}{p}\right)}\mathbf{E}\left(y,\sigma_0,S_0(y),V\right), \tag{289}$$

for all $0 < \sigma < \frac{\sigma_0}{2}$. Now take $y_1, y_2 \in \mathrm{Graph}^{\Sigma(\xi,\mathbf{S}(\xi))}\left(\mathbf{f}_{\mathbf{S}(\xi)}\right) \cap B_{d_g}\left(\xi,\frac{\sigma_0}{2}\right)$, $\sigma = \left\|\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi))}(y_1) - \mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi))}(y_2)\right\|_{d_g}$ and $\beta : [0,1] \to M^n$ the unique distance minimizer geodesic between $\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi))}(y_1)$ and $\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi))}(y_2)$ such that $\beta(0) = \mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi))}(y_1)$ and $\beta(1) = \mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi))}(y_2)$ (note that $\beta([0,1])$ is not necessarily contained in $\Sigma(\xi,\mathbf{S}(\xi))$ since $\Sigma(\xi,\mathbf{S}(\xi))$ is not totally geodesic). Thus

$$B_{d_g}\left(\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi)}\left(\gamma\left(\frac{1}{2}\right)\right),\frac{\sigma}{2}\right) \subset B_{d_g}\left(\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi)}(y_1),\sigma\right) \cap B_{d_g}\left(\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi)}(y_2),\sigma\right). \tag{290}$$

By an analogous argument to the one from the proof of the Lipschitz Approximation Lemma 13.5.2 we have that there exists $\rho_0 := \rho_0(K, r_H) > 0$ such that $\mathcal{L}^m_{d_g}\left(\Sigma(\xi,\mathbf{S}(\xi)) \cap B_{d_g}\left(\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi)}\left(\gamma\left(\frac{1}{2}\right)\right),\frac{\sigma}{2}\right)\right) \geq \omega_m\frac{\sigma^m}{2^m}\left(1+\frac{n^2K}{12(m+2)}\right)$. Therefore,

$$\begin{aligned}&\int_{\Sigma(\xi,\mathbf{S}(\xi))\cap B_{d_g}\left(\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi)}\left(\gamma\left(\frac{1}{2}\right)\right),\frac{\sigma}{2}\right)}\left\|d\mathbf{f}_{\mathbf{S}(\xi)\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi)}(y_1)} - \left(d\mathbf{f}_{\mathbf{S}(\xi)\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi)}(y_2)}\right)_{\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi)}(y_1)}\right\|^2_{\mathrm{HS}_{g_e}} d\mathcal{L}^m_{d_g}(x) \\ &\geq \left\|d\mathbf{f}_{\mathbf{S}(\xi)\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi)}(y_1)} - \left(d\mathbf{f}_{\mathbf{S}(\xi)\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi)}(y_2)}\right)_{\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi)}(y_1)}\right\|^2_{\mathrm{HS}_{g_e}} \omega_m\frac{\sigma^m}{2^m}\left(1+\frac{n^2K}{12(m+2)}\right).\end{aligned} \tag{291}$$

By Lemma 13.1.13, Equations (279) and (289) to (291), and squared triangle inequality we have that

$$\left\|d\mathbf{f}_{\mathbf{S}(\xi)\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi)}(y_1)} - \left(d\mathbf{f}_{\mathbf{S}(\xi)\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi)}(y_2)}\right)_{\mathbf{p}^{\Sigma(\xi,\mathbf{S}(\xi)}(y_1)}\right\|^2_{\mathrm{HS}_{g_e}} \leq \frac{2^{m+2}}{\omega_m\left(1+\frac{n^2K}{12(m+2)}\right)}2\mathbf{C_5}\left(\frac{\sigma}{\sigma_0}\right)^{2\left(1-\frac{m}{p}\right)}2^{m+1}\mathbf{C_2}\delta^{\frac{1}{m+1}}.$$

Due to the arbitrariness of $y_1, y_2 \in \mathrm{Graph}^{\Sigma(\xi,\mathbf{S}(\xi))}\left(\mathbf{f}_{\mathbf{S}(\xi)}\right) \cap B_{d_g}\left(\xi,\frac{\sigma_0}{4}\right)$, we have the following for $y_1, y_2 \in \Sigma(\xi,\mathbf{S}(\xi)) \cap B_{d_g}\left(\xi,\frac{\sigma_0}{4}\right)$ (with a slight abuse of notation):

$$\left\|d\mathbf{f}_{\mathbf{S}(\xi)y_1} - \left(d\mathbf{f}_{\mathbf{S}(\xi)y_2}\right)_{y_1}\right\|_{\mathrm{HS}_{g_e}} \leq \mathbf{C_8}\left(\frac{\|y_1-y_2\|_{d_g}}{\frac{\delta}{8}\rho}\right)^{\left(1-\frac{m}{p}\right)}\delta^{\frac{1}{2(m+1)}}, \tag{292}$$

where $\mathbf{C_8} = \mathbf{C_8}(m, n, p, K, r_H) > 0$. Since $\mathbf{f}_{\mathbf{S}(\xi)}$ is Lipschitz, Equation (292) implies that $\mathbf{f}_{\mathbf{S}(\xi)} \in \mathcal{C}^{1,1-\frac{m}{p}}\left(B_{d_g}(\xi, \varepsilon\sigma) \cap \Sigma(\xi, T(\xi))\right)$, for $\varepsilon := \frac{\delta}{8}$ and $T(\xi) := \mathbf{S}(\xi)$. To conclude the proof, we choose $\delta_0 := \delta_0(m, n, p, K, r_H) > 0$ and $\rho_0 := \rho_0(m, n, p, K, r_H) > 0$ as the minimum value ensuring that all the above estimates hold. □

Lemma 13.6.6. *Let $m < p < \infty$, $\xi \in M^n$. There exist $\delta_0 := \delta_0(m, p, K, r_H) > 0$, $\rho_0 := \rho_0(m, p, K, r_H) > 0$ and $\mathbf{C_0} := \mathbf{C_0}(m) > 0$ such that if $V \in \mathcal{AC}(\xi, \rho, \delta, p)$ for some $0 < \delta < \delta_0$ and some $0 < \rho < \min\{\rho_0, \sqrt{\delta}\}$, then*

$$\frac{\|V\|(B_{d_g}(x, \sigma))}{\omega_m \sigma^m} \leq 1 + \mathbf{C}\left(\delta + \frac{r}{\rho}\right) \leq 2,$$

where $\mathbf{C} := \mathbf{C}(m, p, K, r_H) > 0$, for all $\frac{r}{\rho} \in [0, \mathbf{C_0}]$, $x \in B_{d_g}(\xi, r)$, $\sigma \in]0, \rho - r]$,

Proof. The proof follows as a standard consequence of Theorem 12.3.16. □

Corollary 13.6.7. *Let $m < p < \infty$, $\xi \in M^n$, and $\varepsilon \in]0, 1[$. There exist $\delta_0 := \delta_0(m, n, p, K, r_H, \varepsilon) > 0$ and $\rho_0 := \rho_0(m, n, p, K, r_H) > 0$ such that if $V \in \mathcal{AC}(\xi, \rho, \delta, p)$, for some $0 < \delta < \delta_0$ and for some $0 < \rho < \min\{\rho_0, \sqrt{\delta}\}$, then there exist $T(\xi) \in \mathrm{Gr}(m, T_\xi M^n)$ and a function $\mathbf{f}_{T(\xi)} : B_{d_g}(\xi, \varepsilon\rho) \cap \Sigma(\xi, T(\xi)) \to \mathbb{R}^{n-m}$ such that $\mathbf{f}_{T(\xi)}$ is of class $\mathcal{C}^{1,1-\frac{m}{p}}$, $\mathbf{f}_{T(\xi)}(\xi) = 0$, $d\mathbf{f}_{T(\xi)_\xi} \equiv 0$,*

$$\mathrm{supp}(\|V\|) \cap B_{d_g}(\xi, \varepsilon\rho) = \mathrm{Graph}^{\Sigma(\xi, T(\xi))}\left(\mathbf{f}_{T(\xi)}\right) \cap B_{d_g}(\xi, \varepsilon\rho),$$

and the estimate of the scaling invariant $C^{1,1-\frac{m}{p}}$-norm of $\mathbf{f}_{T(\xi)}$ depend only on m, n, p, K, r_H and ε.

Proof. The proof of this Corollary is a standard combination of Theorem 13.6.5 with Lemma 13.6.6. □

Theorem 13.6.8. *Let $m, n \in \mathbb{N}$ and $p \in \mathbb{R}$ such that $1 \leq m < n$ and $m < p < \infty$; in case $m = 1$, we require that $p \geq 2$, (M^n, g) an n-dimensional complete connected Riemannian manifold with differentiable structure of class $\mathcal{C}^3$ and with abstract metric tensor g of class $\mathcal{C}^2$, $\xi \in M^n$, and $\varepsilon \in]0, 1[$. There exist $\delta_0 := \delta_0(\xi, m, n, p, K, r_H, \varepsilon) > 0$ and $\rho_0 := \rho_0(\xi, m, n, p, K, r_H) > 0$ such that, if $V \in \mathbf{V}_m(M^n)$ satisfies that*

$$\Theta^m(\|V\|, x) \geq d > 0, \tag{293}$$

for $\|V\|$-a.e. $x \in B_{d_g}(\xi, \rho)$,

$$\frac{\|V\|(B_{d_g}(\xi, \rho))}{\omega_m \rho^m} \leq d(1 + \delta). \tag{294}$$

and

$$\left|\delta_g V(X)\right| \leq \frac{d^{\frac{1}{p}}\delta}{\rho^{1-\frac{m}{p}}} \|X\|_{L^{\frac{p}{p-1}}(M^n, \|V\|)}, \tag{295}$$

for all $X \in \mathfrak{X}^0_c(M^n)$ such that $\mathrm{supp}(X) \subset B_{d_g}(\xi, \rho)$, for some $0 < \delta < \delta_0$, for some $0 < d < \infty$, and for some $0 < \rho < \min\{\rho_0, \sqrt{\delta}\}$, then there exist $T(\xi) \in \mathrm{Gr}(m, T_\xi M^n)$ and a function $\mathbf{f}_{T(\xi)} : B_{d_g}(\xi, \varepsilon\rho) \cap \Sigma(\xi, T(\xi)) \to \mathbb{R}^{n-m}$ such that $\mathbf{f}_{T(\xi)}$ is of class $\mathcal{C}^{1,1-\frac{m}{p}}$, $\mathbf{f}_{T(\xi)}(\xi) = 0$, $d\mathbf{f}_{T(\xi)_\xi} \equiv 0$,

$$\mathrm{supp}(\|V\|) \cap B_{d_g}(\xi, \varepsilon\rho) = \mathrm{Graph}^{\Sigma(\xi, T(\xi))}\left(\mathbf{f}_{T(\xi)}\right) \cap B_{d_g}(\xi, \varepsilon\rho),$$

and the estimate of the scaling invariant $C^{1,1-\frac{m}{p}}$*-norm of* $\mathbf{f}_{T(\xi)}$ *depend only on* m, n, p, K, r_H *and* ε.

Proof. The conditions presented in Theorem 13.6.8 also depend on a parameter d, as given by Equations (293) to (295). However, since we can always modify the density of the varifold without changing its support, we will restrict ourselves to varifolds in $\mathcal{AC}(\xi, \rho, \delta, p)$, that is, when $d = 1$.

Note that for a fixed $\xi \in M^n$, there exists $\rho_0(\xi) > 0$ such that for all $x \in B_{d_g}(\xi, \rho_0)$, the following conditions hold:

$$|\sec_g(x)| \leq K, \quad \text{and} \quad r_H(M^n, g)(x) > 0,$$

for some $K \geq 0$.

This holds because g being of class $\mathcal{C}^2$ implies that $\sec_g(x)$ and $r_H(M^n, g)(x)$ are continuous functions. Furthermore, the conditions $r_H(M^n, g)(\xi) > 0$ ensure the desired properties in a neighborhood of ξ. Thus, by combining this with Corollary 13.6.7 we have the desired result. □

14 ALLARD'S INTERIOR ε-REGULARITY THEOREM IN ALEXANDROV SPACES

We recall some basic definitions and properties of Alexandrov spaces. For synthetic notions of curvature and definitions of CAT and CBB spaces, we refer to [3, 57] and the references therein.

Definition 14.0.1. Let (X, d) be a complete locally compact metric space, and let $\overline{\mathcal{K}}(X, d), \underline{\mathcal{K}}(X, d) \in \mathbb{R}$ satisfy $\underline{\mathcal{K}}(X, d) \leq \overline{\mathcal{K}}(X, d)$. We say that:

1. (X, d) is an Alexandrov space with curvature bounded above by $\overline{\mathcal{K}}(X, d)$ if it is CAT($\overline{\mathcal{K}}(X, d)$), and with curvature bounded below by $\underline{\mathcal{K}}(X, d)$ if it is CBB($\underline{\mathcal{K}}(X, d)$);

2. $(X, d, \mathcal{H}^n_d)$ is non-collapsed if there exists a constant $V_0(X, d) > 0$ such that $\inf_{x\in X} \mathcal{H}^n_d(B_d(x, 1)) \geq V_0(X, d)$.

Remark 14.0.2. We recall several properties of Alexandrov spaces (X, d) with curvature bounded both from above and below that will be used repeatedly throughout this section:

1. (X, d) admits a $C^{3,\alpha}$ differentiable structure together with a $C^{1,\alpha}$ Riemannian metric g, for every $\alpha \in (0, 1)$, whose induced distance coincides with d; see [22, Theorem 14.1].

2. The $C^{1,\alpha}$ structure can be approximated by smooth Riemannian metrics; see [79, Theorem 3.1]. More precisely, there exists a sequence of smooth metrics $g_i \in C^\infty$ on X converging to g in the $C^{1,\alpha}$ topology. If d_i denotes the distance induced by g_i, then each (X, d_i) is CAT($\overline{\mathcal{K}}(X, d_i)$) and CBB($\underline{\mathcal{K}}(X, d_i)$), with

$$\underline{\mathcal{K}}(X, d) \leq \liminf_{i\to\infty} \underline{\mathcal{K}}(X, d_i) \leq \limsup_{i\to\infty} \overline{\mathcal{K}}(X, d_i) \leq \overline{\mathcal{K}}(X, d).$$

3. By [29, Theorem 4.7, p. 47], (X, d) admits a positive lower bound on its injectivity radius $\mathrm{inj}(X, d)$, depending only on n, $\underline{\mathcal{K}}(X, d)$, $\overline{\mathcal{K}}(X, d)$, and $V_0(X, d)$. Consequently, by [13, Main Lemma 2.2, p. 433], it also admits a positive lower bound on its harmonic radius $r_H(X, d)$, depending only on n, $\underline{\mathcal{K}}(X, d)$, $\overline{\mathcal{K}}(X, d)$, and $V_0(X, d)$.

We close this section by stating a complete version of Theorem 1.3.1. Unlike the statement given in the introduction, we make explicit here that $\mathrm{supp}(\|V\|)$ is not merely a submanifold on a neighborhood of ξ, but is in fact the normal graph of a $C^{1,1-\frac{m}{p}}$ function — the precise analogue of the classical Euclidean result. We also state the precise dependence of all constants on the geometric data.

Theorem 14.0.3. *Let $m, n \in \mathbb{N}$ and $p \in \mathbb{R}$ such that $1 \leq m < n$ and $m < p < \infty$; in case $m = 1$, we require that $p \geq 2$, (X, g) be an n-dimensional non-collapsed Alexandrov space with curvature bounded both from above and below by $\overline{\mathcal{K}}(X, g), \underline{\mathcal{K}}(X, g)$, respectively, $\xi \in X$, and $\varepsilon \in]0, 1[$. Then there exist $\delta_0 := \delta_0(m, n, p, \underline{\mathcal{K}}(X, g), \overline{\mathcal{K}}(X, g), V_0(X, g), \varepsilon) > 0$ and $\rho_0 := \rho_0(m, n, p, \underline{\mathcal{K}}(X, g), \overline{\mathcal{K}}(X, g), V_0(X, g)) > 0$ such that, if $V \in \mathbf{V}_m(X)$ satisfies*

$$\Theta^m(\|V\|, x) \geq d > 0,$$

for $\|V\|$-a.e. $x \in B_{d_g}(\xi, \rho)$,

$$\frac{\|V\|(B_{d_g}(\xi, \rho))}{\omega_m \rho^m} \leq d(1 + \delta),$$

and

$$\left|\delta_g V(Y)\right| \leq \frac{d^{\frac{1}{p}}\delta}{\rho^{1-\frac{m}{p}}} \|Y\|_{L^{\frac{p}{p-1}}(X, \|V\|)},$$

for all $Y \in \mathfrak{X}^1_c(X)$ such that $\operatorname{supp}(Y) \subset B_{d_g}(\xi, \rho)$, for some $0 < \delta < \delta_0$, for some $0 < d < \infty$, and for some $0 < \rho < \min\{\rho_0, \sqrt{\delta}\}$, then there exist $T(\xi) \in \operatorname{Gr}(m, T_\xi X)$ and a function $\mathbf{f}_{T(\xi)} : B_{d_g}(\xi, \varepsilon\rho) \cap \Sigma(\xi, T(\xi)) \to \mathbb{R}^{n-m}$ such that $\mathbf{f}_{T(\xi)}$ is of class $\mathcal{C}^{1,1-\frac{m}{p}}$, $\mathbf{f}_{T(\xi)}(\xi) = 0$, $d\mathbf{f}_{T(\xi)_\xi} \equiv 0$,

$$\operatorname{supp}(\|V\|) \cap B_{d_g}(\xi, \varepsilon\rho) = \operatorname{Graph}^{\Sigma(\xi, T(\xi))}\left(\mathbf{f}_{T(\xi)}\right) \cap B_{d_g}(\xi, \varepsilon\rho),$$

and the estimate of the scaling invariant $C^{1,1-\frac{m}{p}}$-norm of $\mathbf{f}_{T(\xi)}$ depend only on $m, n, p, \underline{\mathcal{K}}(X, d), \overline{\mathcal{K}}(X, d)$, $V_0(X, d)$ and ε.

Proof of Theorem 14.0.3. Let (X, d) be a metric space with bounded curvature and (X, g_t) a sequence of infinitely differentiable Riemannian manifolds given by Item 2 of Remark 14.0.2, such that (X, g_t) converge to (X, g) with respect to the $\mathcal{C}^{1,\alpha}$ topology.

By Theorem 13.6.8 we have that there exist $\delta_0^t := \delta_0^t(m, n, p, \overline{\mathcal{K}}(X, d_t), \underline{\mathcal{K}}(X, d_t), r_H(X, d_t), \varepsilon) > 0$ and $\rho_0^t := \rho_0^t(m, p, \overline{\mathcal{K}}(X, d_t), \underline{\mathcal{K}}(X, d_t), r_H(X, d_t)) > 0$. By the uniformly dependence of the parameter by the proofs obtained here and Item 2 of Remark 14.0.2, we can define

$$\delta_0 := \delta_0(m, n, p, \overline{\mathcal{K}}(X, d), \underline{\mathcal{K}}(X, d), r_H(X, d), \varepsilon) := \inf_{t \leq t_0} \delta_0^t > 0$$

and

$$\rho_0 := \rho_0(m, n, p, \overline{\mathcal{K}}(X, d), \underline{\mathcal{K}}(X, d), r_H(X, d)) := \inf_{t \leq t_0} \rho_0^t > 0,$$

where t_0 is chosen so that $r_H(X, d_t)$, $\overline{\mathcal{K}}(X, d_t)$, and $\underline{\mathcal{K}}(X, d_t)$ are uniformly bounded over $t \leq t_0$, ensuring the infimum above are finite. Suppose that $V \in \mathbf{V}_m(X)$ satisfies

$$\Theta^m(\|V\|, x) = \lim_{r \downarrow 0} \frac{\|V\|(B_{d_g}(x, r))}{\omega_m r^m} \geq 1, \tag{296}$$

for $\|V\|$-a.e. $x \in B_{d_g}(\xi,\rho)$,

$$\frac{\|V\|(B_{d_g}(\xi,\rho))}{\omega_m \rho^m} \leq 1+\delta, \tag{297}$$

and

$$\left|\delta_g V(Y)\right| \leq \frac{\delta}{\rho^{1-\frac{m}{p}}} \|Y\|_{L^{\frac{p}{p-1}}(X,\|V\|)}, \tag{298}$$

for all $Y \in \mathfrak{X}_c^0(X)$ such that $\operatorname{supp}(Y) \subset B_{d_g}(\xi,\rho)$, for some $0<\delta<\delta_0$ and $0<\rho<\min\{\rho_0,\sqrt{\delta}\}$.

Now note that since V satisfies Equations (296) to (298), by Remark 13.3.3, we have that $V \in \mathbf{RV}_m(B_{d_g}(\xi,\delta\rho))$, that is, there exist a countable m-rectifiable set $\Gamma \subset M^n$ and $\theta \in L^1_{loc}(\Gamma,]0,\infty[)$ such that $V = V(\Gamma,\theta,g)$.

Secondly, note that since $\|g_t - g\|_{C^0} \to 0$, when $t \downarrow 0$, we have, for all t sufficiently small, for all $y \in X$, and for all $r>0$, that

$$B_{d_{g_t}}\left(y, \frac{r}{1+O(t)}\right) \subset B_{d_g}(y,r) \subset B_{d_{g_t}}(y,(1+O(t))r). \tag{299}$$

Therefore, by Equation (299), we obtain, for $\|V\|$-a.e. $x \in B_{d_g}(\xi,\rho)$ and for all t sufficiently small, that

$$\begin{aligned}\Theta^m(\|V\|,x)\,(1-O(t)) &\leq \liminf_{r\downarrow 0} \frac{\|V\|(B_{d_{g_t}}(x,r))}{\omega_m r^m} \leq \\ &\leq \limsup_{r\downarrow 0} \frac{\|V\|(B_{d_{g_t}}(x,r))}{\omega_m r^m} \leq \Theta^m(\|V\|,x)\,(1+O(t))\,.\end{aligned} \tag{300}$$

By Equation (300) we have, for $\|V\|$-a.e. $x \in B_{d_g}(\xi,\rho)$ and for all t sufficiently small, that

$$\left|\limsup_{r\downarrow 0} \frac{\|V\|(B_{d_{g_t}}(x,r))}{\omega_m r^m} - \liminf_{r\downarrow 0} \frac{\|V\|(B_{d_{g_t}}(x,r))}{\omega_m r^m}\right| < O(t),$$

which implies that $\Theta^m_t(\|V\|,x) := \lim_{r\downarrow 0} \frac{\|V\|(B_{d_{g_t}}(x,r))}{\omega_m r^m}$ exists for $\|V\|$-a.e. $x \in B_{d_g}(\xi,\rho)$ and for all t sufficiently small. Furthermore, by Equations (296) and (300), we also have that

$$\Theta^m_t(\|V\|,x) \geq \Theta^m(\|V\|,x)\,(1-O(t)) \geq 1-O(t),$$

for $\|V\|$-a.e. $x \in B_{d_g}(\xi,\rho)$ and for all t sufficiently small. Now define

$$\widetilde{V}_t := V\left(\Gamma, \frac{\theta}{1-O(t)}, g\right)$$

and note that for $\tilde{\rho} < \rho$ we have

$$\Theta^m_t(\|\widetilde{V}_t\|,x) = \frac{\Theta^m_t(\|V\|,x)}{1-O(t)} \geq 1, \tag{301}$$

for $\|\widetilde{V}\|$-a.e. $x \in B_{d_{g_t}}(\xi,\tilde{\rho})$ and for all t sufficiently small. Take $\tilde{\rho} < \frac{\rho}{1+O(t)}$ for all t sufficiently small and $|\rho - \tilde{\rho}|$ arbitrarily small. By Equation (297) we have that there exists $1 > \tilde{\delta} > \delta$ such that $|\tilde{\delta} - \delta|$ is arbitrarily small and

$$\frac{\|\widetilde{V}_t\|(B_{d_{g_t}}(\xi,\tilde{\rho}))}{\omega_m \tilde{\rho}^m} \leq \frac{1}{1+O(t)} \frac{\|V\|(B_{d_{g_t}}(\xi,\tilde{\rho}))}{\omega_m \tilde{\rho}^m} \leq \frac{1}{1-O(t)} \frac{\rho^m}{\tilde{\rho}^m}(1+\delta) \leq 1+\tilde{\delta}. \tag{302}$$

Note that since $\|g_t - g\|_{\mathcal{C}^0} \to 0$ and $\|\partial g_t - \partial g\|_{\mathcal{C}^0} \to 0$, when $t \downarrow 0$, we have, for all t sufficiently small, for all $y \in X$, for all $S \in \mathrm{Gr}(m, T_yX)$ and all $Y \in \mathfrak{X}^0_c(X)$, that

$$\begin{aligned}
\left|\mathrm{div}^{g_t}_S(Y)(y) - \mathrm{div}^{g}_S(Y)(y)\right| &= \left|\sum_{i=1}^m \sum_{j=1}^m g_t^{ij}(y) \left\langle \nabla^{g_t}_{\tau_i} Y(y), \tau_j \right\rangle_{g_{t_y}} - g^{ij} \left\langle \nabla^{g}_{\tau_i} Y(y), \tau_j \right\rangle_{g_y}\right| \\
&= \left|\sum_{i=1}^m \sum_{j=1}^m \sum_{k=1}^n g_t^{ij}(y) g_{t,kj}(y) \left(\frac{\partial X^k}{\partial \tau_i}(y) + \sum_{l=1}^n X^l(y) \Gamma^k_{t,il}(y)\right) - g^{ij} g_{kj}(y) \left(\frac{\partial X^k}{\partial \tau_i}(y) + \sum_{l=1}^n X^l(y) \Gamma^k_{il}(y)\right)\right| \\
&= \left|\sum_{i=1}^m \sum_{l=1}^n X^l(y) \left(\Gamma^i_{t,il}(y) - \Gamma^i_{il}(y)\right)\right| \\
&\leq \max_{i,l} \left|\Gamma^i_{t,il}(y) - \Gamma^i_{il}(y)\right| m\sqrt{n} \|X(y)\|_{g_e} \\
&= O(t) \|X(y)\|_{g_y},
\end{aligned} \tag{303}$$

where $\{\tau_1, \ldots, \tau_m\}$ is a basis for S and $\{\tau_1, \ldots, \tau_n\}$ a basis for T_yX.

For all $Y \in \mathfrak{X}^0_c(X)$ such that $\mathrm{supp}(Y) \subset B_{d_g}(\xi,\rho)$ and for all t sufficiently small, by Hölder inequality, Equations (297), (298) and (303) we have that

$$\begin{aligned}
|\delta_{g_t} \widetilde{V}_t(Y)| &\leq \frac{1}{1-O(t)} |\delta_g V(Y)| + \left| \int_{B_{d_g}(\xi,\rho)} O(t) \|Y(y)\|_{g_y} d\|V_t\|(y) \right| \\
&\leq \frac{\delta}{\rho^{1-\frac{m}{p}}} \|Y\|_{L^{\frac{p}{p-1}}(X,\|V_t\|)} + O(t) \|Y\|_{L^{\frac{p}{p-1}}(X,\|V_t\|)} \\
&\leq \frac{\tilde{\delta}}{\tilde{\rho}^{1-\frac{m}{p}}} \|Y\|_{L^{\frac{p}{p-1}}(X,\|V_t\|)}.
\end{aligned} \tag{304}$$

Therefore, we have proven, in Equations (301), (302) and (304), that $\widetilde{V}_t \in \mathcal{AC}(\xi,\tilde{\rho},\tilde{\delta},p)$, for some $0 < \tilde{\delta} < \delta^t_0$ and $0 < \tilde{\rho} < \rho^t_0$, for t arbitrarily small. Thus, by Theorem 13.6.8 there exist $T(\xi) \in \mathrm{Gr}(m, T_\xi X)$ and a function $\mathbf{f}_{T(\xi)} : B_{d_{g_t}}(\xi, \varepsilon\tilde{\rho}) \cap \Sigma(\xi, T(\xi)) \to \mathbb{R}^{n-m}$ such that $\mathbf{f}_{T(\xi)}$ is of class $\mathcal{C}^{1,1-\frac{m}{p}}$, $\mathbf{f}_{T(\xi)}(\xi) = 0$, $d\mathbf{f}_{T(\xi)_\xi} \equiv 0$,

$$\mathrm{supp}(\|\widetilde{V}_t\|) \cap B_{d_{g_t}}(\xi, \varepsilon\tilde{\rho}) = \mathrm{Graph}^{\Sigma(\xi,T(\xi))}\left(\mathbf{f}_{T(\xi)}\right) \cap B_{d_{g_t}}(\xi, \varepsilon\tilde{\rho}), \tag{305}$$

and the estimate of the scaling invariant $C^{1,1-\frac{m}{p}}$-norm of $\mathbf{f}_{T(\xi)}$ depend only on $m, n, p, \overline{\mathcal{K}}(X,d_t), \underline{\mathcal{K}}(X,d_t)$, $r_H(X,d_t)$ and ε

Since $\mathrm{supp}(\|\widetilde{V}_t\|) = \mathrm{supp}(\|V\|)$ by construction, taking limits as $t \downarrow 0$ in the scaling invariant $\mathcal{C}^{1,1-\frac{m}{p}}$-norm of $\mathbf{f}_{T(\xi)}$ — which is possible by the uniform dependence of the constants on the approximation parameter — yields the desired conclusion, up to the dependence on $r_H(X,d)$ in place

of $V_0(X,d)$. The latter is addressed by Item 3 of Remark 14.0.2, which allows one to replace the lower bound on $r_H(X,d)$ by the non-collapsing constant $V_0(X,d)$, yielding precisely the dependence stated in Theorem 14.0.3. □

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